Exploring Cell Types with the Allen Institute

Leloir Institute, Buenos Aires

August 13th, 2026

About us/

Lauren Alfiler

Education & Engagement Specialist

Allen Institute

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Allen Institute

Rachel Hostetler

Scientist I

Allen Institute

Daniela Di Bella

Principal Investigator

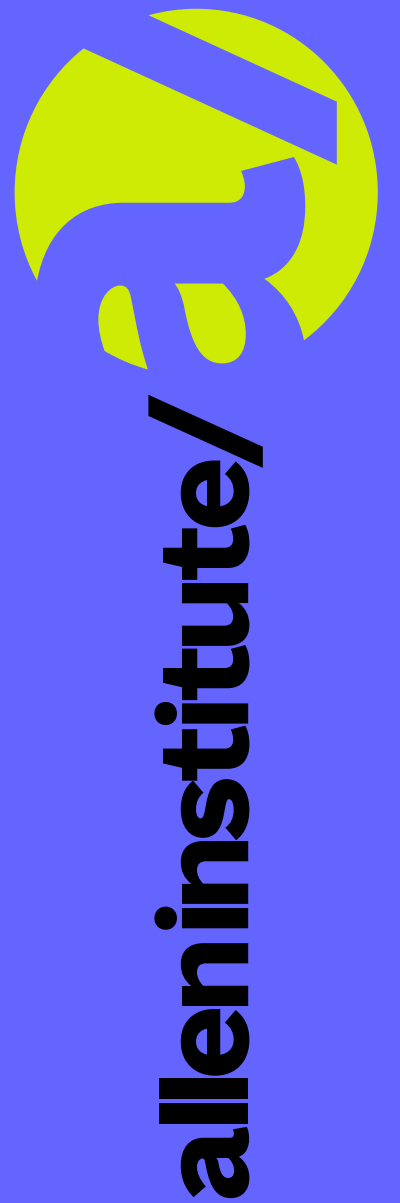
Leloir Institute

August 13th morning agenda/

- **10:00-10:30 |** Allen Institute introduction (10 min Q&A) – Lauren
- **10:30-11:30 |** Intro to scRNAseq (15 min Q&A) – Daniela
- **11:30-11:45 |** Break
- **11:45-1:00 |** Overview of Allen mouse atlases (15 min Q&A) – Rachel

About Allen Institute/

Lauren Alfiler



Allen Institute/

- **Founding date**: September 16, 2003
- **Founders**: Paul G. Allen; Jody Allen
- **Employees**: ~900
  - ~300 with PhDs.
- **Non-Profit (academic flavored industry)**
  - No university affiliation, no classes
  - No degrees, Grad Students, Post-Docs
  - No traditional lab structure with a single PI
- **Foundational Science**

understanding life
advancing health

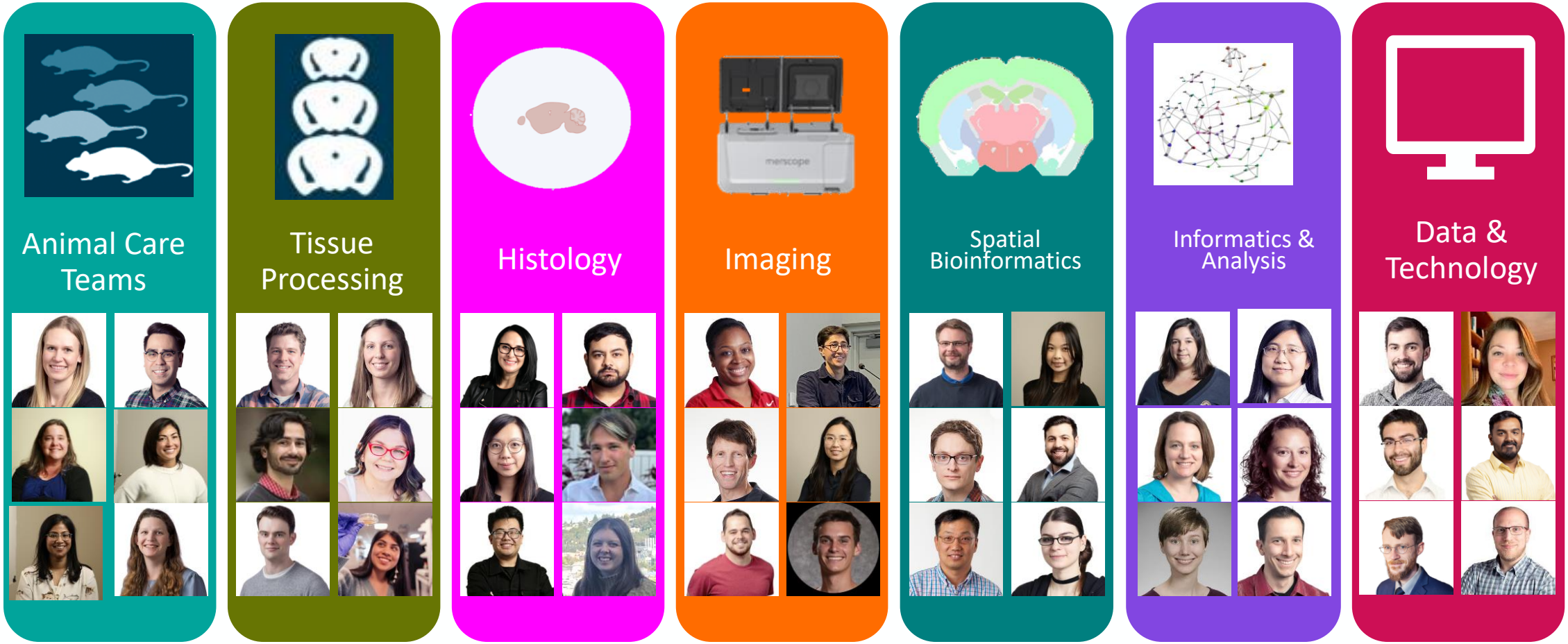
Allen Institute funding & structure

- Majority funded by private funding, particularly ”Fund for Science & Technology” – part of Paul G. Allen’s estates
- We also have public funding, which is primarily NIH grants
  - Note: no K series (training), a lot of U series (cooperative agreements)
- **We are privileged in our generous funding support therefore we want to ensure we are serving the scientific community with having open scientific resources, data & tools**

- Common academic structure: 1 PI with lab of 3-10 researchers, all focus on project related to PI’s interest
- Allen Institute structure: multiple teams who all work on the same big project. There is no single PI.
  - Projects are determined by multiple scientific leaders, along with input from our scientific advisory council
  - “Pipeline” is a common word you will hear to describe how we do our work
  - Cross-team communication & organization are essential in order to work within our structure. Simple lab notebooks are not enough documentation

team science: example pipeline: spatial transcriptomics

MANY more folks within a team, and teams are not pictured!

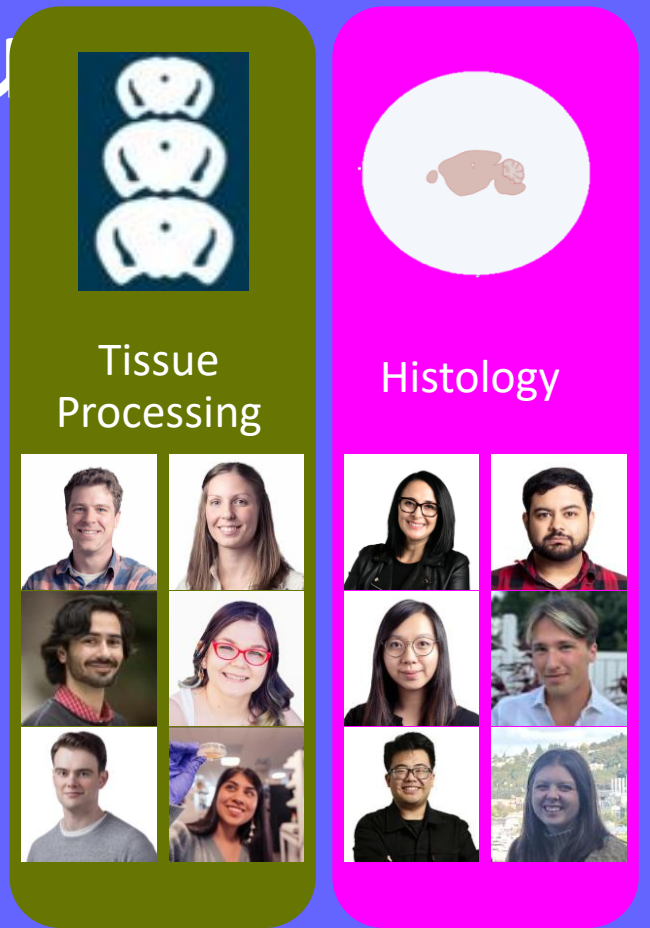


Additional teams:

- Project Management
- Scientific Support
- Research Compliance
- Finance
- Shipping & Receiving

work behind the data: tissu processing histology

I am presenting on behalf of these amazing teams – who support multiple pipelines for the Allen Institute!

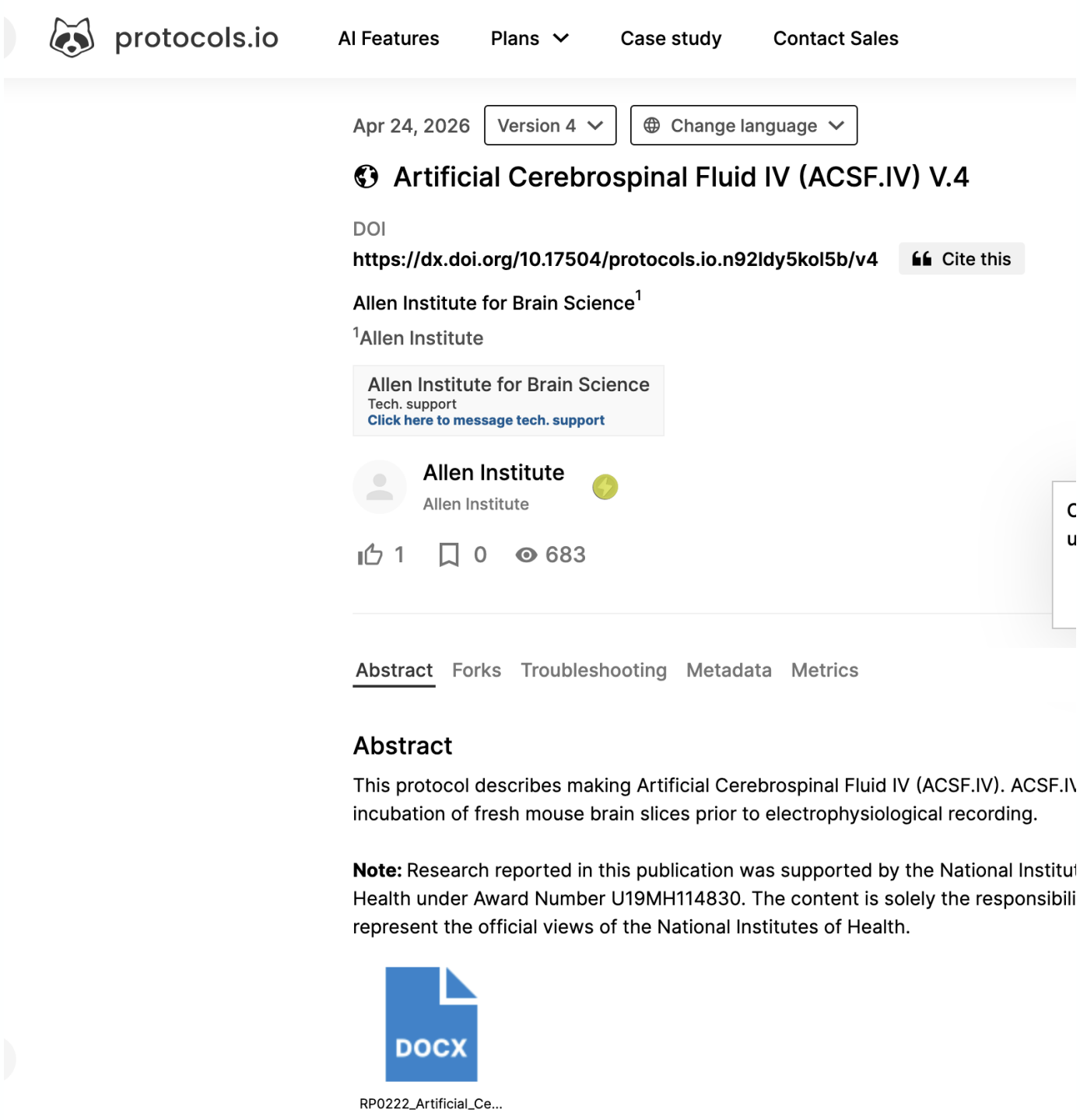


tissue collection

- Collected with ACSF (Artificial Cerebrospinal Fluid)
  - ACSF mimics important properties of CSF, which serves to protect, nourish, and facilitate cellular brain function
    - Energy in the form of sugar
    - Positive (+) and Negative (-) charged ions in the form of salts
    - Carbogen bubbling (95% O2, 5%CO2): oxygenates the tissue
  - Why do we use artificial CSF for research purposes?
    - Allows us to reliably obtain healthy brain slices for patch clamp electrophysiology
    - Maintains tissue health so we can take accurate recordings over long periods of time

Hepes long term holding chamber

Human brain slice in mesh well



Full Protocol on how to make ACSF on our protocols.io page

ACSF Preparation

+all other reagents we request!

General Information

If you have questions or having problems with filling out Infopath forms, PowerApps, or accessing our SharePoint, please directly contact Alice Pom. For any questions regarding requests, please e-mail Alice and !Histology.

Please see our **troubleshoot folder** (link also available on the sidebar) in case your Infopath or SharePoint issue was encountered before. If you do not see an option to submit a new request and you are in the appropriate library, your permissions may need to be updated to gain Requestor level. Please e-mail or Teams message Alice Pom.

Reagent Requests

For reoccurring reagent requests, if we notice that your order hasn't been picked up while we are prepping the next round of your order, we will contact you before **canceling your order request.** When your order request is canceled, please submit a new form. Due to the current nature of back-orders for supplies, we cannot spend resources on fulfilled orders that do not get picked up. Thank you!

Please allow at least one week for delivery. Delivery times may vary based on available on-hand resources.

Equipment Usage Information

Troubleshooting & How To Make Requests

Reagent Request
Active as of 2/4/22

IVSCCh Section Staining

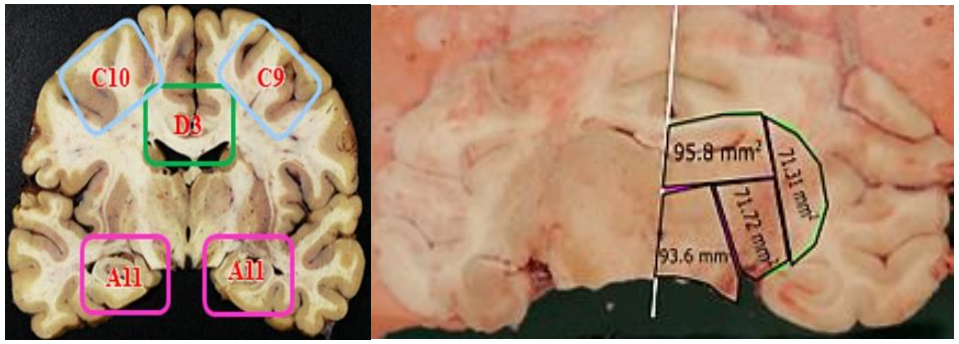
Cryostat Sectioning Request

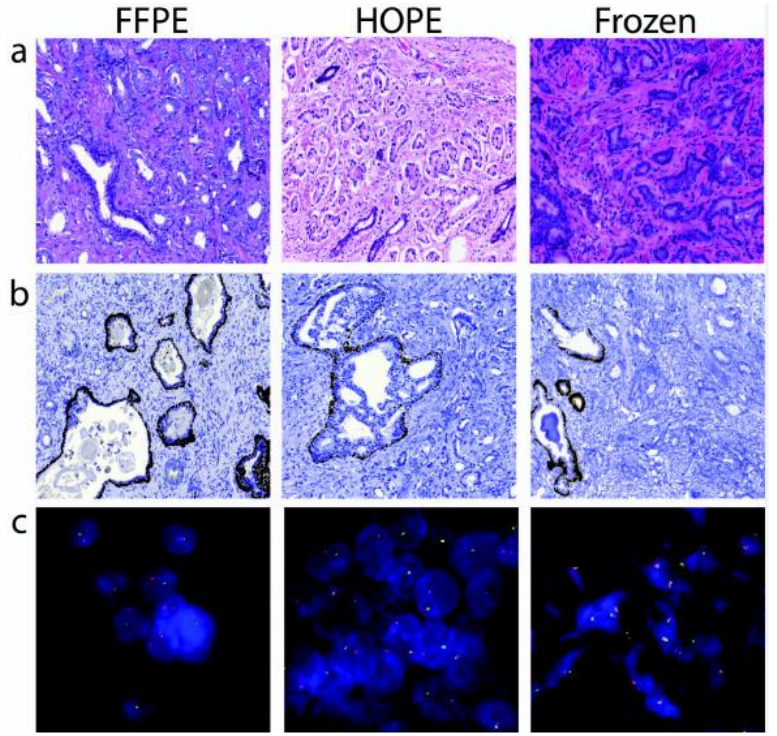
fresh frozen vs fixed tissue

- Fresh Frozen Tissue Samples
  - Native biological state is maintained
  - DNA and RNA are well-preserved
  - Ice crystals can damage tissue
  - Energy-heavy storage at -80°C

- Fixation
  - Protects tissue from degradation via enzymes or bacteria
  - Optimal for protein-based work
  - Cellular structure and morphology is preserved
  - Stabilizes tissue
  - Can reduce staining capabilities

- In general, we use:
  - Fresh-Frozen: single-cell RNAseq, spatial transcriptomics & patch-seq
  - Fixed-Frozen: testing for protein-work and other projects*
    - **we used to use it for DNA applications but no longer do*





chemical fixatives

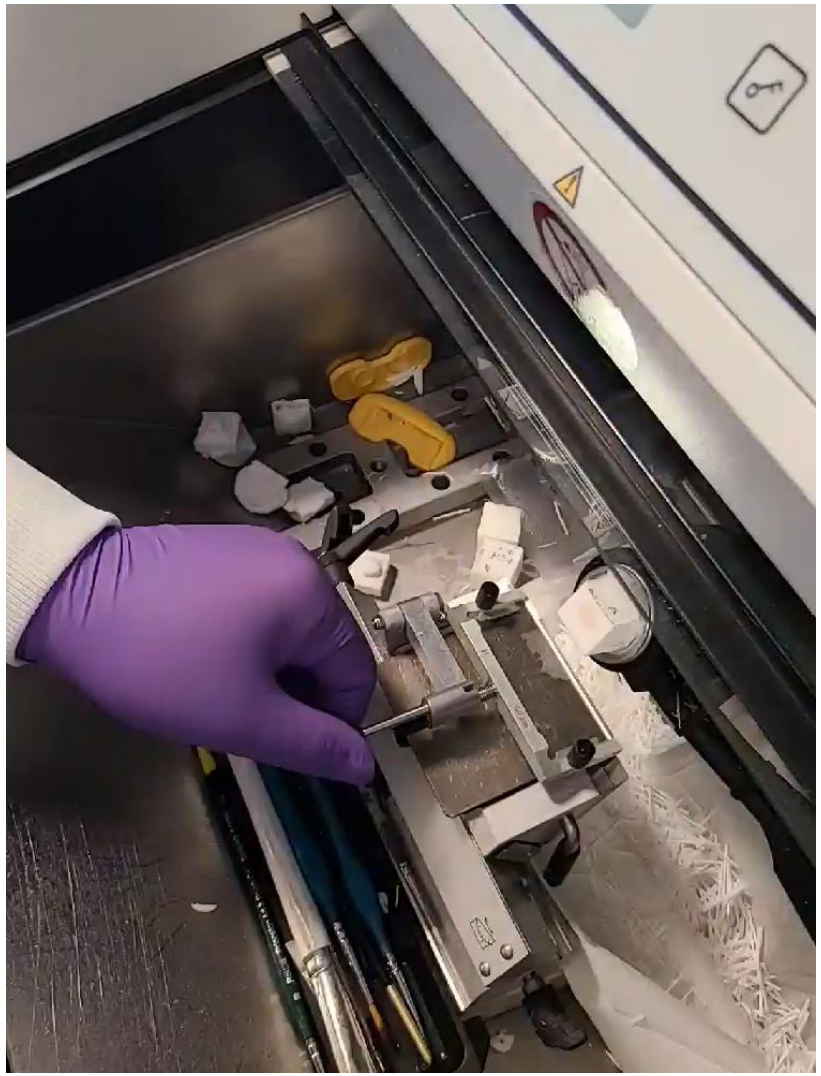
- Things to consider
  - Penetration speed
  - Desired processing techniques
  - Storage capabilities

- Cross-linking fixative: creates mesh-like structure within tissues by creating covalent bonds to cross-link proteins
  - Paraformaldehyde (PFA)
  - Glutaraldehyde

- Denaturation fixative: replaces water within proteins, causing hydrogen bonds to break and protein denaturation
  - Ethanol
  - Methanol
  - Acetone

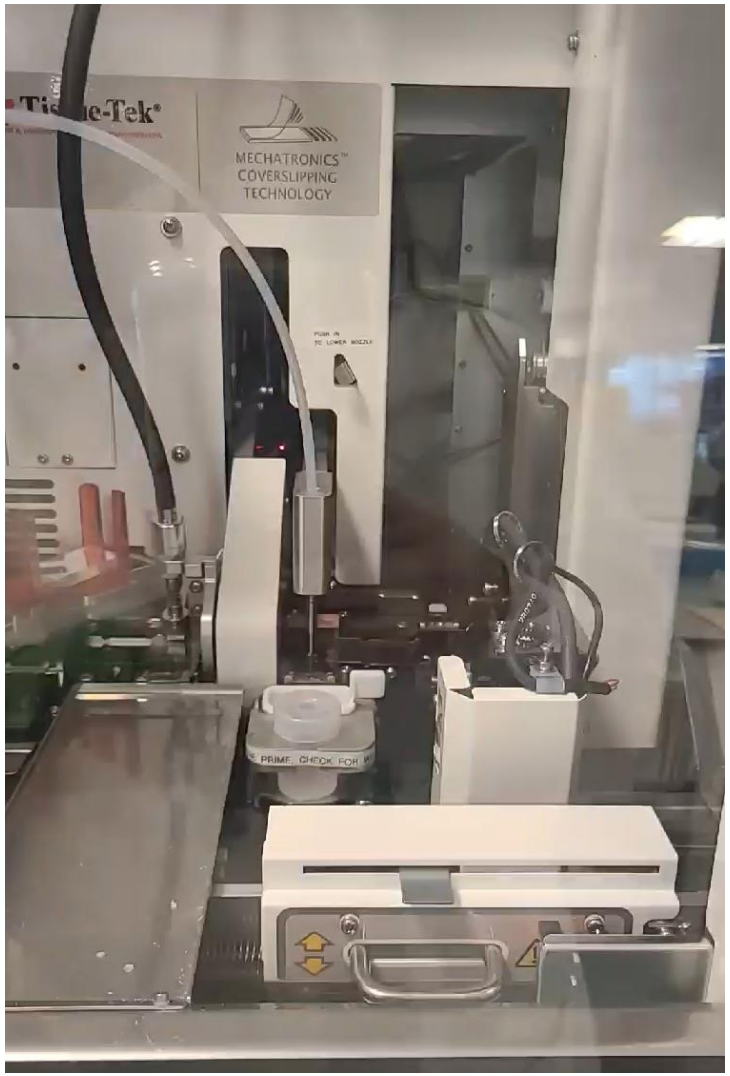
Automation at Allen Institute

sectioning - cryostat

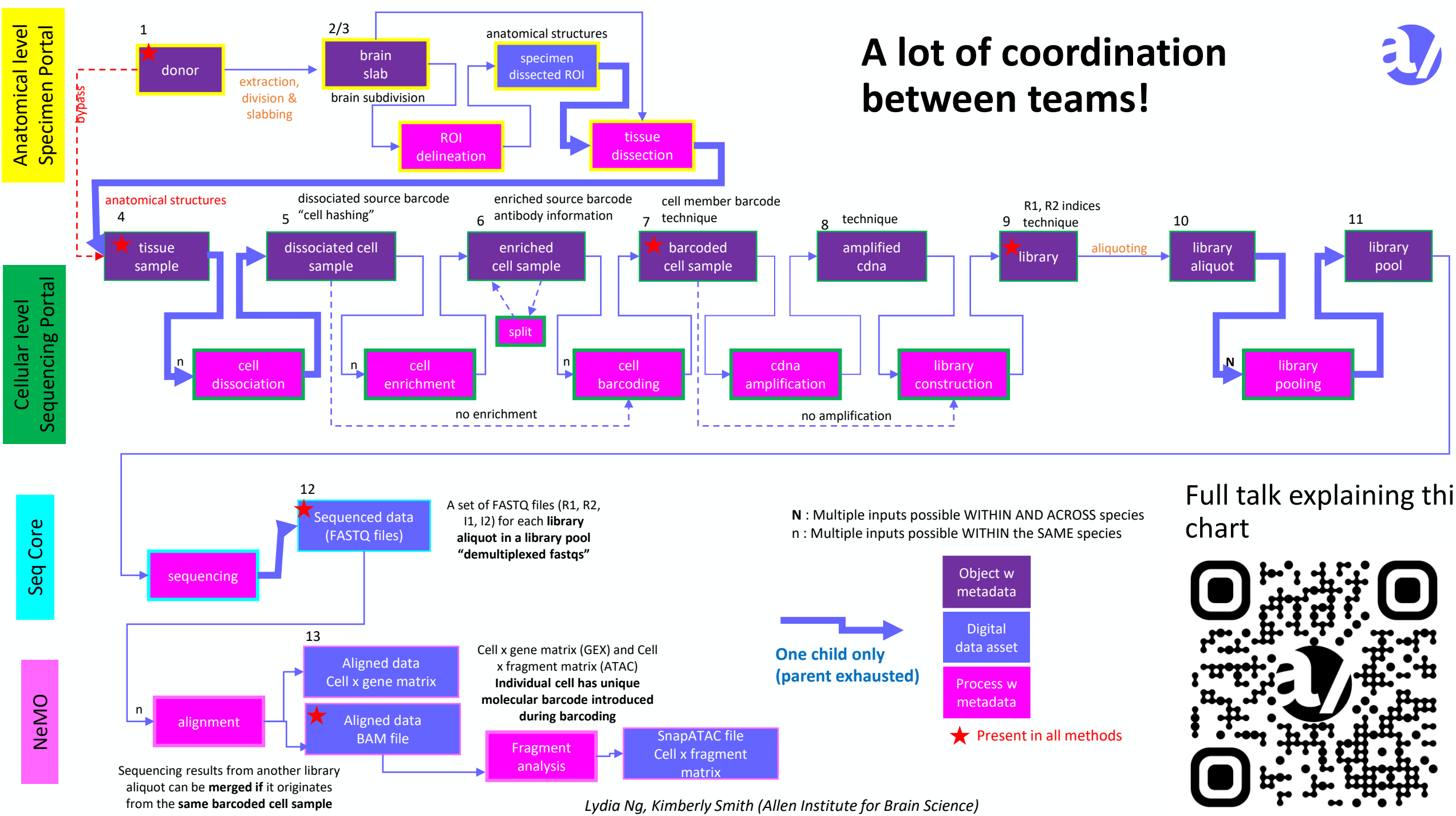


staining - IHC

cover slipping



A lot of coordination between teams!



Full talk explaining this chart

Lydia Ng, Kimberly Smith (Allen Institute for Brain Science)

Careers

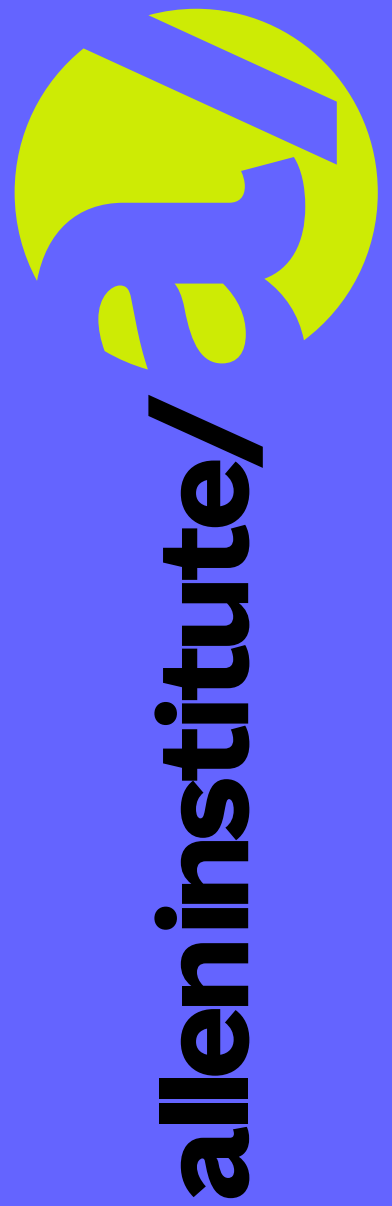
- We hire straight out of graduate school – no post-doc required!
  - This is typically a Scientist I
  - The average “Scientist I” salary is about $80k-$100k USD
  - Each position will be clear if it supports Visa or not - it is position dependent
- Hiring is year-round, although sometimes more open spots during the start of the new year
- We often look for specific skill sets to fill open positions; the skill set depends on the position.

- *NOTE: We don’t have any undergraduate internships available for international students, and as of now, we do not have any PhD-level internships*

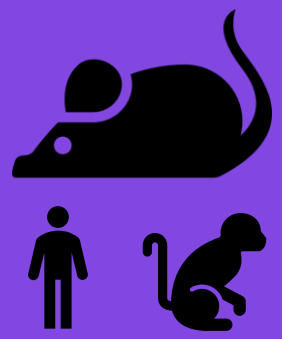
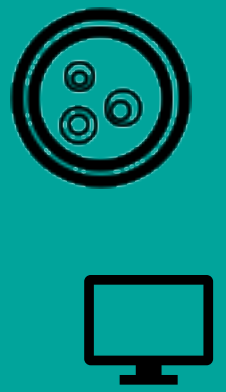
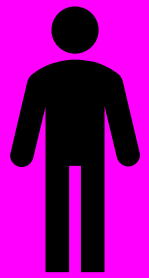
All careers will be posted on:
alleninstitute.org/careers

More than neuroscience!

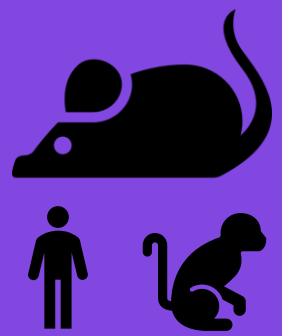
We will be mainly focusing on neuroscience but want to highlight other scientific resources



All of Allen Institute's focuses

	brain science	cell science	immunology	neural dynamics	synthetic biology	brain health
general goal	building blueprint of the healthy brain	understand how cells becomes organs	create the immune system reference	explore the brain's circuits and activity	develop a way for cell to record history	accelerate human brain research in disease
model organism						

What we will cover during this workshop

	brain science	cell science	immunology	neural dynamics	synthetic biology	brain health
general goal	building blueprint of the healthy brain	understand how cells becomes organs	create the immune system reference	explore the brain's circuits and activity	develop a way for cell to record history	accelerate human brain research in disease
model organism						

I am going to briefly go over these

	brain science	cell science	immunology	neural dynamics	synthetic biology	brain health
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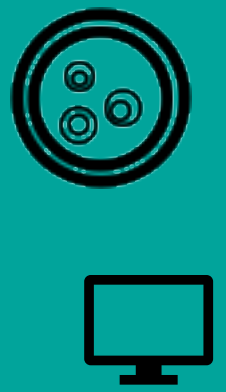
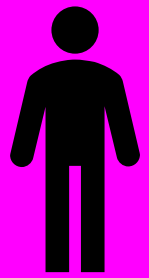
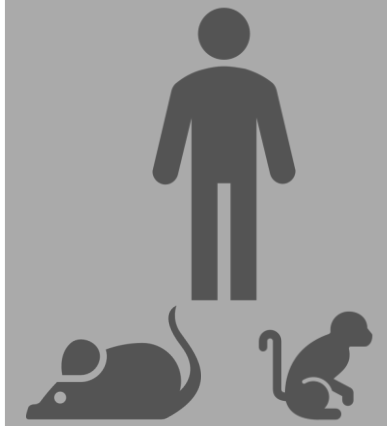
model organism						

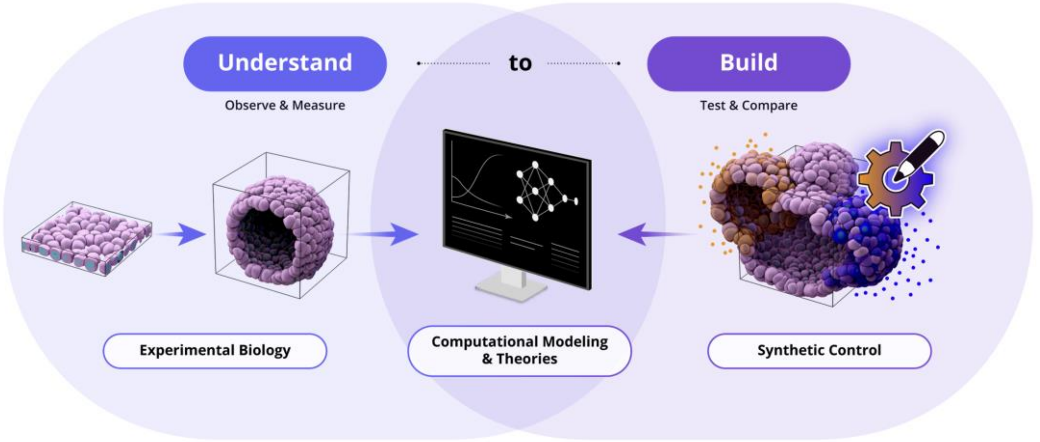
cell science

Focus on uncovering the design principles of multicellular programs essential for life

- Human pluripotent stem cells with various structures that are GFP-tagged
- New focus: on cell dynamics - understanding how cells go from flat 2D to 3D shape

Popular resources

- **Allen Cell Catalog:** purchase hiPSC lines – we don't profit it from it so as low cost as we could
- **Allen Cell & Structure Segmenter:** python kit for cell segmenting
- **Timelapse Feature Explorer**: track colonies of nuclei and their features (height, volume, etc)
- **BioFile Finder:** helps organize your metadata – can be used for any type of data!



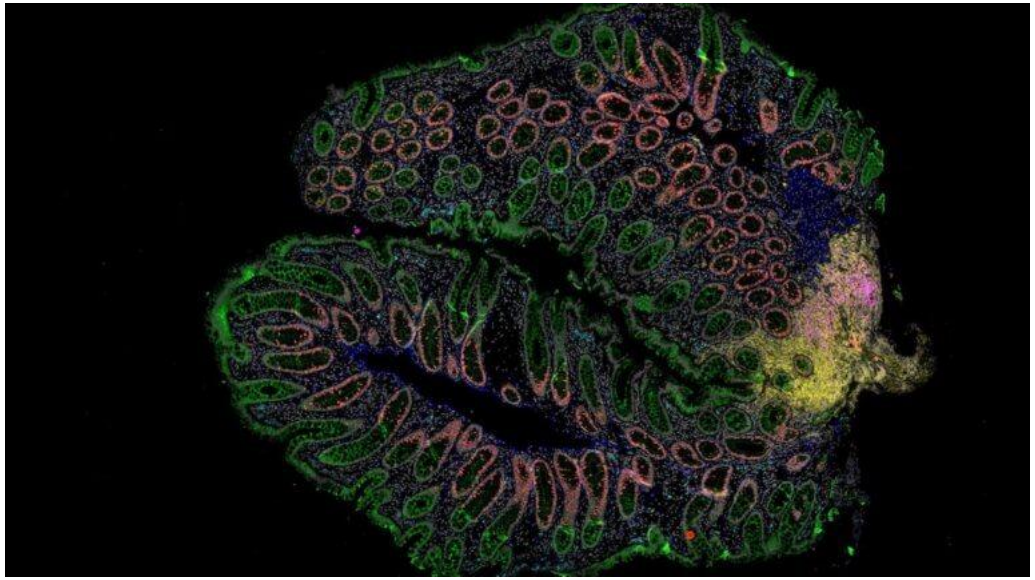
allencell.org for all the resources ->

look in the labs/
cell science

immunology

Focus on creating a reference of what the human immune system is

- Popular resources
  - **Human Immune Health Atlas:** single-cell RNAseq data of 108 healthy people, ranging from pediatric to young adult to older adults (Gong, et al 2025)
  - **TEA-seq:** multiomics approach of sc-RNAseq + epitopes + chromatin accessibility (Swanson, et al. 2021)



explore.allenimunology.org for all the resources ->

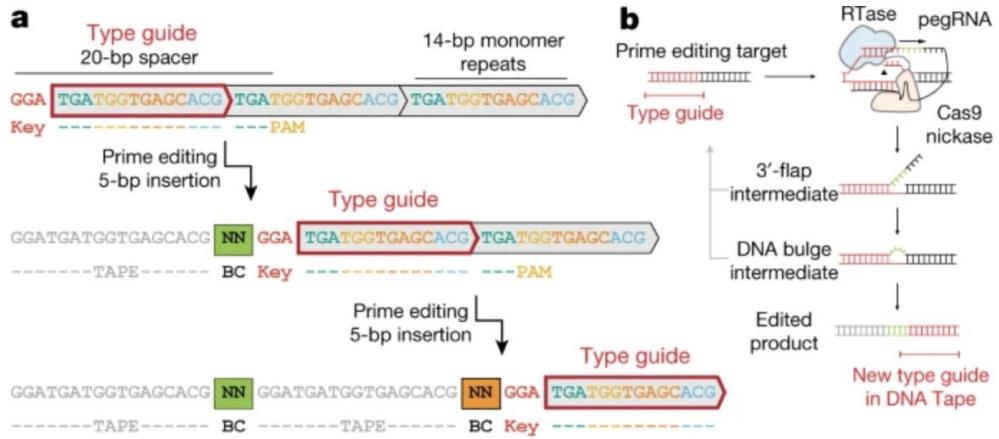
look in the labs/
immunology

synthetic biology

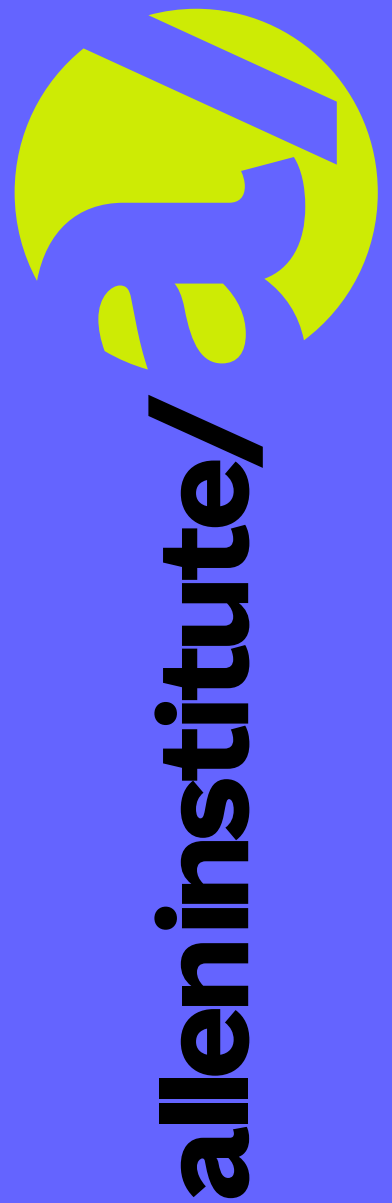
Focus on engineering cells to record their histories as they divide, differentiate, and mature

- Launched in 2023 with partnership with the University of Washington and the Chan Zuckerberg Institute– lead by Dr. Jay Shendure
- General idea: building a **“DNA typewriter”** where there is **“DNA Tape”** that is inserted into a cell’s genome where there are CRISPR-Cas9 target sites, therefore the cell will insert various guide RNAs that represent some event it experienced
    - Paper explaining the technology: Choi, J., Chen, W., Minkina, A. *et al.* A time-resolved, multi-symbol molecular recorder via sequential genome editing. (2022). https://doi.org/10.1038/s41586-022-04922-8
- *No official data & resources released yet*

Fig. 1: Sequential genome editing with DNA Typewriter.



Neuorscience datasets we won’t have time to talk about

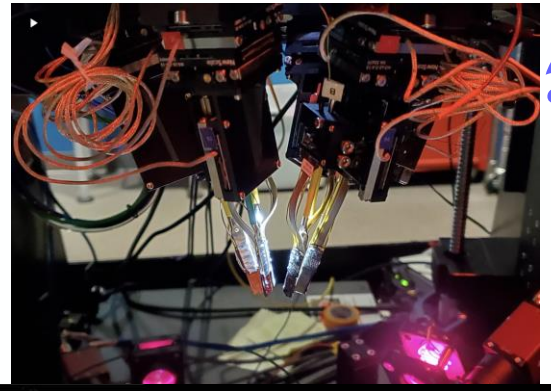


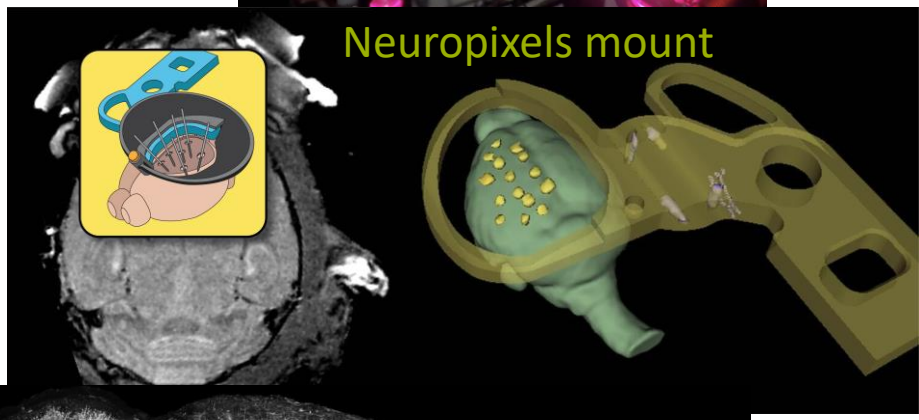
neural dynamics

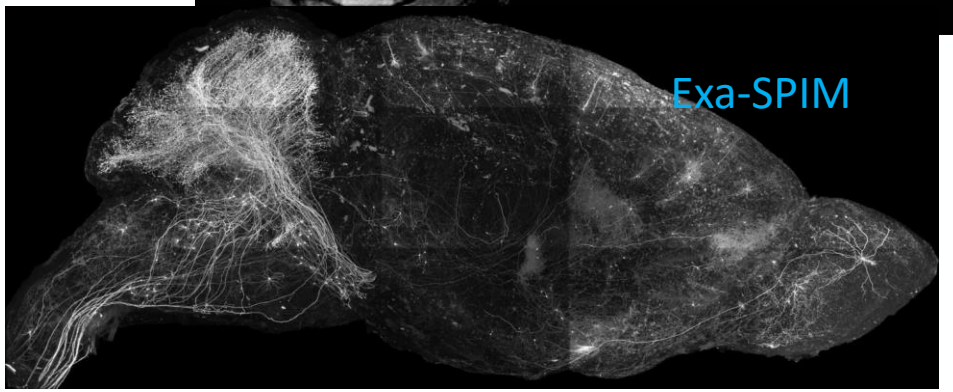
Focus on understanding how large groups of cells connect into neural circuits, process information, and generate cognition and behavior.

- Popular resources
  - **OpenScope:** scientists submit project proposal using Allen Institute Neuropixel/2P/Ca+ Imaging pipeline
  - **Neuropixels:** 24 probes for cell-type-specific electrophysiology
  - **ExaSPIM:** Brain-wide single-neuron reconstructions
  - **Their schematics!** They are very open with the equipment and blueprints for everything they use & custom builds

Neuropixels



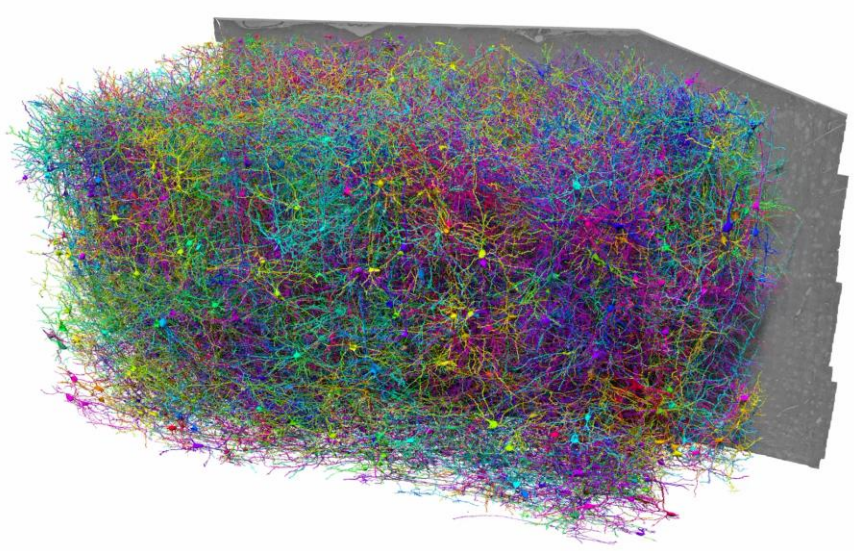




VR foraging platform

look in the labs/
neural dynamics

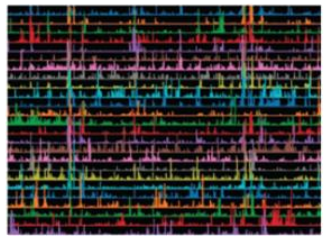
Machine Intelligence from Cortical Networks (MICrONS)



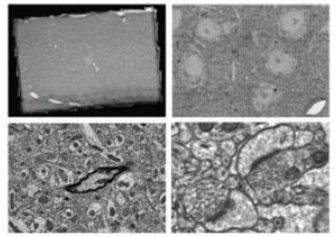
EM of mm^3 mouse of mouse cortex – fully annotated & functional data

all data: microns-explorer.org

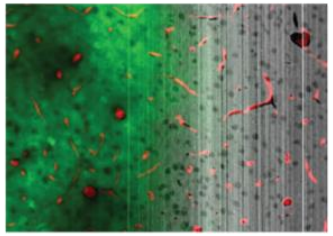
Data types available as data resource



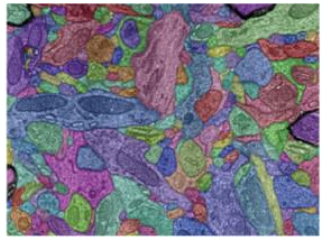
FUNCTIONAL DATA



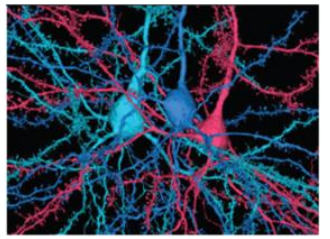
ELECTRON MICROSCOPY IMAGERY



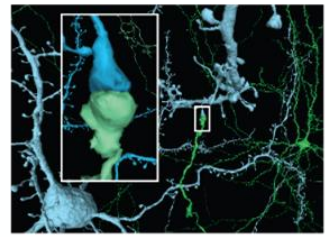
FUNCTIONAL -STRUCTURAL CO-REGISTRATION



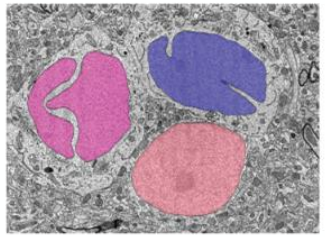
CELL SEGMENTATION



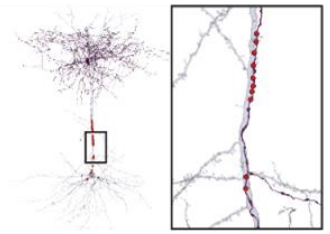
CELL MESHES



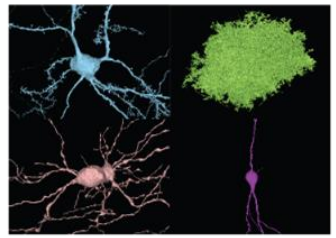
SYNAPSE CONNECTIVITY



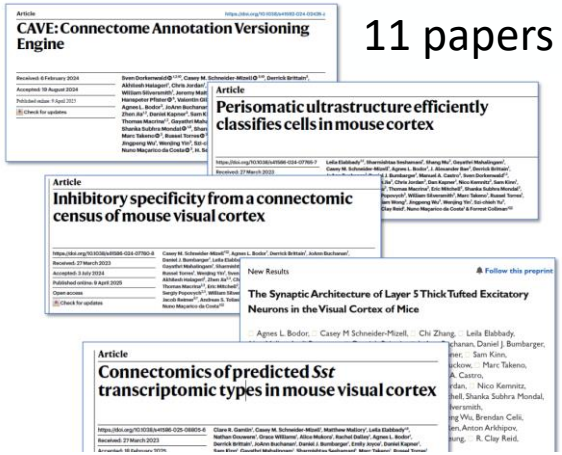
NUCLEUS SEGMENTATION



PROOFREADING STATUS

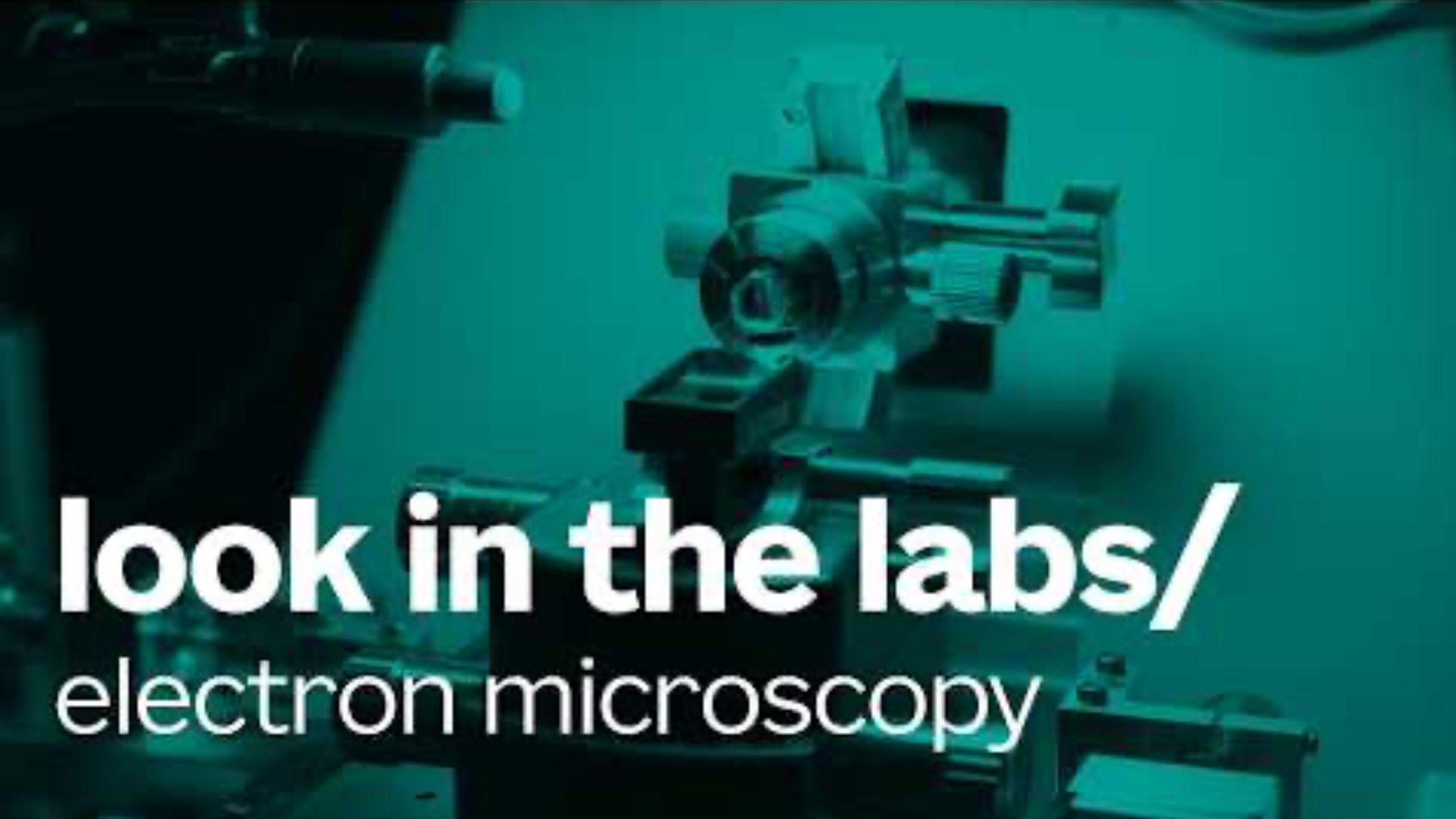


CELL TYPES



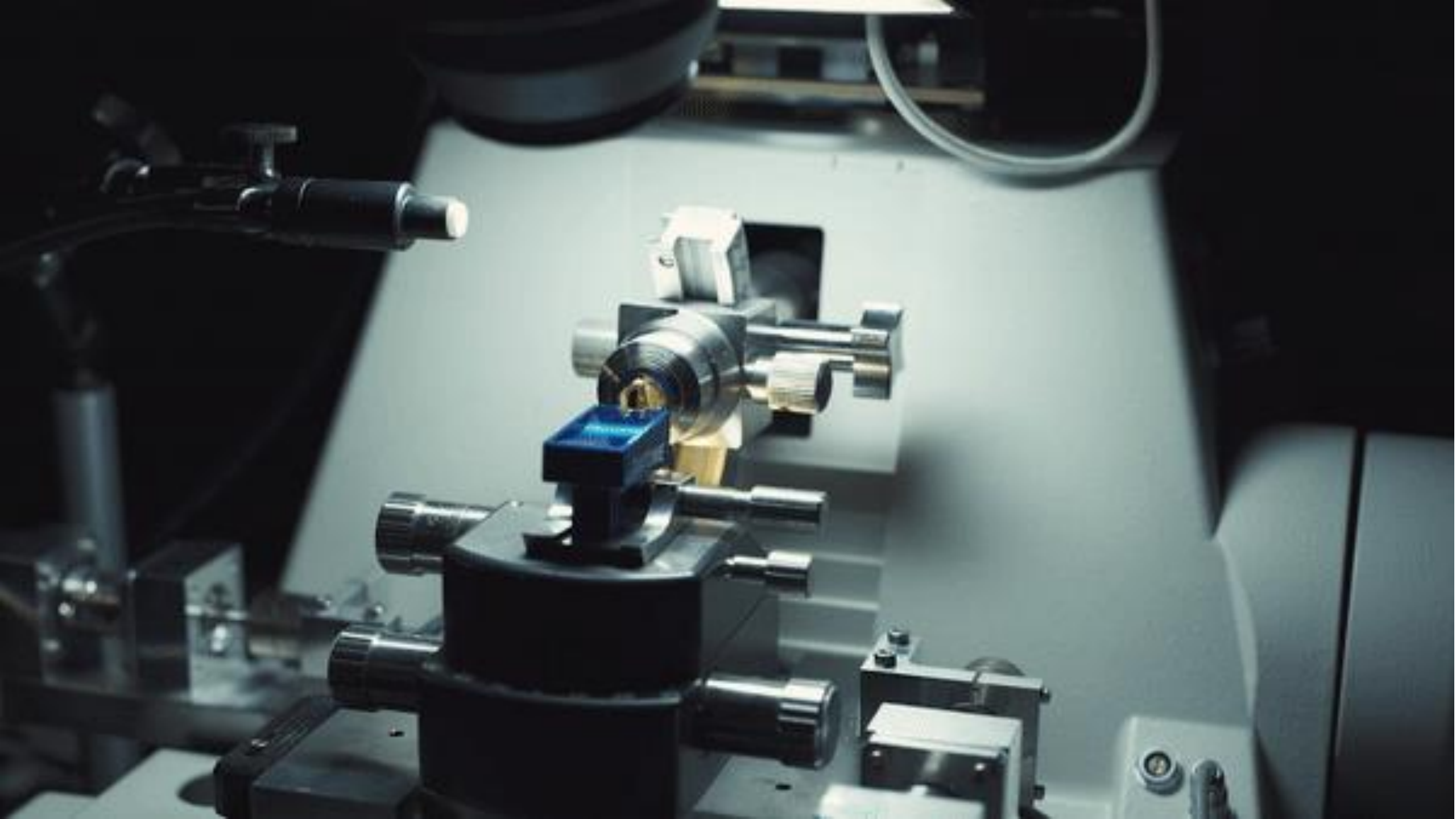
11 papers in total

Flagship Paper: “Functional connectomics spanning multiple areas of visual cortex”
The MiCRONS Consortium (2025)



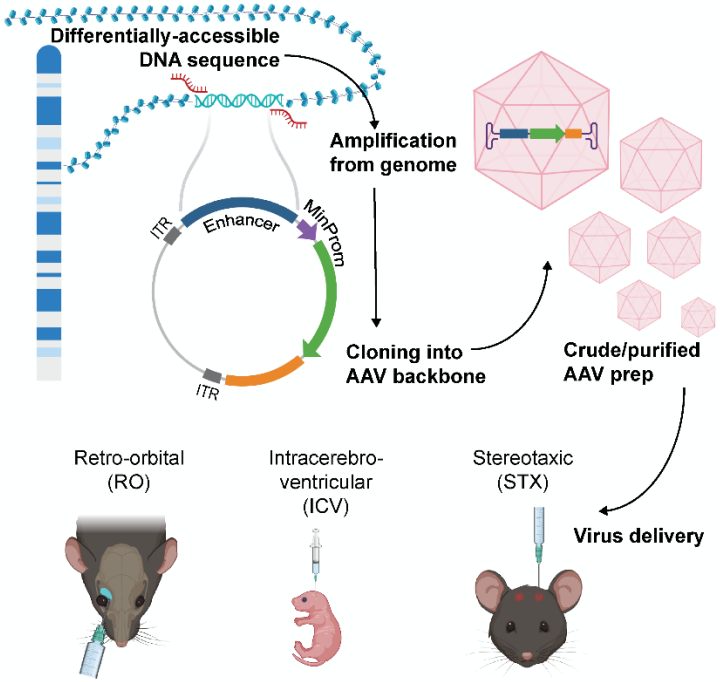
look in the labs/

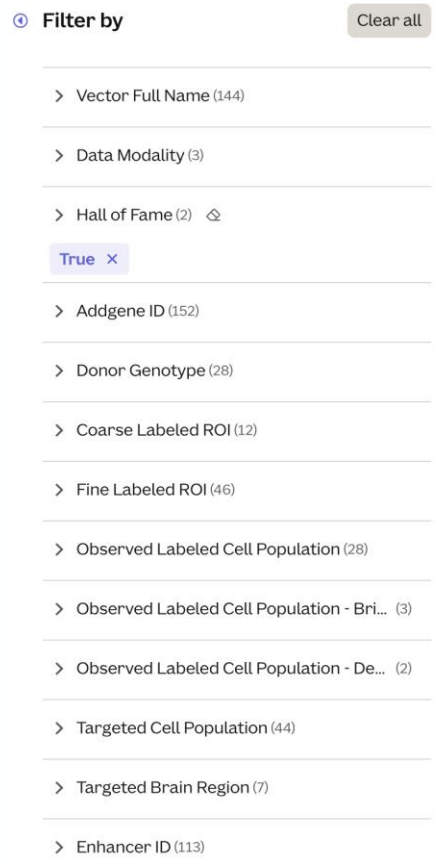
electron microscopy

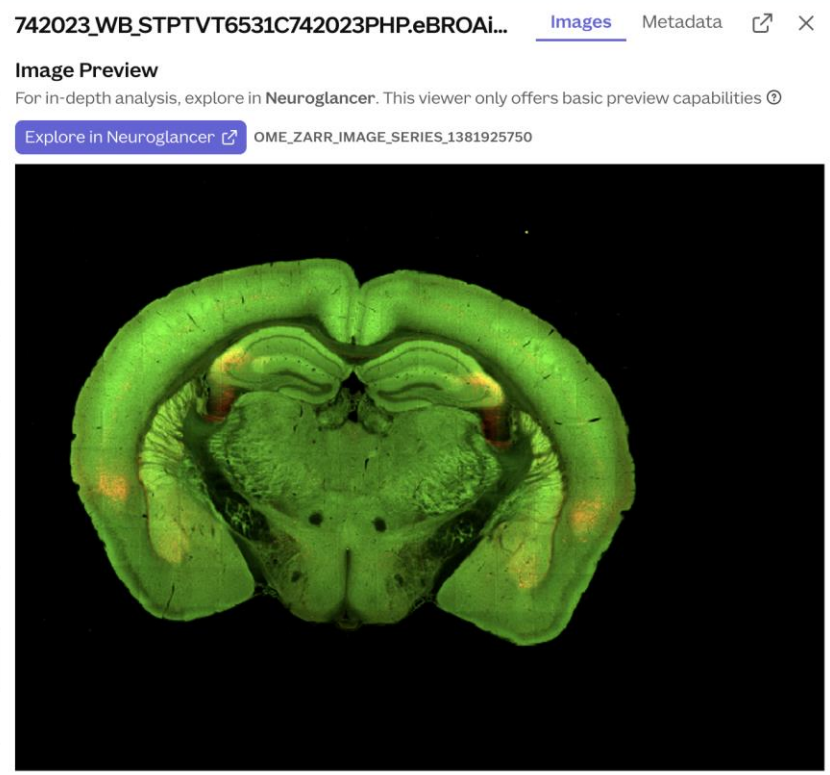


Genetic Tools Atlas

We have created in-house enhancer AAVs to target specific cell type populations – you can order directly on Addgene – continuously adding more

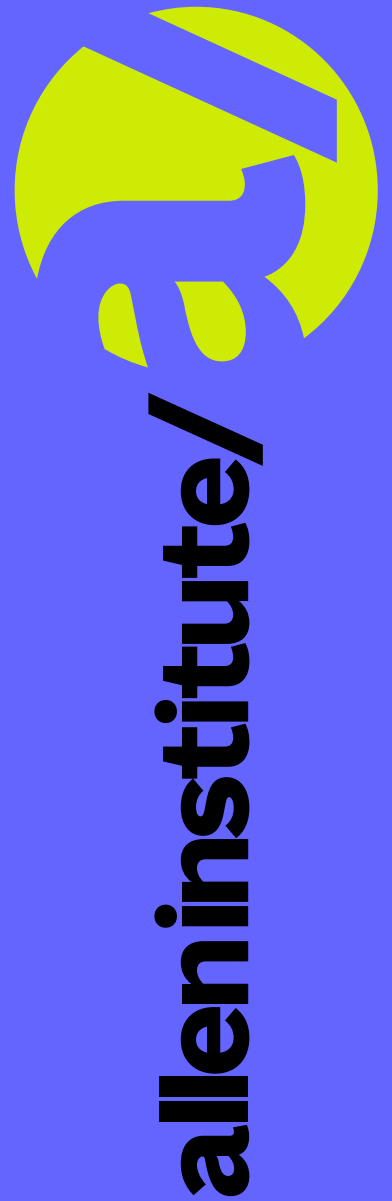






There is a lot data

Here is overview of the data with viewers



Is there a viewer for data of…

Mouse, Healthy, Adult

	Isocortex	Olfactory areas	Hippo-campal formation	Cortical subplate	Striatum	Pallidum	Thalamus	Hypo-thalamus	Midbrain	Pons	Medulla	Cere-bellum	Fiber tracts
Spatial Transcriptomics	X	X	X	X	X	X	X	X	X	X	X	X	X
scRNAseq	X	X	X	X	X	X	X	X	X	X	X	X	X
PatchSeq	VIS & MO only												
EM	VIS Only		X		X	X							

Mouse, Healthy, Aged

	Isocortex	Olfactory areas	Hippo-campal formation	Cortical subplate	Striatum	Pallidum	Thalamus	Hypo-thalamus	Midbrain	Pons	Medulla	Cere-bellum	Fiber tracts
scRNAseq	X		X	X	X	X	X	X	X	X			

Mouse, Healthy, Developing

	Isocortex	Olfactory areas	Hippo-campal formation	Cortical subplate	Striatum	Pallidum	Thalamus	Hypo-thalamus	Midbrain	Pons	Medulla	Cere-bellum	Fiber tracts
scRNAseq	VIS Only												

Is there a viewer for data of…

Human, Healthy

	Cerebral cortex	Cerebral nuclei	Thalamus	Hypothalamus	Midbrain	Pons	Medulla oblongata	Cerebellum	White matter
Spatial Transcriptomics		X							
scRNAseq	X	X	X	X	X	X	X	X	X
PatchSeq	M1 Only								

Human, Alzheimer's Disease

	Cerebral cortex	Cerebral nuclei	Thalamus	Hypothalamus	Midbrain	Pons	Medulla oblongata	Cerebellum	White matter
Spatial Transcriptomics	MTG & PFLC Only								
scRNAseq*	MTG Only								

**more available raw data only*

Human, Parkinson's Disease (data from Michael J Fox Foundation)

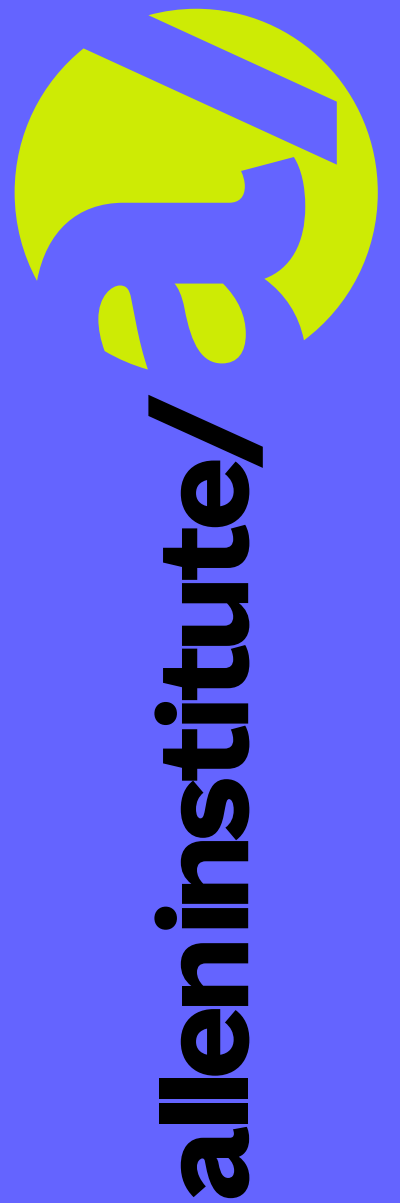
	Cerebral cortex	Cerebral nuclei	Thalamus	Hypothalamus	Midbrain	Pons	Medulla oblongata	Cerebellum	White matter
scRNAseq	X	X			X				

Is there a viewer for data of…

Macaque, Healthy

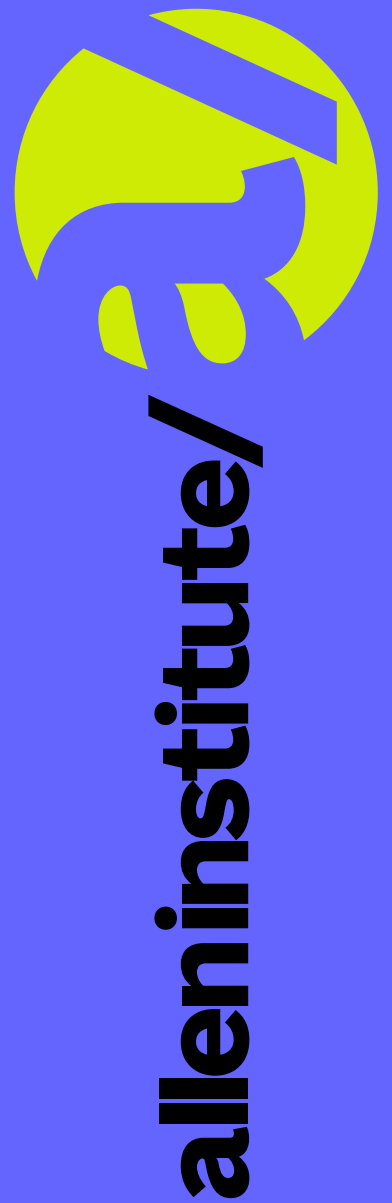
	Cerebral cortex	Cerebral nuclei	Thalamus	Hypothalamus	Midbrain	Pons	Medulla oblongata	Cerebellum	White matter
Spatial Transcriptomics	M1 Only	X	X		X				
scRNAseq		X	X		X				

Questions?



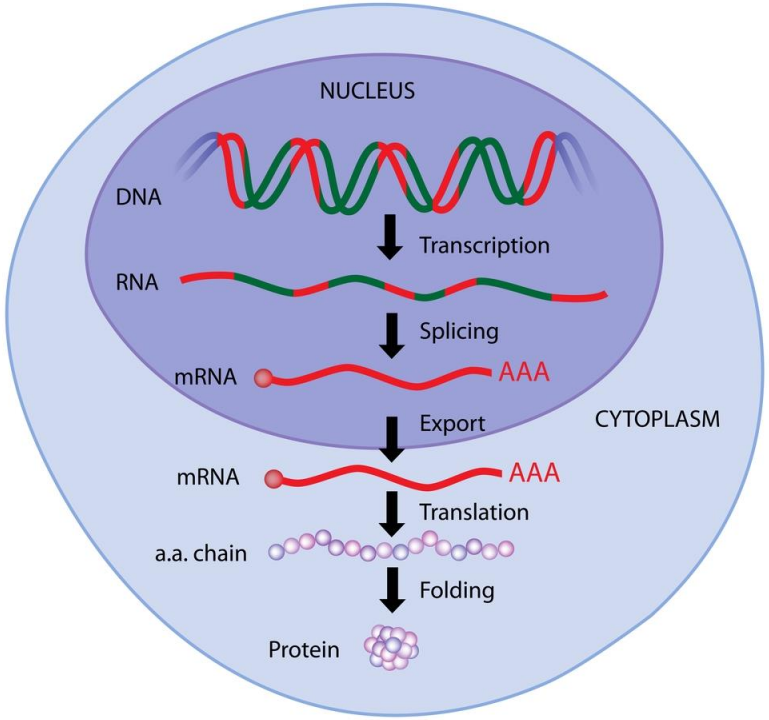
Intro to scRNAseq/

Daniela Di Bella



Why measure gene expression?

- Gene expression reveals changes in cell states responding to different conditions:
  - Disease, development, activity, infection, and others



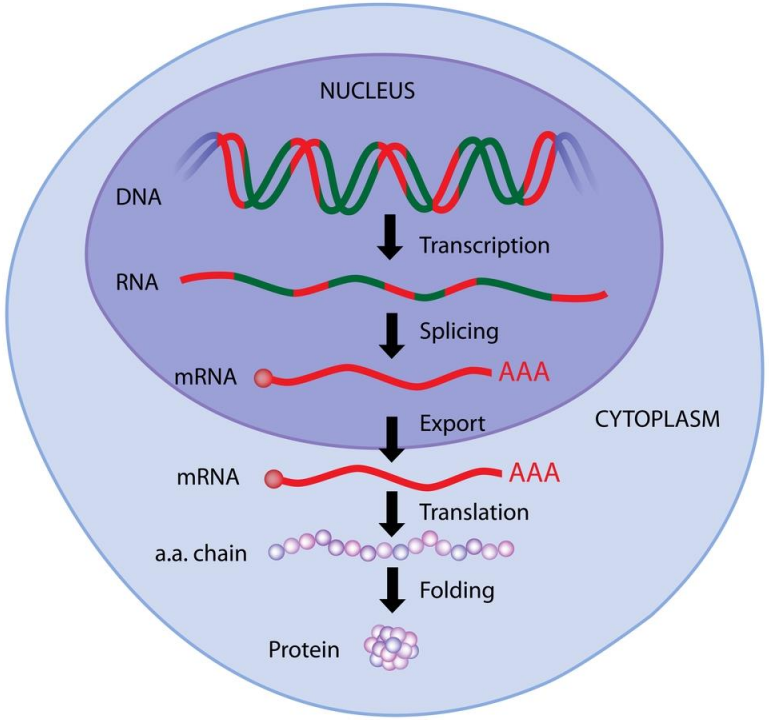
Central dogma of molecular biology

Tools to assess gene expression

- qPCR
- In situ hybridization
- Northern blot

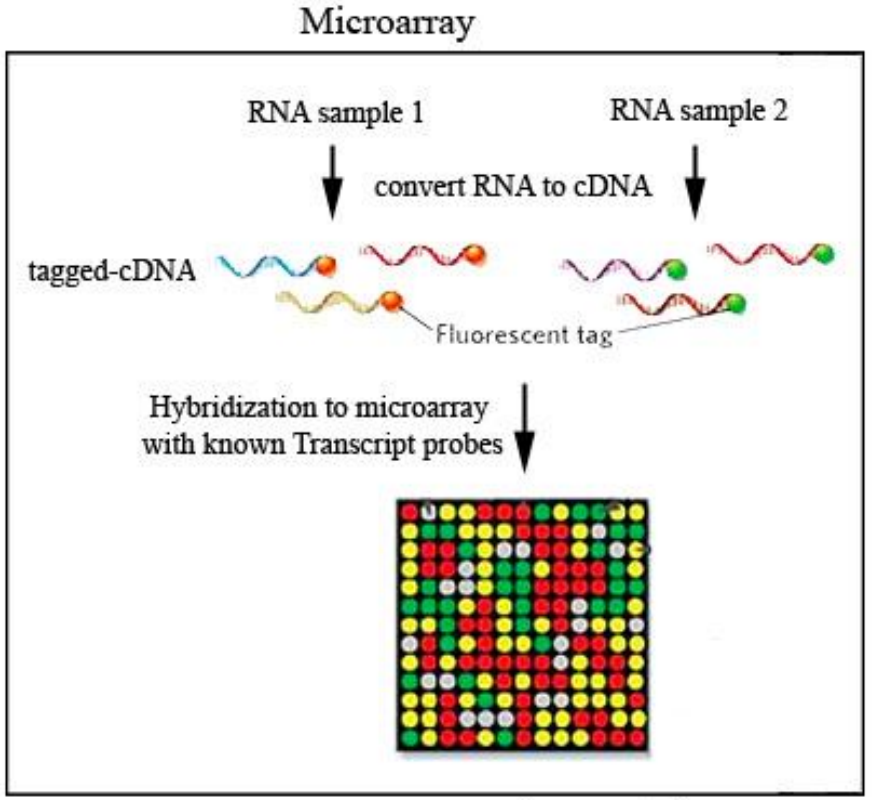
- Immunohistochemistry
- Western blot
- Mass spectrometry

- **RNA or Proteins?** Proteomics limitations
  - No amplification
  - Dynamic range
  - Post-translational modifications
  - Diversity



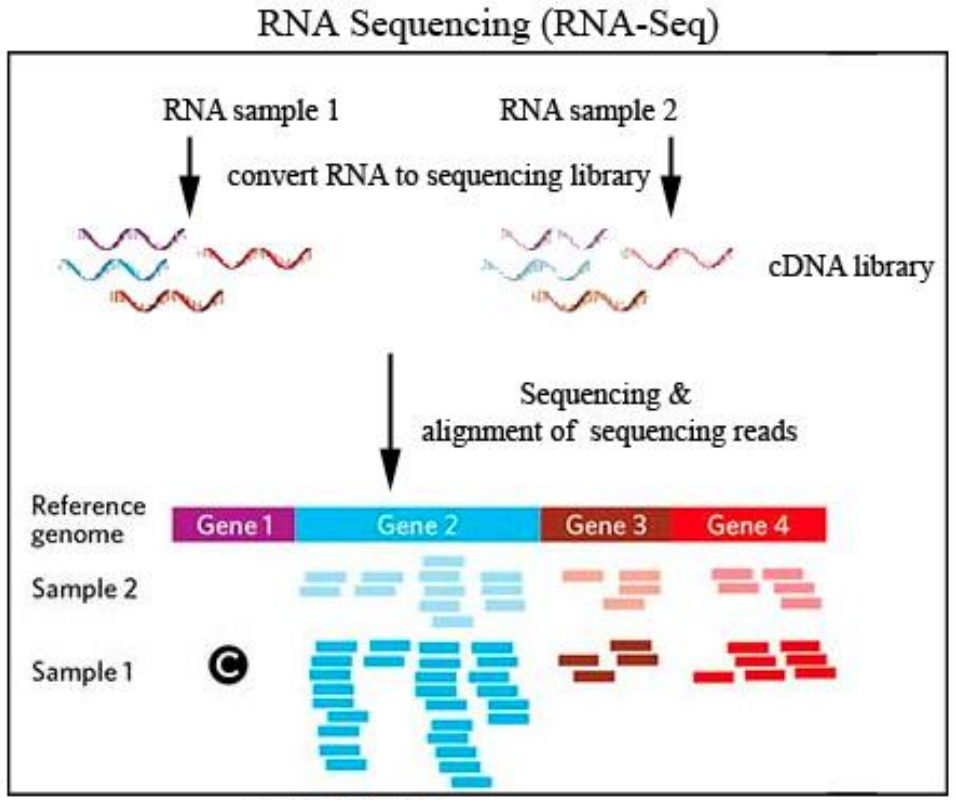
Central dogma of molecular biology

Tools to assess gene expression



relative intensity
=
expression levels

Low sensitivity
Low dynamic range
known transcript only
No alternative splicing information
lower cost



High sensitivity
High dynamic range
Novel transcripts sequences identified
structural variation & alternative splicing revealed
unlimited sample comparisons

Sequencing Reads
=
expression levels

Transcriptomics: Bulk vs single cell

bulk
RNA sequencing

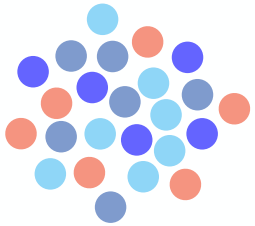
single cell
RNA sequencing

Detects changes in individual cells (cell types) within heterogenous samples

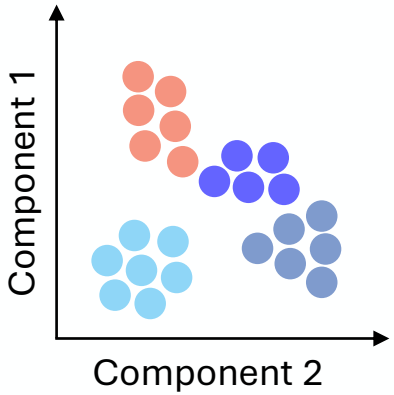
Shalek and Regev (2016)

Why single cell-RNA sequencing?

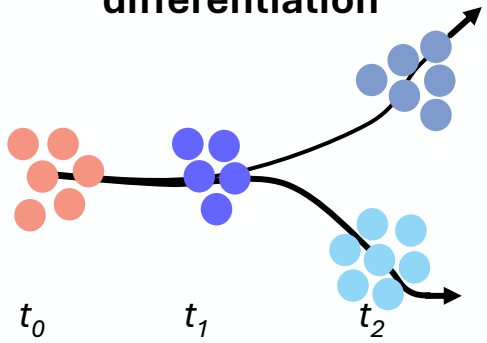
Studying heterogeneity



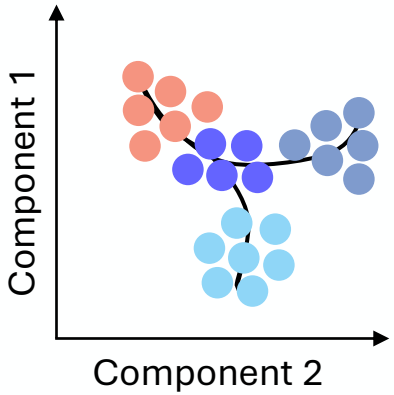
heterogeneous tissue



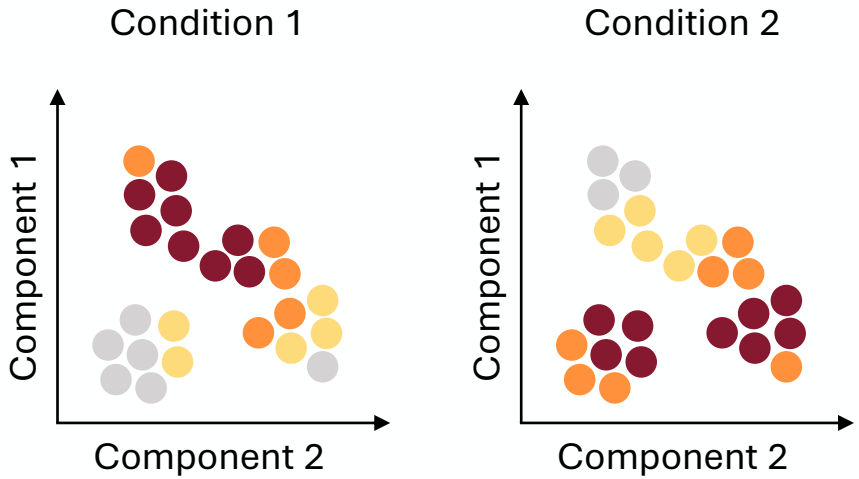
Trajectory analysis of cell differentiation



cell differentiation

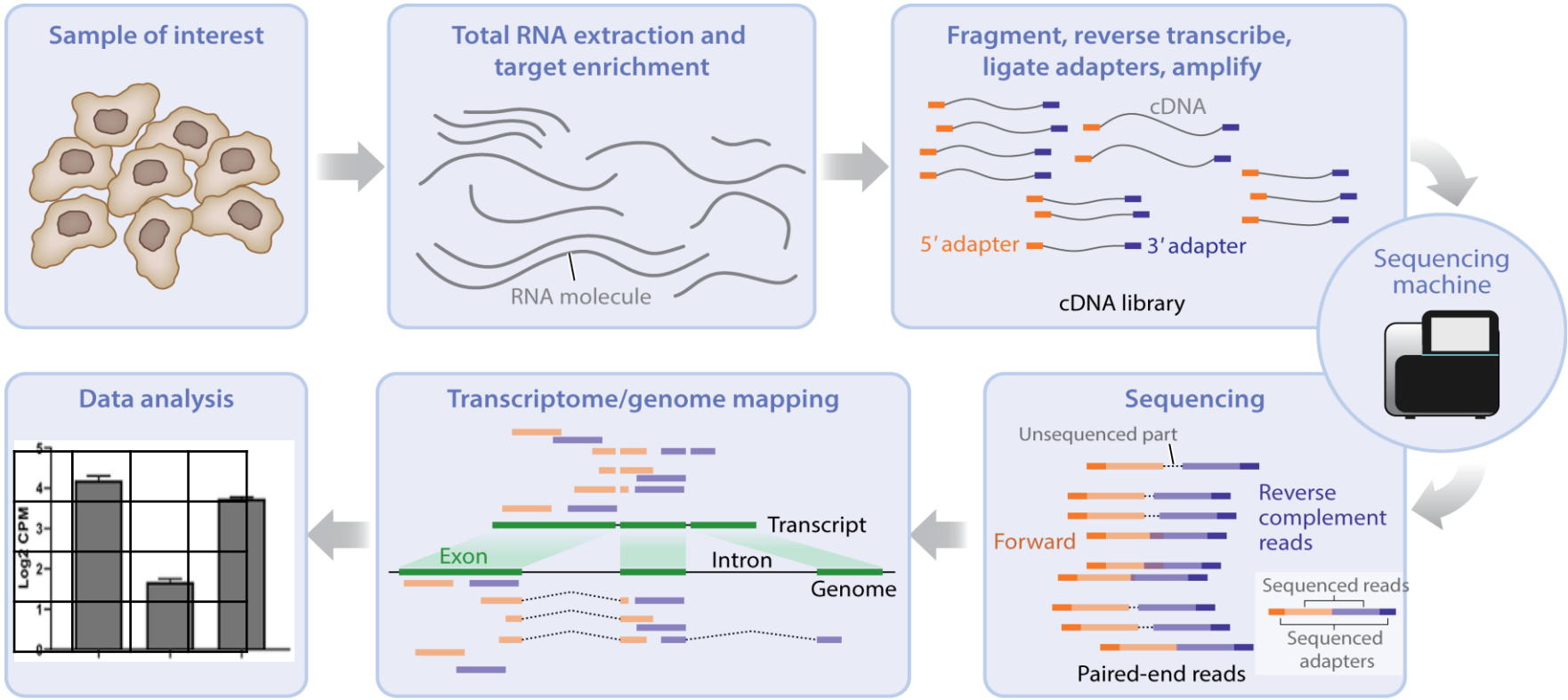


Compositional analysis



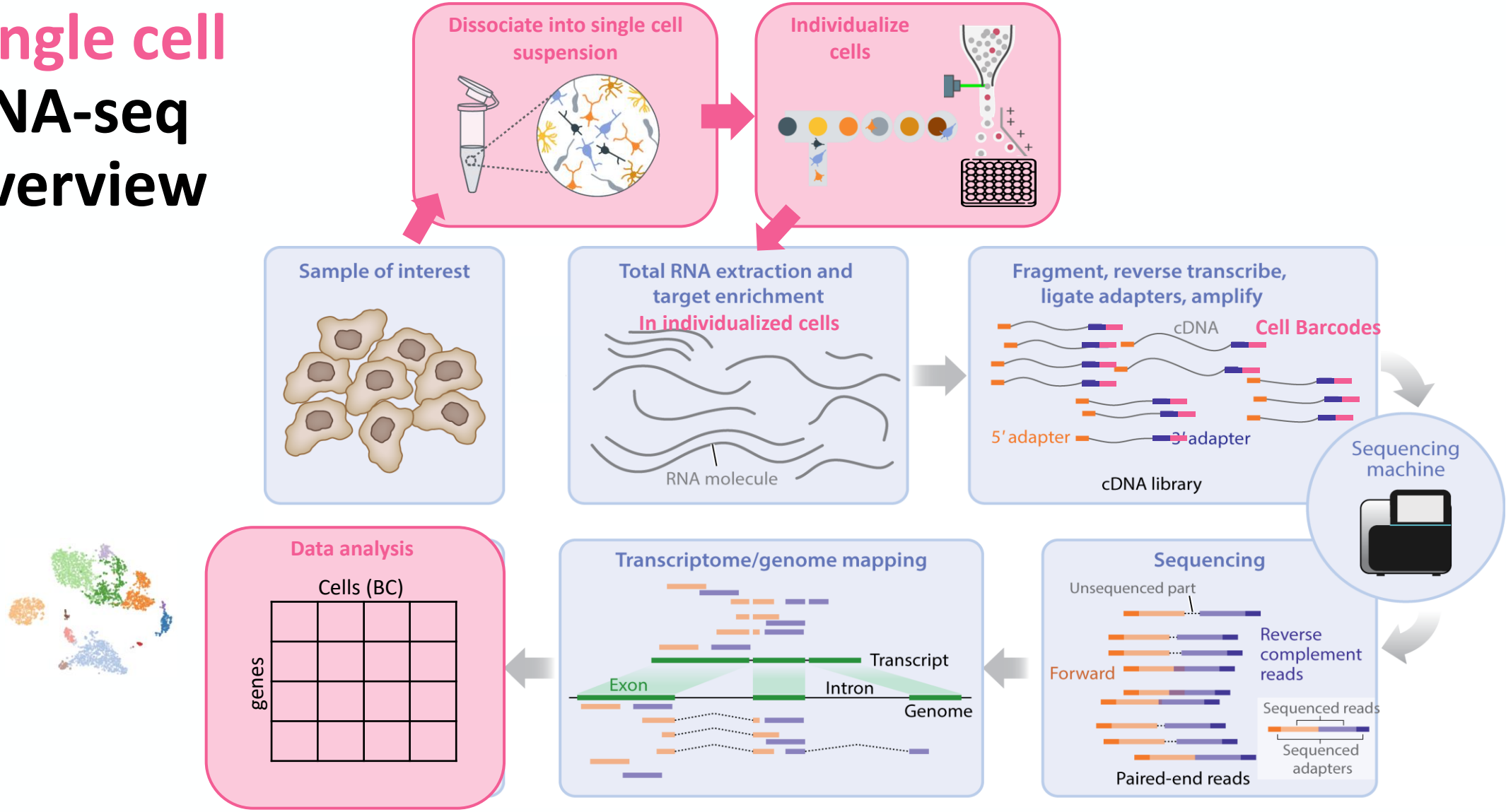
Liu and Trapnell, F1000 Research 2016
Luecken and Theis, Mol Systems Biology 2019

RNA sequencing overview



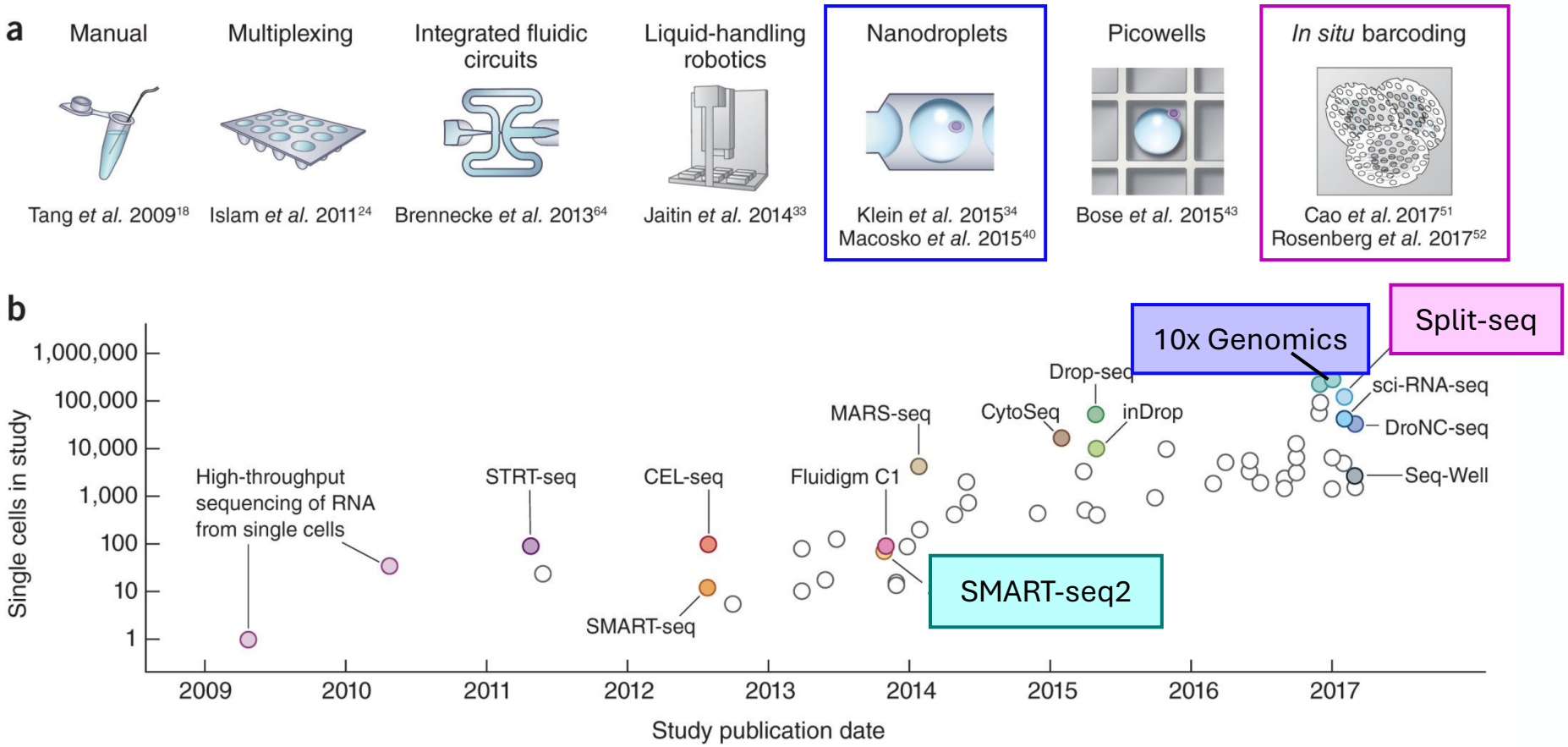
Van den Berge 2019, Annual Rev Biomed Data Sci

Single cell RNA-seq overview



Van den Berge 2019, Annual Rev Biomed Data Sci

How to get single cells (+ a timeline)

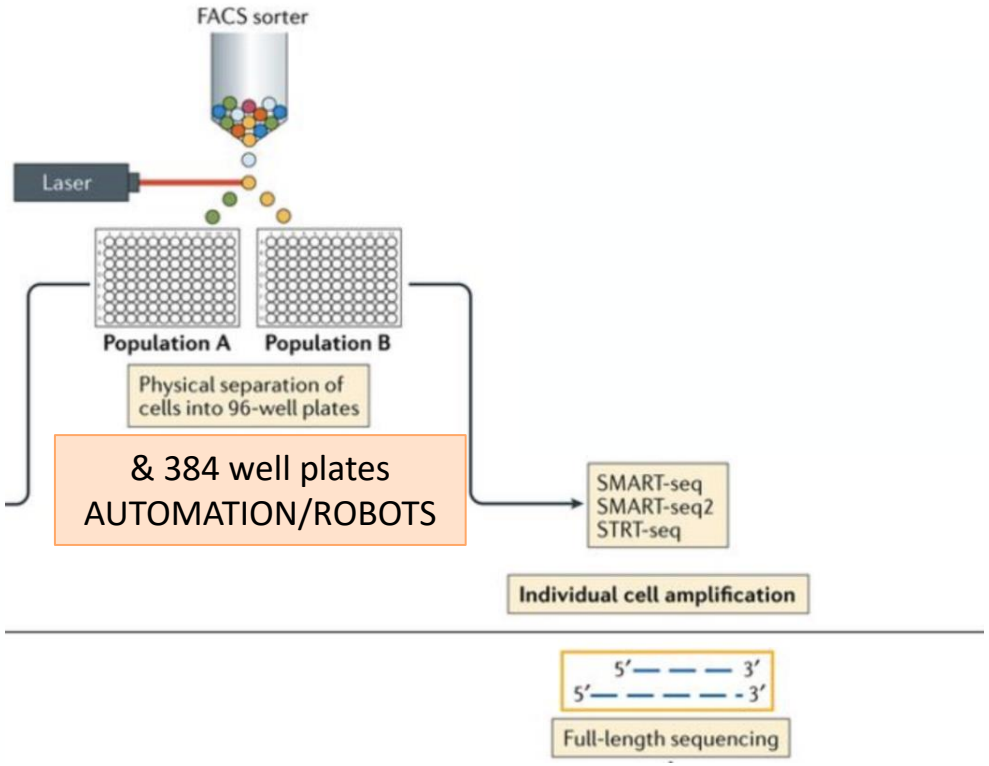


Svensson et al., Nature Protocols, 2018

	Droplet based	Combinatorial BCs	Plate based	Droplet + probes
Cell numbers	High	HighER	low	HighER
Gene coverage	3' end or 5’ end	Full length	Full length	-
Gene capture	Shallow		Deep	Hybridization
Cost / equipment	Equipment needed	Lower when >> cells	automation	Equipment needed
Isoform information	-	Some isoform info	Full isoform information	-
Sequencing needed	$$	$$$	$$$$$	$$
Tissue requirements	Fresh/ frozen (nuclei)	Prop. fixed & frozen	Fresh/frozen	Fixed (archive!)
	10X GENOMICS	PARSE BIOSCIENCES	TaKaRa **Smart-seq2 & 3**	10X GENOMICS FLEX

Plate based & full length – Smart-seq2 & 3

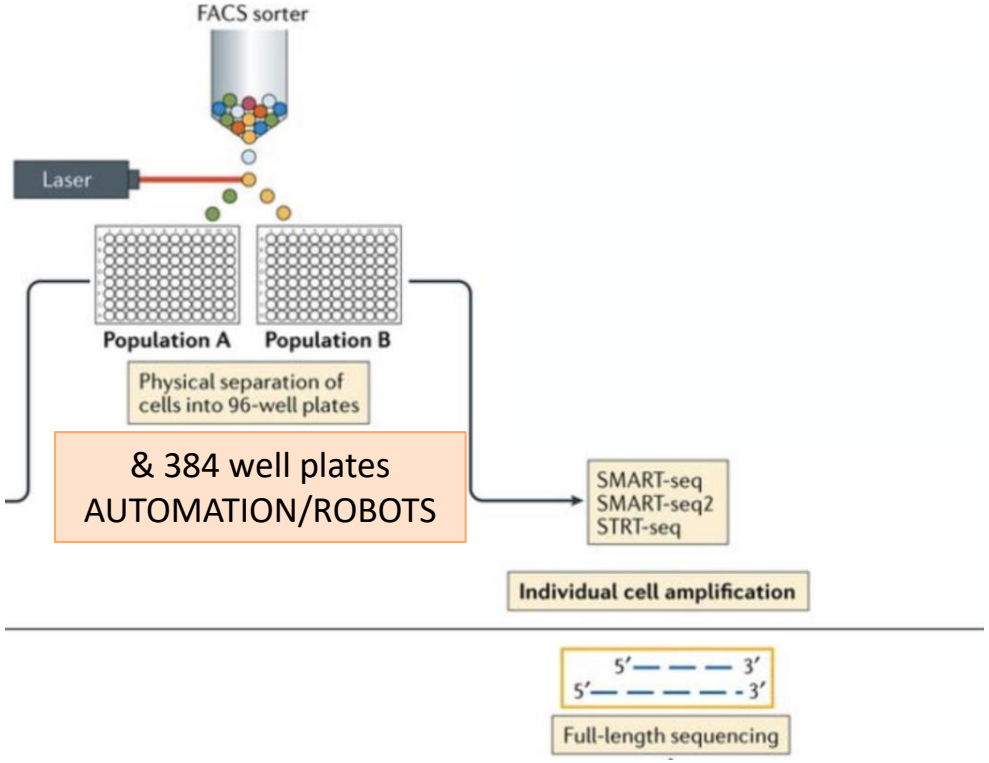
Full length gene coverage
10^2 – 10^3 cells/assay
FACS sorted
Depth, isoforms
No UMI control (only in ss3)

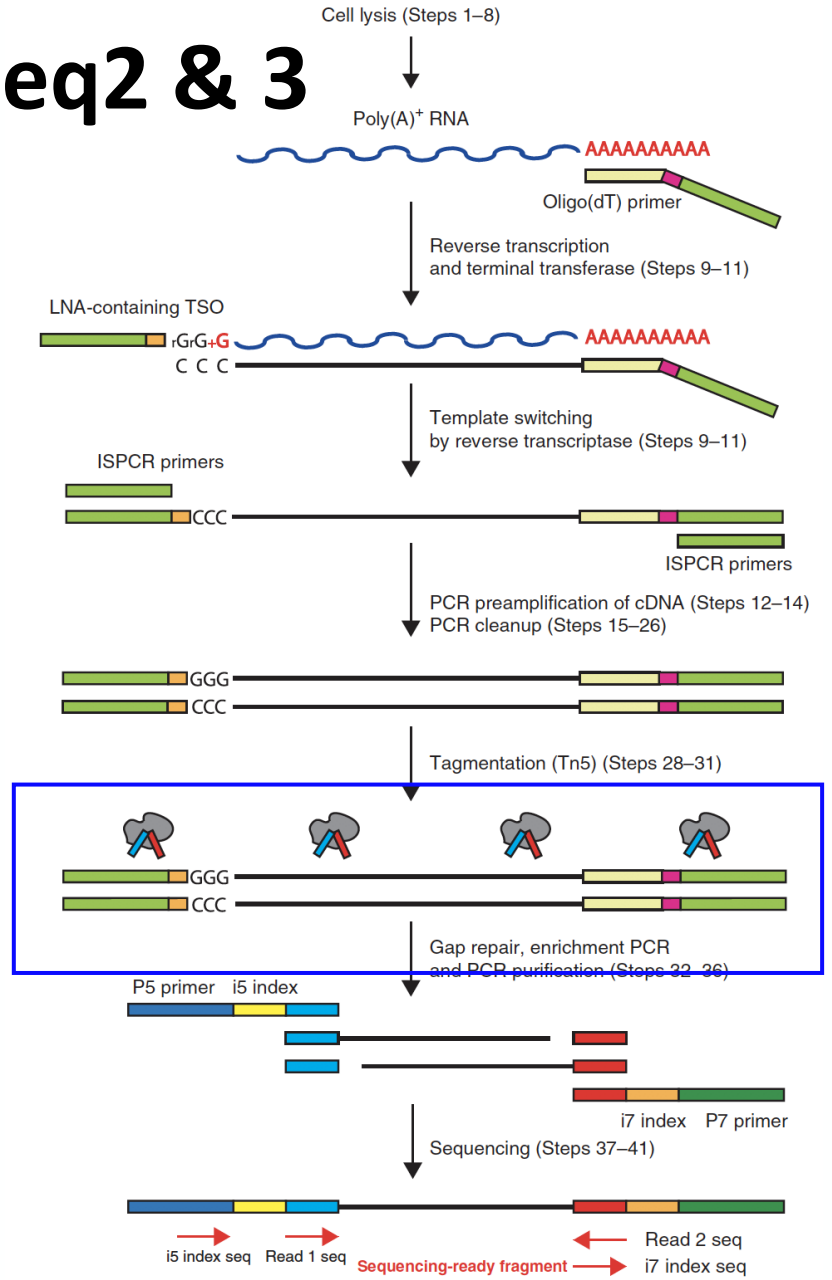


Picelli et al., Nature Protocols, 2014
Papalexi et al., Nature Reviews Immunology, 2017

Plate based & full length – Smart-seq2 & 3

Full length gene coverage
10^2 – 10^3 cells/assay
FACS sorted
Depth, isoforms
No UMI control (only in ss3)

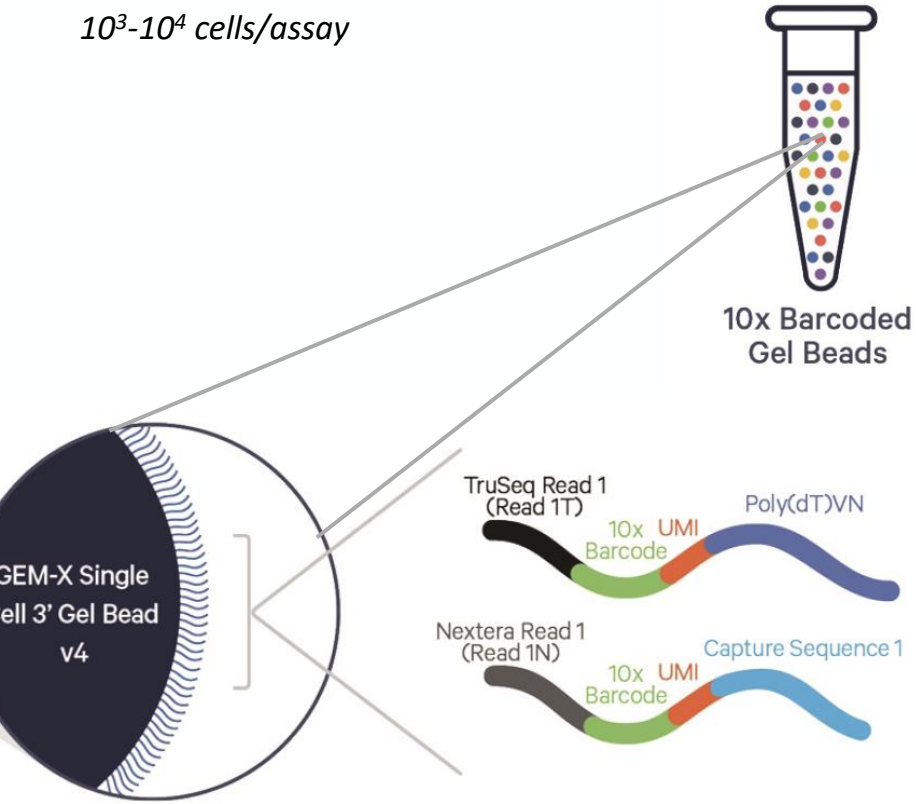




Picelli et al., Nature Protocols, 2014
Papalexi et al., Nature Reviews Immunology, 2017

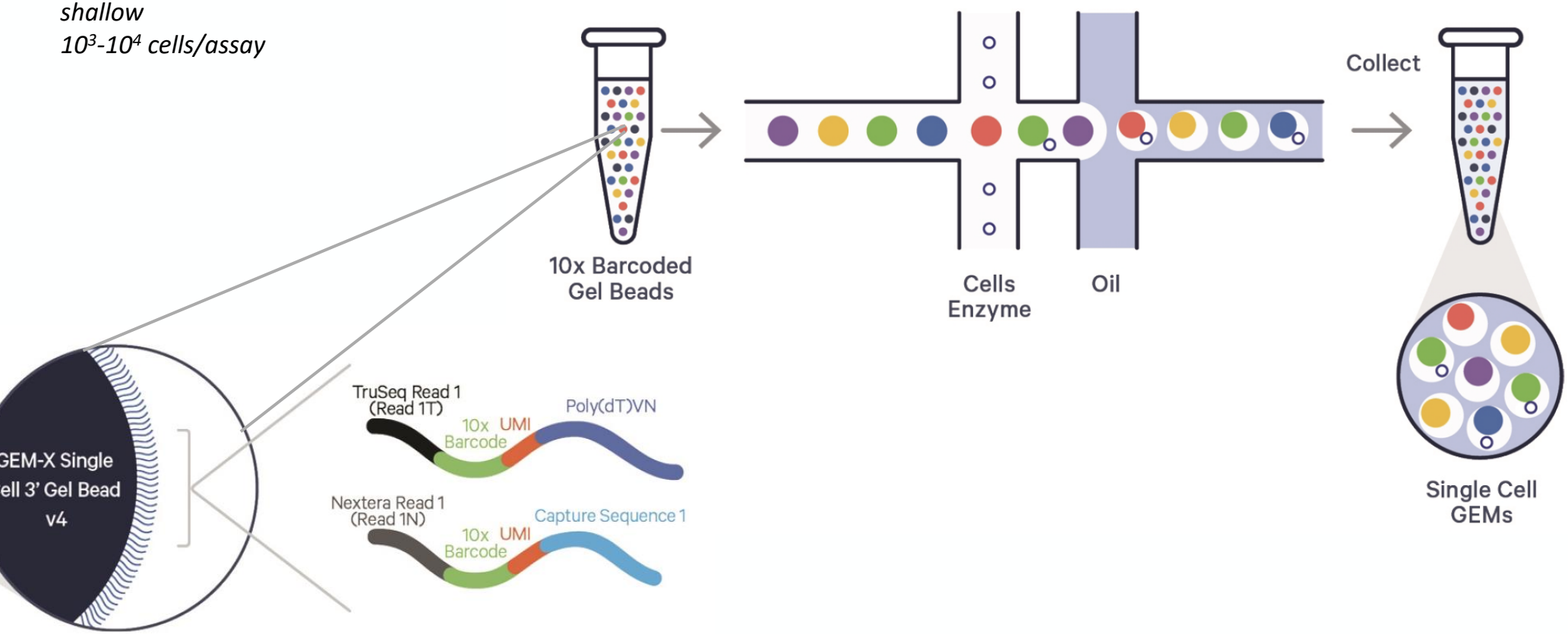
Droplet based – 10x genomics

3' end */ 5' end gene coverage*
shallow
10^3-10^4 cells/assay



Droplet based – 10x genomics

***3' end** / 5' end gene coverage*
shallow
10^3-10^4 cells/assay



Droplet based – 10x genomics

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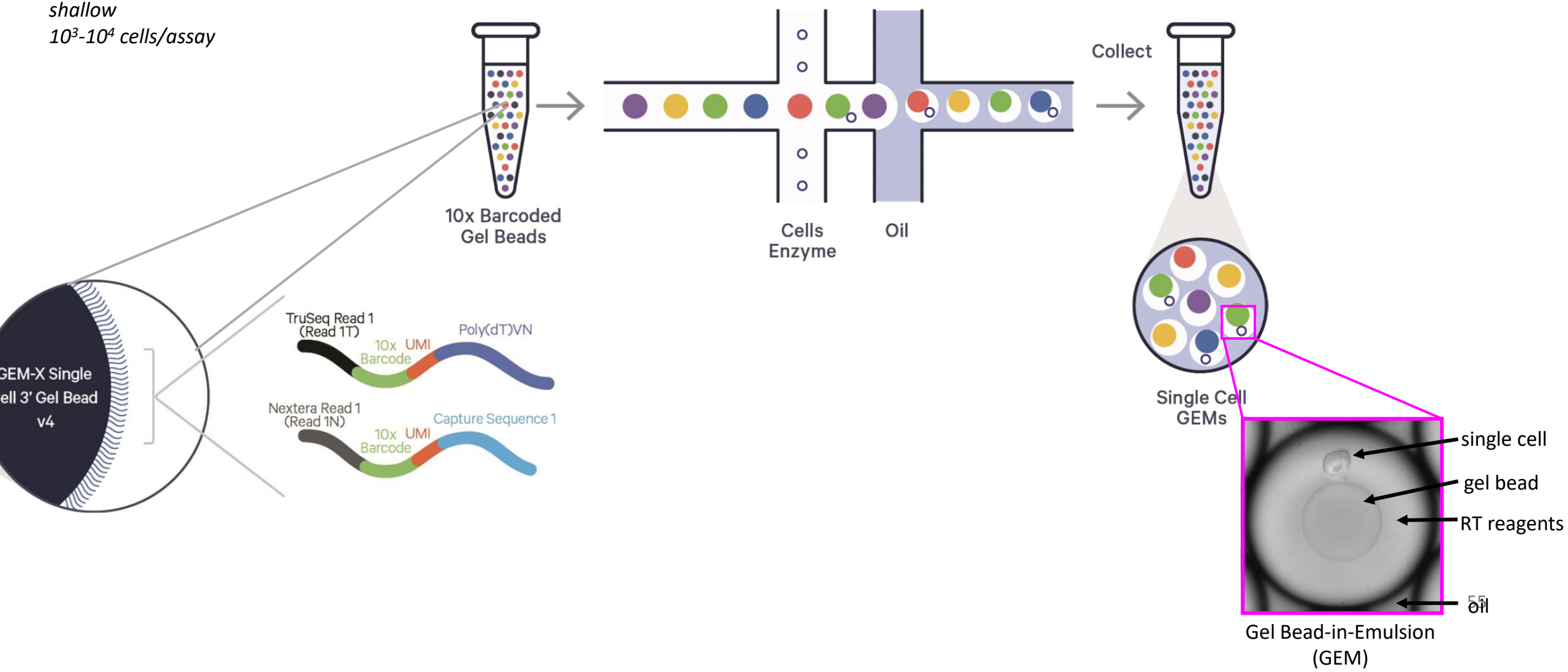

Droplet based – 10x genomics

3' end */ 5' end gene coverage*
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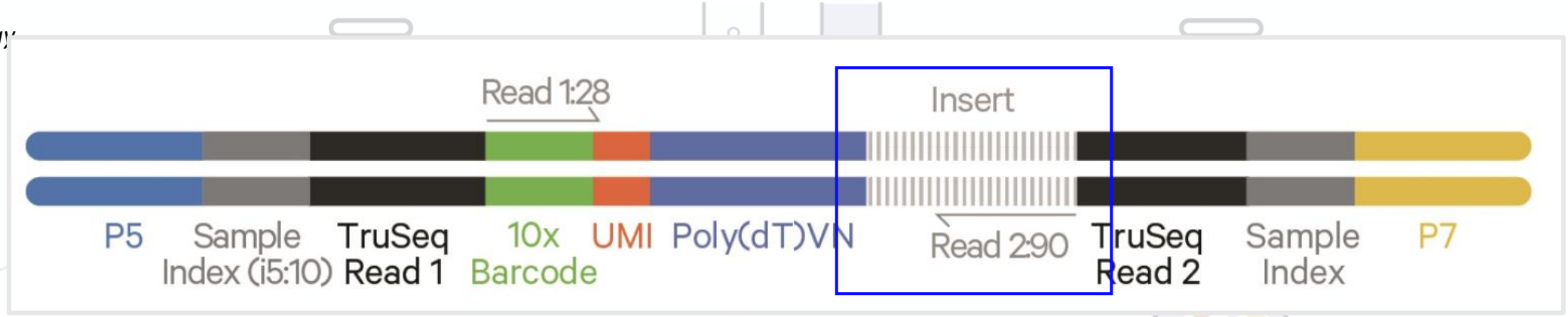
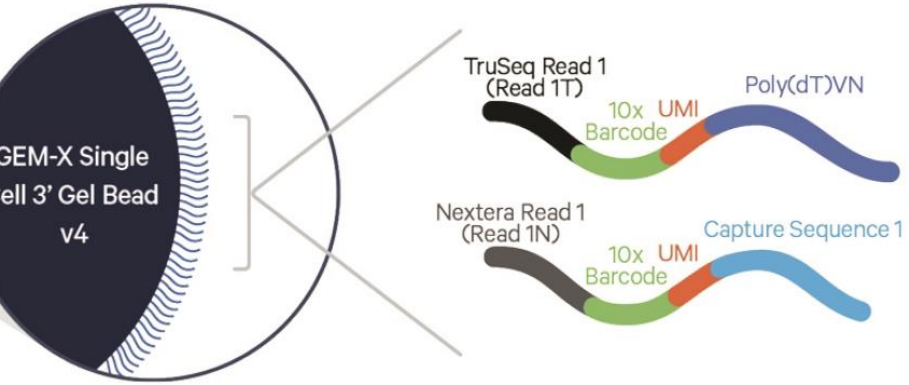
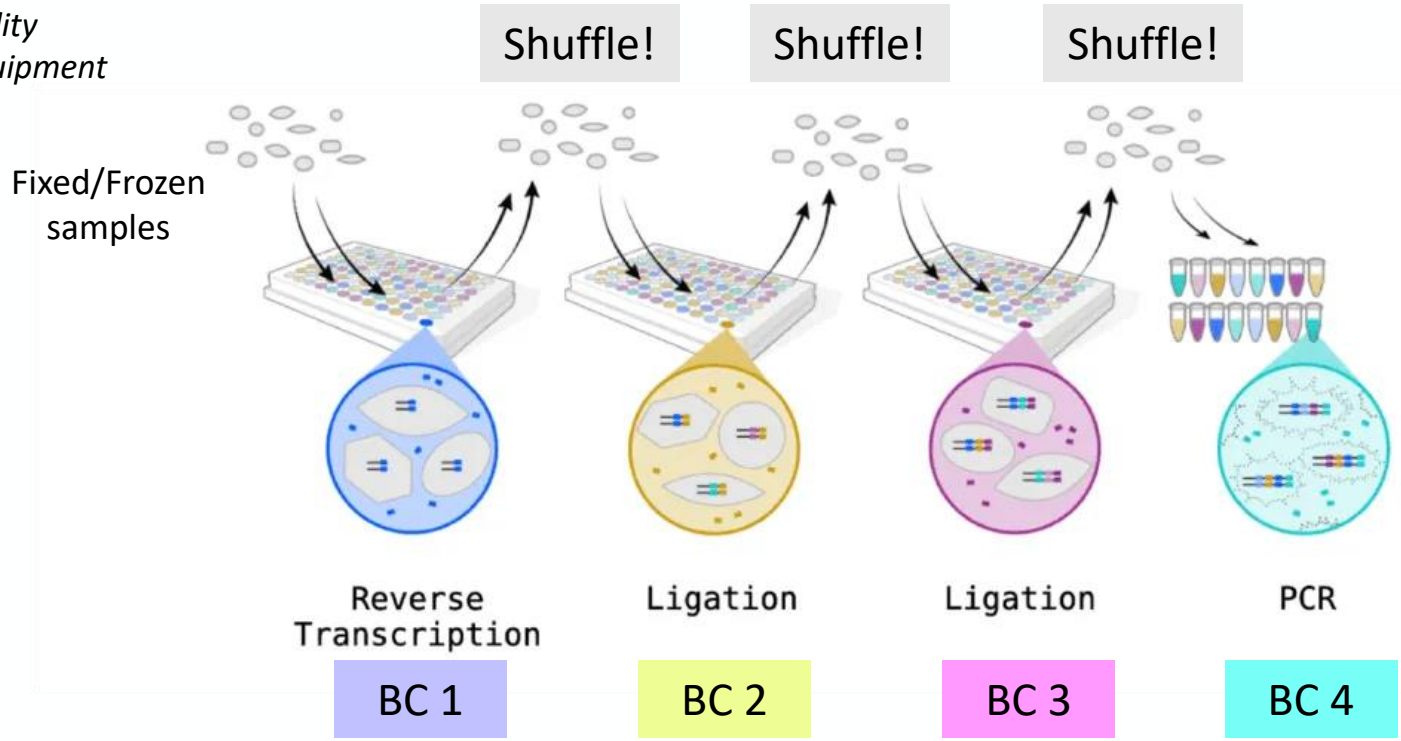
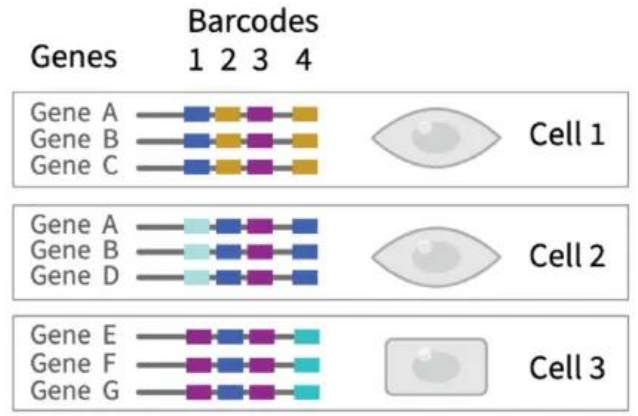




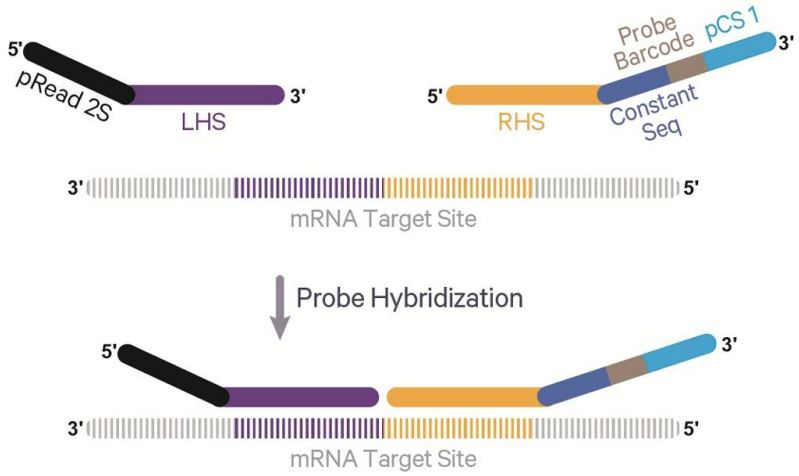
Combinatorial barcoding - Parse

Full length
10^5 – 10^6 cells/assay
Proprietary fixation & freeze
Flexibility
No equipment





Probe based – 10x genomics FLEX

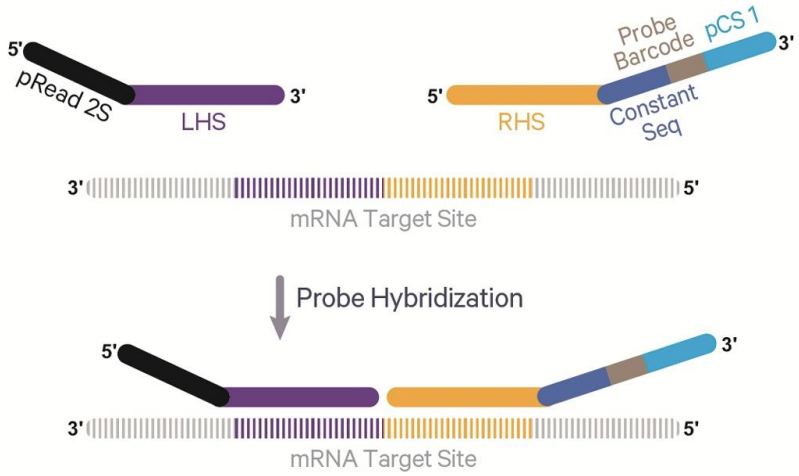


HYBRIDIZE PROBES (Inside permeabilized cells)

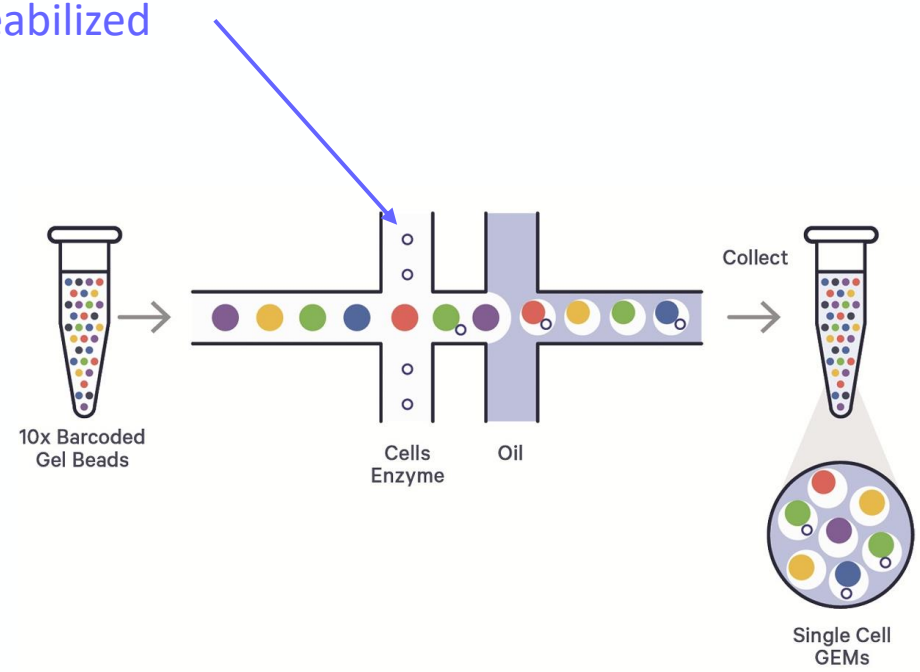
Probes!!!
Limited species
10^5 – 10^6 cells/assay
Fixed, fresh, archive
Flexibility

Probe based – 10x genomics FLEX

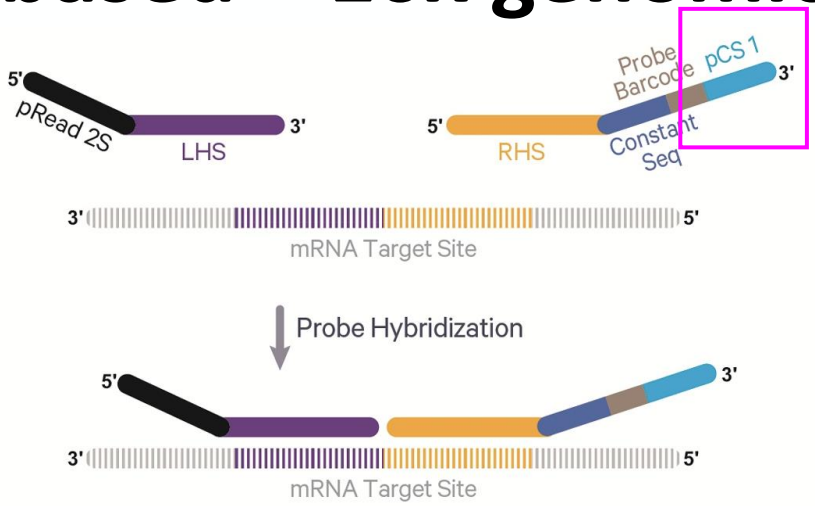
Probes!!!
Limited species
10^5 – 10^6 cells/assay
Fixed, fresh, archive
Flexibility



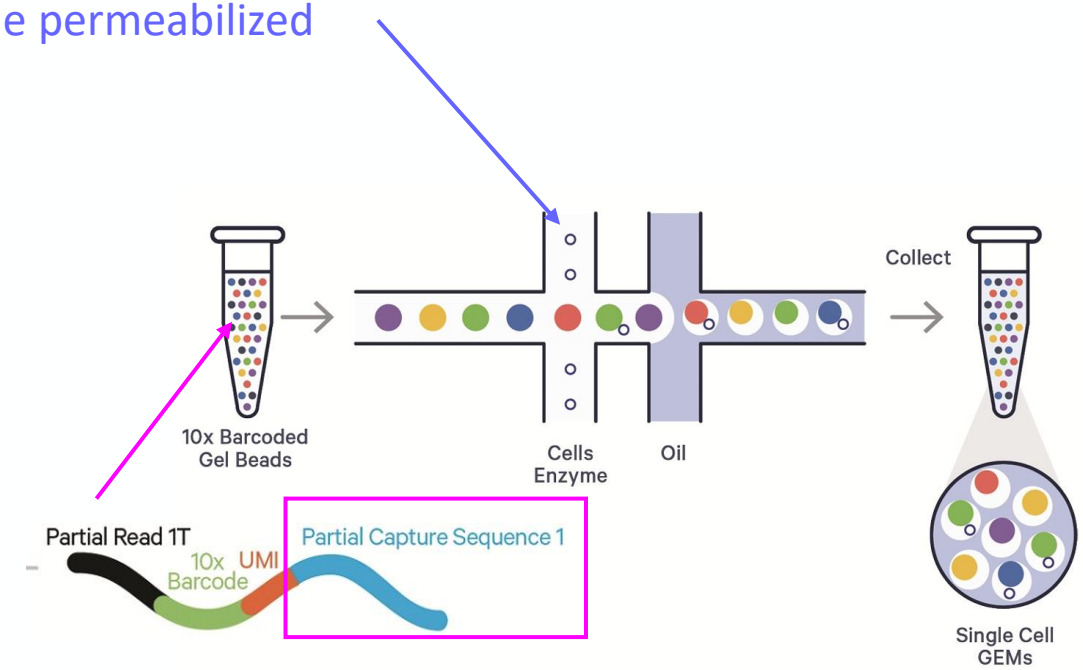
HYBRIDIZE PROBES
(Inside permeabilized cells)



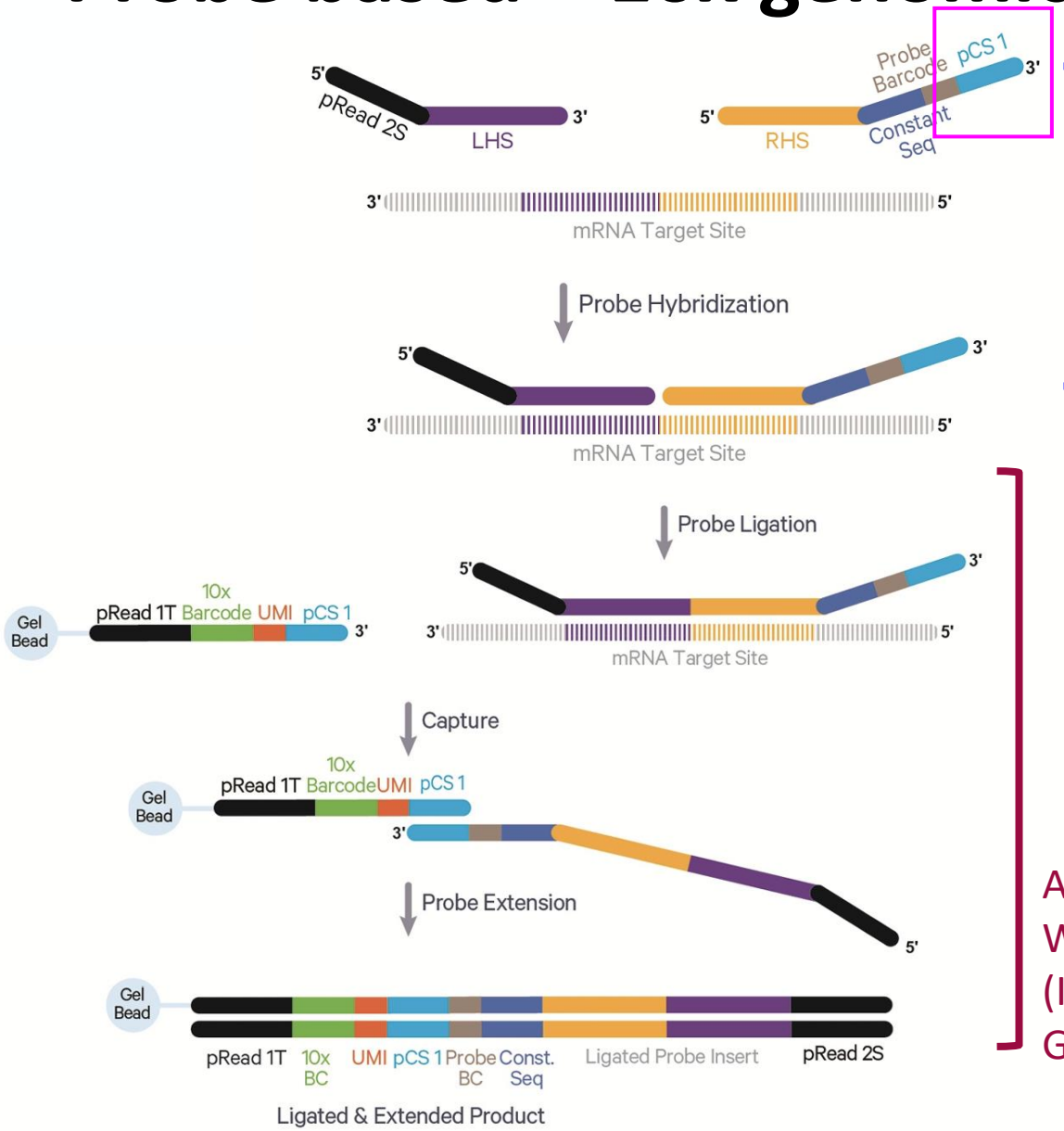
Probe based – 10x genomics FLEX



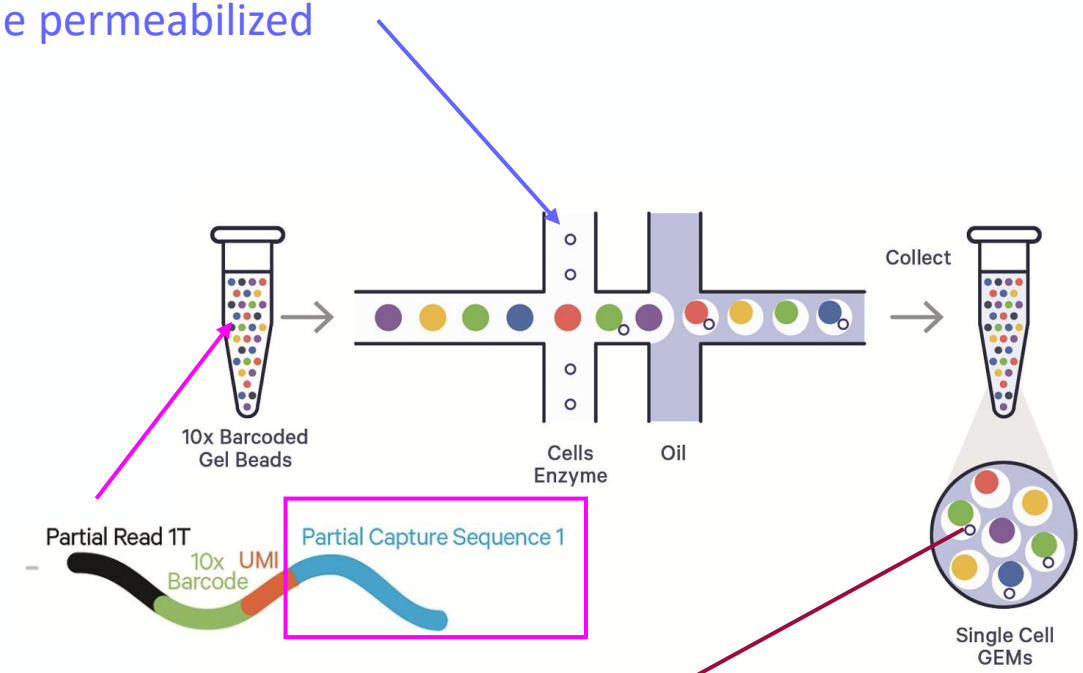
HYBRIDIZE PROBES (Inside permeabilized cells)



Probe based – 10x genomics FLEX

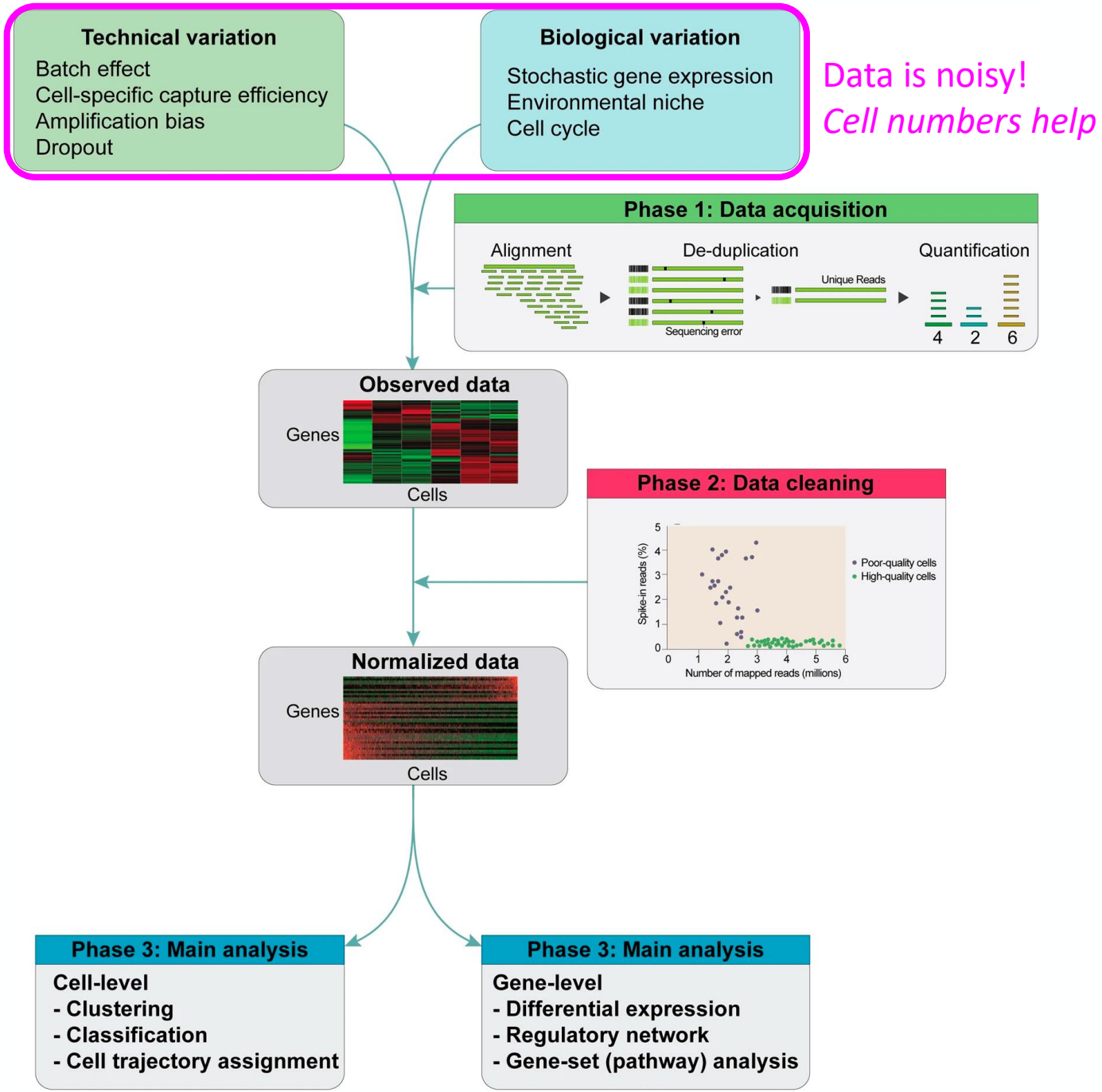


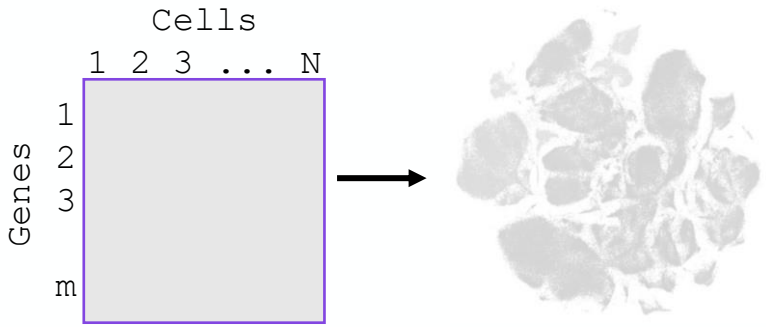
HYBRIDIZE PROBES
(Inside permeabilized cells)



ASSOCIATE **BOUND PROBES** WITH CELL BARCODES
(Inside individual cells in GEMs)

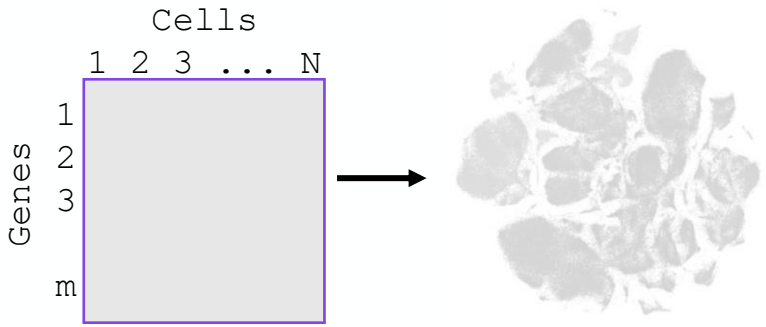
Data analysis Challenges & Limitations

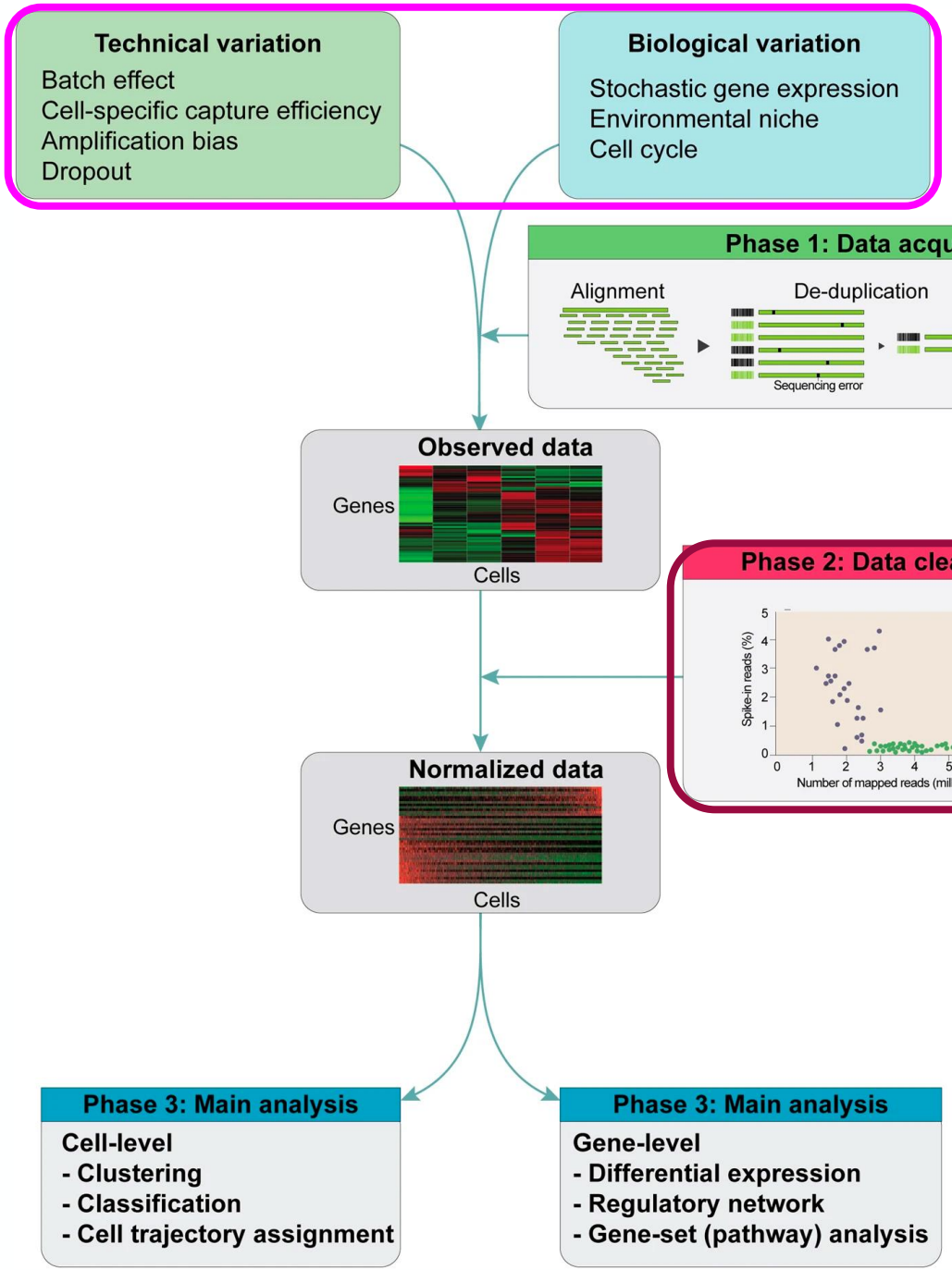




Luecken and Theis, Mol Systems Biology 2019

Data analysis Challenges & Limitations



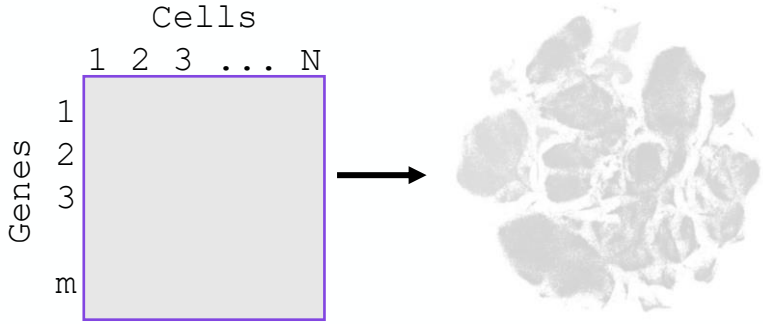


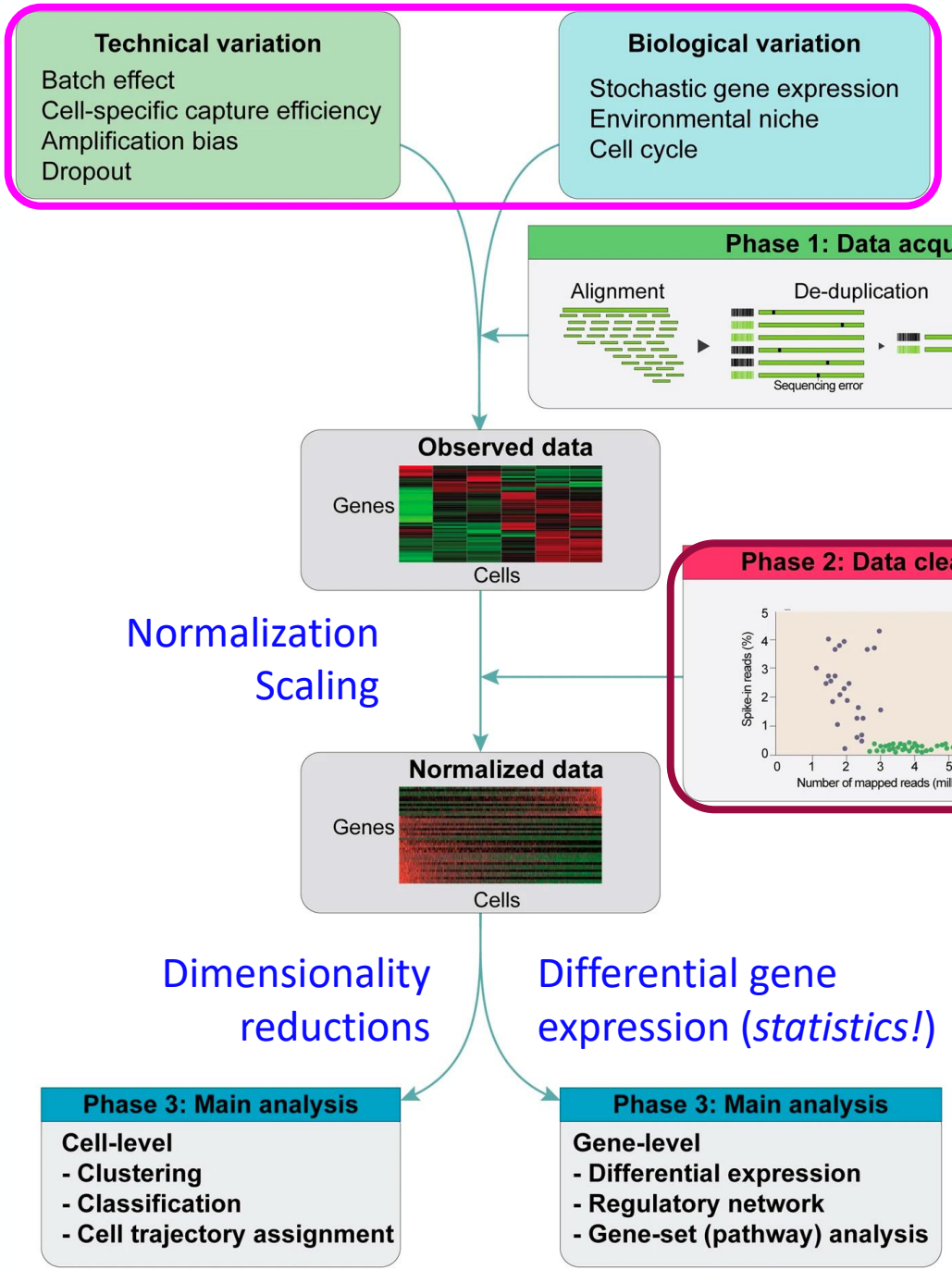
Data is noisy!
Cell numbers help

QC:
genes/cell
% mito reads
Sequencing depth matters

Luecken and Theis, Mol Systems Biology 2019

Data analysis Challenges & Limitations





Luecken and Theis, Mol Systems Biology 2019

Data analysis Challenges & Limitations

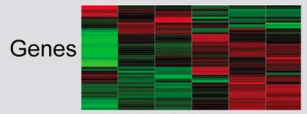
Technical variation
Batch effect
Cell-specific capture efficiency
Amplification bias
Dropout

Biological variation
Stochastic gene expression
Environmental niche
Cell cycle

Data is noisy!
Cell numbers help

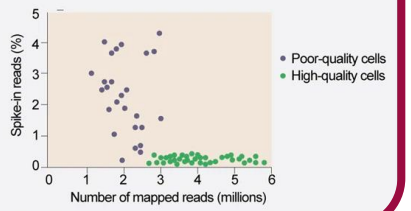
Phase 1: Data acquisition

Observed data



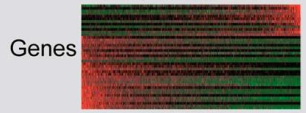
Normalization
Scaling

Phase 2: Data cleaning



QC:
genes/cell
% mito reads
Sequencing depth matters

Normalized data

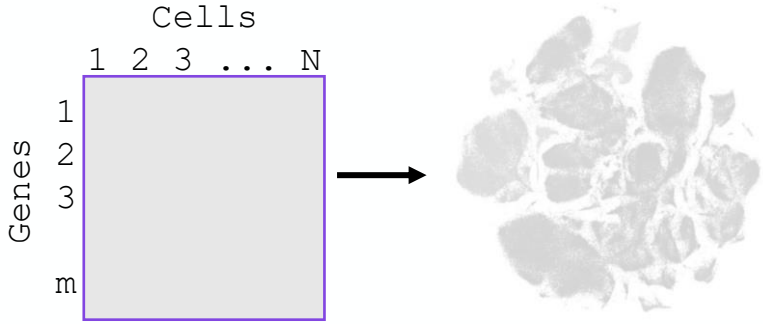


Dimensionality reductions

Differential gene expression (*statistics!*)

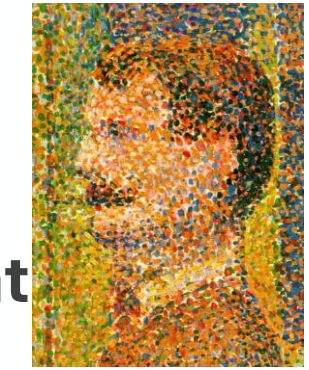
Phase 3: Main analysis
Cell-level
- Clustering
- Classification
- Cell trajectory assignment

Phase 3: Main analysis
Gene-level
- Differential expression
- Regulatory network
- Gene-set (pathway) analysis



Luecken and Theis, Mol Systems Biology 2019

Seurat
Satija Lab



Transcriptomics: Bulk vs single cell vs spatial

bulk
RNA sequencing

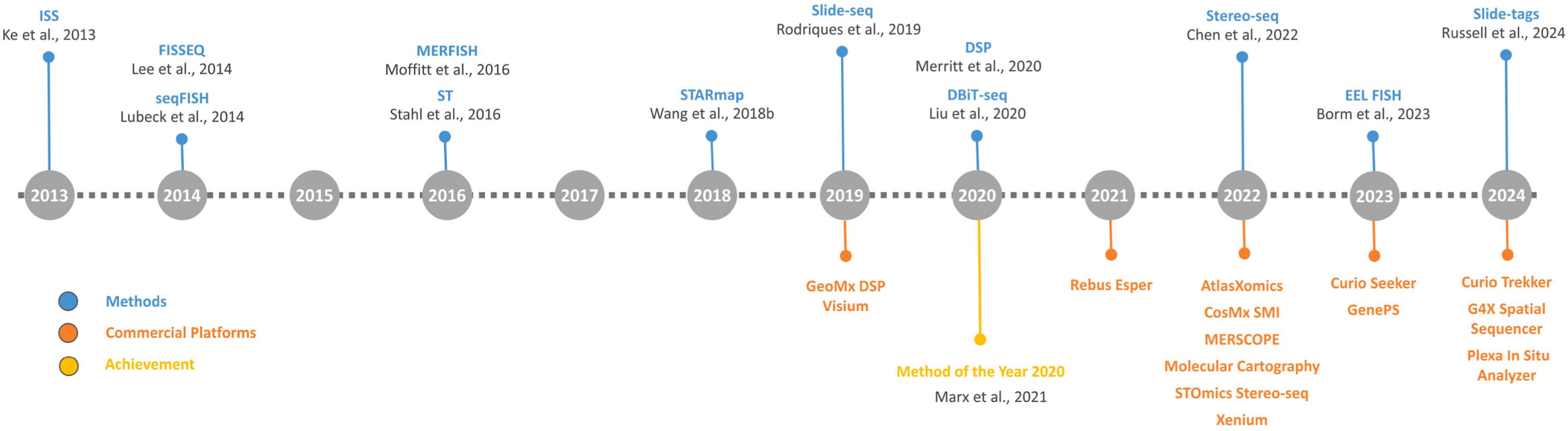
single cell
RNA sequencing

Spatial
Transcriptomics

Detects changes in individual cells (cell types) within their actual position in heterogenous samples

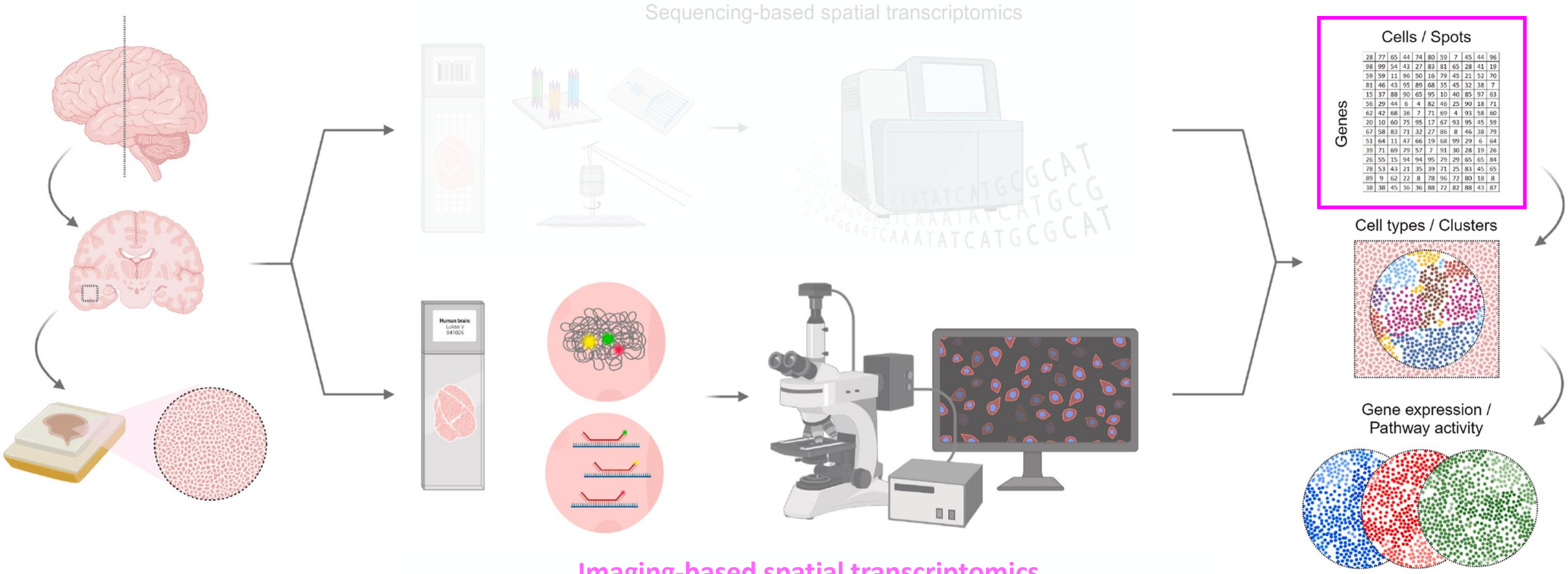
Shalek and Regev (2016)

New developments



Valhrach *et al* (2024)

Sequencing vs Imaging



Imaging-based spatial transcriptomics

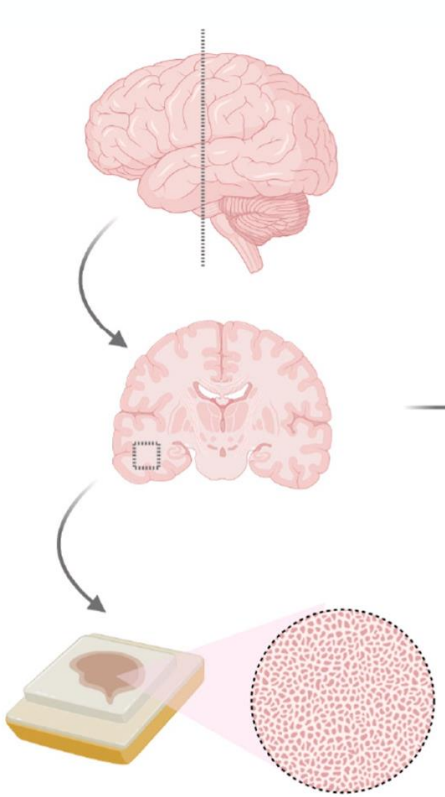
Challenge: go beyond 4 channels in the microscope!

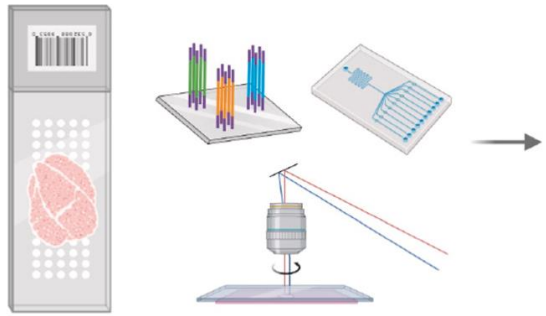
Predesigned gene panels – Cellular resolution

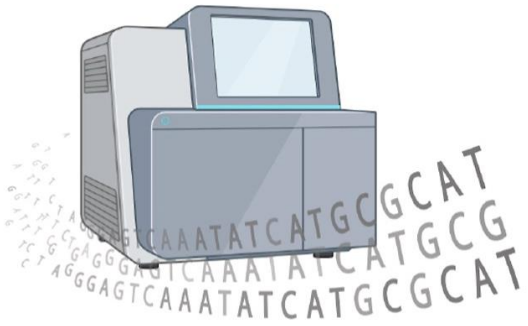
Merfish (Vizgen) - 10x Xenium – StarMap

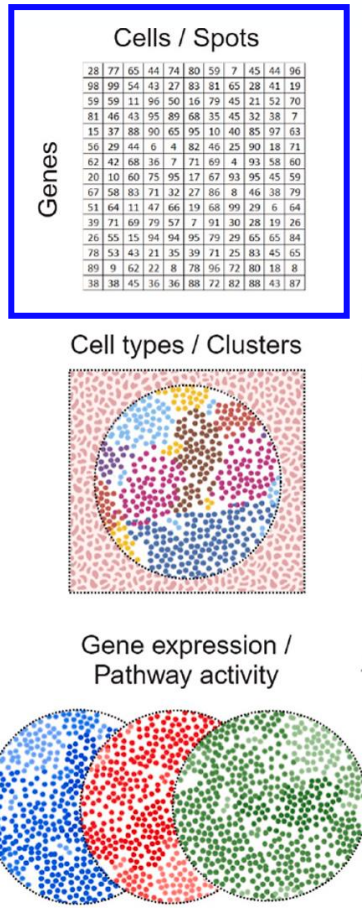
Valhrach *et al* (2024)

Sequencing vs Imaging









Sequencing-based spatial transcriptomics

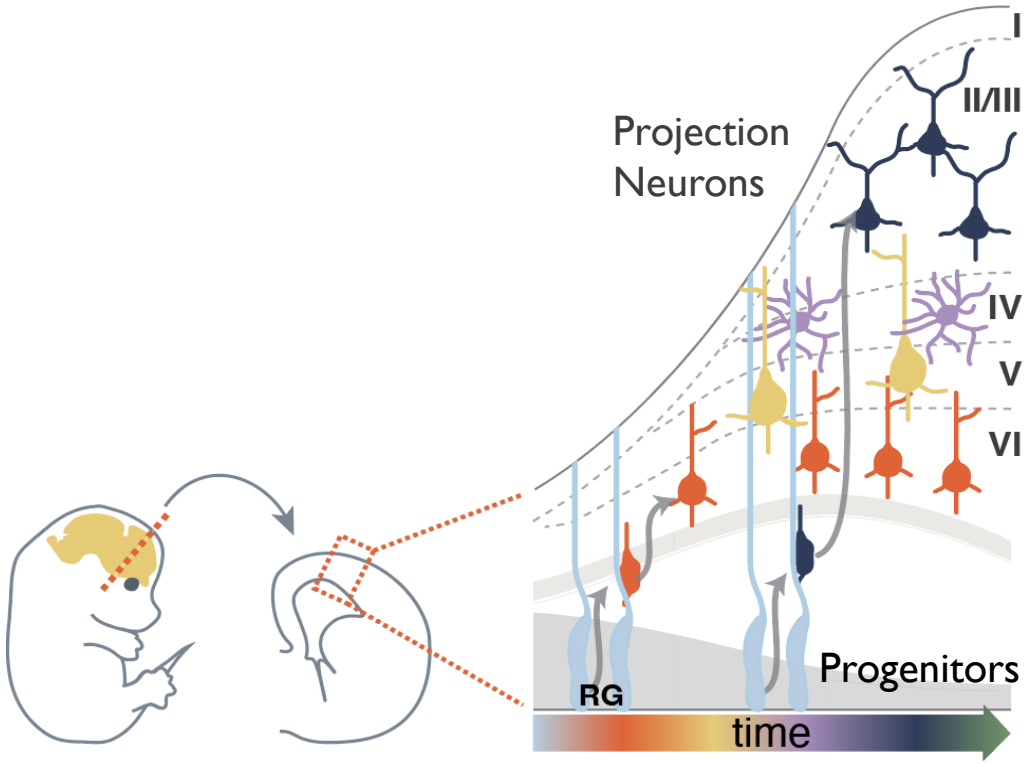
Challenge: how many sequencing/capture spots fit in a tissue?

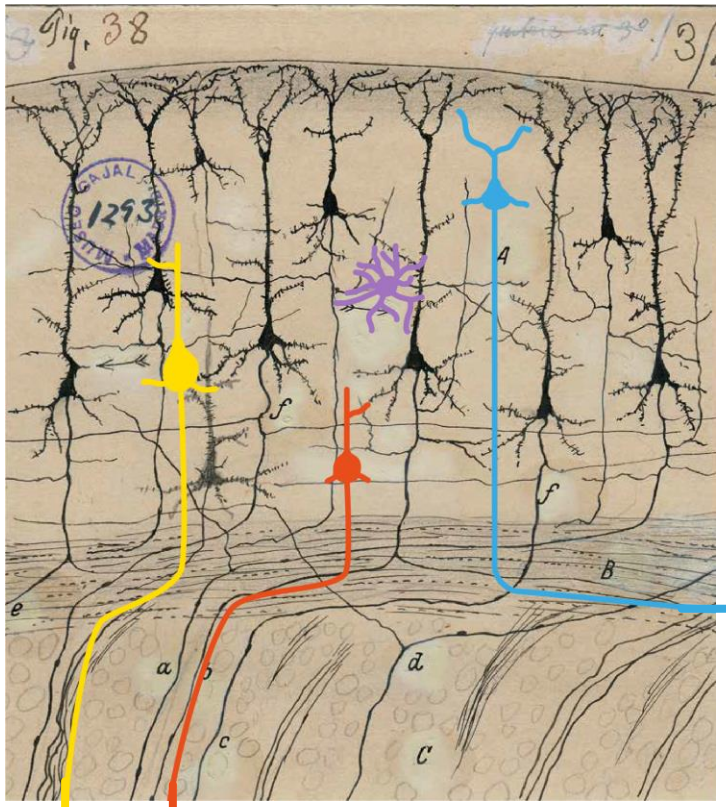
Species and gene panel agnostic – Less spatial resolution

10x Visium – Slidetags (Curio) - Stereoseq – Illumina Stratamap

Valhrach *et al* (2024)

NEURONAL DIVERSITY IN THE CEREBRAL CORTEX





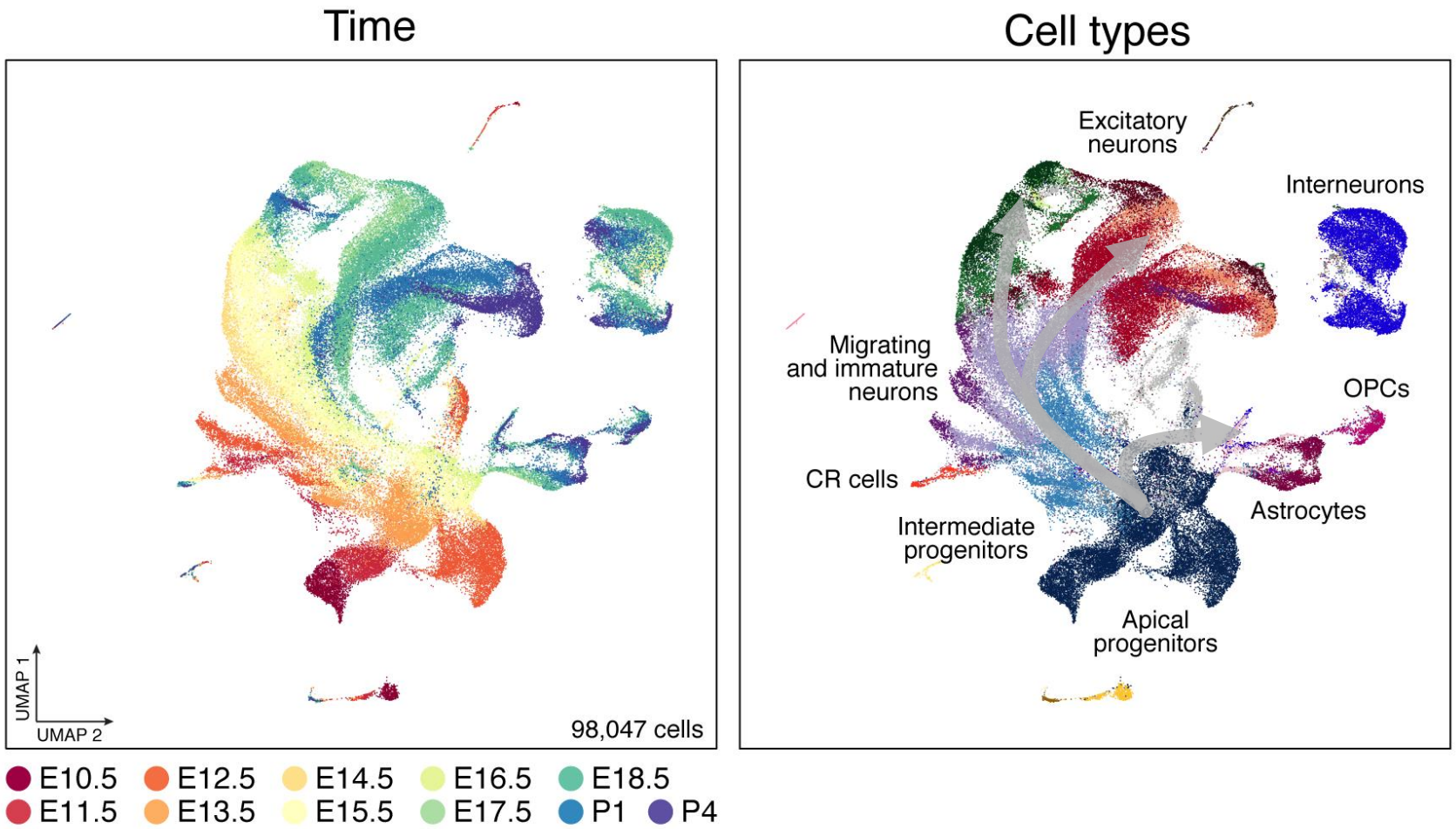
To the contralateral cortex

To targets outside of the cortex

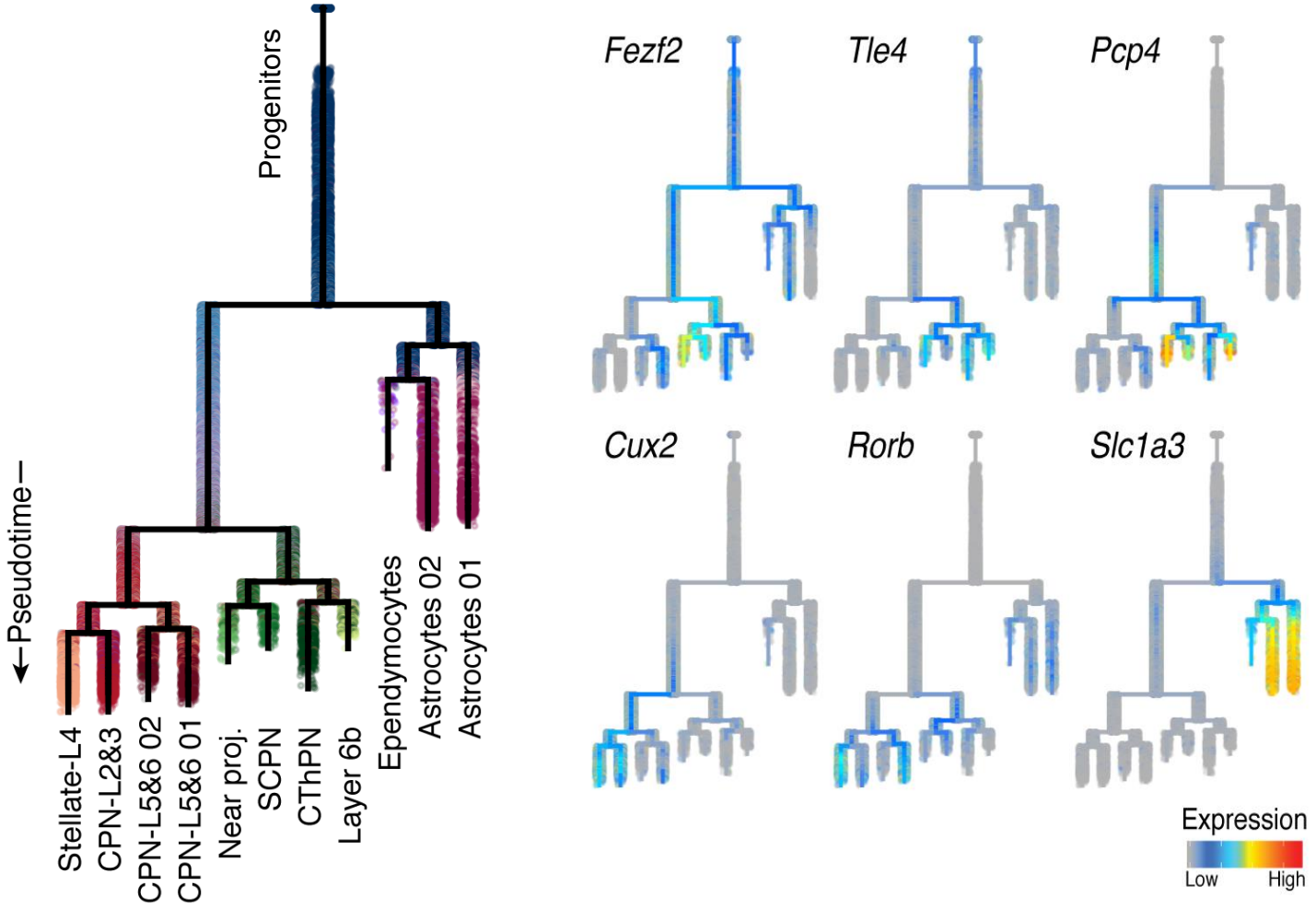
Di Bella, Dominguez-Iturza*, et al., 2023*

Ramón y Cajal, 1887-1913

Molecular trajectories in the developing cerebral cortex



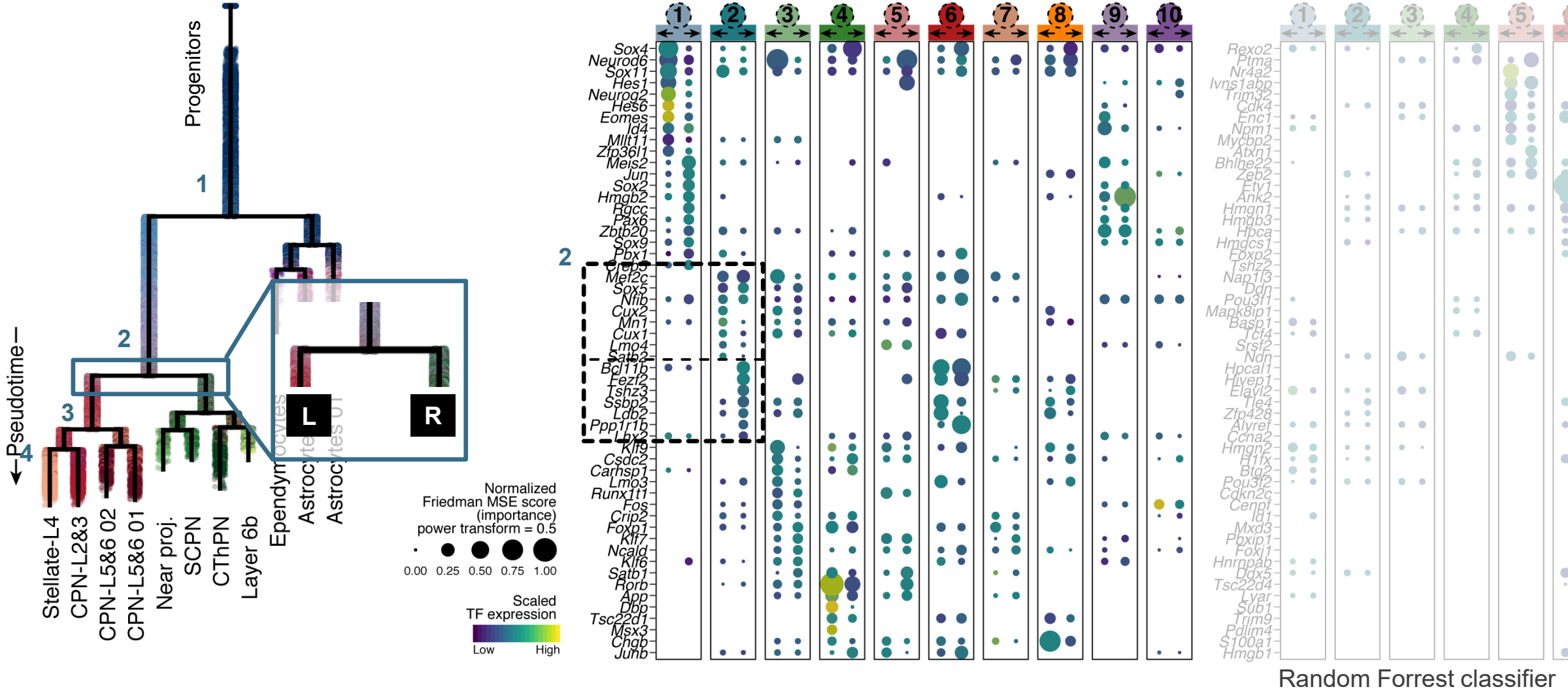
Branching-tree model of corticogenesis



URD – Schier & Regev labs – *Farrell 2018, Science*

Di Bella, Habibi*, et al., 2021. Nature*

Inferred molecular codes of identity divergence



Random Forrest classifier

Discovery of identity regulators in the developing cortex

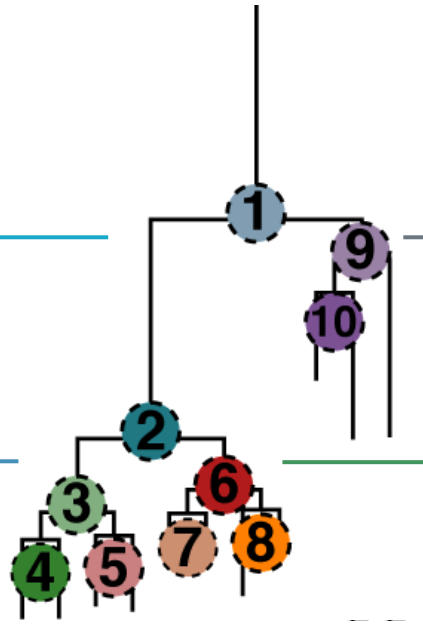
Candidate regulators

Neurogenesis

Sox4, Neurod6, Sox11, Hes1, Neurog2, Hes6, Eomes, Id4, Mllt11, Zfp36l1

Gliogenesis

Meis2, Jun, Sox2, Hmgb2, Rgcc, Pax6, Zbtb20, Sox9, Pbx1, Creb5



CPN specification

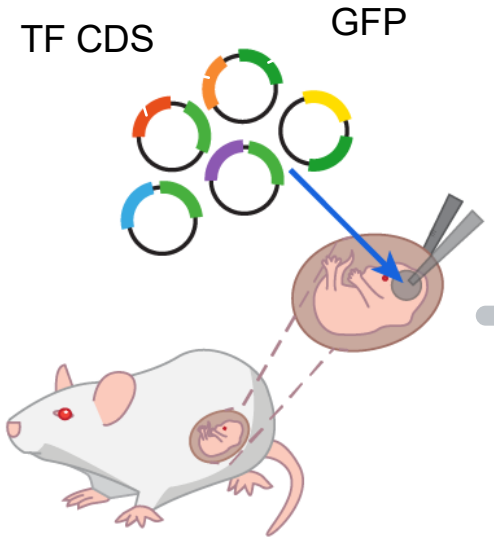
Cux2, Mn1, Zbtb20, Cux1, Pbx1, Lmo4, Gas7, Ppp1r16b, Zeb2, Cul4a

CFuPN specification

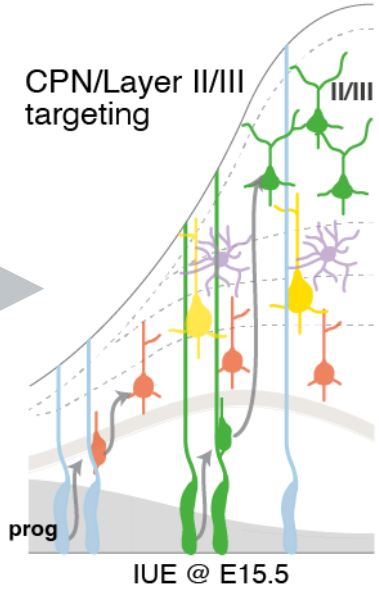
Bcl11b, Fezf2, Tshz3, Ssbp2, Ldb2, Ppp1r1b, Lhx2, Lmo1, Lmo3, Nfe2l3

A functional validation of cell identity regulators

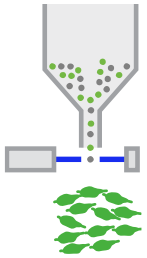
pooled gain of function *in vivo* screen



in utero electroporation (pooled ~30 candidates to promote ***heterochronic*** fate)

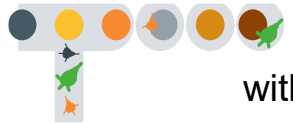


Target Neuron subtypes by birthdate



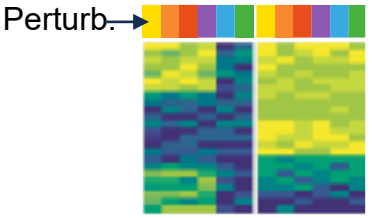
sort GFP+ cells

Readout: change in transcriptional identity

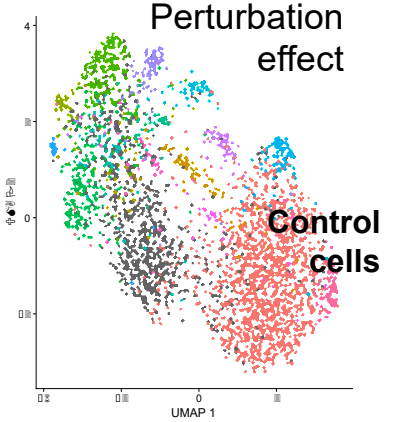


scRNA-seq with perturbation detection

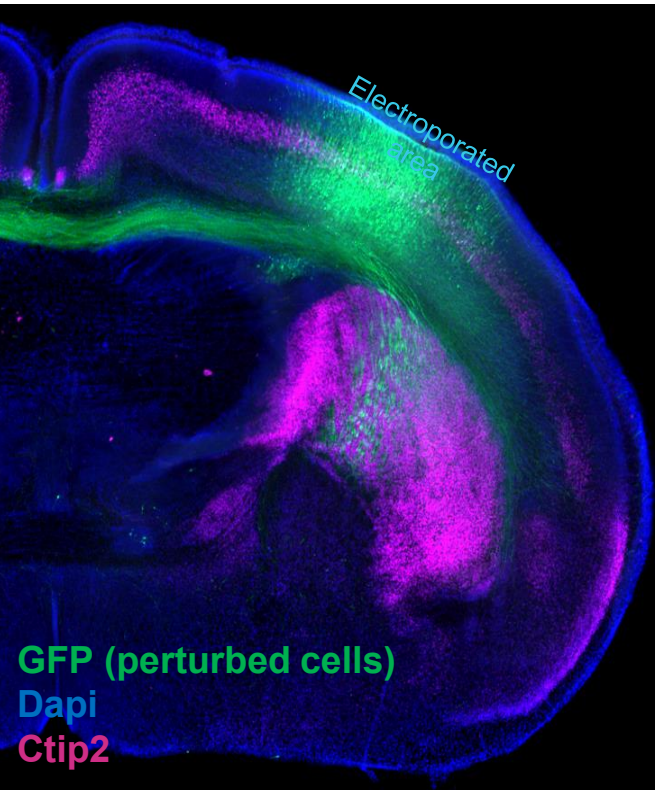
DGE across cell types and perturbations

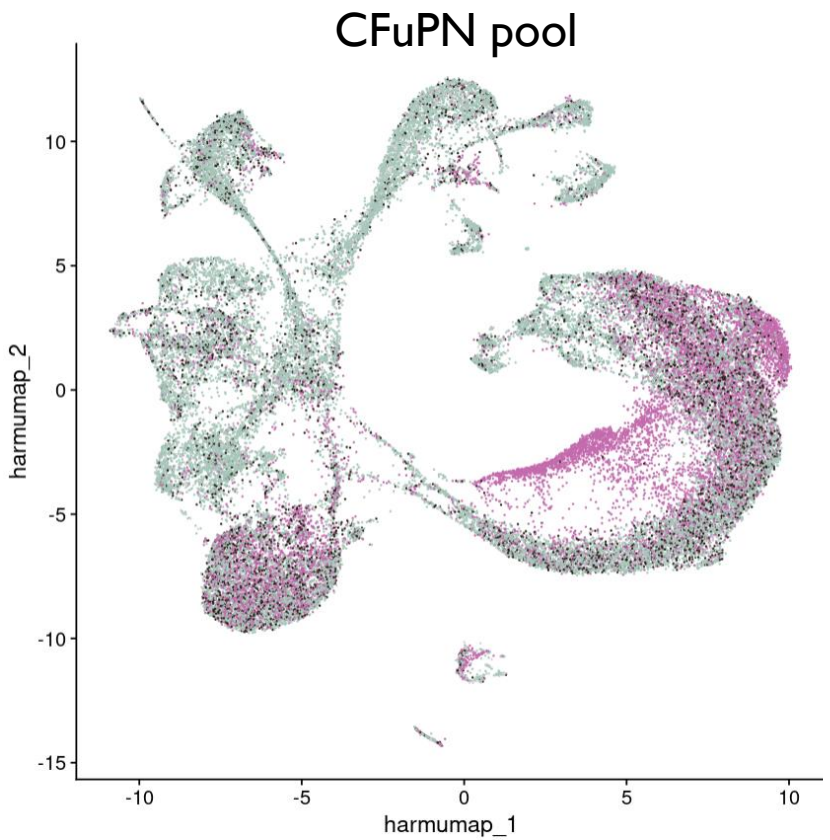


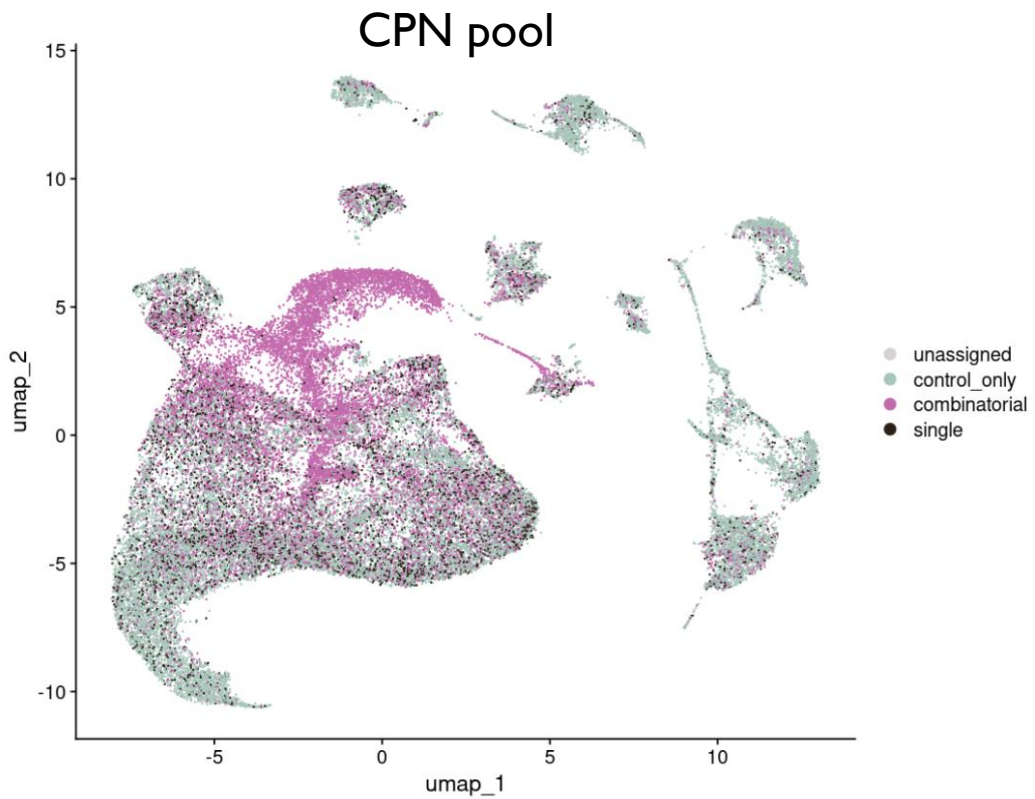
Perturbation effect



Pooled *in vivo* screen of fate determinants







Preliminary data

THANK YOU!

Paola Arlotta
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Nuria Dominguez
Ana Uzquiano

HSCRB
HARVARD DEPARTMENT OF STEM CELL & REGENERATIVE BIOLOGY

BROAD INSTITUTE

NIH The BRAIN Initiative

Di Bella Lab
Lucas Arbués
Angeles Estrada
Candela Pastore

Aviv Regev
Ehsan Habibi

INSTITUTO LELOIR FUNDACIÓN

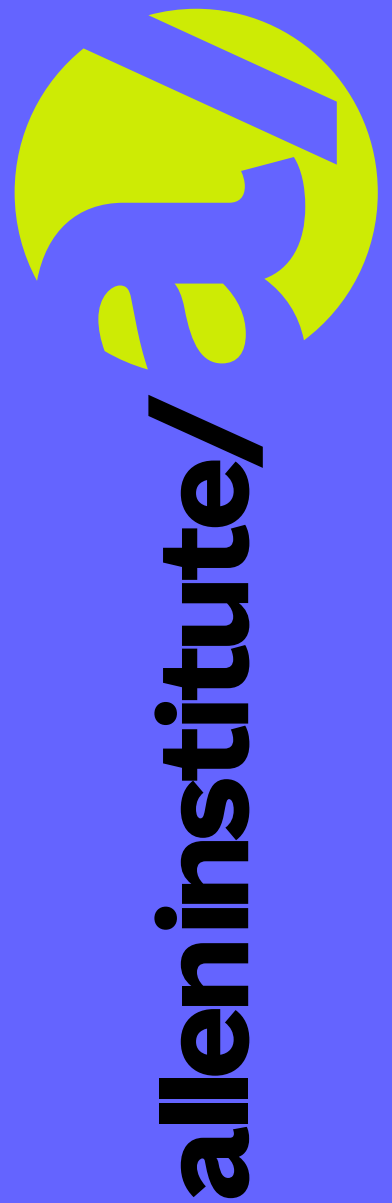
CONICET

THE PEW CHARITABLE TRUSTS

wellcome

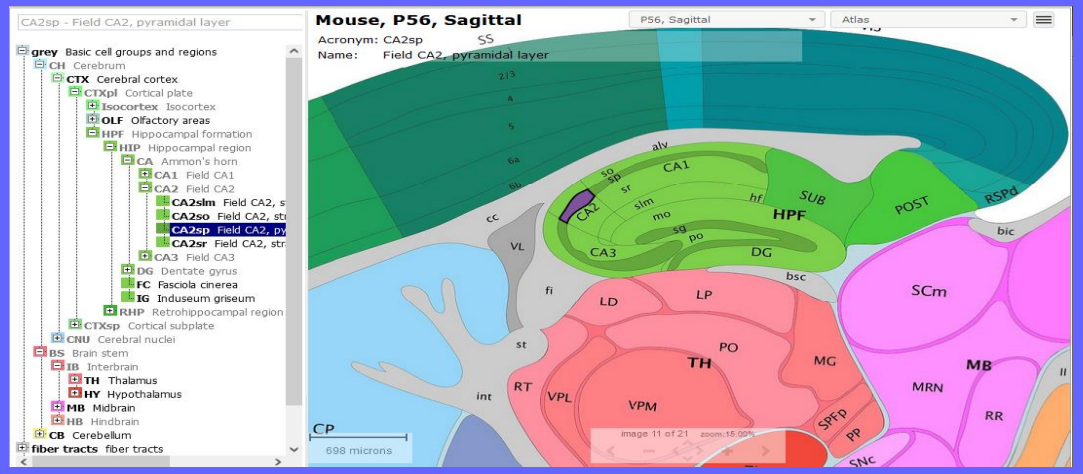
Mouse atlases overview/

Rachel Hostetler

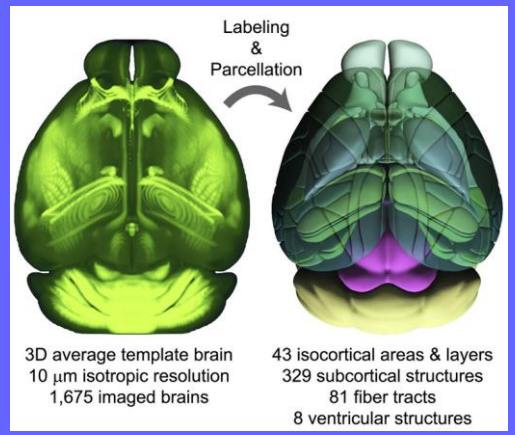


Historical Allen mouse atlases/

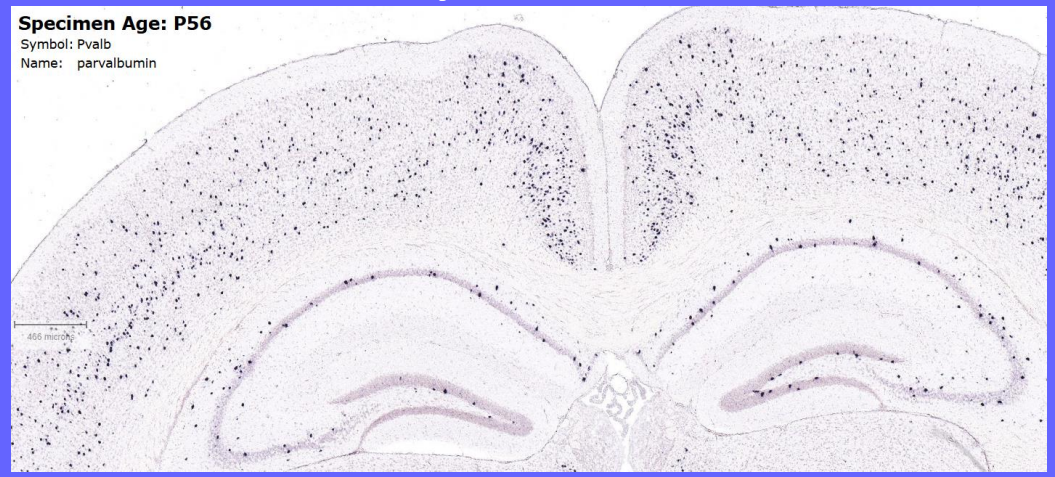
Allen Brain Reference Atlas



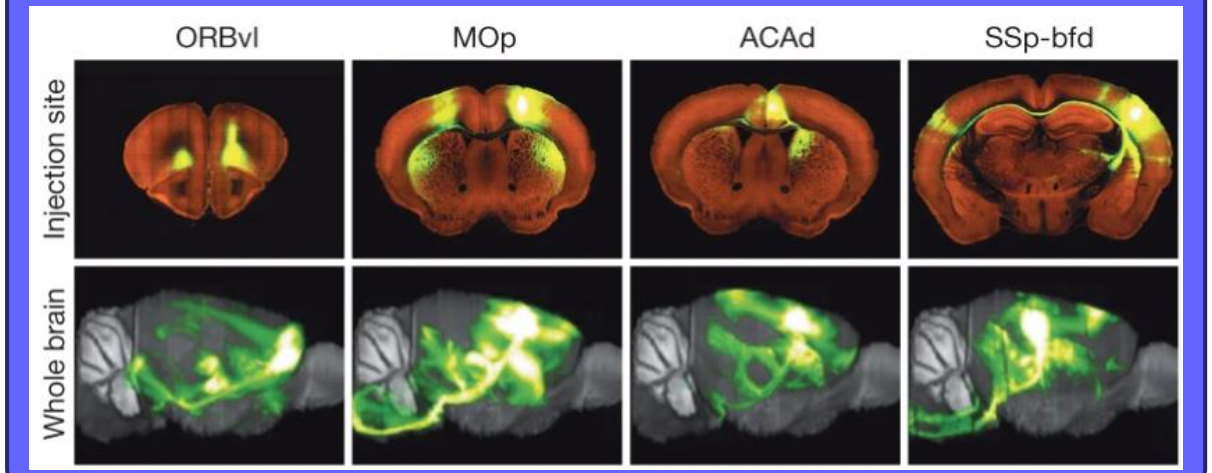
Allen Common Coordinate Framework (CCF)



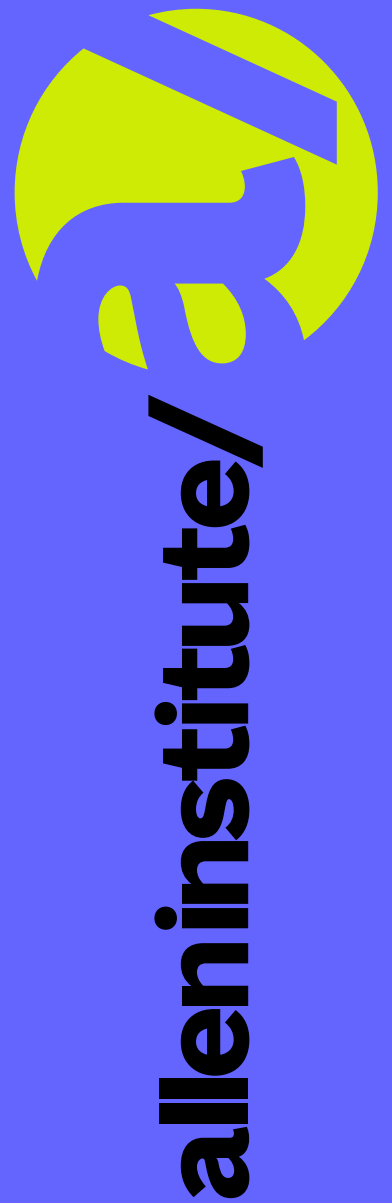
in situ hybridization Atlas



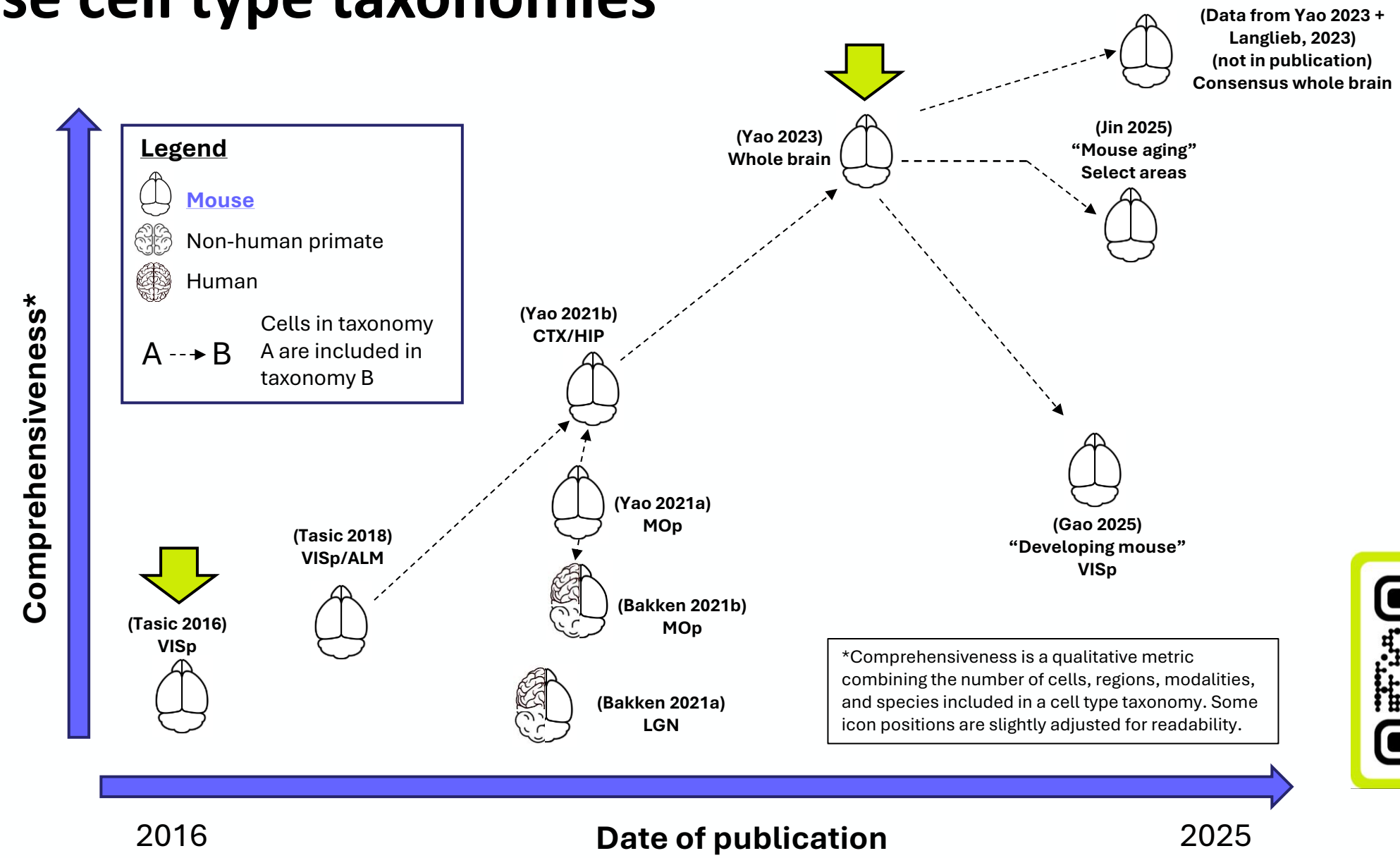
Allen Connectivity Atlas



Single cell transcriptomic atlases



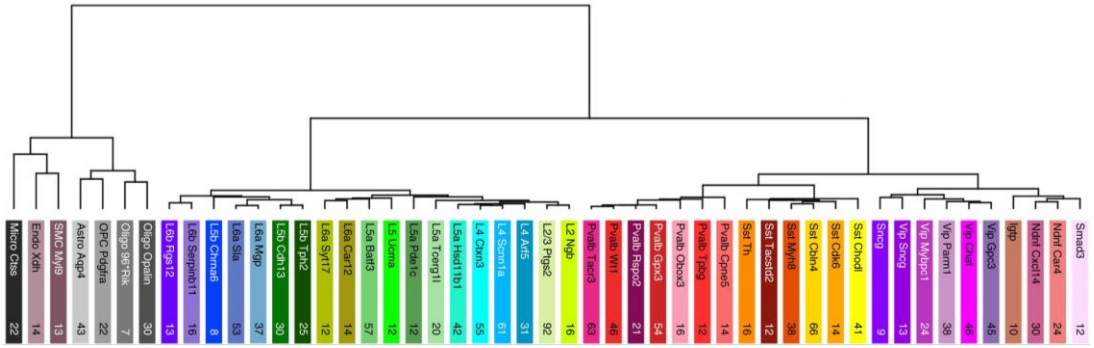
mouse cell type taxonomies



https://brain-map.org/our-research/cell-types-taxonomies/taxonomies-index

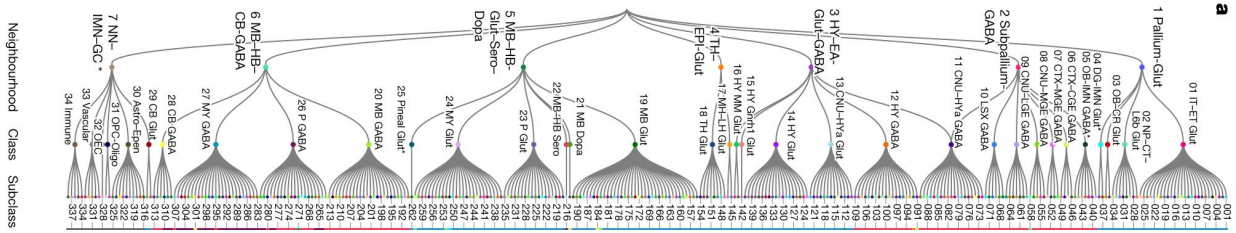
Tasic et al., 2016 → Yao et al., 2023

~ 1,600 cells
49 cell types
Visual cortex

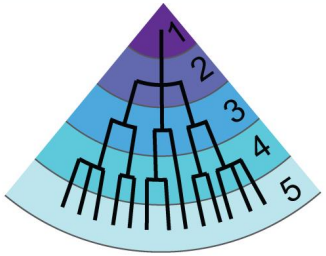


→

~4 million cells
5,322 cell types
Whole mouse brain

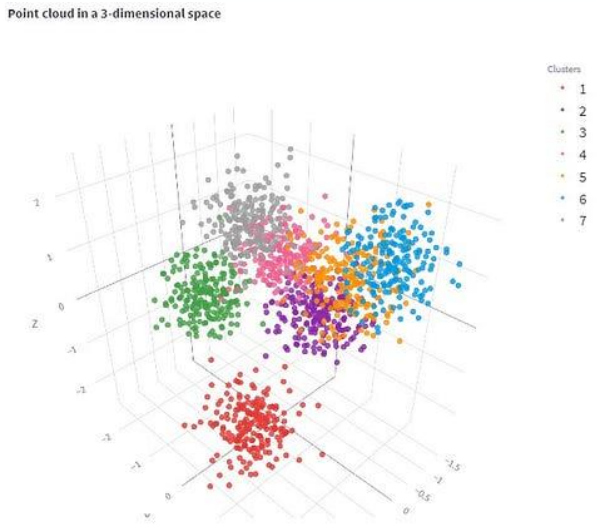


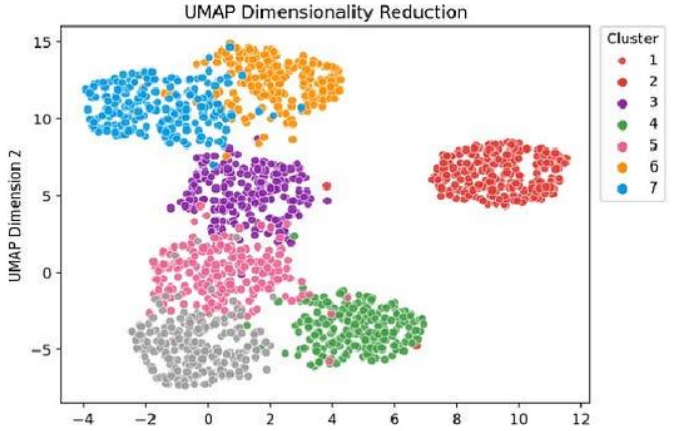
Only first 3 levels shown



- Level 1: Neighborhood (n=7)
- Level 2: Class (n=34)
- Level 3: Subclass (n=338)
- Level 4: Supertype (n=1201)
- Level 5: Cluster (n=5322)

UMAP (uniform manifold approximation and projection) plot

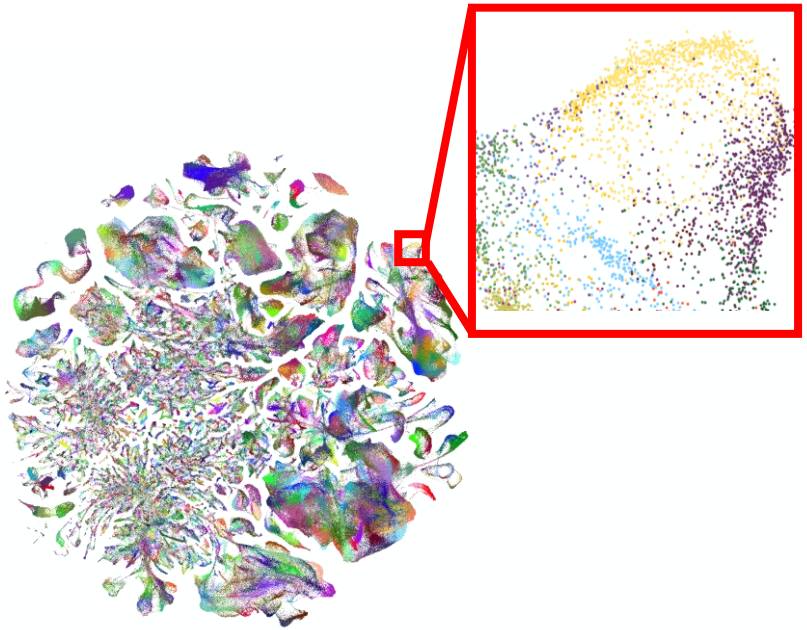




# of transcripts in each cell

	Gene 1	Gene N
Cell 1	…	…
Cell N	…	…

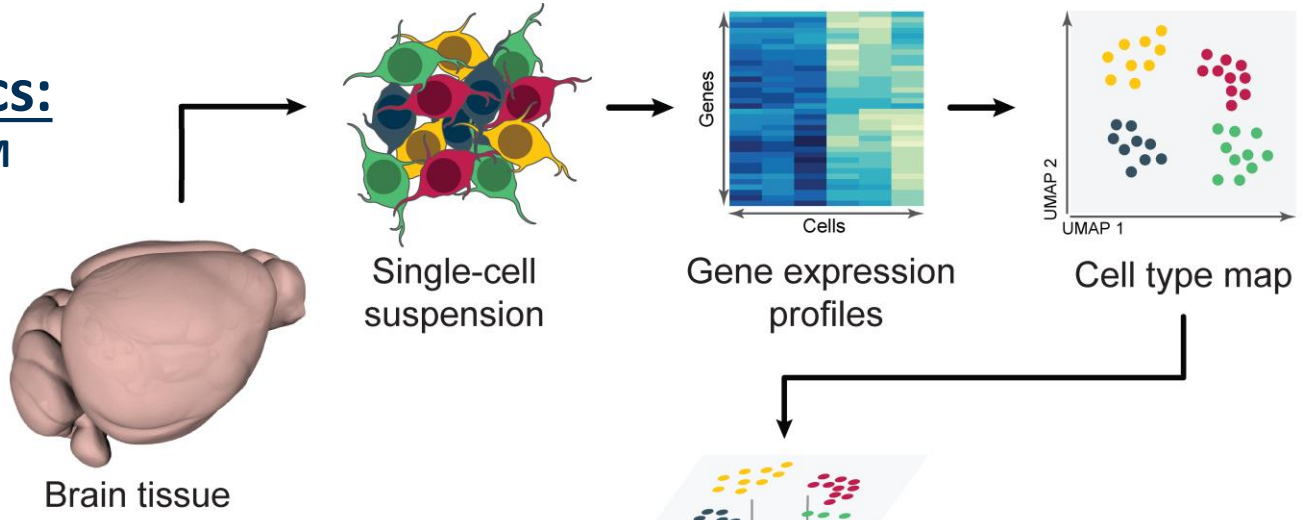
4+ million cells
30,000+ genes



whole mouse brain atlas/ combining single cell and spatial transcriptomics

Single cell transcriptomics:

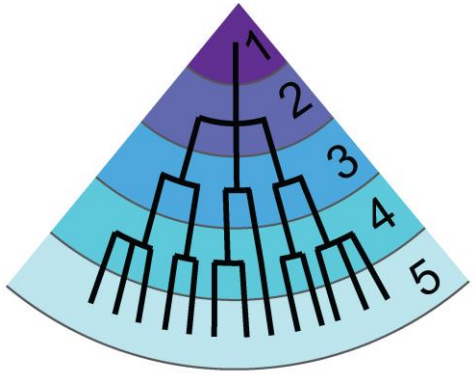
whole transcriptome profiling of ~4M individual cells



Spatial transcriptomics:

~4M cells profiled using a 500 gene panel

Hierarchical cell type taxonomy



- Level 1: Neighborhood (n=7)
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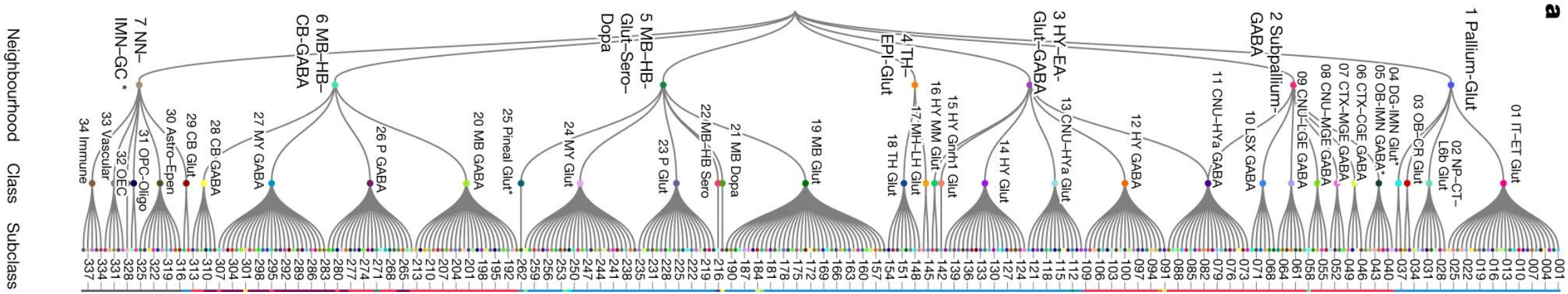
- Level 4: Supertype (n=1201)
- Level 5: Cluster (n=5322)

Yao et al., 2023

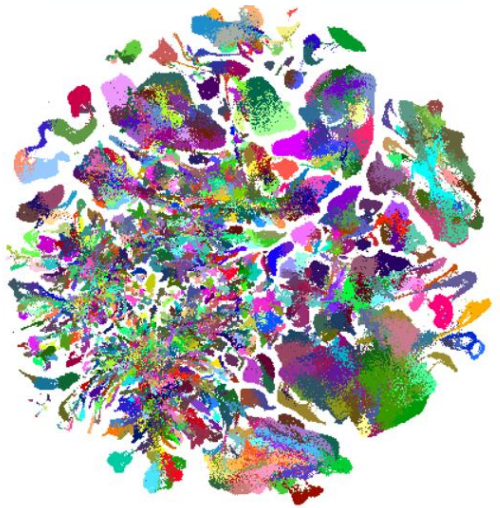
whole mouse brain taxonomy



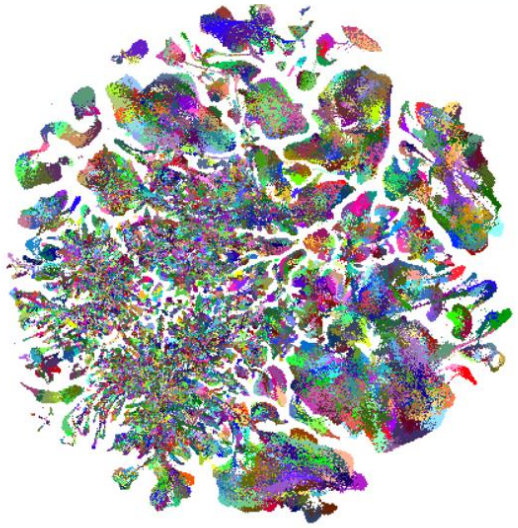
34 classes

338 subclasses

1,201 supertypes



5,322 clusters



Plots made with ABC Atlas

Yao et al., 2023

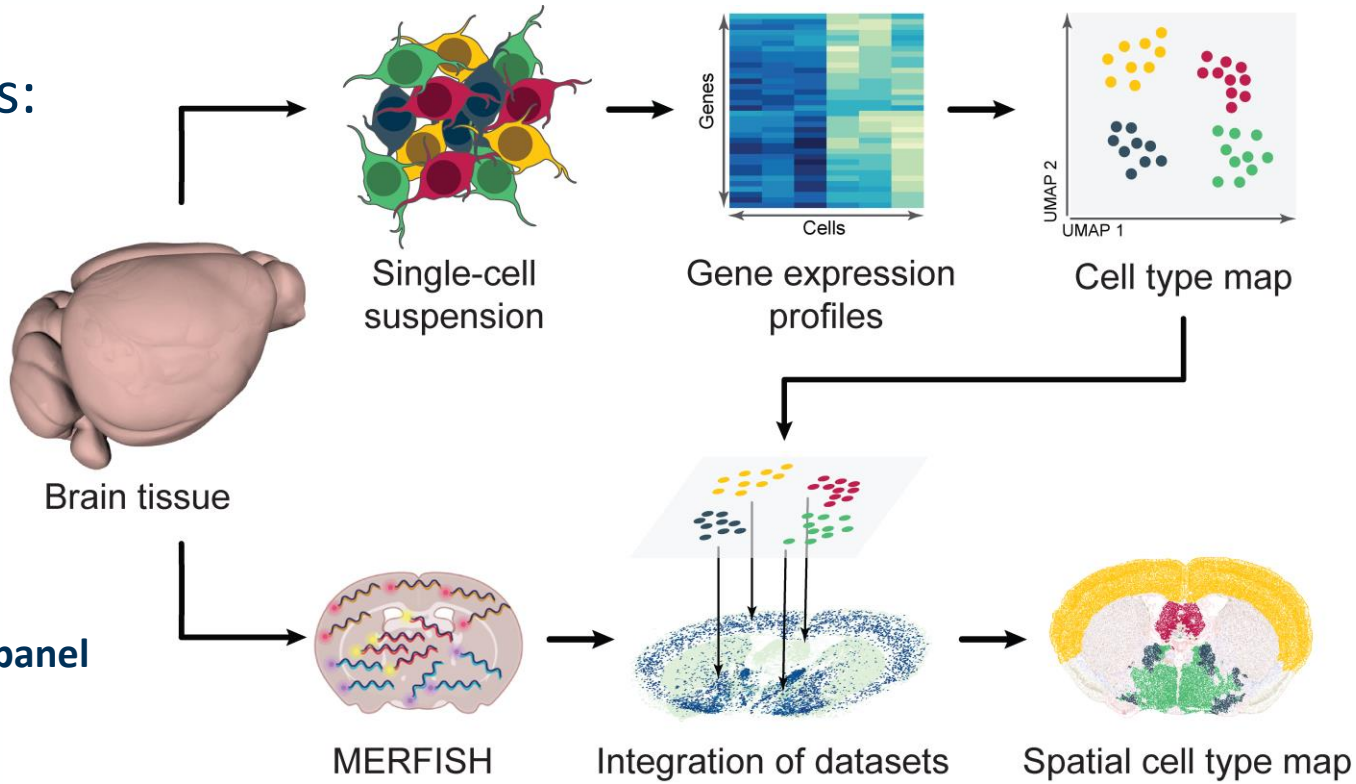
whole mouse brain atlas/ combining single cell and spatial transcriptomics

Single cell transcriptomics:

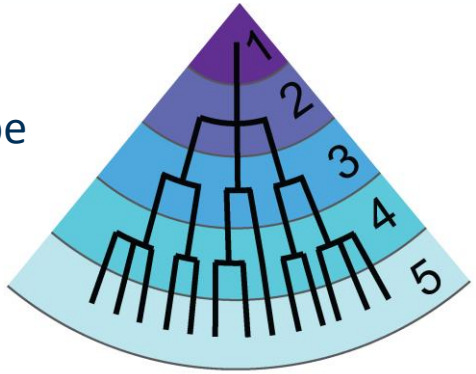
whole transcriptome profiling of ~4M individual cells

Spatial transcriptomics:

~4M cells profiled using a 500 gene panel



Hierarchical cell type taxonomy



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- Level 3: Subclass (n=338)
- Level 4: Supertype (n=1201)
- Level 5: Cluster (n=5322)

Yao et al., 2023

MERFISH (multiplexed error robust *in situ* hybridization)

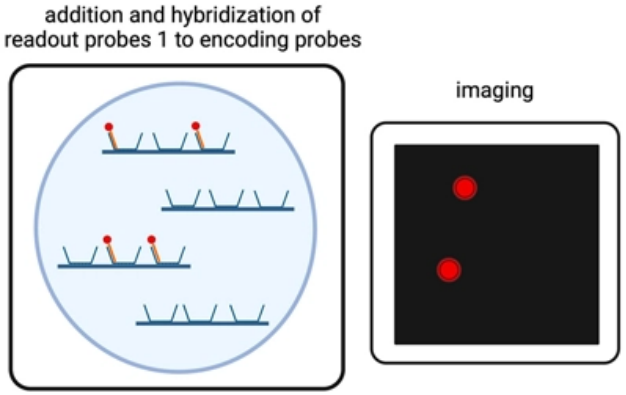
1. Gene panel design

Each gene gets:

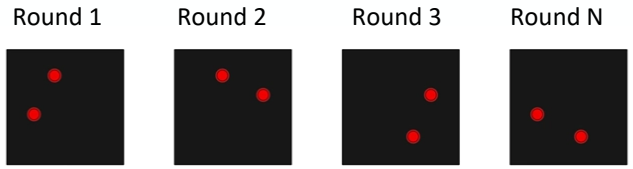
Probe =

& Barcode = 0100110…..1

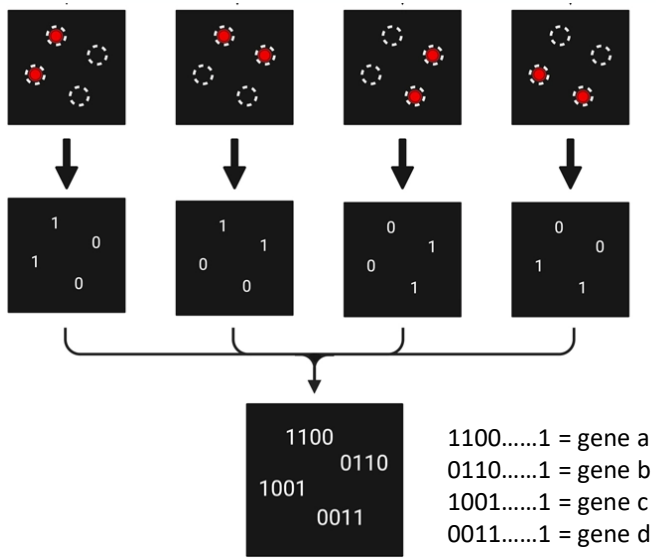
2. ISH and imaging



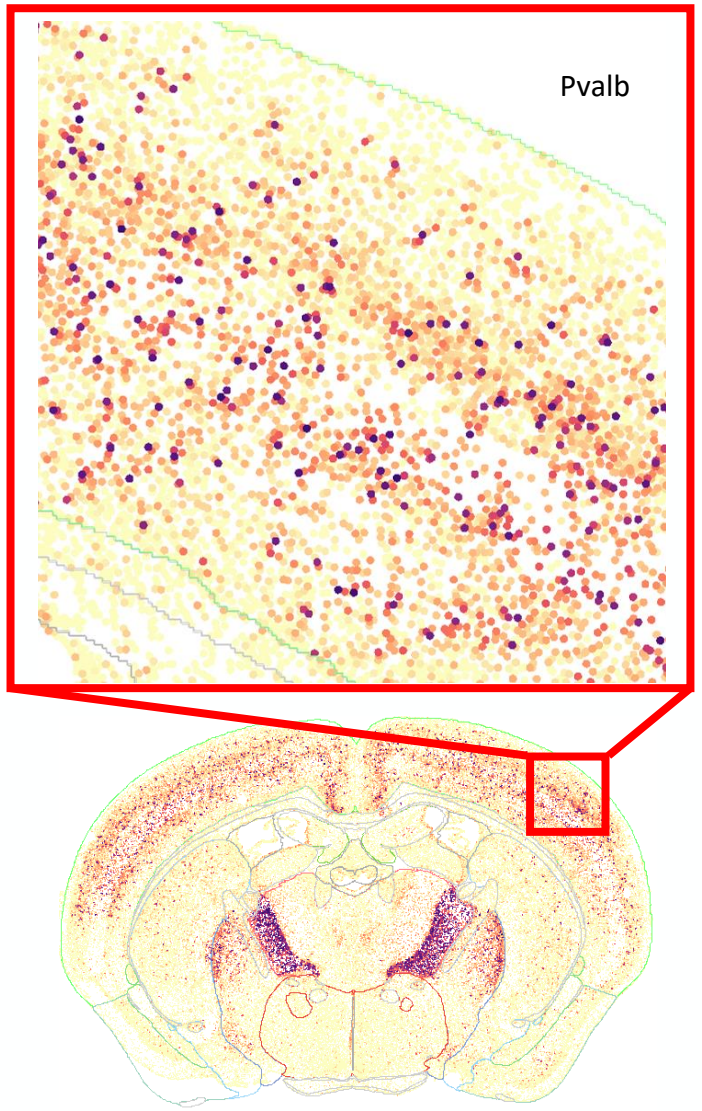
3. Sequential imaging



4. Barcode identification



Yao et al., 2023



x 500 genes

Step 1-4 figures adapted from Fortner and Bucur, 2024

MERFISH (multiplexed error robust *in situ* hybridization)

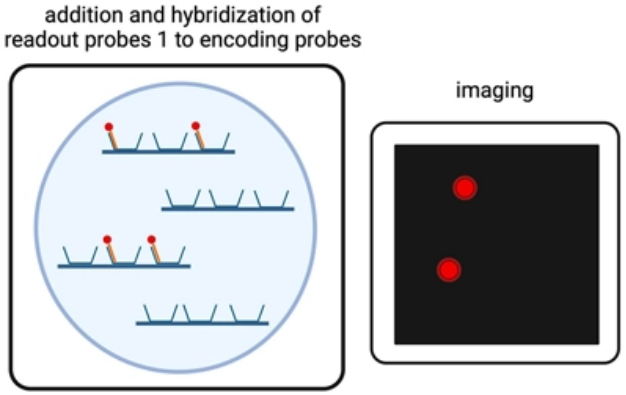
1. Gene panel design

Each gene gets:

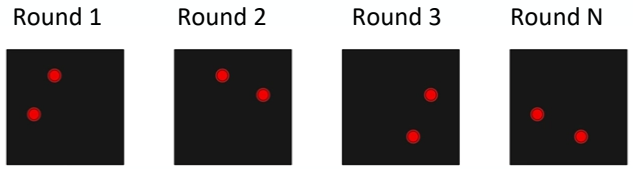
Probe =

& Barcode = 0100110…..1

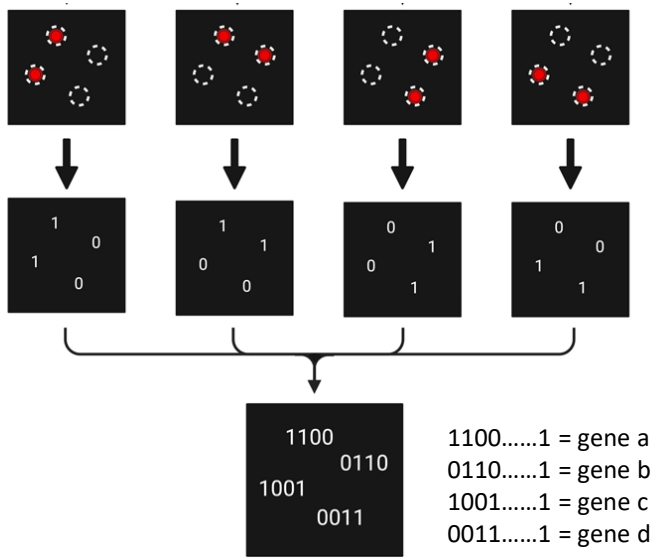
2. ISH and imaging



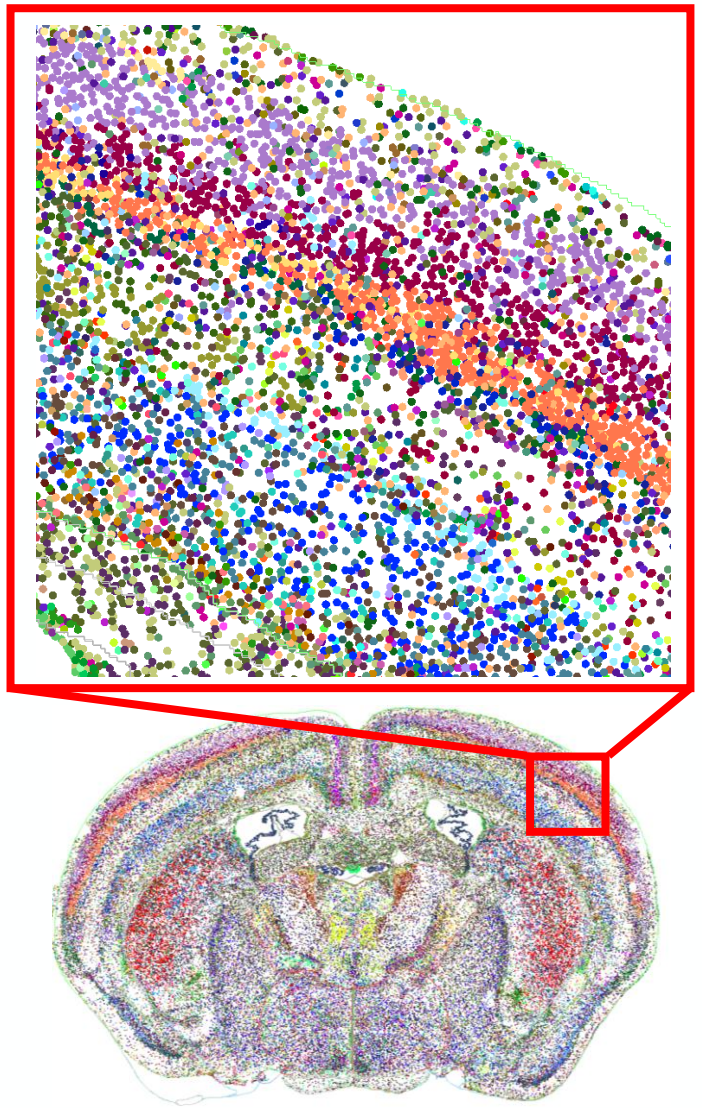
3. Sequential imaging



4. Barcode identification



Yao et al., 2023



5,322 cell types

Step 1-4 figures adapted from Fortner and Bucur, 2024

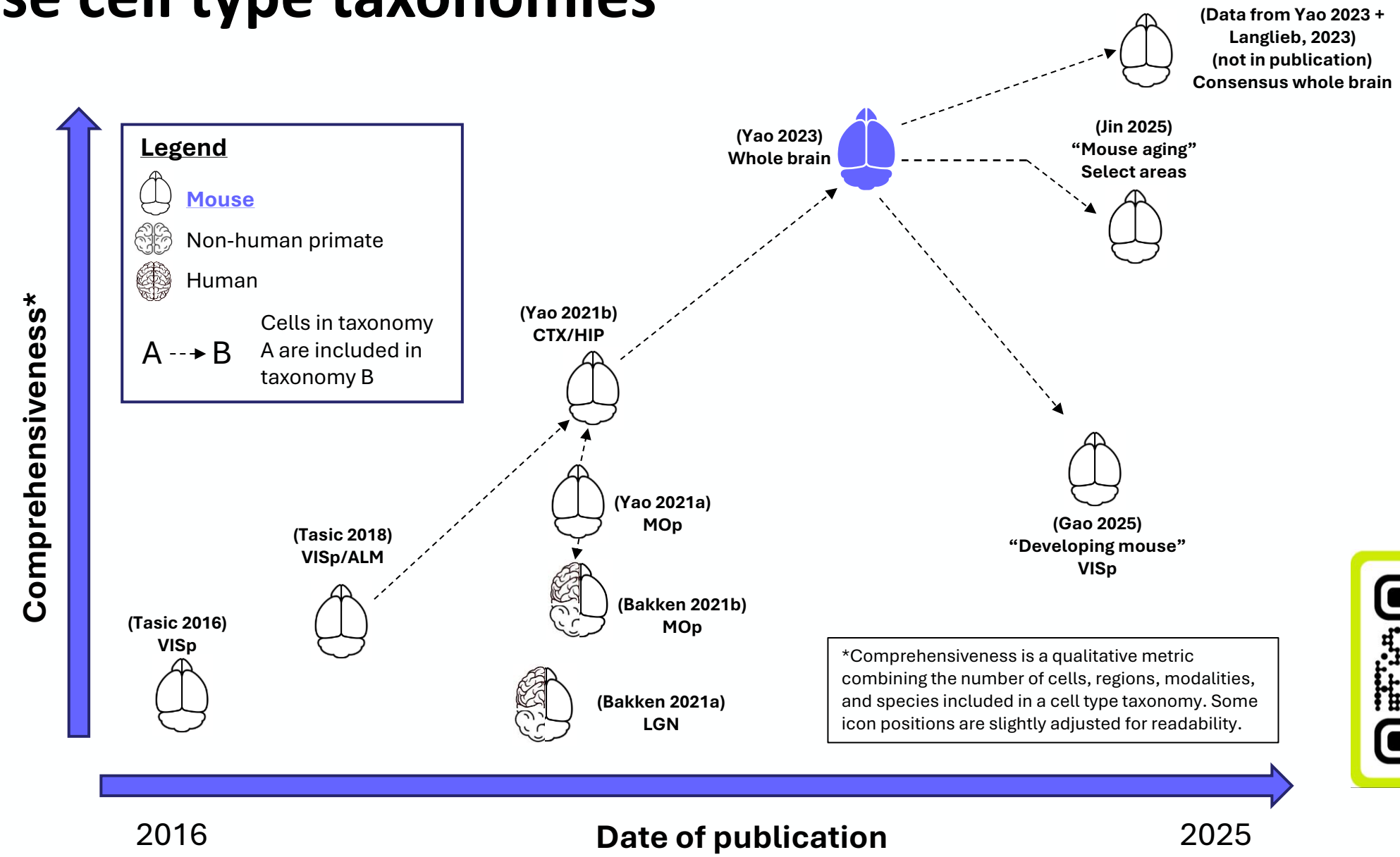
whole mouse brain data

Data	Paper	ABC Atlas Dataset Name	# of genes	# of donors	Plane of Section
10X scRNAseq Mouse Whole-Brain Transcriptomic Cell Type Atlas – Hongkui Zeng, Allen Institute	Yao et al., 2023 – Nature	"10x ScRNAseq whole brain"	31,113	317	
MERFISH Zeng Mouse Whole-Brain Transcriptomic Cell Type Atlas – Hongkui Zeng, Allen Institute	Yao et al., 2023 – Nature	"MERFISH-C57BL6J-638850" "MERFISH-C57BL6J-638850 with Imputed genes"	500 8460 (scRNAseq)	1	Coronal
MERFISH Zhuang MERFISH Whole Mouse Brain – Xiaowei Zhuang, Harvard	Zhang et al., 2023 – Nature	"Zhuang-ABCA-1/2/3/4"	1,122	4	Coronal and Sagittal

Both MERFISH datasets were integrated with the 10x scRNAseq data and have the same taxonomy in the ABC Atlas (34 classes, 338 subclasses, 1,201 supertypes and 5,322 clusters)

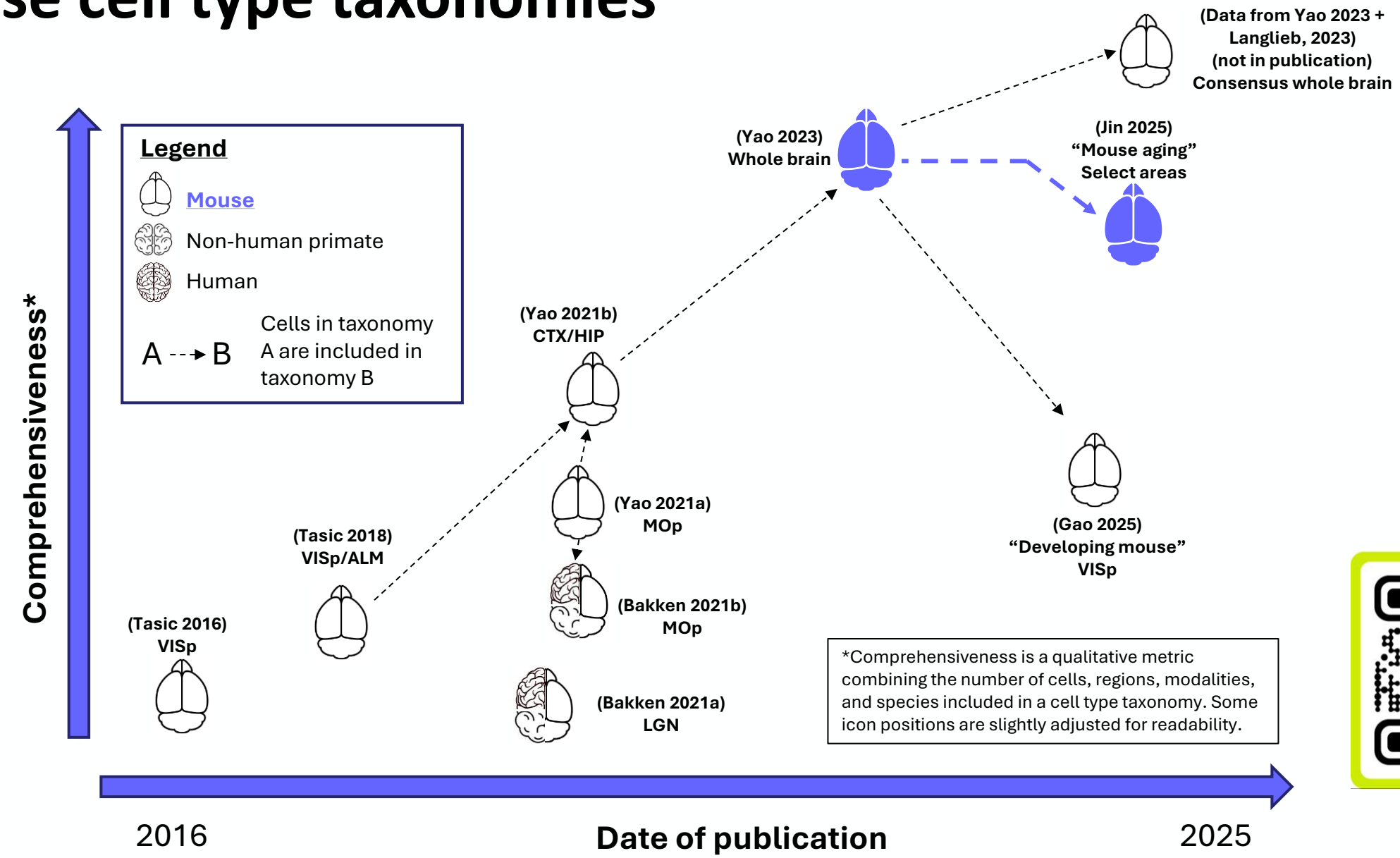
Yao and Zhang papers are part of a 10-paper Nature package, published December 2023 nature.com/collections/fgihbeccbd

mouse cell type taxonomies



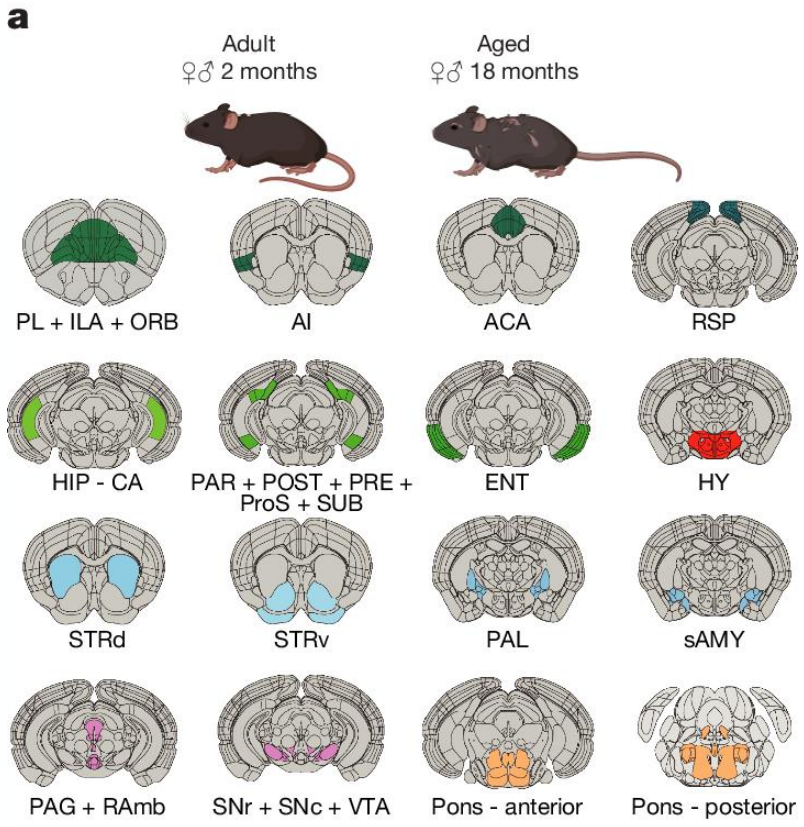
https://brain-map.org/our-research/cell-types-taxonomies/taxonomies-index

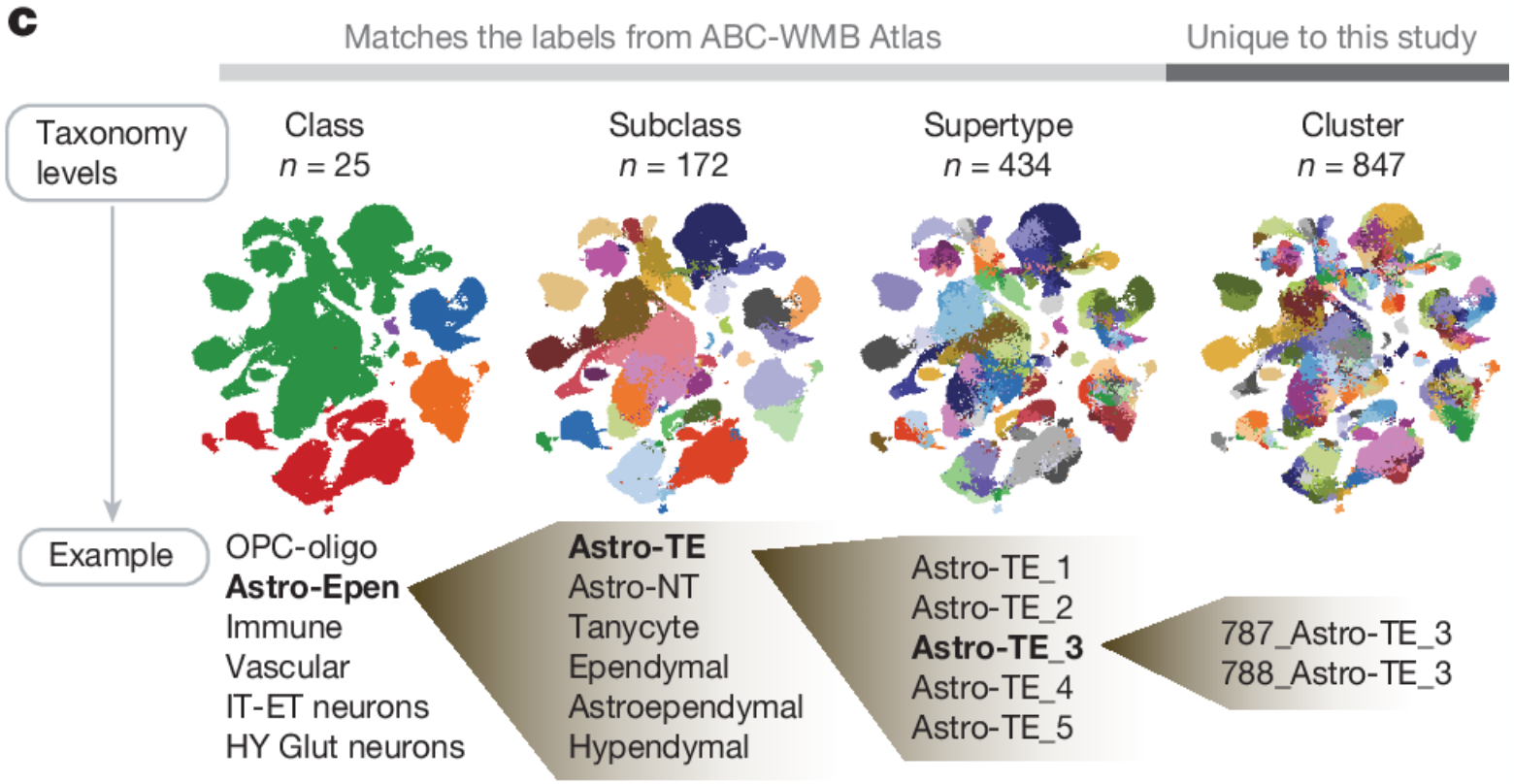
mouse cell type taxonomies



https://brain-map.org/our-research/cell-types-taxonomies/taxonomies-index

aging mouse taxonomy



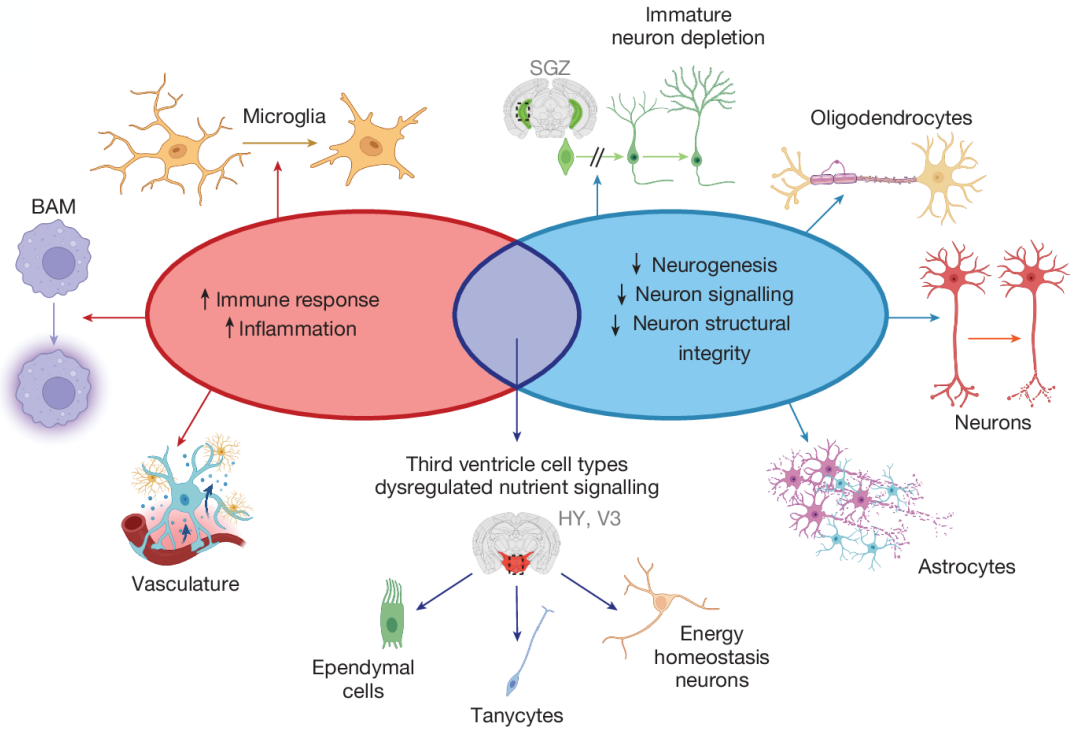


Jin et al., 2025

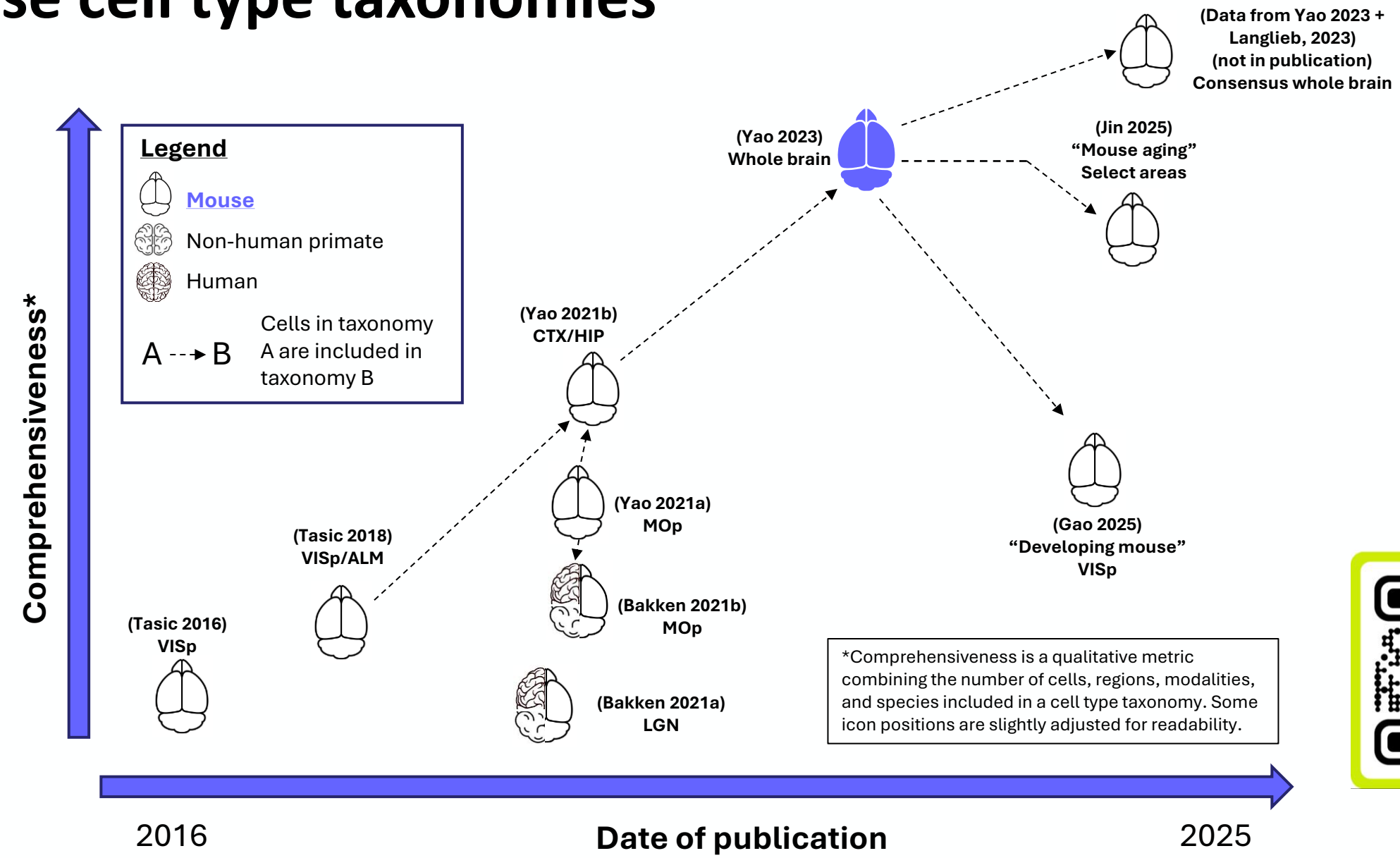
Class, subclass, and supertypes mapped to Whole Mouse Brain Taxonomy

aging mouse data

Data	Paper	ABC Atlas Dataset Name	# of genes	# of donors
10x scRNAseq Cellular and Molecular Characterization of the Aged Mouse Brain	Jin et al., 2025 – Nature	"10x-scRNAseq-aged-adult"	30,824	44 aged (18 months), 64 young adult mice (from whole mouse brain data Yao et al., 2023)

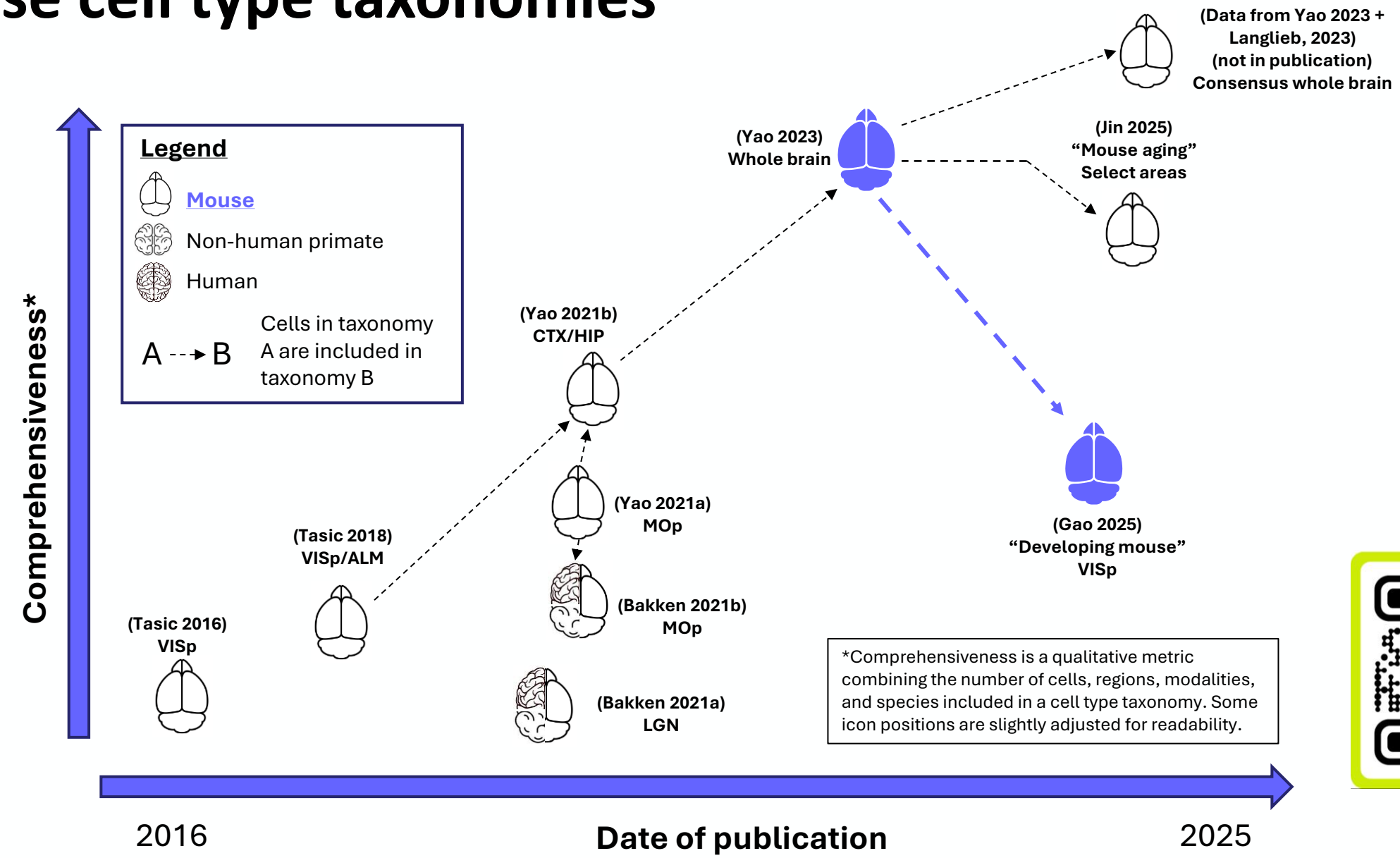


mouse cell type taxonomies



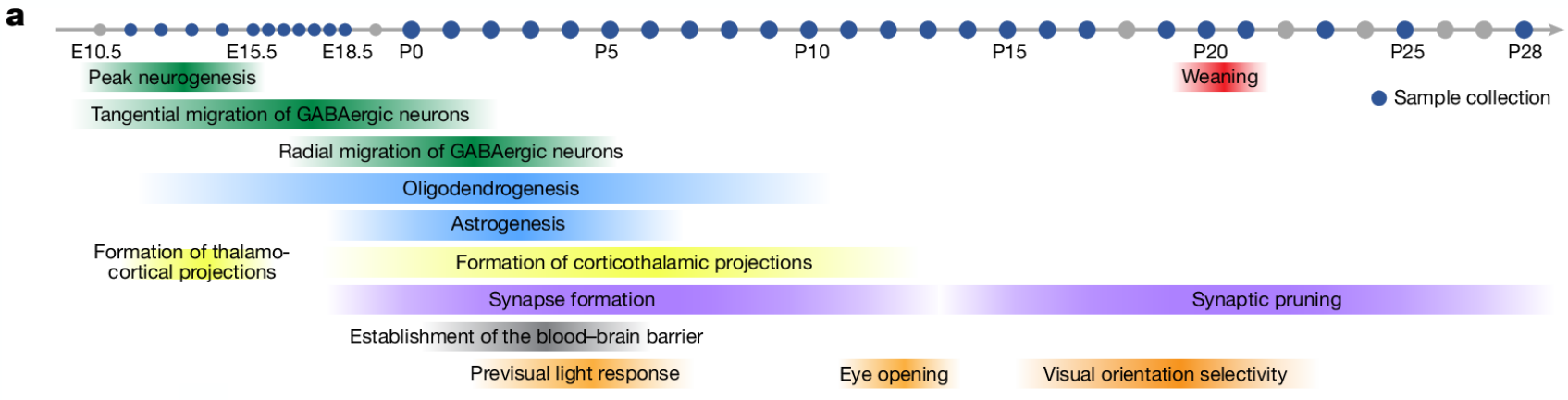
https://brain-map.org/our-research/cell-types-taxonomies/taxonomies-index

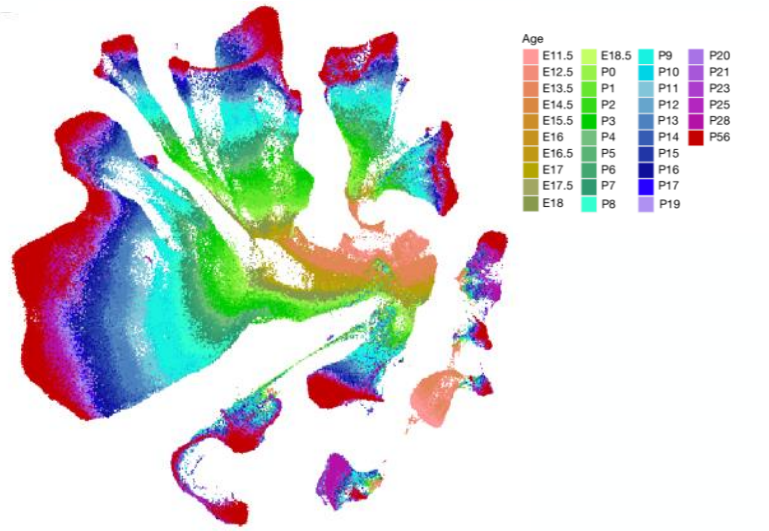
mouse cell type taxonomies

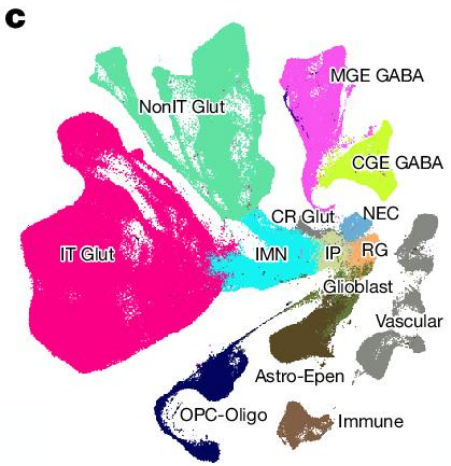


https://brain-map.org/our-research/cell-types-taxonomies/taxonomies-index

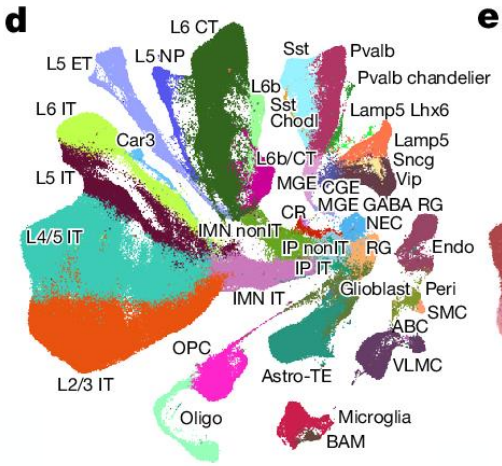
developing mouse visual cortex taxonomy



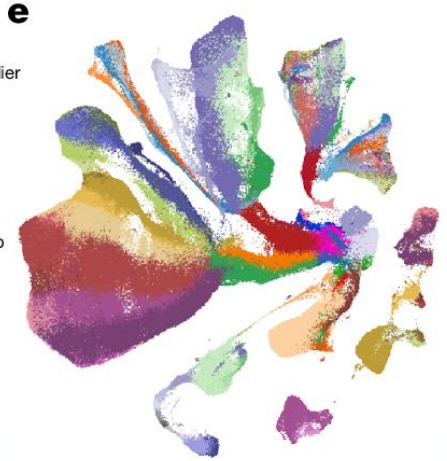




15 classes



40 subclasses

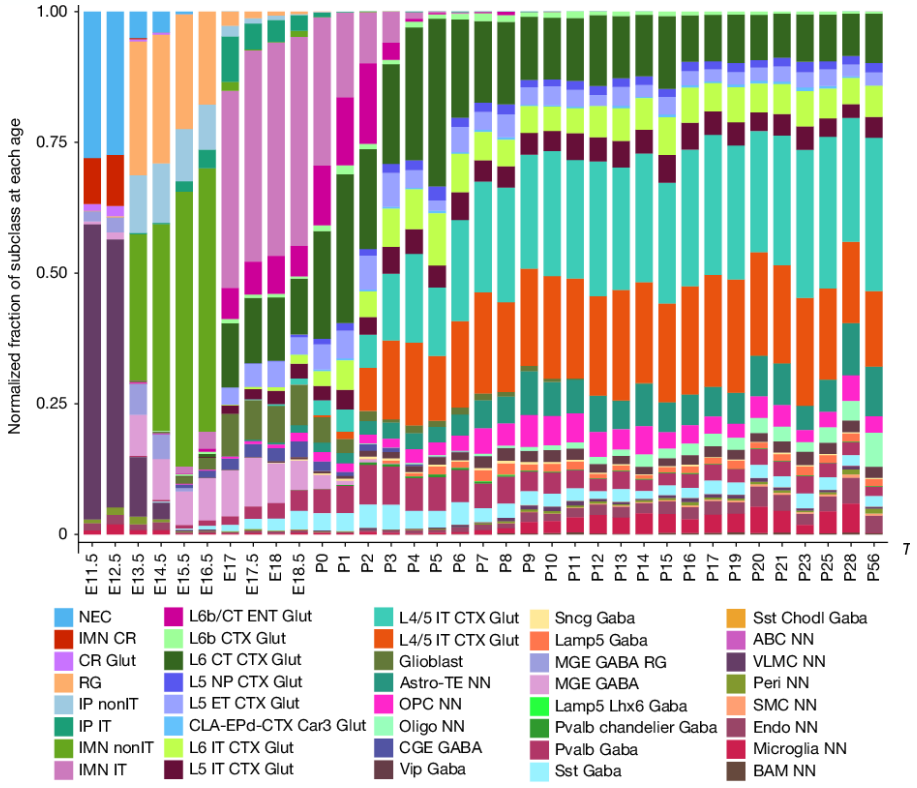


148 clusters

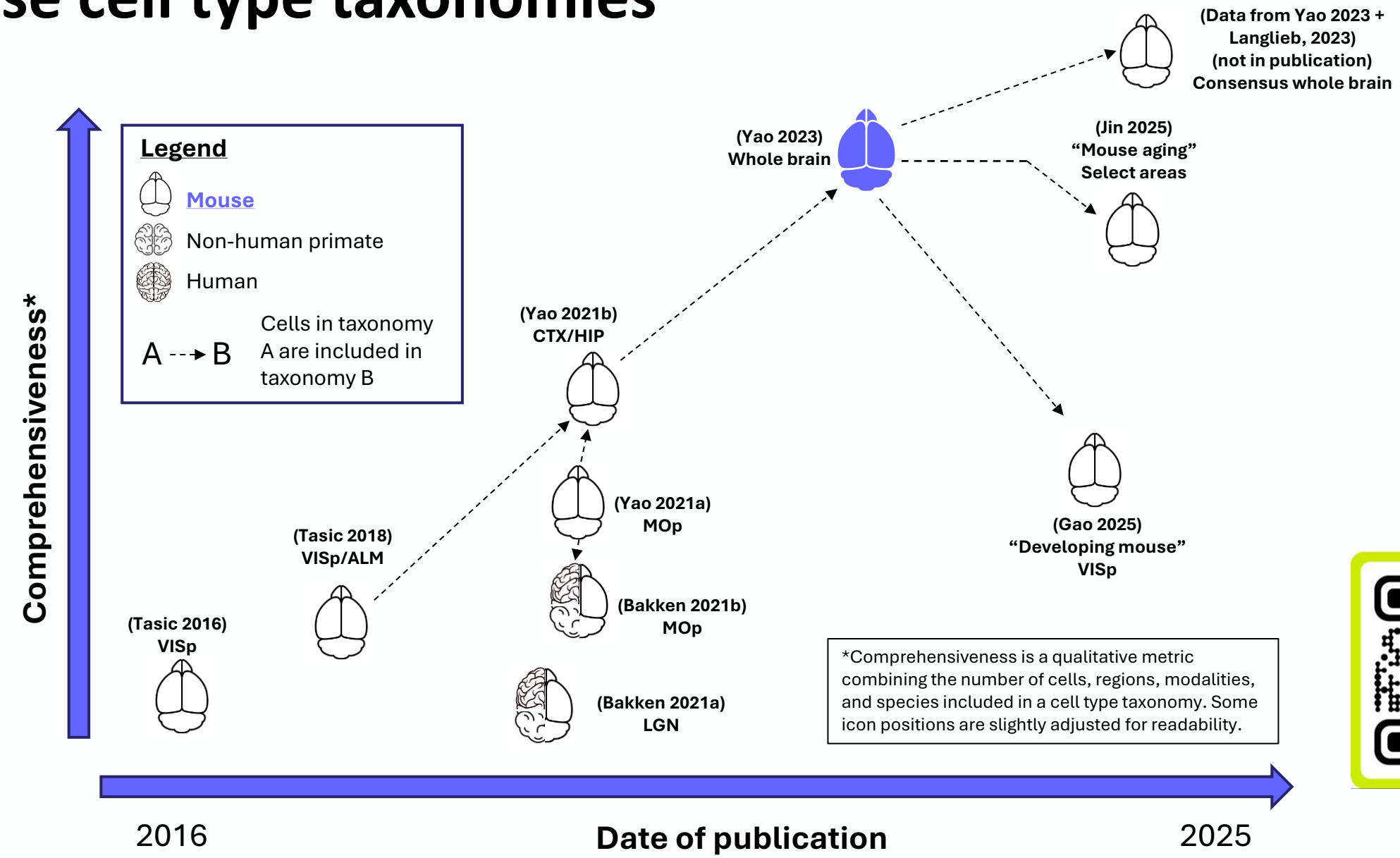
714 subclusters

developing mouse visual cortex data

Data	Paper	ABC Atlas Dataset Name	# of genes	# of donors
10x scRNAseq Developing Mouse - Visual Cortex	Gao et al., 2025 – Nature	“Dev-Mouse-Vis-Cortex-taxonomy”	32,285	53 (from E11.5 to young adult P56 (from WMB Zeng data))

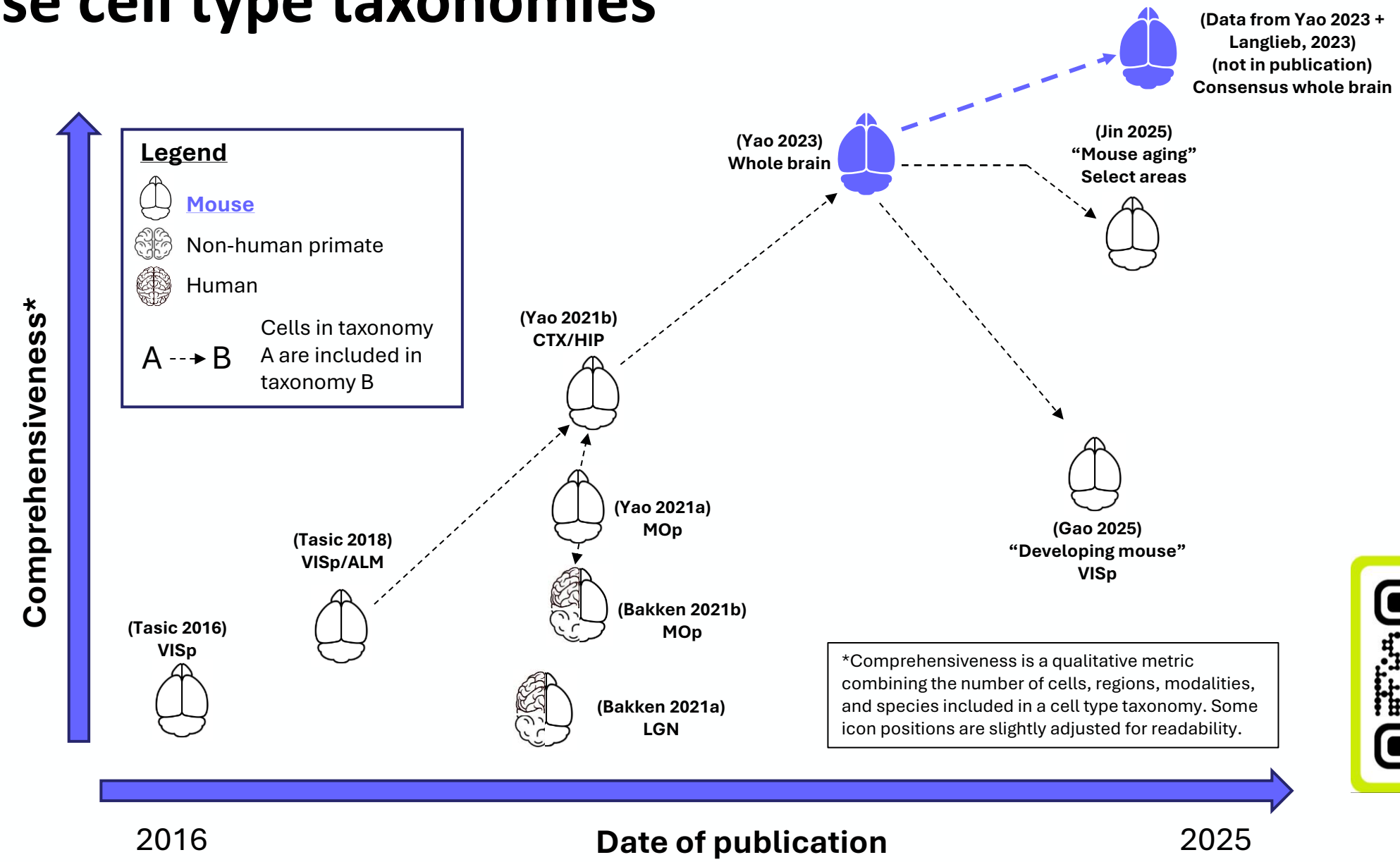


mouse cell type taxonomies



https://brain-map.org/our-research/cell-types-taxonomies/taxonomies-index

mouse cell type taxonomies

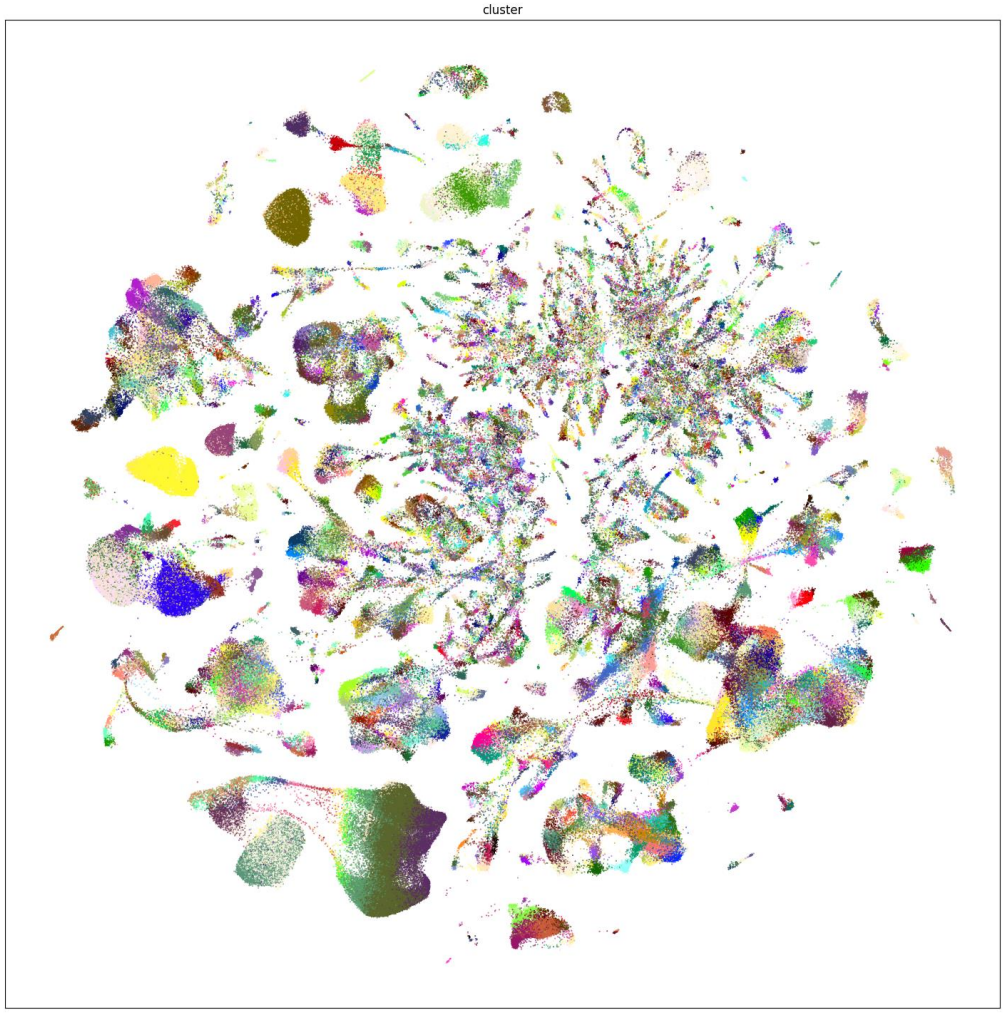


https://brain-map.org/our-research/cell-types-taxonomies/taxonomies-index

consensus whole mouse brain

43 classes
414 subclasses
1,386 supertypes
6,721 clusters

~7 million cells



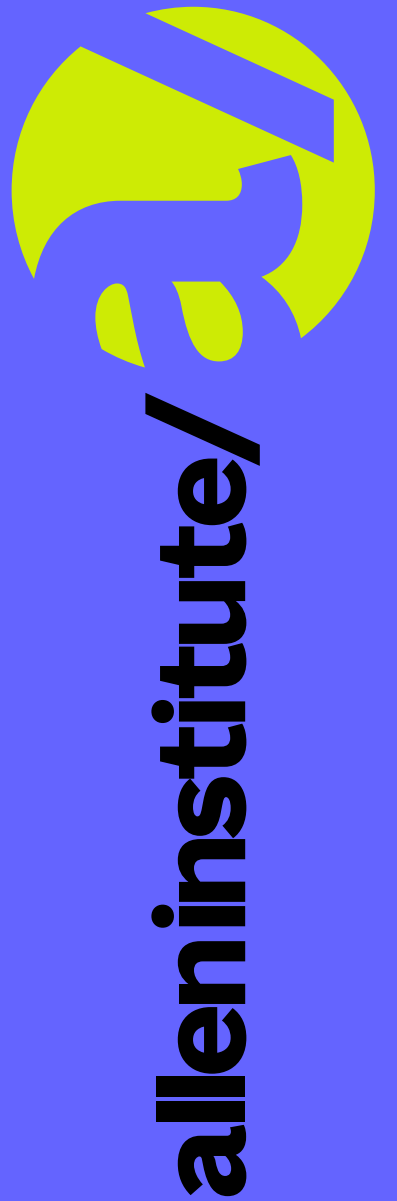
WMB Zeng **WMB Macosko** **6,721 clusters**

consensus whole mouse brain

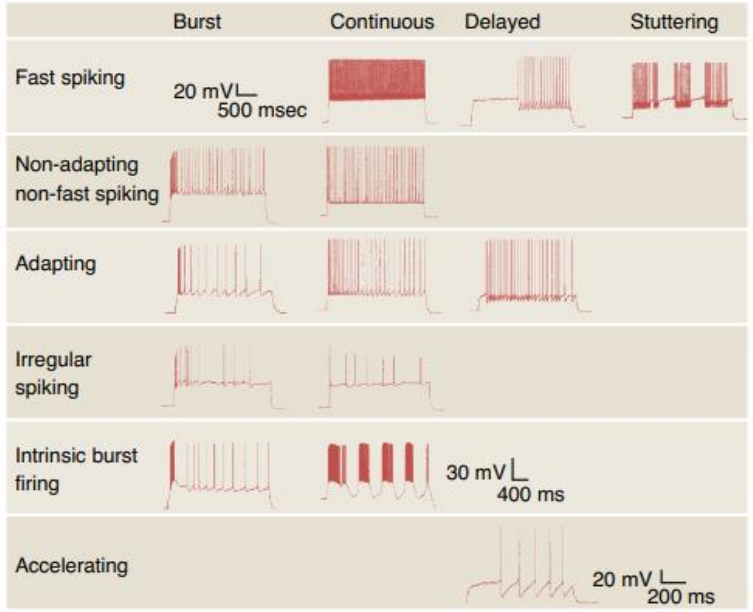
Data	Paper	ABC Atlas Dataset Name	# of genes	# of donors
10x scRNAseq Mouse Whole-Brain Transcriptomic Cell Type Atlas – Hongkui Zeng, Allen Institute	Yao et al., 2023 – Nature	"10x ScRNAseq whole brain"	31,113	317
Consensus Whole Mouse Brain Macosko (snRNAseq)	Langlieb et al., 2023	N/A (coming soon!)	21,205	55

Yao and Langlieb papers are part of a 10-paper Nature package, published December 2023
nature.com/collections/fgihbeccbd

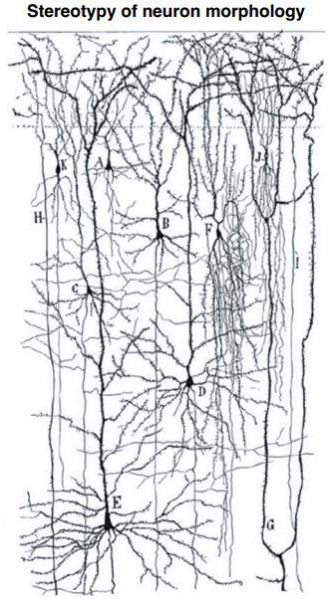
Mouse patch-seq datasets



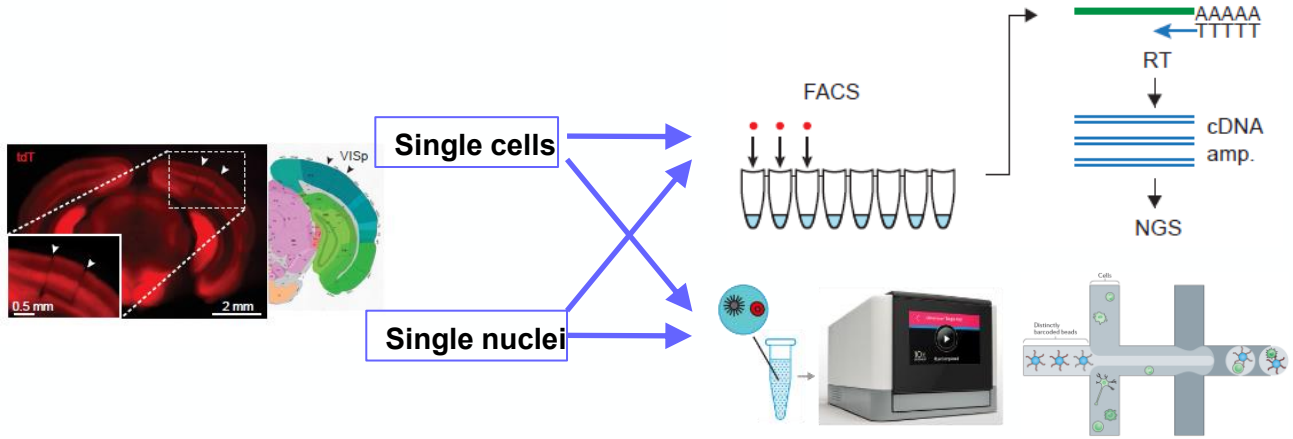
multi-modal characterization



Ascoli et al (2008)



Ramón y Cajal (1904),
Yuste et al (2020)

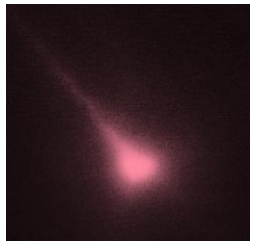


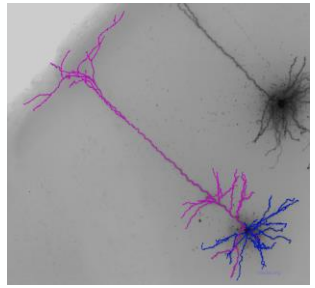
SMART-Seq v4
(full-length, deep sequencing, no multiplexing)

10X Chromium
(Droplet, 3' end, shallow sequencing, multiplexing)

patch-seq steps

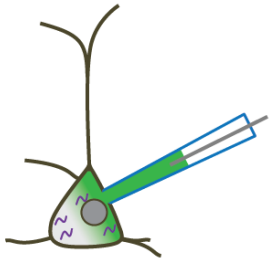
patch-seq in real life

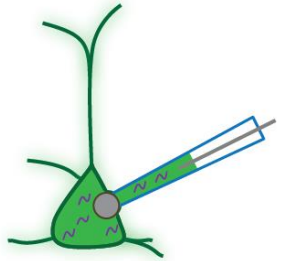


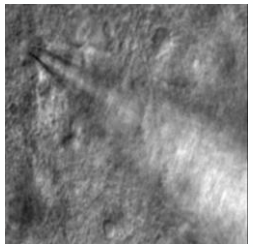


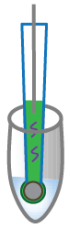
Biocytin imaging and morphological analysis

Patch clamp recording

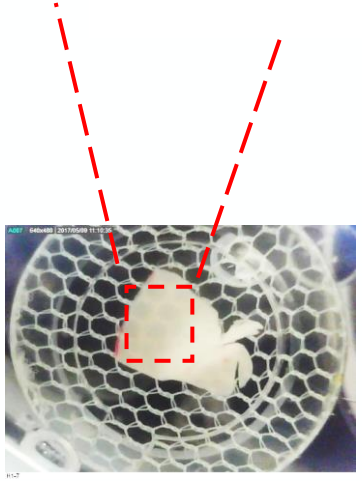








Cytosol/nuclear extraction



~300 um thick brain slices

Ephys feature collection

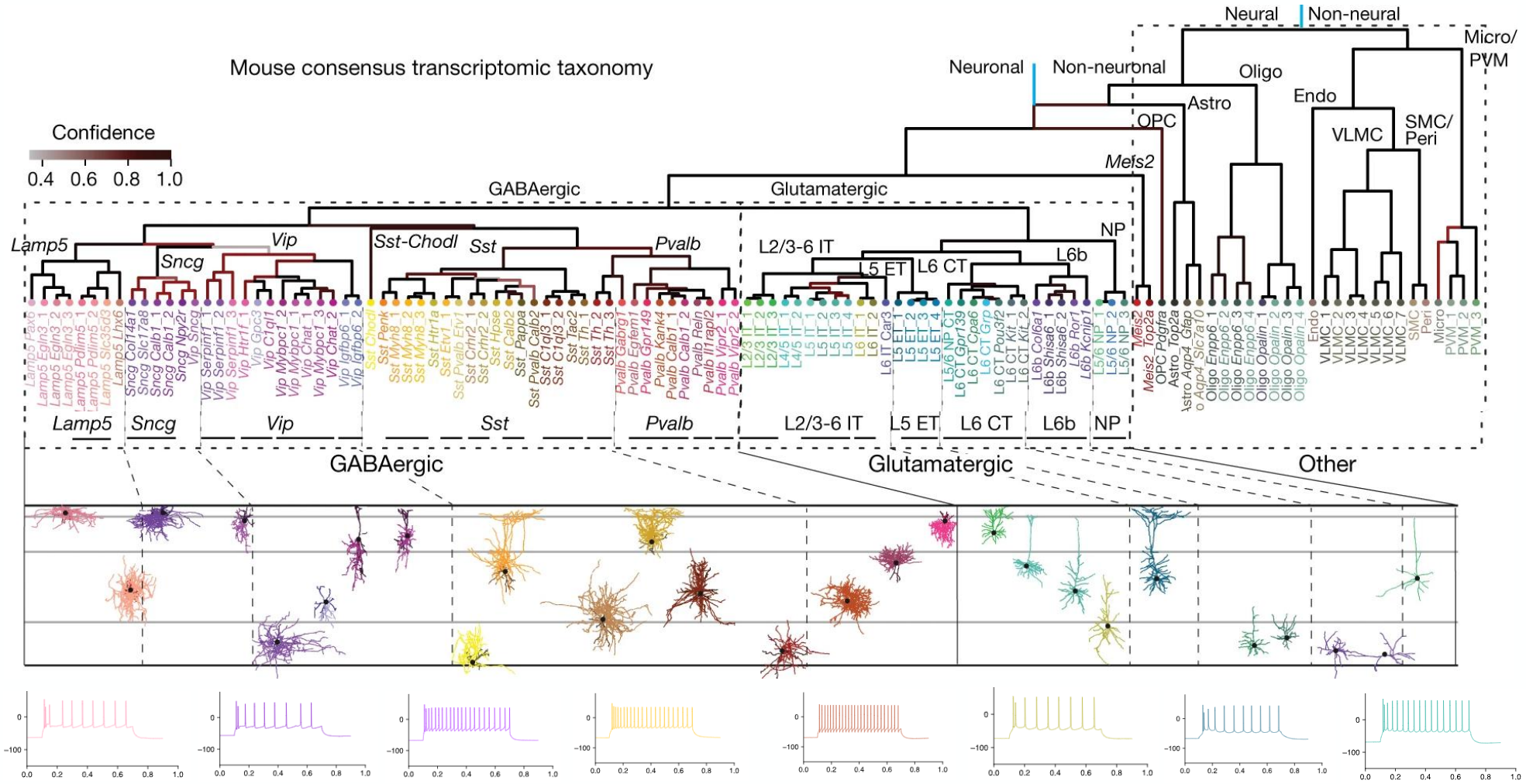
look in the labs/
patch-seq pipeline

patch-seq mouse motor cortex taxonomy

Gene Expression

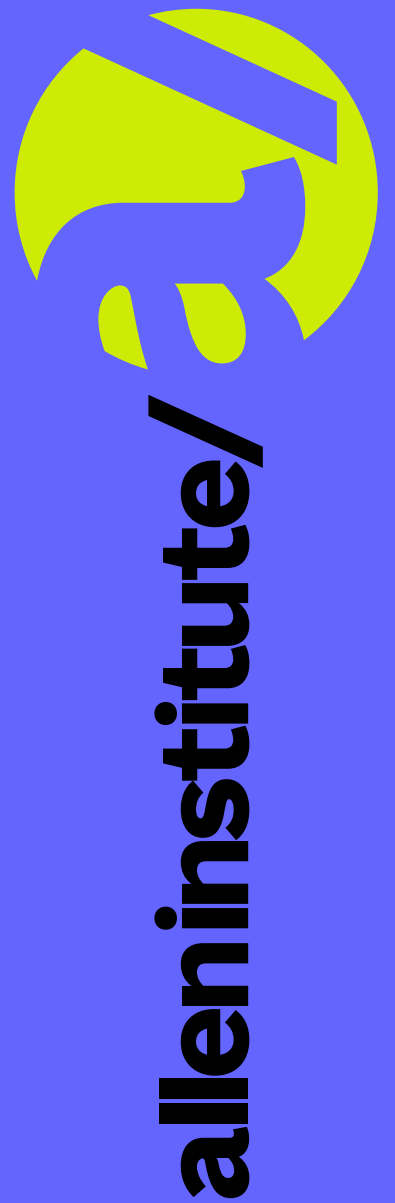
Morphology

Electrophysiology



BICCN et al (2021)

Data access



Is there a viewer for data of…

Mouse, Healthy, Adult

	Isocortex	Olfactory areas	Hippo-campal formation	Cortical subplate	Striatum	Pallidum	Thalamus	Hypo-thalamus	Midbrain	Pons	Medulla	Cere-bellum	Fiber tracts
Spatial Transcriptomics	X	X	X	X	X	X	X	X	X	X	X	X	X
scRNAseq	X	X	X	X	X	X	X	X	X	X	X	X	X
PatchSeq	VIS & MO only												

Mouse, Healthy, Aged

	Isocortex	Olfactory areas	Hippo-campal formation	Cortical subplate	Striatum	Pallidum	Thalamus	Hypo-thalamus	Midbrain	Pons	Medulla	Cere-bellum	Fiber tracts
scRNAseq	X		X	X	X	X	X	X	X	X			

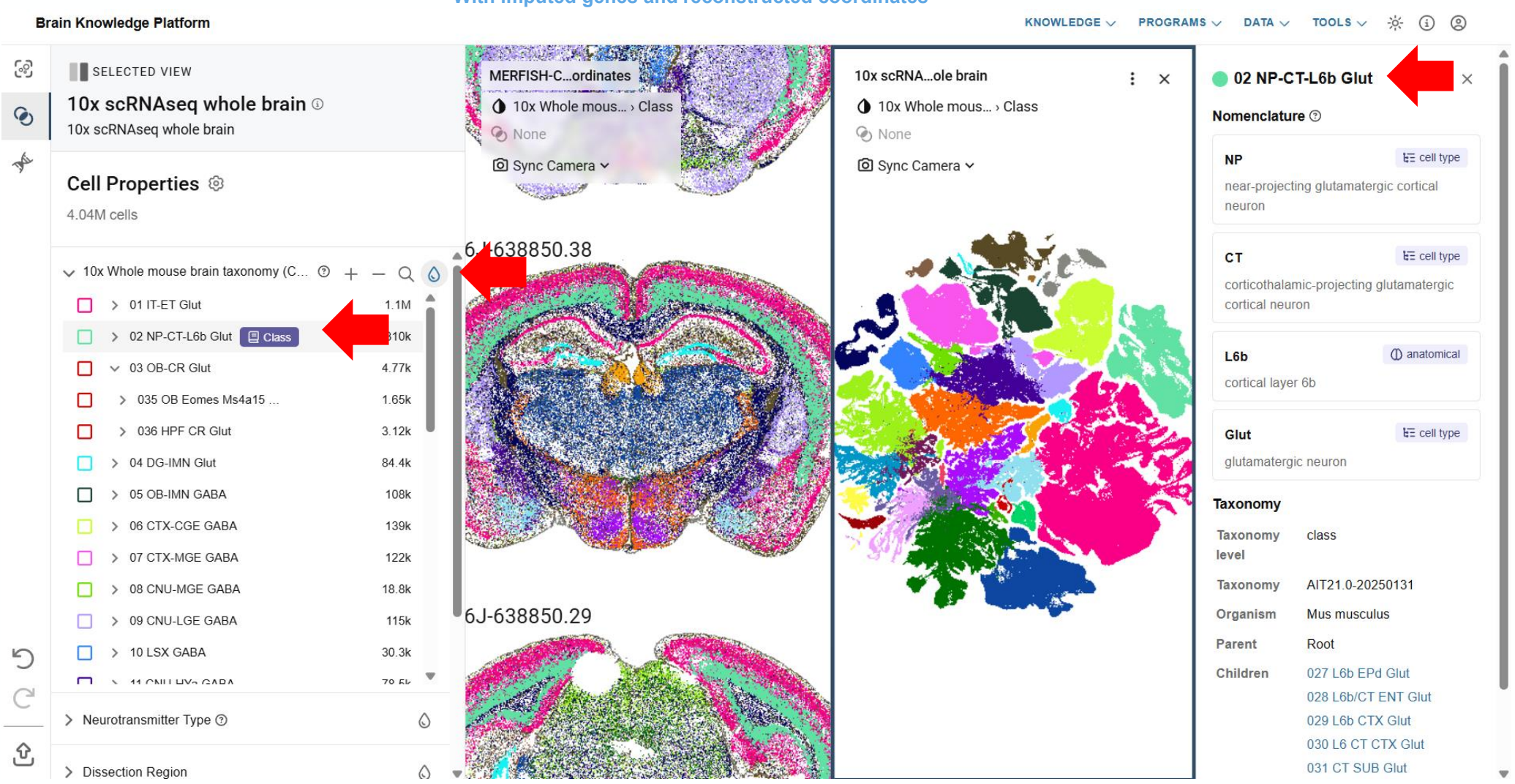
Mouse, Healthy, Developing

	Isocortex	Olfactory areas	Hippo-campal formation	Cortical subplate	Striatum	Pallidum	Thalamus	Hypo-thalamus	Midbrain	Pons	Medulla	Cere-bellum	Fiber tracts
scRNAseq	VIS Only												

Using the Allen Brain Cell Atlas

MERFISH-C57BL6J-638850
With imputed genes and reconstructed coordinates

10x ScRNAseq whole brain

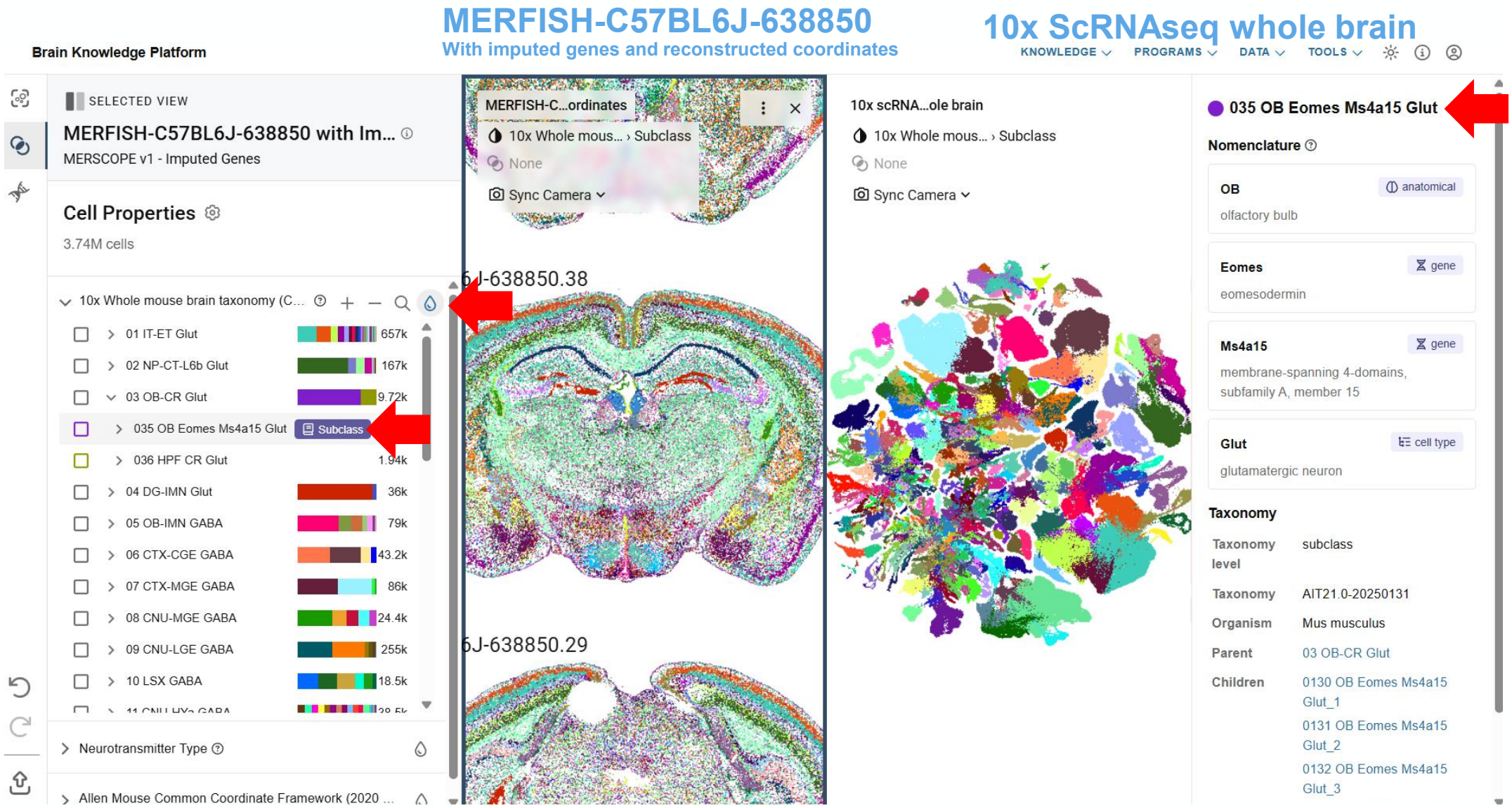


brain-map.org/atlases-and-data/bkp/abc-atlas

7 neighborhoods,
34 classes,
338 subclasses,
1,201 supertypes,
5,322 clusters

Tip: Click on the purple "Class" button to learn more about acronyms in the cell class name

Viewing cell subclasses



brain-map.org/atlases-and-data/bkp/abc-atlas

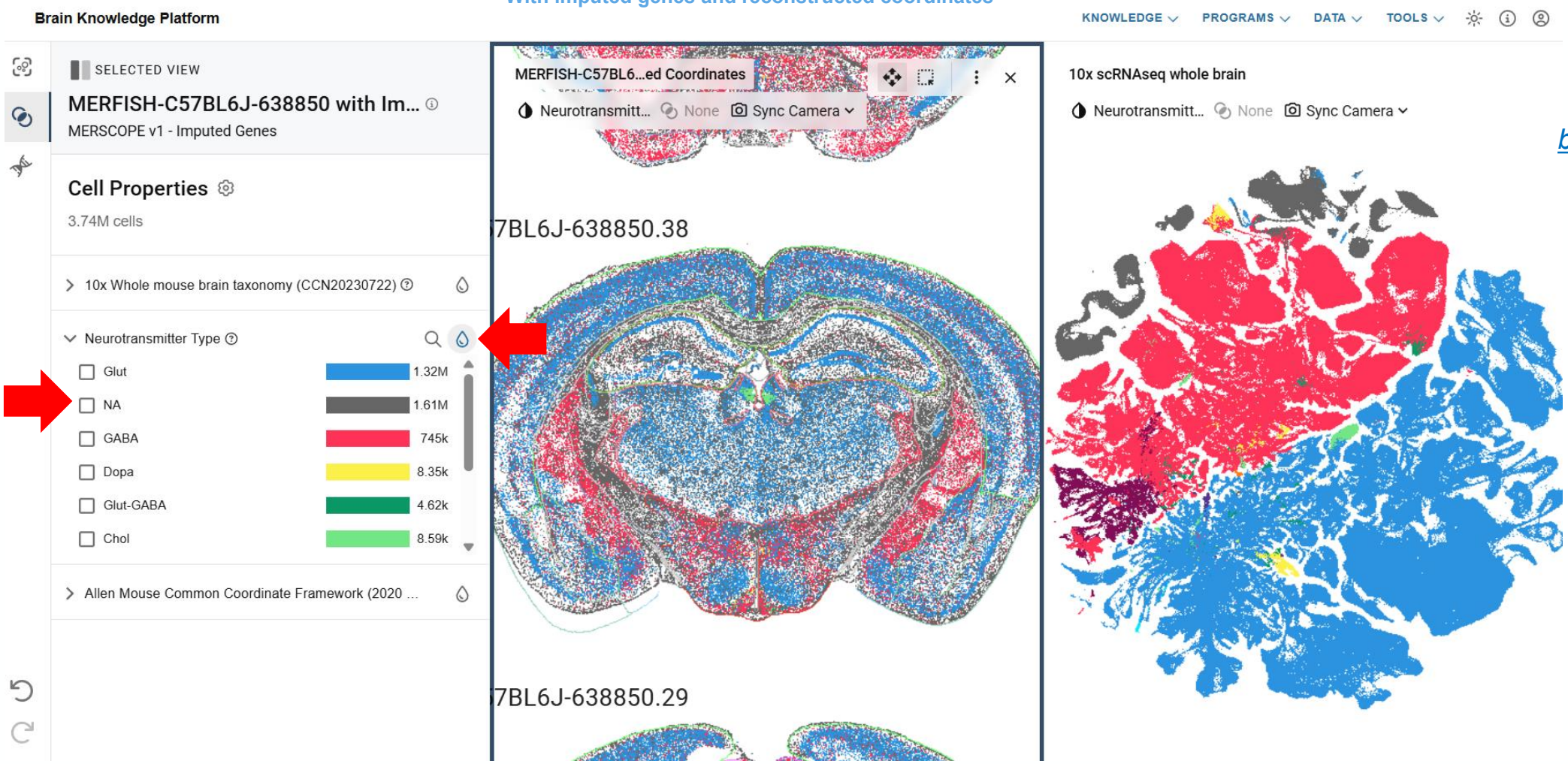
7 neighborhoods,
34 classes,
338 subclasses,
1,201 supertypes,
5,322 clusters

Tip: Click on the purple “Subclass” button to learn more about acronyms in the cell subclass name

Viewing neurotransmitters:

MERFISH-C57BL6J-638850
With imputed genes and reconstructed coordinates

10x ScRNAseq whole brain

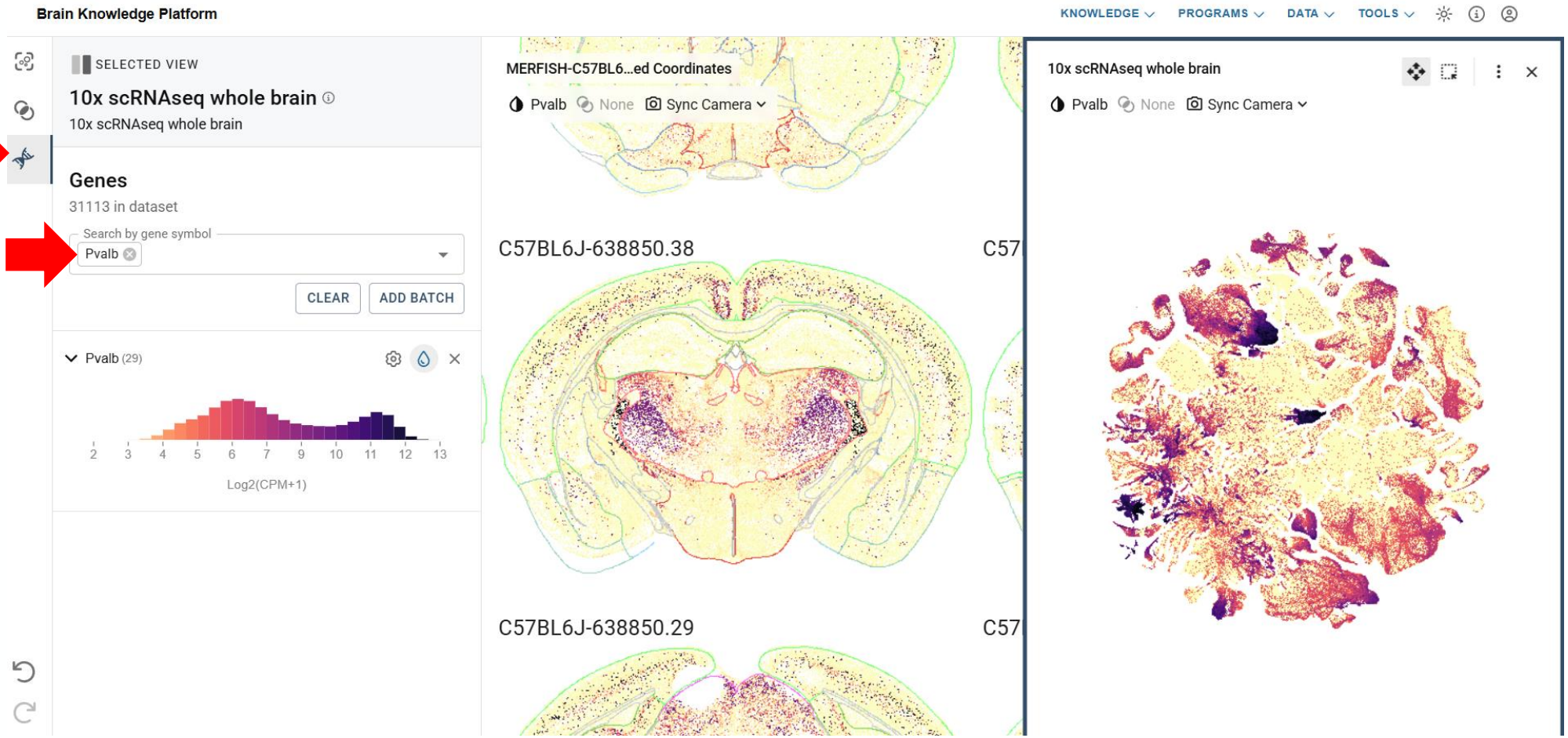


brain-map.org/atlases-and-data/bkp/abc-atlas

Viewing genes:

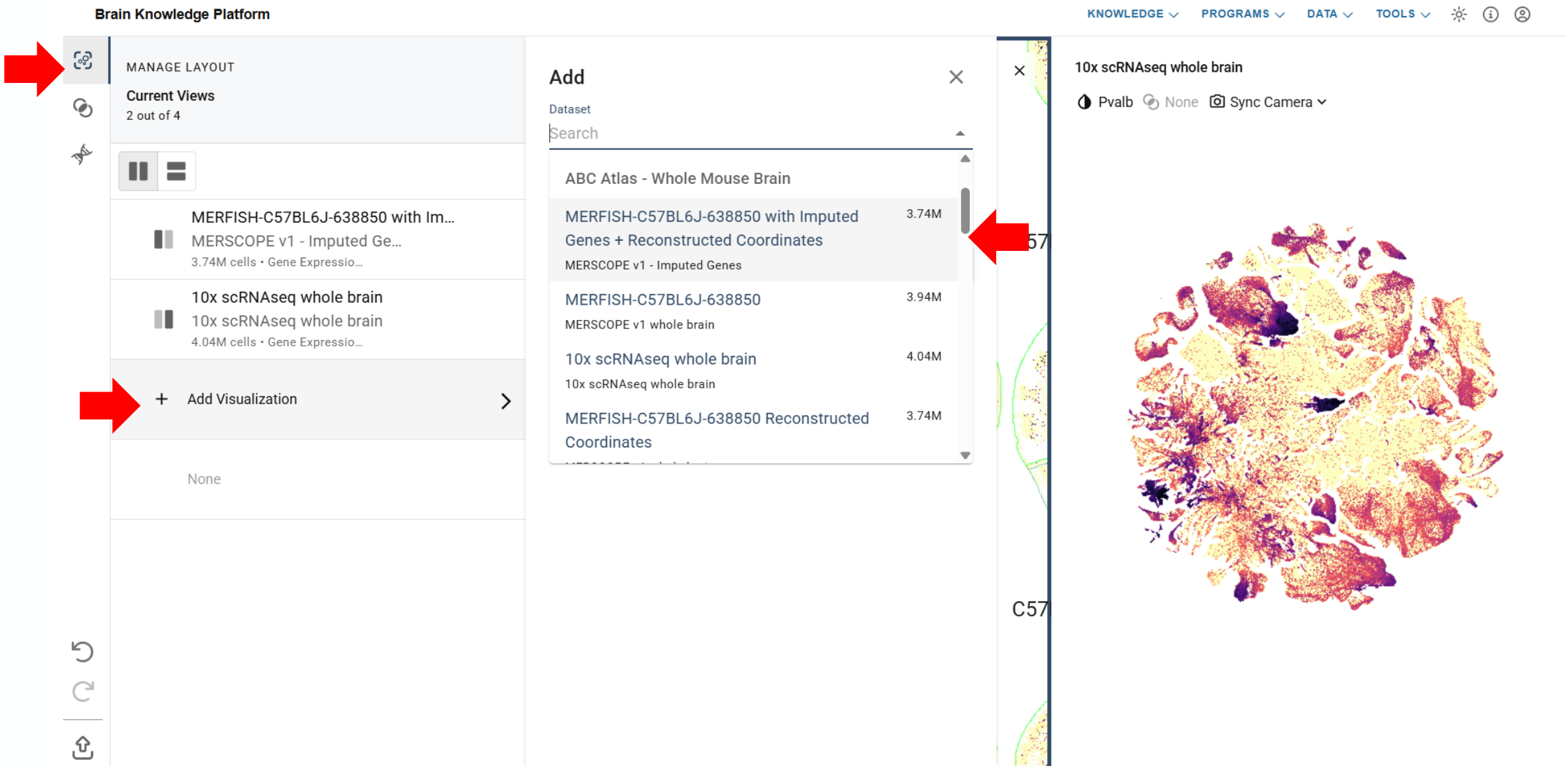
MERFISH-C57BL6J-638850
With imputed genes and reconstructed coordinates

10x ScRNAseq whole brain

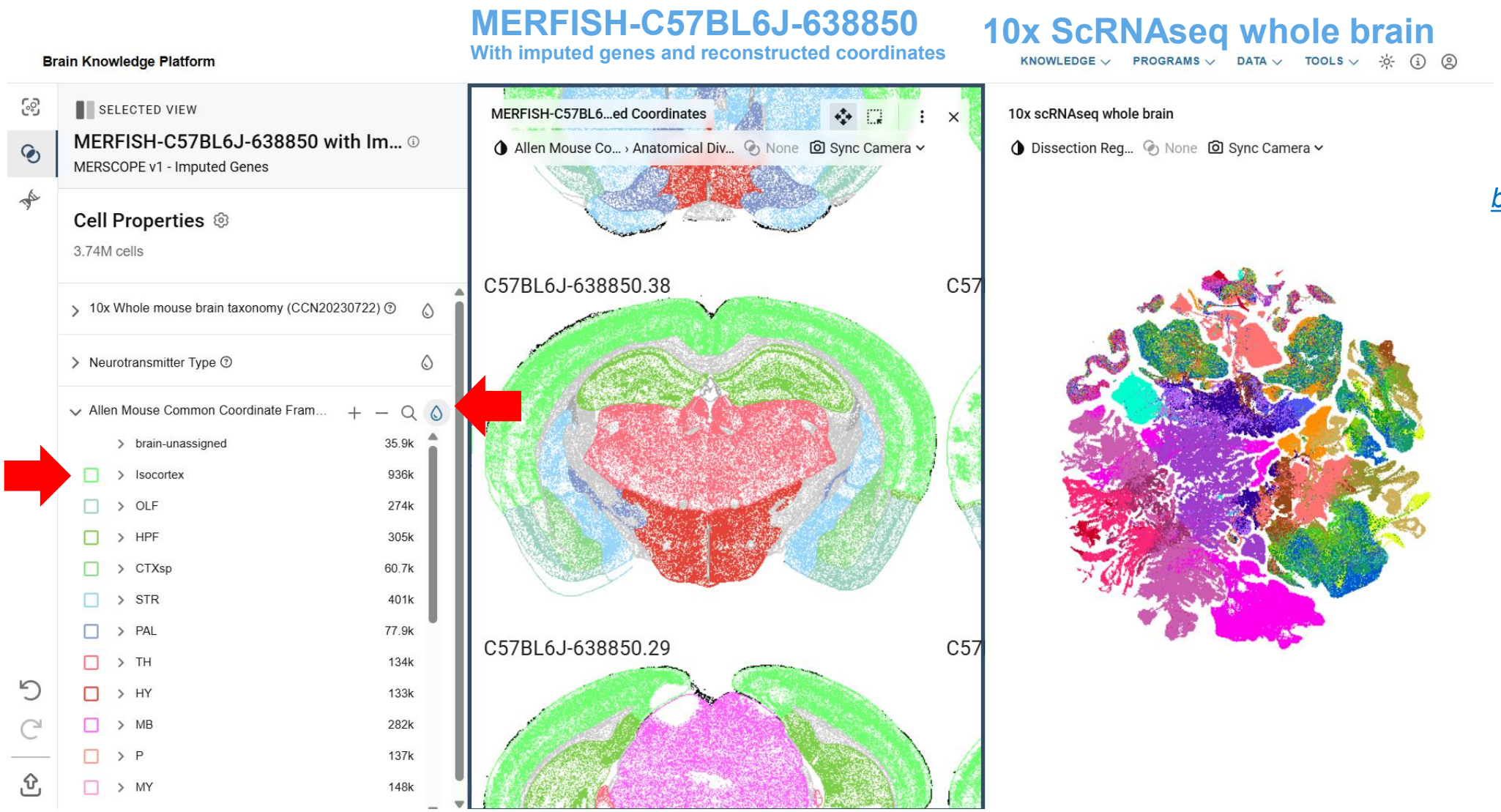


brain-map.org/atlases-and-data/bkp/abc-atlas

Viewing different datasets:



Viewing dissection/anatomical regions:



brain-map.org/atlases-and-data/bkp/abc-atlas

Tip: Have “Whole Mouse Brain Anatomical Annotations” open to better understand anatomical acronyms

brain-map.org/help-and-community/guide-cell-types

Use cases for ABC Atlas

“Are genes X and Y co-expressed in the same cell type?”
“Where is gene X expressed in the brain?”
“Is gene X expressed in a specific cell type?”
“Where is cell type X in the brain?”
“What is the spatial relationship between cell type X and cell type Y?”

However,

- MERFISH data in ABC Atlas is based on cell segmentation; cell-to-cell or synaptic contacts are not shown.
  - *“Does cell type X synapse onto cell type Y?”* cannot be answered
- MERFISH data in ABC Atlas shows centroids of cells, not cell shape or size
  - *“Is the soma of cell type X larger than the soma of cell type Y?”* cannot be answered

Accessing ABC Atlas data programmatically

Search Ctrl + K

Allen Brain Cell Atlas - Data Access

INTRODUCTION

Getting started

Accessing 10x RNA-seq gene expression data

Using cells selected in the ABC Atlas with Gene Expression data.

DATA

Allen Brain Cell Atlas - Data Access

The Allen Brain Cell Atlas (ABC Atlas) aims to empower researchers worldwide to explore and analyze multiple whole-brain datasets simultaneously. As the Allen Institute and its collaborators continue to add new modalities, species, and insights to the ABC Atlas, this groundbreaking platform will keep growing, opening up endless possibilities for groundbreaking discoveries and breakthroughs in neuroscience. With the ABC Atlas, researchers everywhere can gain new insights into the brain's complex workings, advancing our understanding of this amazing organ in ways we never thought possible. The ABC Atlas can be accessed at here and is the primary method of interacting with these data. Any questions or issues associated with the ABC Atlas are best directed to the Allen Institute Community Forum. This repository is intended for users who wish to download the ABC Atlas data and processes it locally.

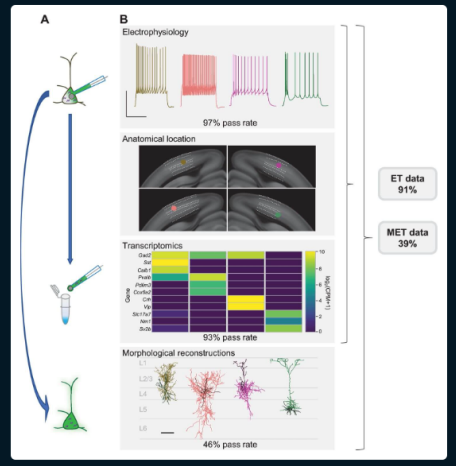
Data associated with the ABC Atlas is hosted on Amazon Web Services (AWS) in an S3 bucket as a AWS Public Dataset, arn:aws:s3:::allen-brain-cell-atlas. No account or login is required for access. ***The purpose of this repo is to provide an overview of the available data, how to download and use it through example use cases.***

The Early-Winter 2026 public beta data release includes:

- **Mouse whole-brain transcriptomic cell type atlas (Hongkui Zeng)**
- **Consensus Whole Mouse Brain cell type atlas (Evan Macosko, Hongkui Zeng)**
- **Aging Mouse transcriptomic cell type atlas (Hongkui Zeng)**
- **Developing Mouse - Visual Cortex (Hongkui Zeng)**

Viewing and accessing patch-seq data

Protocol

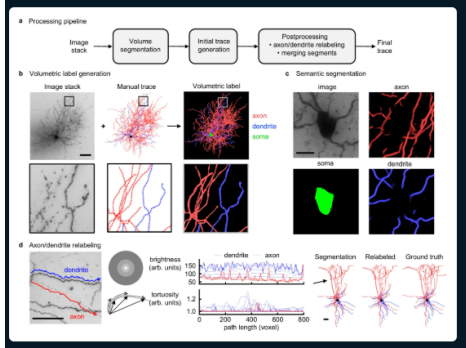


Scaled, high fidelity electrophysiological, morphological, and transcriptomic cell characterization

We have optimized the Patch-seq technique to efficiently collect high-quality electrophysiological, morphological and transcriptomic data. Here we describe this optimized technique as well as the publicly available tools that can be used to generate comparable data.

Patch-seq Analysis Tools (GitHub) | Community Forums

Lee, Budzillo et al. eLife 2021

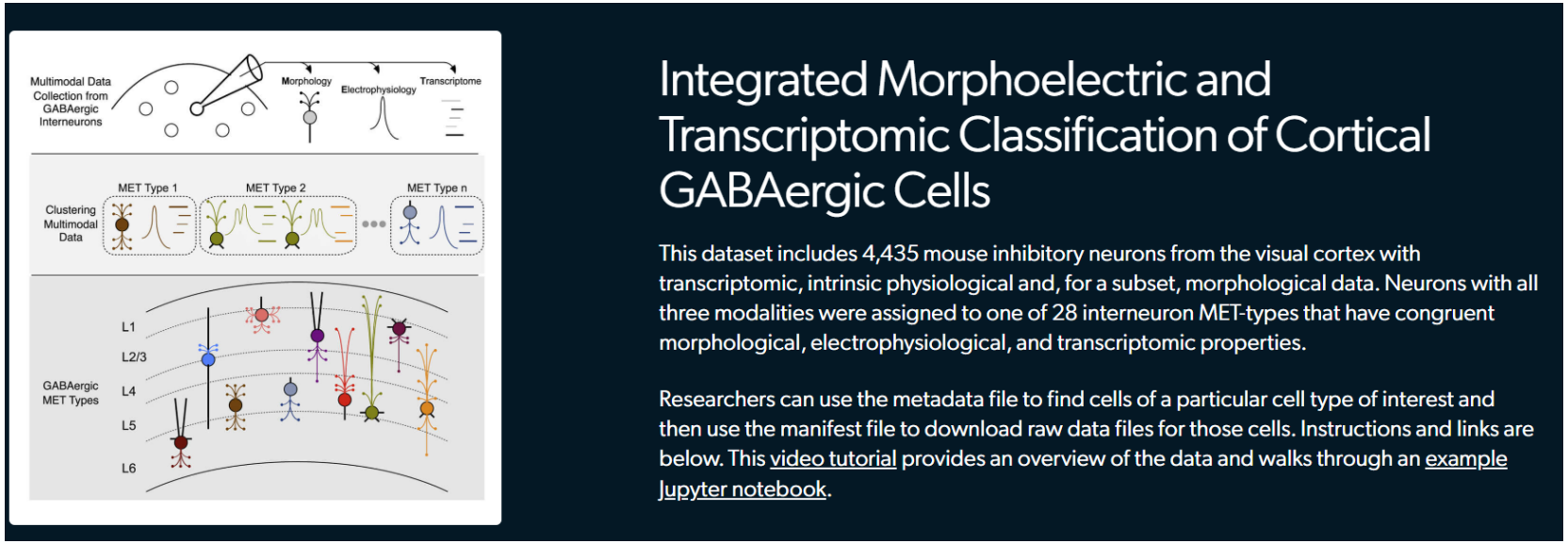


High-throughput analysis of dendrite and axonal arbors reveals transcriptomic correlates of neuroanatomy

Introduces an automated pipeline for high-throughput reconstruction and analysis of dendritic and axonal arbors from brightfield-imaged neurons, enabling large-scale morphological characterization across transcriptomically defined cell types.

Analysis Code (GitHub) | Gliko, Mallory et al. Nature Communications 2024

Viewing and accessing patch-seq data

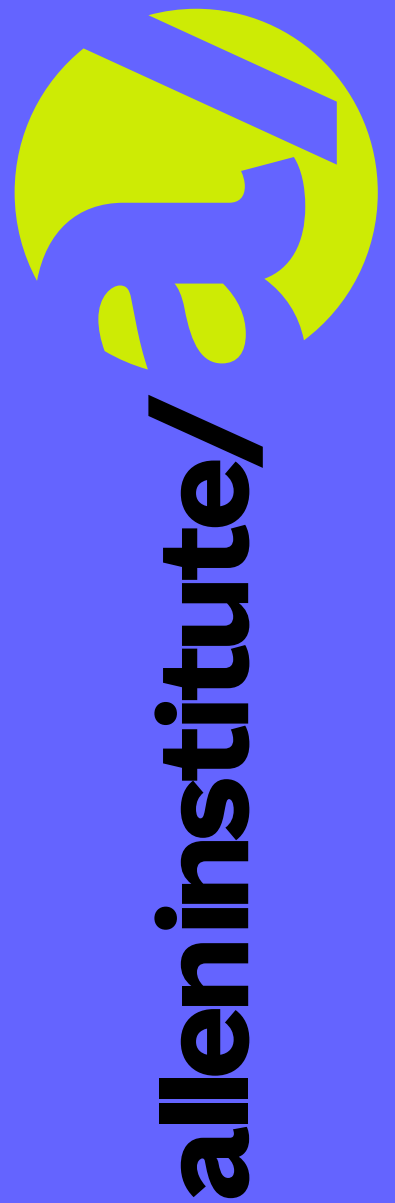


Access more cell type presentations:

YouTube

Slides

Questions?



August 13th afternoon agenda/

- **2:00-3:00 |** Guided exploration of data and tools – Rachel
- **3:00-3:15 |** Coffee break
- **3:15-4:00 |** Open exploration time – hands-on help from us
- **4:00-4:30 |** Group examples of data/tool use
- **4:30-5:00 |** Group discussion and Q&A

Guided exploration

Exploring Cell Types with the Allen Institute

Leloir Institute, Buenos Aires

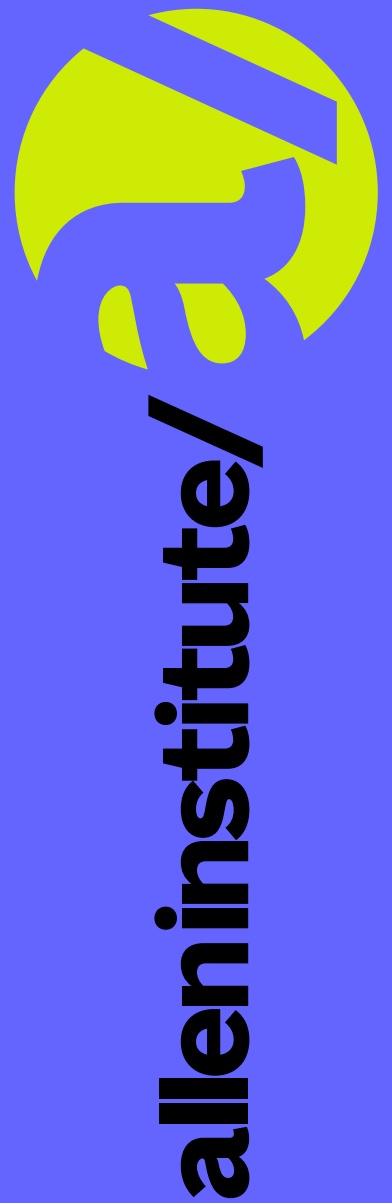
August 14th, 2026

August 14th morning agenda/

- **10:00-11:15 |** Primate basal ganglia (15 min Q&A) – Rachel
- **11:15-11:30 |** Break
- **11:30-12:30 |** Seattle Alzheimer's Disease Brain Cell Atlas (15 min Q&A) – Mariano
- **12:30-1:00 |** Conclusions and resources (10 min Q&A) – Lauren

Primate basal ganglia data/

Rachel Hostetler



Evolutionary Genomics Team

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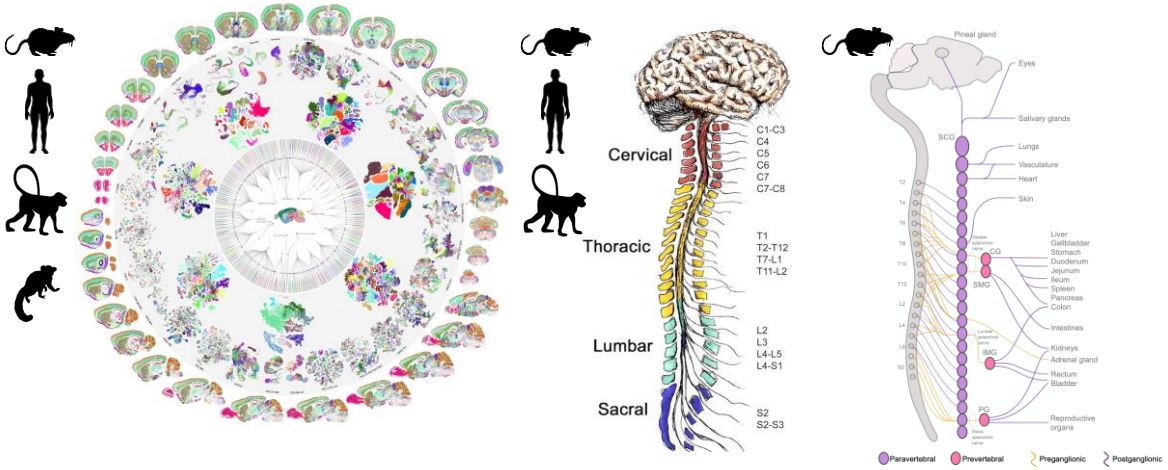
Alma Dubuc (Yale)
Matthew Schmitz
Nelson Johansen
Niklas Kempynck (KU Leuven)
Morgan Wirthlin
Yuanyuan Fu
Saroja Somasundaram

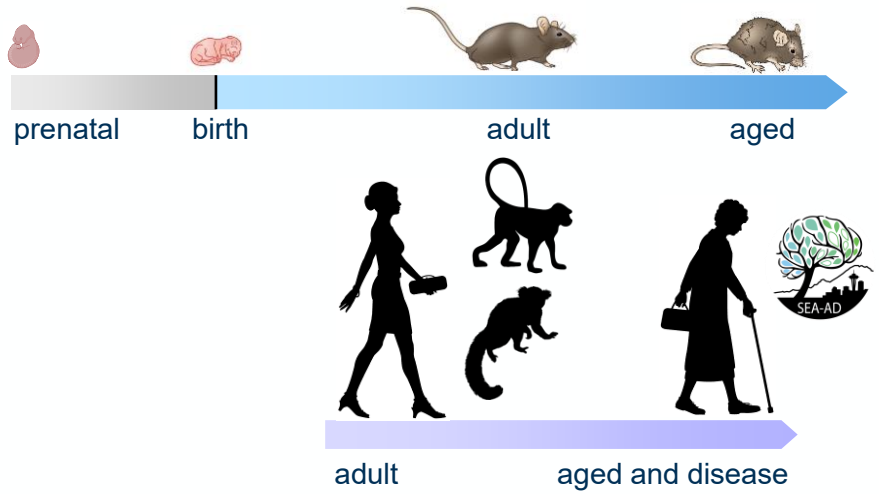
(Left to right)

Building foundational molecular cell type atlases

Across the brain, spinal cord and peripheral nervous system

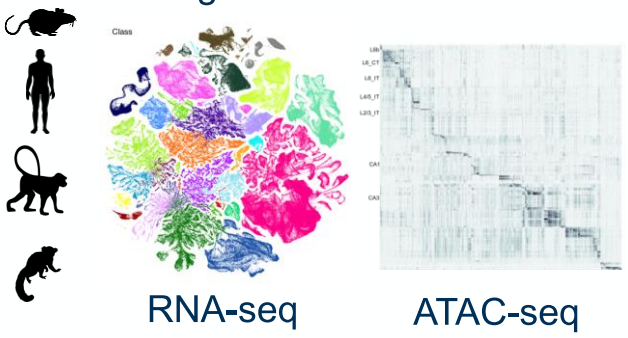


Across time and in disease

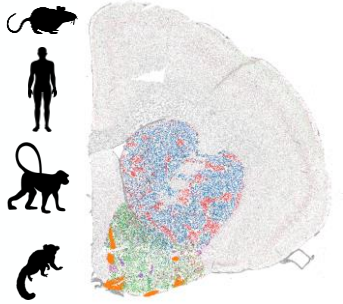


Across multiple data modalities and with spatial distributions

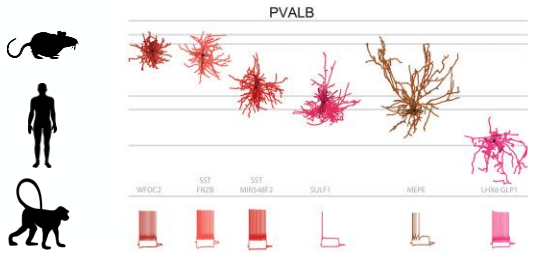
Single cell multiomics



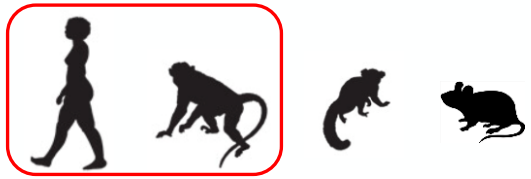
Spatial transcriptomics



Morphology and physiology



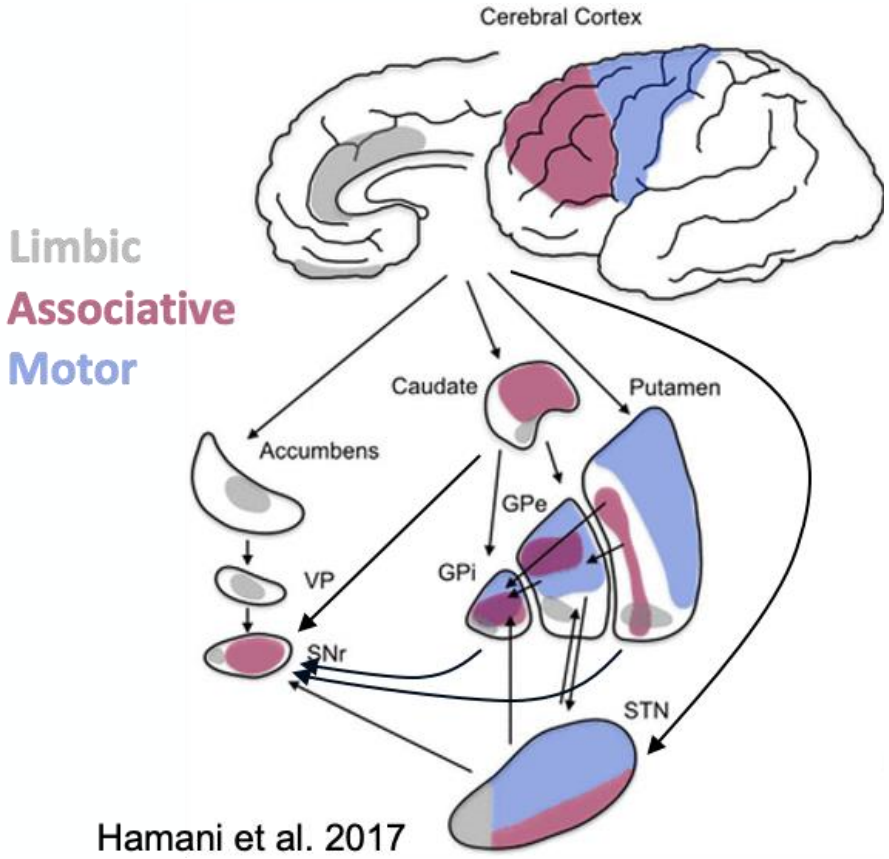
Profiling cellular diversity in the basal ganglia (BG)



Single-nucleus **Multiome** ATAC + Gene Expression
10X Genomics

CaB	27,777	28,385
CaH	35,964	40,430
CaT	35,130	24,755
GPe	36,440	14,361
GPi	49,399	16,684
NAC	71,986	55,134
Pu	103,017	121,514
SN/SN-VTA	24,437	19,287
STH	32,675	23,789

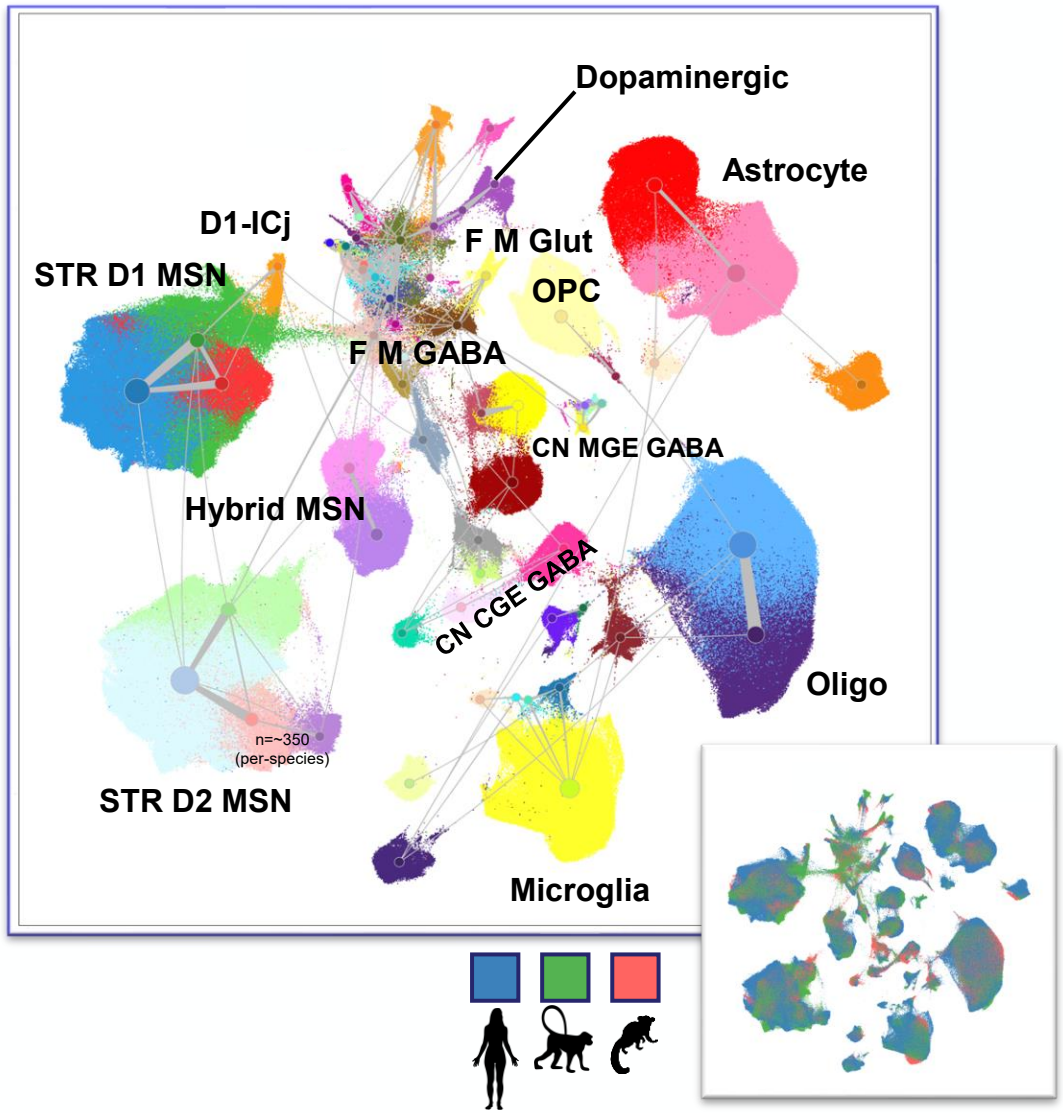
>1.7 million nuclei total

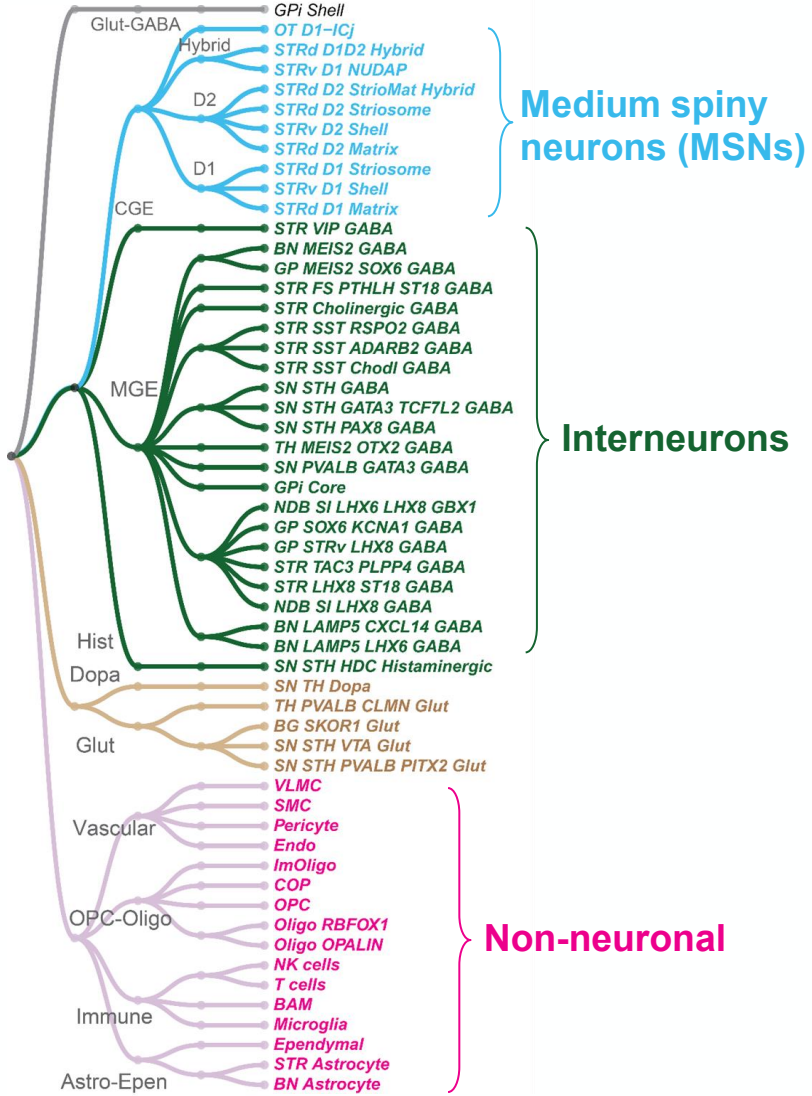


Hamani et al. 2017

Human and non-human primate consensus BG taxonomy

Cross-species integration





Distinct localization and functional properties of MSN types

Dorsal Striatum *(MSNs)*

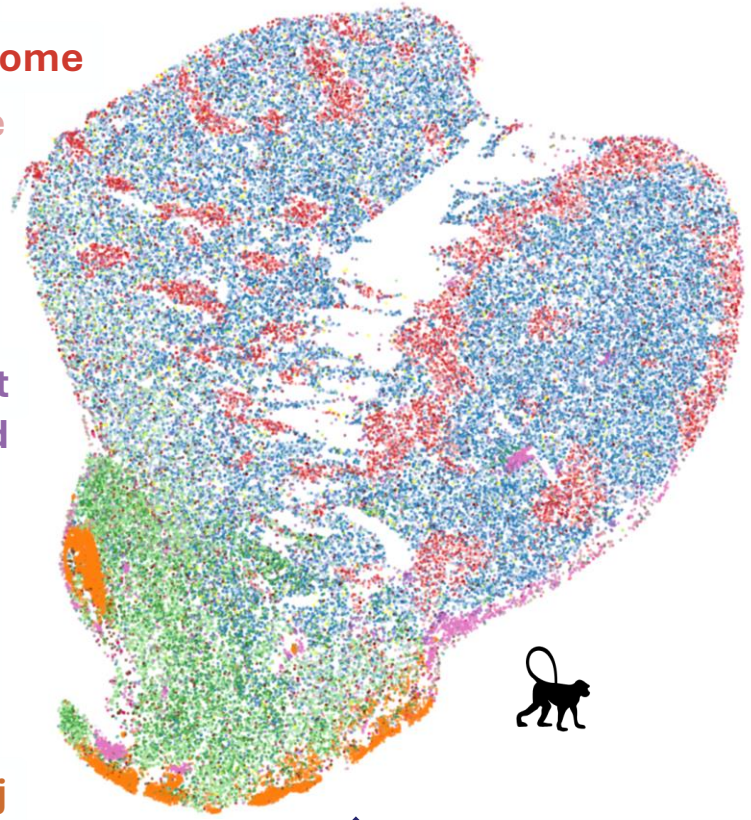
- STRd D1 Striosome
- STRd D2 Striosome
- STRd D1 Matrix
- STRd D2 Matrix
- STRd D1D2-Hybrid
- STRd D2-StrioMat Hybrid

Ventral Striatum *(MSNs)*

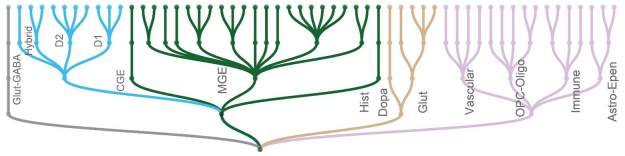
- STRv D1 MSN
- STRv D2 MSN
- D1-NUDAP

Islands of Calleja *(Granular Cells)*

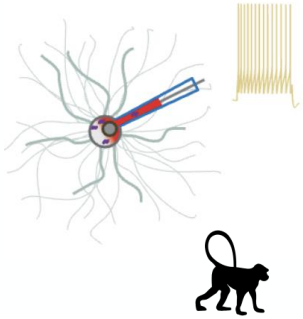
- OT D1-ICj

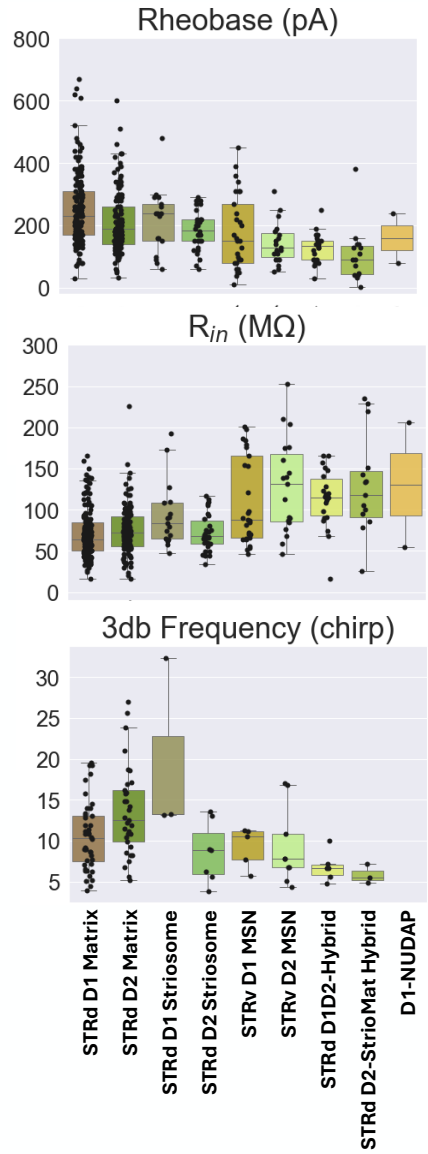


Spatial mapping



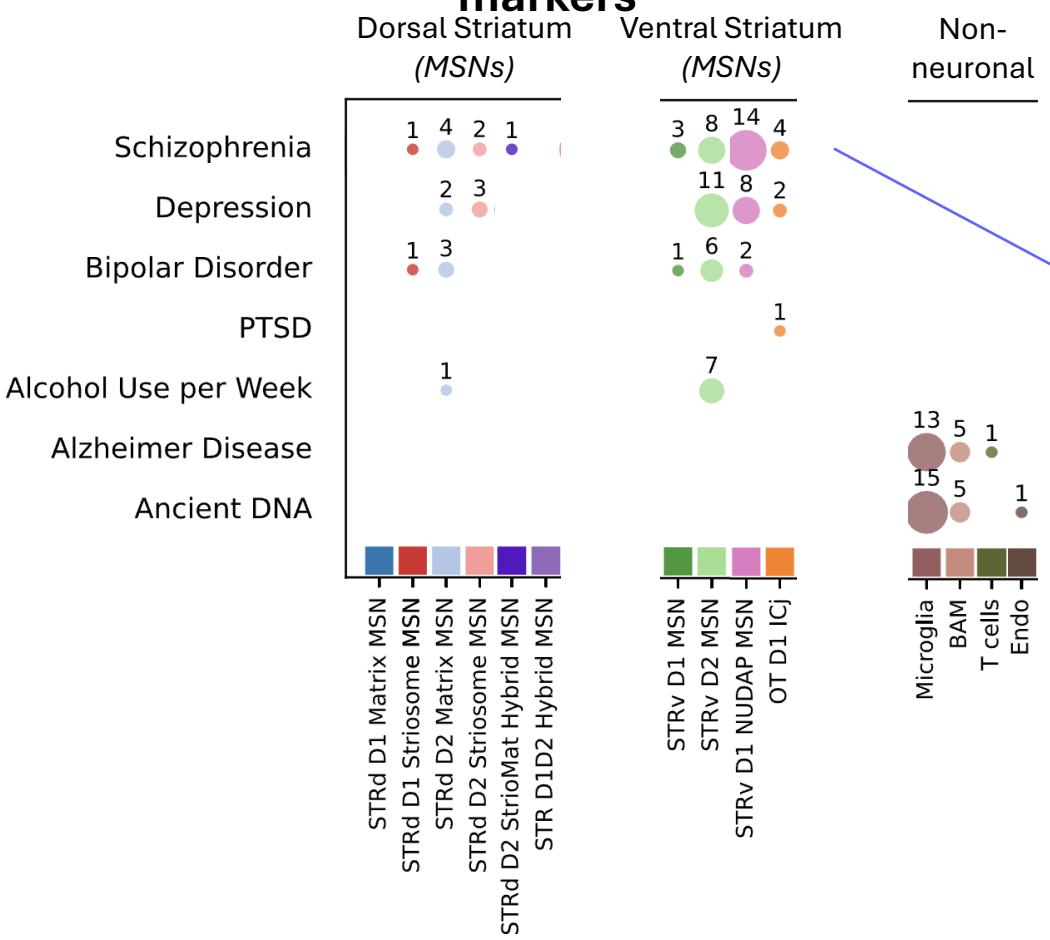
Patch-seq



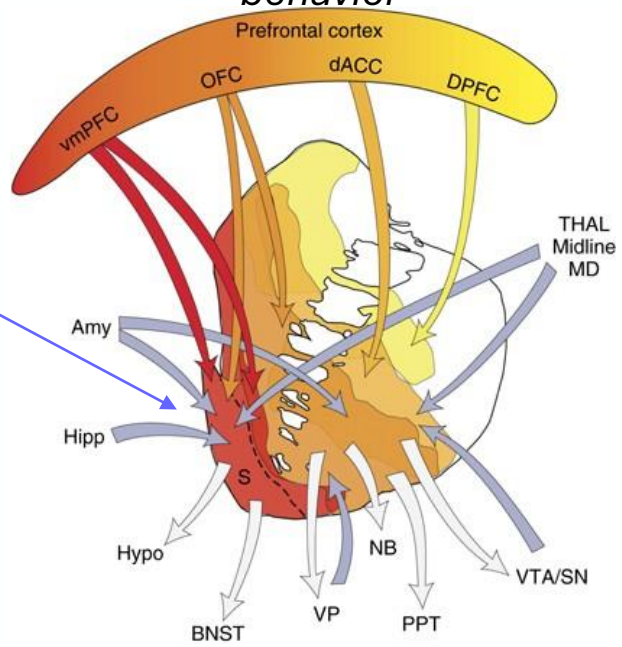


Psychiatric disease risk is linked to ventral MSN types

Disease-associated SNPs are enriched near cell type markers



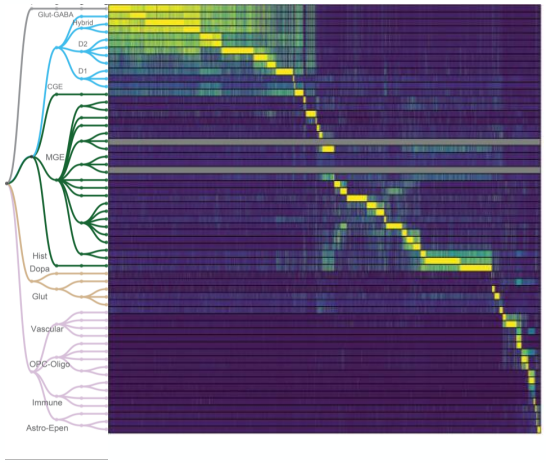
Ventral striatum integrates reward, motivation, and emotion, guiding adaptive behavior

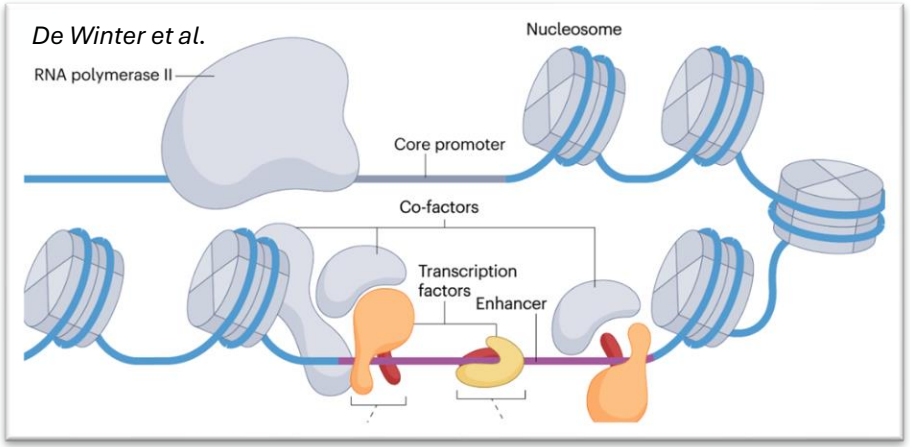


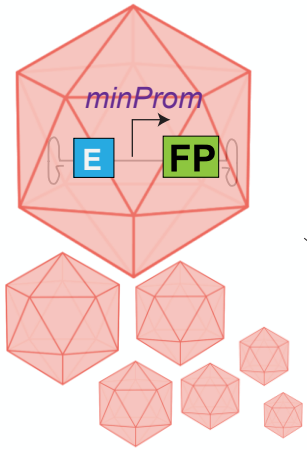
Haber & Knutson 2010

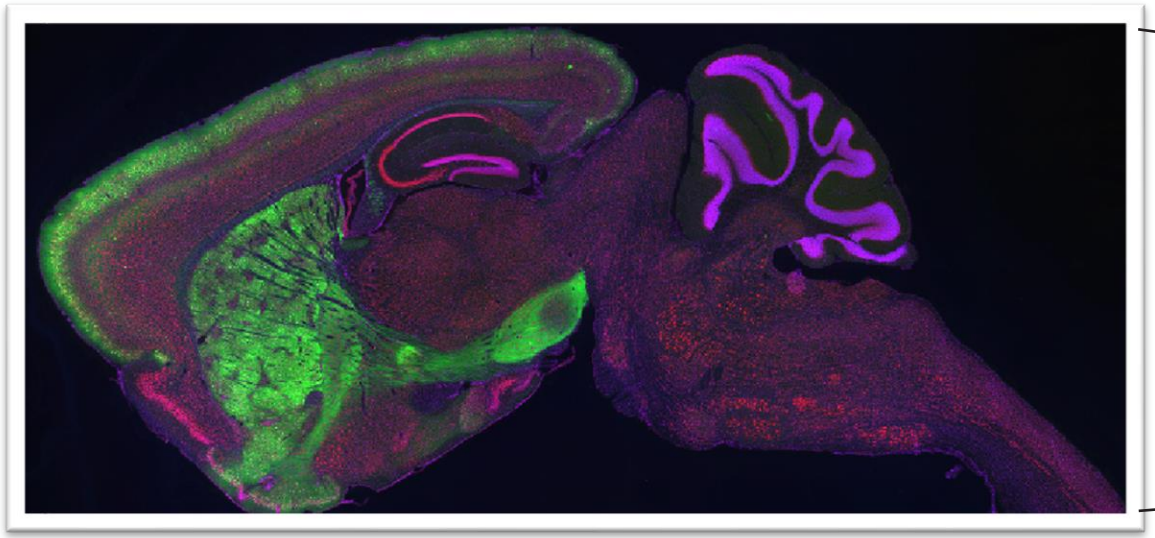
Enhancers can drive cell type-specific genetic tools

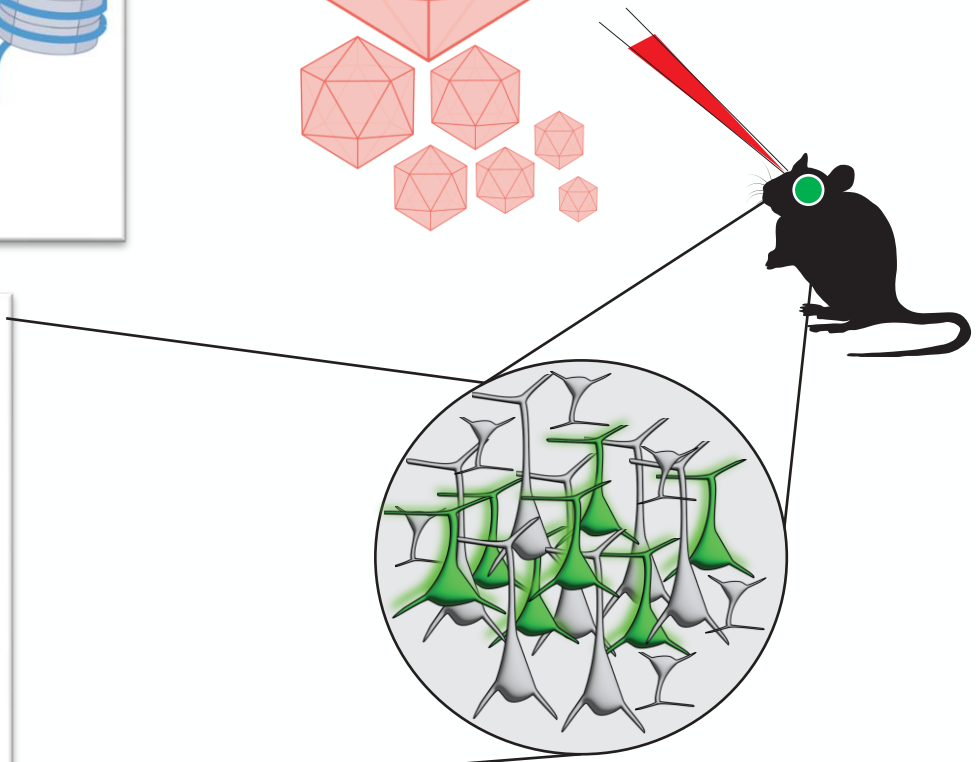
Some candidate enhancers are restricted to specific cell types





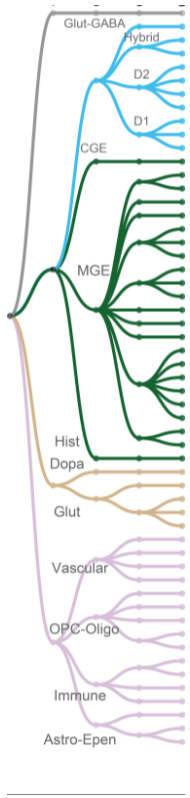






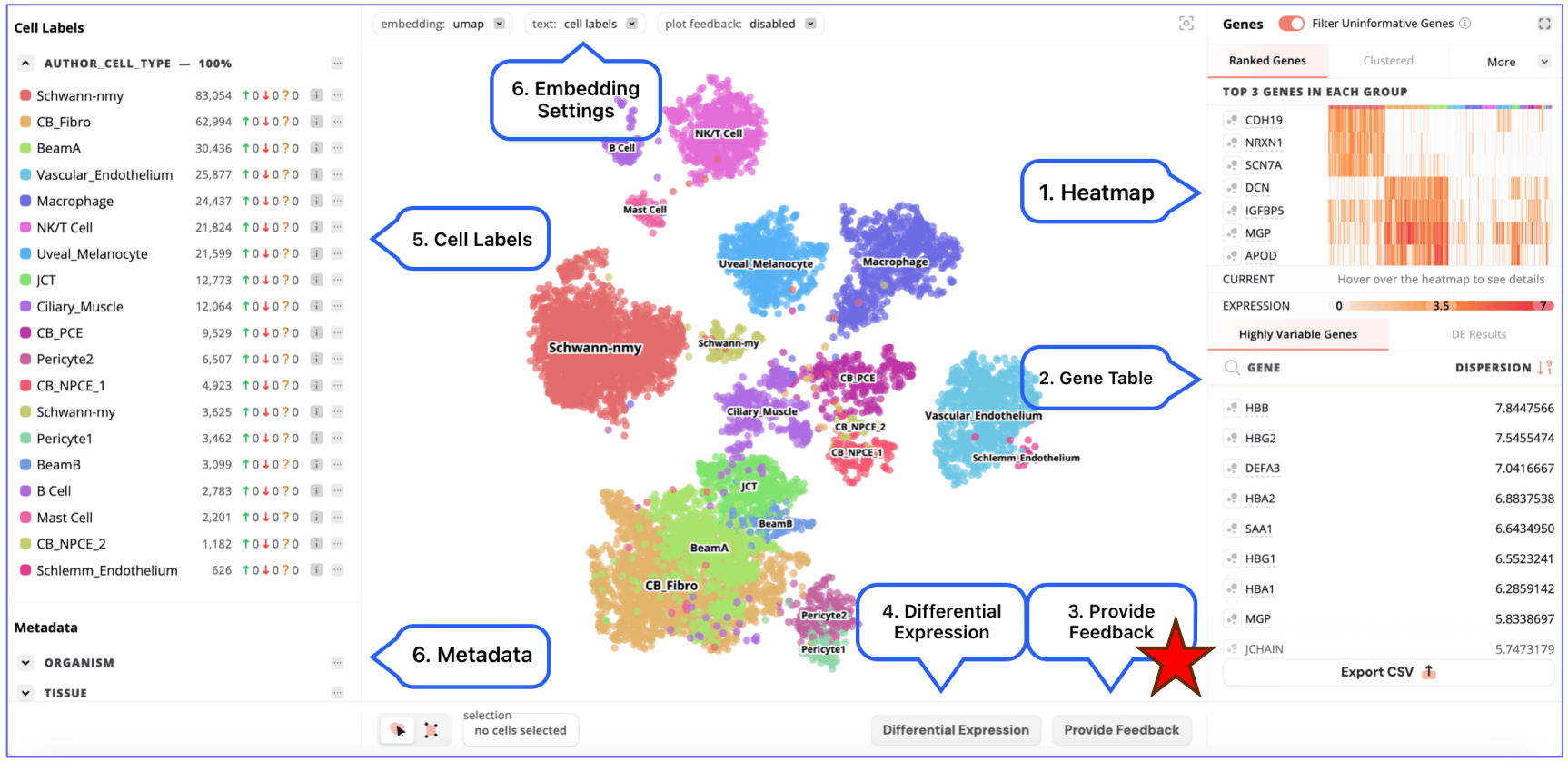
https://portal.brain-map.org/genetic-tools/genetic-tools-atlas

Community refinement of a consensus brain atlas



Cell Annotation Platform

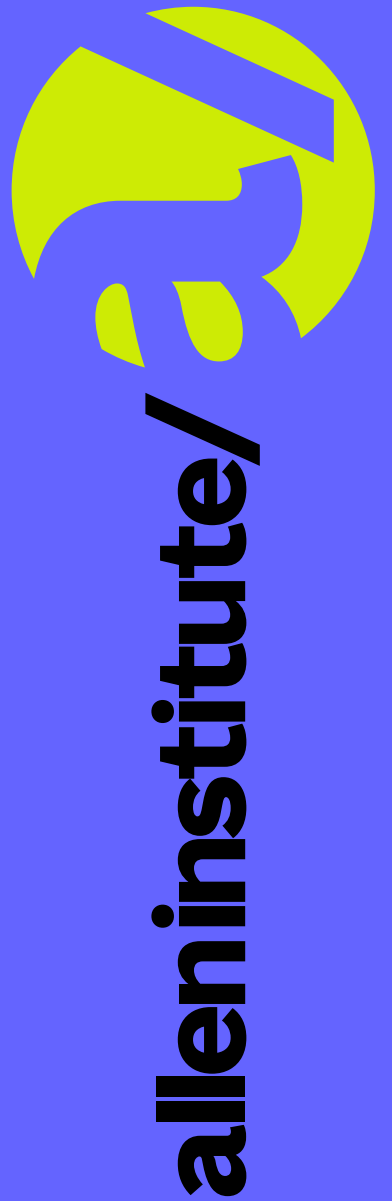
HUMAN CELL ATLAS



An effort by the Villani, Kharchenko, Gehlenborg, Theis, Marioni, and Regev labs.

https://celltype.info/project/637

Patch-seq

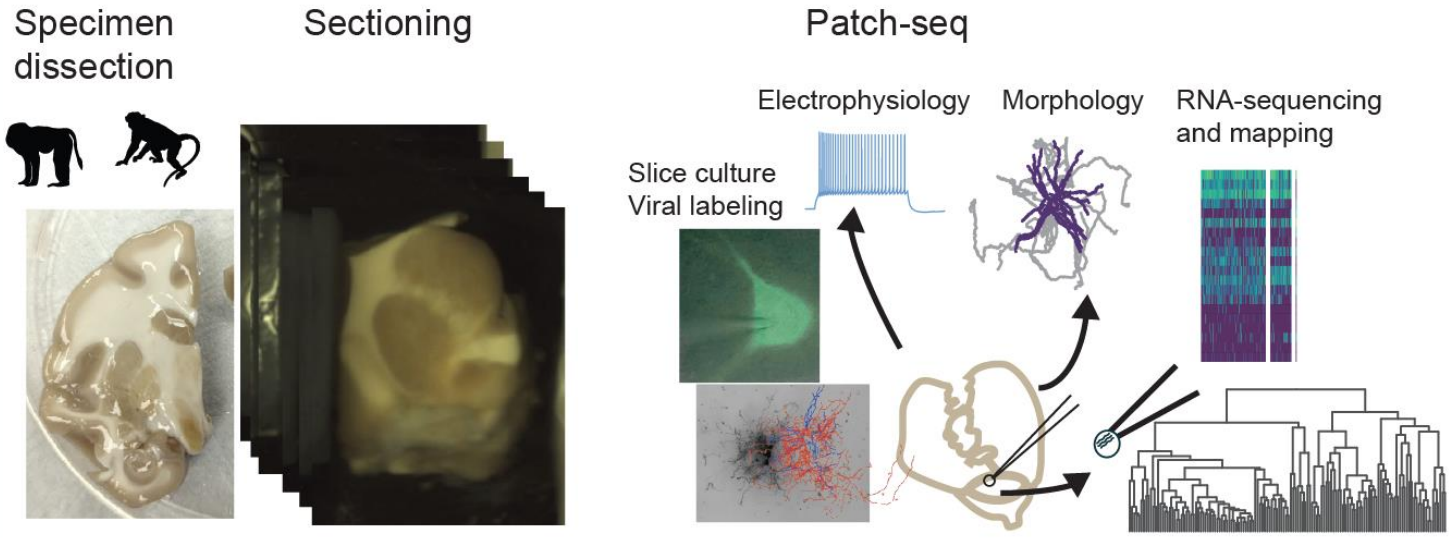


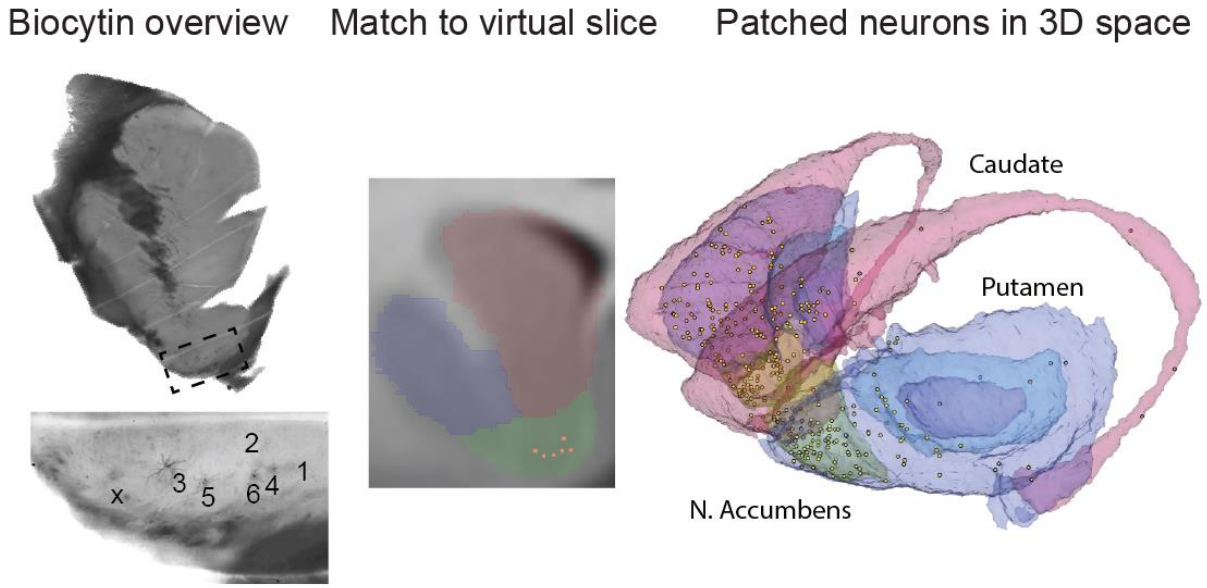
Morphoelectric Diversity and Specialization of Neuronal Cell Types in the Primate Striatum

Xiao-Ping Liu, Rachel Dalley, Nelson Johansen, Agata Budzillo, Julian Thijssen, Jeremy A. Miller, Sarah Walling-Bell, Scott Sawchuk, Lauren Alfiler, Julia Andrade, Angela Ayala, Stuard Barta, Kyla Berry, Darren Bertagnolli, Ashwin Bhandiwad, Madeline Bixby, Krista Blake, Krissy Brouner, Trangthanh Cardenas, Tamara Casper, Anish Bhaswanth Chakka, Thomas Chartrand, Scott Daniel, Nicholas Donadio, Nadezhda I. Dotson, Tom Egdorf, Rachel Enstrom, Yuanyuan Fu, Amanda Gary, Jeff Goldy, Melissa Gorham, Kristen Hadley, Alvin Huang, Avery C Hunker, Atlas Jordan, Zoe C. Juneau, Matthew Jungert, Madhav Kannan, Inkar Kapen, Shannon Khem, Megan Koch, Katalin Kocsis, Rana Kutsal, Gabriela Leon, Matt Mallory, Jocelin Malone, Audrey McCutcheon, Medea McGraw, Kamiliam Nasirova, Dakota Newman, Lindsay Ng, Ximena Opitz-Araya, Alana Oyama, Elliot Phillips, Christina Alice Pom, Lydia Potekhina, Ramkumar Rajanbabu, Shea T. Ransford, Ingrid Redford, Christine Rimorin, Dana Rocha, Augustin Ruiz, Naz Taskin, Michael Tieu, Éva Tóth, Jessica Trinh, Marshall VanNess, Sara Vargas, Maria Camila Vergara, Morgan Wirthlin, Jeanelle Ariza, Trygve Bakken, Nick Dee, Nikolai Dembrow, Song-Lin Ding, Chris English, Samantha D. Hastings, Tim Jarsky, Lauren Kruse, Scott F. Owen, Kimberly A. Smith, Jack Waters, Shenqin Yao, Gábor Tamás, Boudewijn Lelieveldt, Nathan W. Gouwens, Hongkui Zeng, Staci A. Sorensen, Brian R. Lee, Jonathan T. Ting, Ed S. Lein, Brian E. Kalmbach

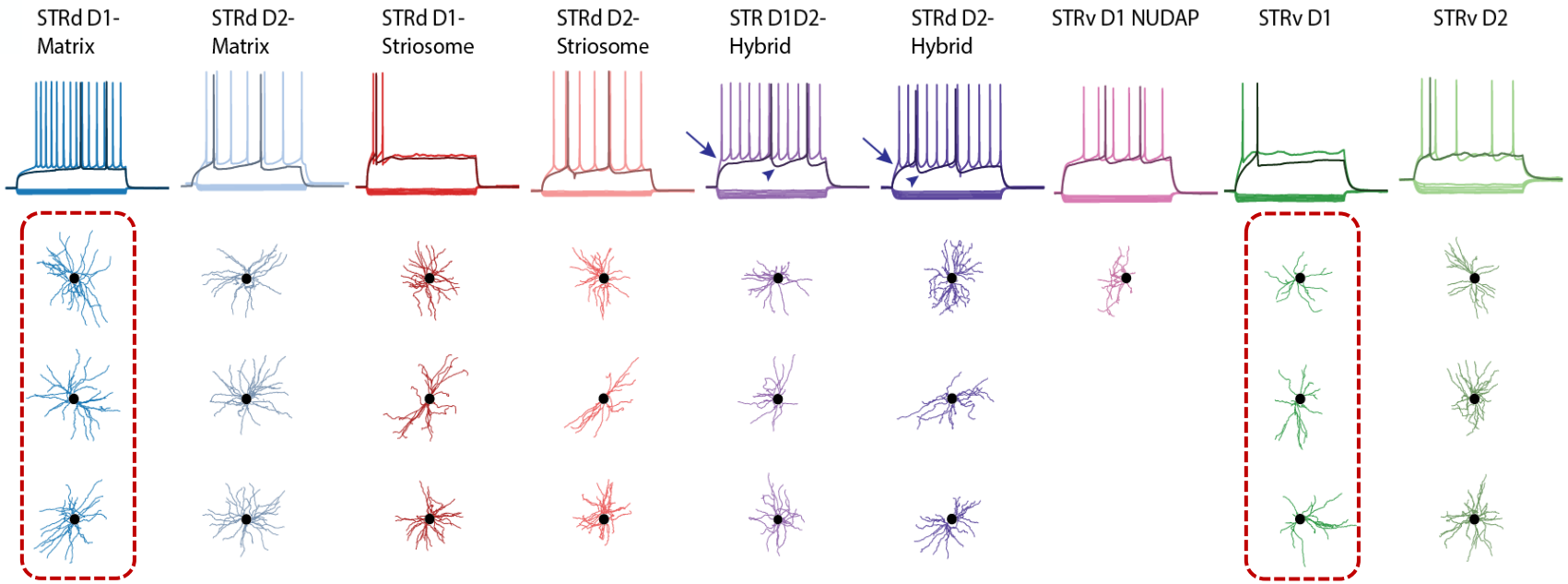
doi: https://doi.org/10.64898/2026.02.26.708019

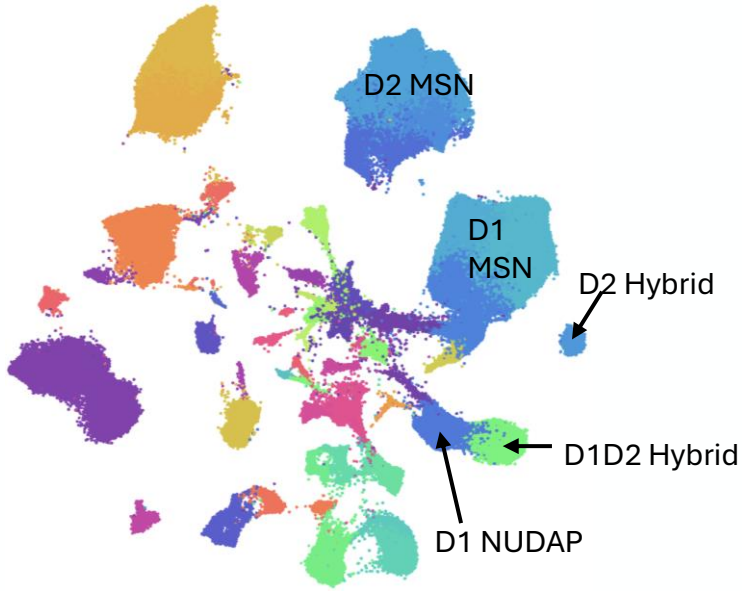
Experimental Preparation





Medium Spiny Neuron (MSN)-like cell types exhibit systematic variability





Liu et al., 2026

- **D1D2 Hybrid MSNs**: also called "eccentric" MSNs
  - Poorly understood functionally
  - Proportion of eccentric MSNs may be higher than in primates than rodents (Buyukkahraman et al., 2025)
- **D1-NUDAP MSNs**: also within the "third island", form spatial clusters in the ventral striatum
  - High expression of mu-opioid receptor (OPRM1)
- **D2 StrioMat Hybrid MSNs ("D2 Hybrid")**: distinct set of MSNs co-expressing striosomal and matrix markers.

Summary

- Medium spiny neurons have underappreciated and multiplexed phenotypic diversity, mirroring transcriptomic variation, but with substantial overlap between groups.
- We should move beyond a binary model of medium spiny neurons. D1/D2 Hybrid MSNs and D2 Hybrid MSNs have unique properties that warrant further investigation.
- Islands of Calleja granule cells frequently have a bursting response in primates.
- A spatial gradient can be seen in several morphological and electrophysiological properties and may be concordant across multiple subclasses of neurons.
- TAC3 interneurons have divergent properties in primates and are unable to sustain firing under sustained depolarization.
- Numerous other differences exist between homologous groups in rodent and primate.

Acknowledgements

Patch-seq team
Brian Kalmbach
Brian Lee
Staci Sorensen
Rachel Dalley
Xiao-Ping Liu
Jonathan Ting
Tim Jarsky
Sarah Walling-Bell
Nathan Gouwens
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Tissue processing (Nick Dee)
Histology (Jeanelle Ariza)
Imaging (Sam Hastings)
Morphology (Rachel Dalley)
Ephys Core (Brian Lee)

Neuroanatomy
Song-Lin Ding

HMBA Spatial Tx Teams
Stephanie Seeman
Jennie Close
Brian Long
Madeleine Hewitt
Meghan Turner

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Ed Lein – Human Cell Types
Hongkui Zeng, Rui Costa
and the late Paul Allen

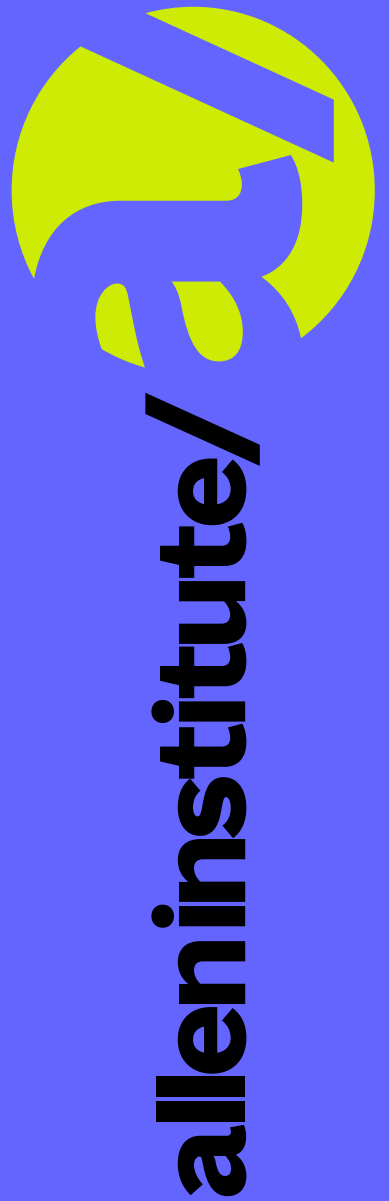
External collaborators
Andreas Tolias-Stanford
Xiaolong Jiang-BCM
Huib Mansvelder –VU
Chris de Kock-VU
Gabor Tamas- University of Szeged
Julian Thijssen (Delft University)
Boudewijn Lelieveldt (Delft University)

WaNPRC staff
Chris English – UW/WaNPRC Veterinary staff

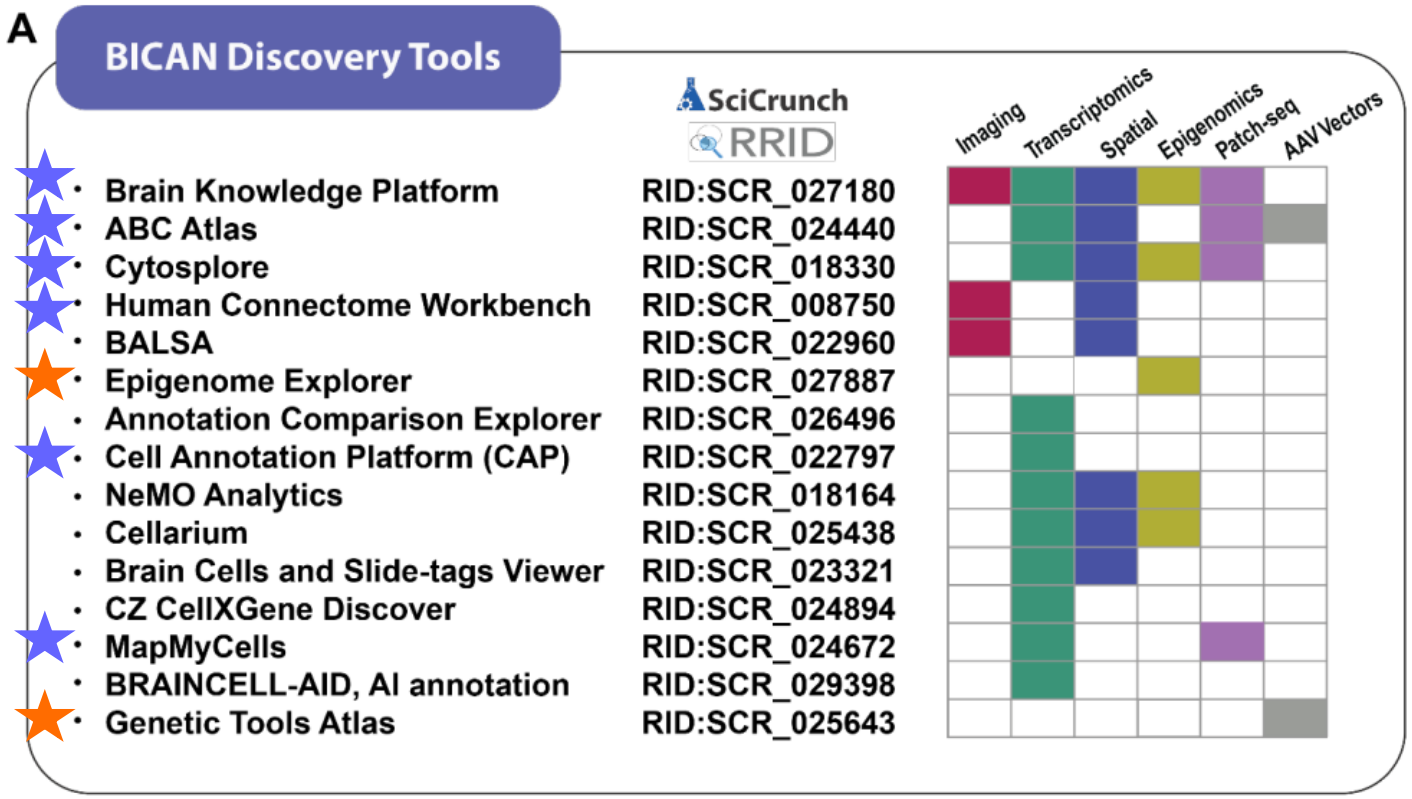
Squirrel Monkey
Tom Reh

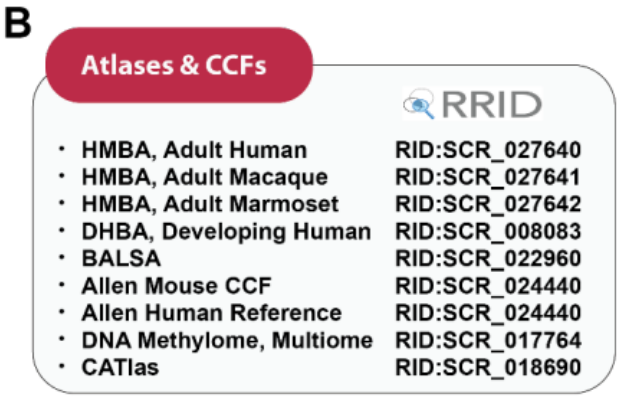
Funding: ***Lein, Zeng:*** *UM1MH130981,* ***Washington National Primate Research Center:*** *P51OD010425 and U42OD011123*

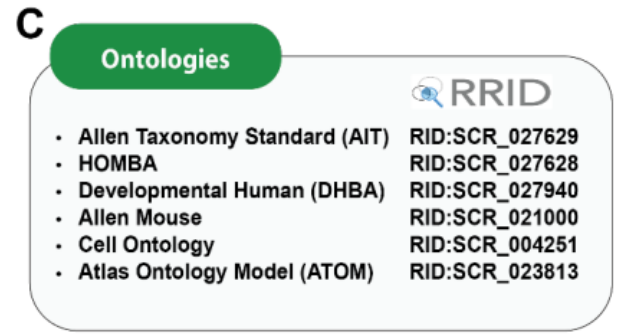
Data and tool access



BICAN discovery tools for basal ganglia





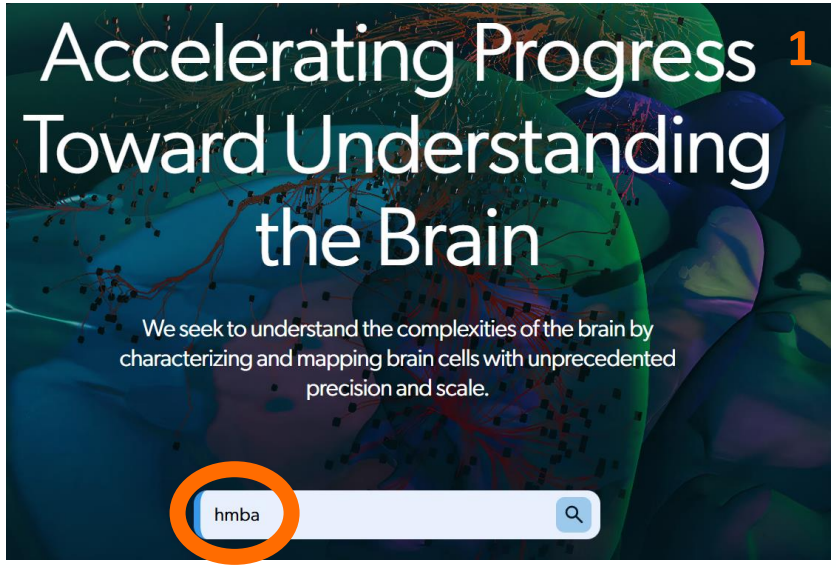


A Community Standard Multispecies Cell Atlas of the Basal Ganglia

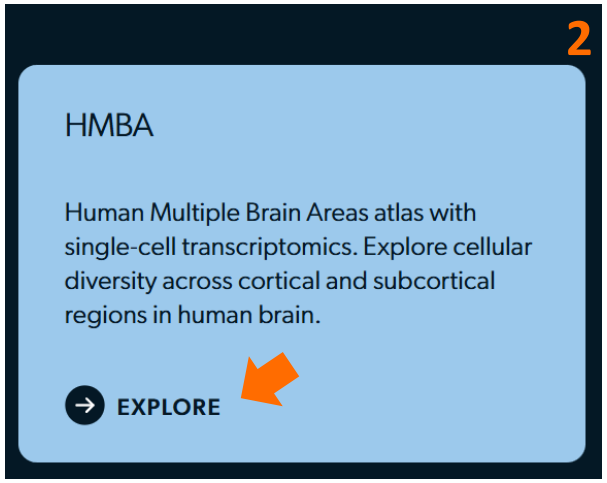
The BRAIN Initiative Cell Atlas Network (BICAN), Joseph R. Ecker, Michael Hawrylycz, Ed Lein, Bing Ren, Carol Thompson, Hongkui Zeng, Owen White, Guo-Qiang Zhang

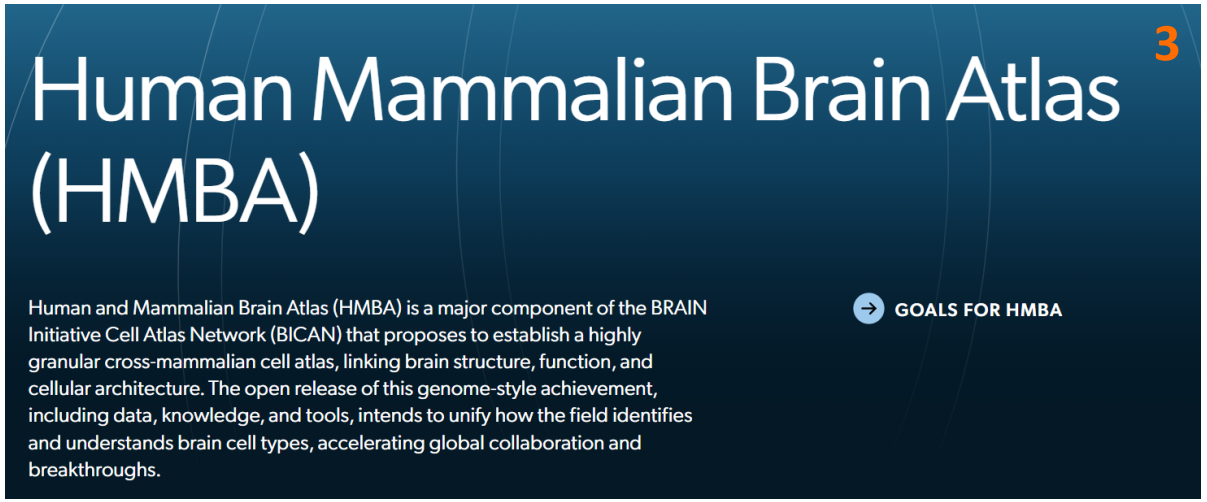
doi: https://doi.org/10.64898/2026.04.14.717814

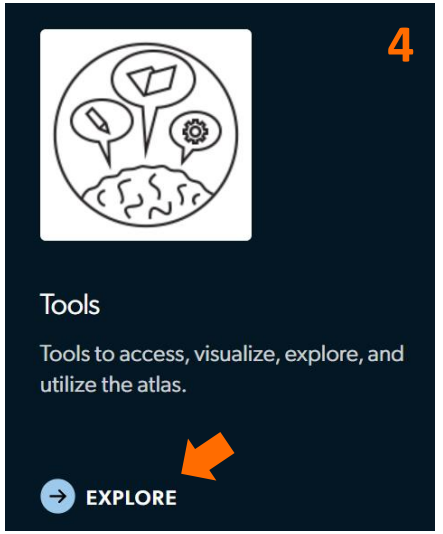
Finding tools on brain-map.org



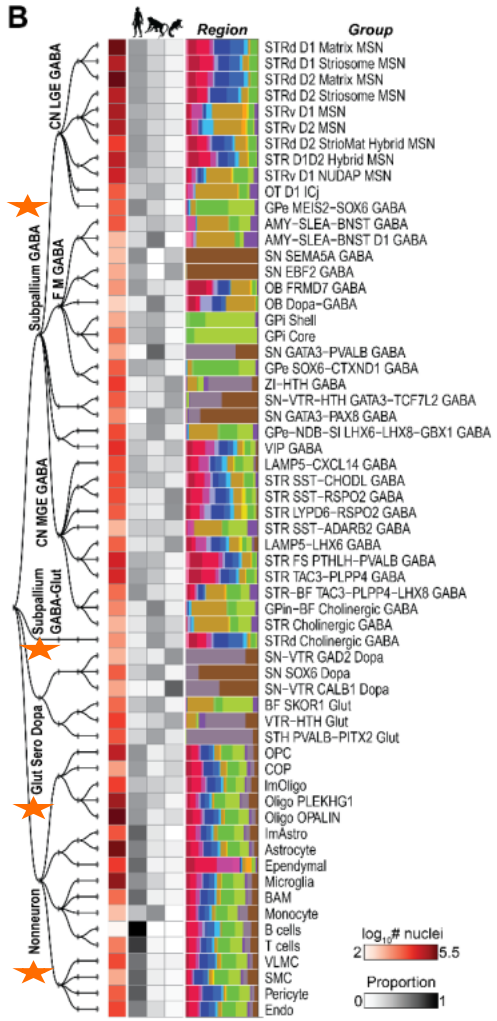
- Cross-species consensus atlas of the basal ganglia
- A cross-species spatial transcriptomic atlas of the human and non-human primate basal ganglia
- Spatial patterning of transcriptional and regulatory programs in the primate subcortex
- Morphoelectric diversity and specialization of neuronal cell types in the primate striatum
- Circuit specific specialization of human basal ganglia astrocytes
- Optimized methods for human postmortem brain preparation and 3D CCF mapping
- Towards human and non-human primate common coordinate frameworks using unified structural ontologies
- A consensus spinal cord cell type atlas across mouse, macaque, and human
- Human Neocortical Glutamatergic Neurons Revealed Through Multimodal Profiling
- Aging selectively affects morpho-electric features of intratelencephalic-projecting pyramidal neurons in human brain
- Gene expression and morphological features drive high prevalence of subthreshold membrane potential resonance in human cortical interneurons
- Fast Optimization of Robust Transcriptomics Embeddings using Probabilistic Inference Autoencoder Networks for multi-Omics

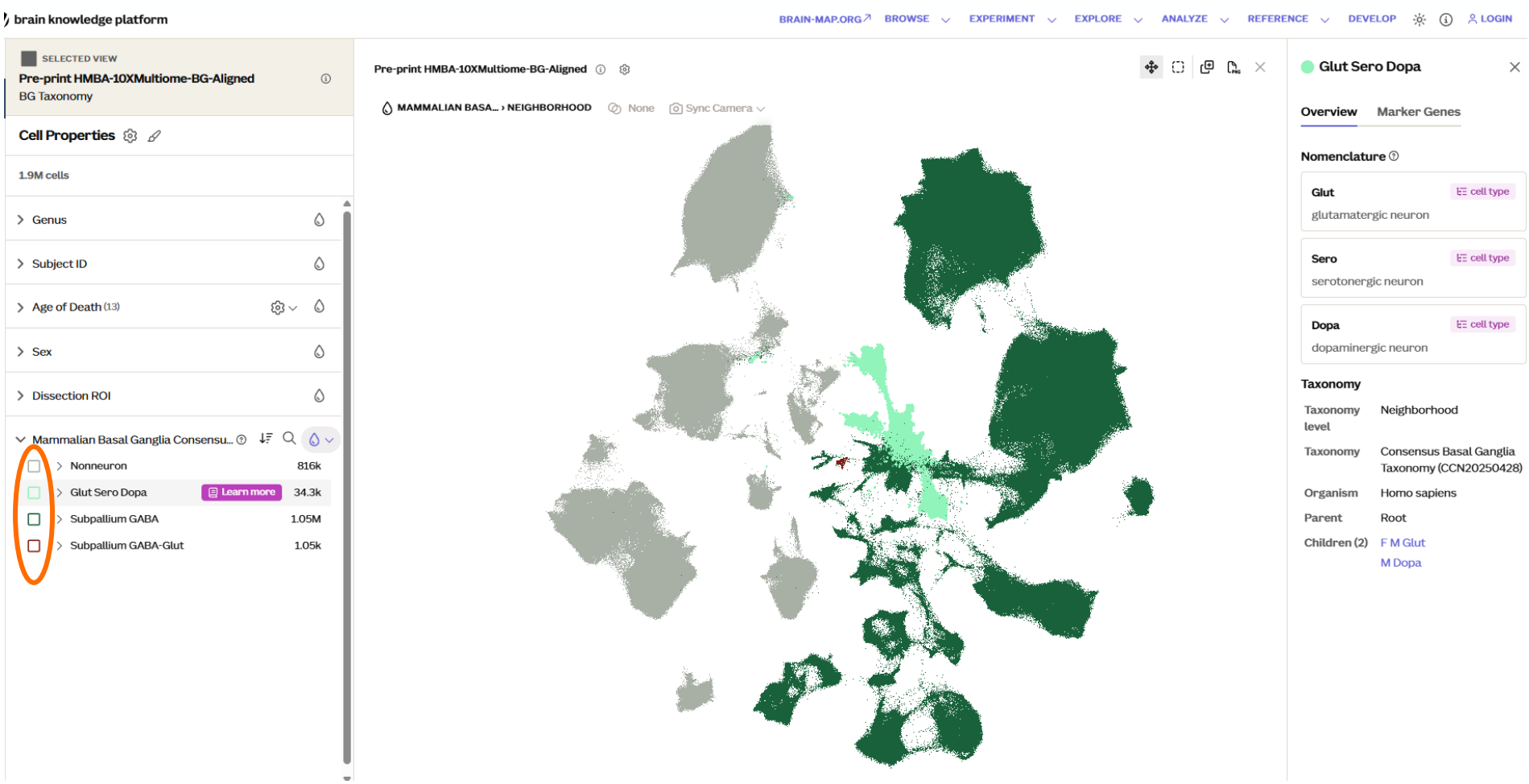






ABC Atlas: cross-species taxonomy data



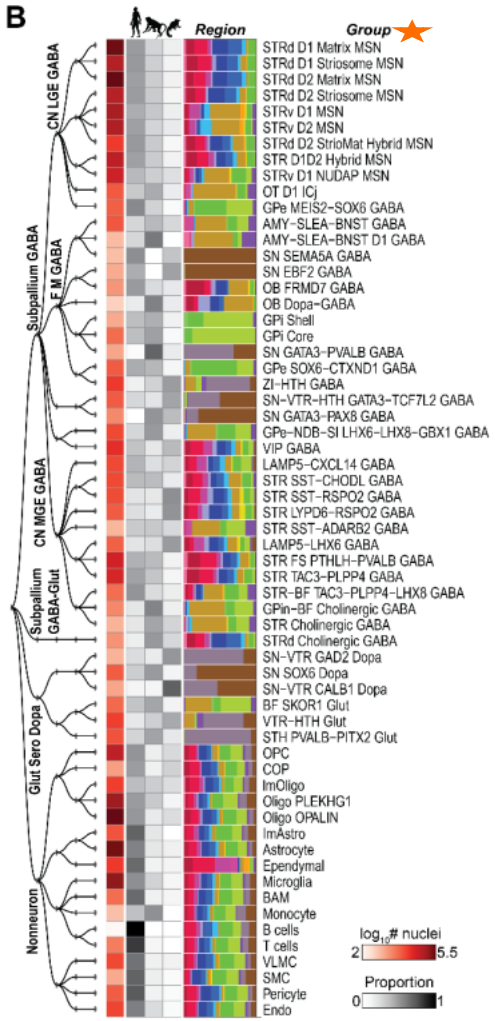


Cross-species consensus atlas of the primate basal ganglia

Johansen and Fu et al., 2025

YouTube tutorial at “Mapping the Cellular Architecture of the Mammalian Basal Ganglia”

ABC Atlas: cross-species taxonomy data



Cross-species consensus atlas of the primate basal ganglia

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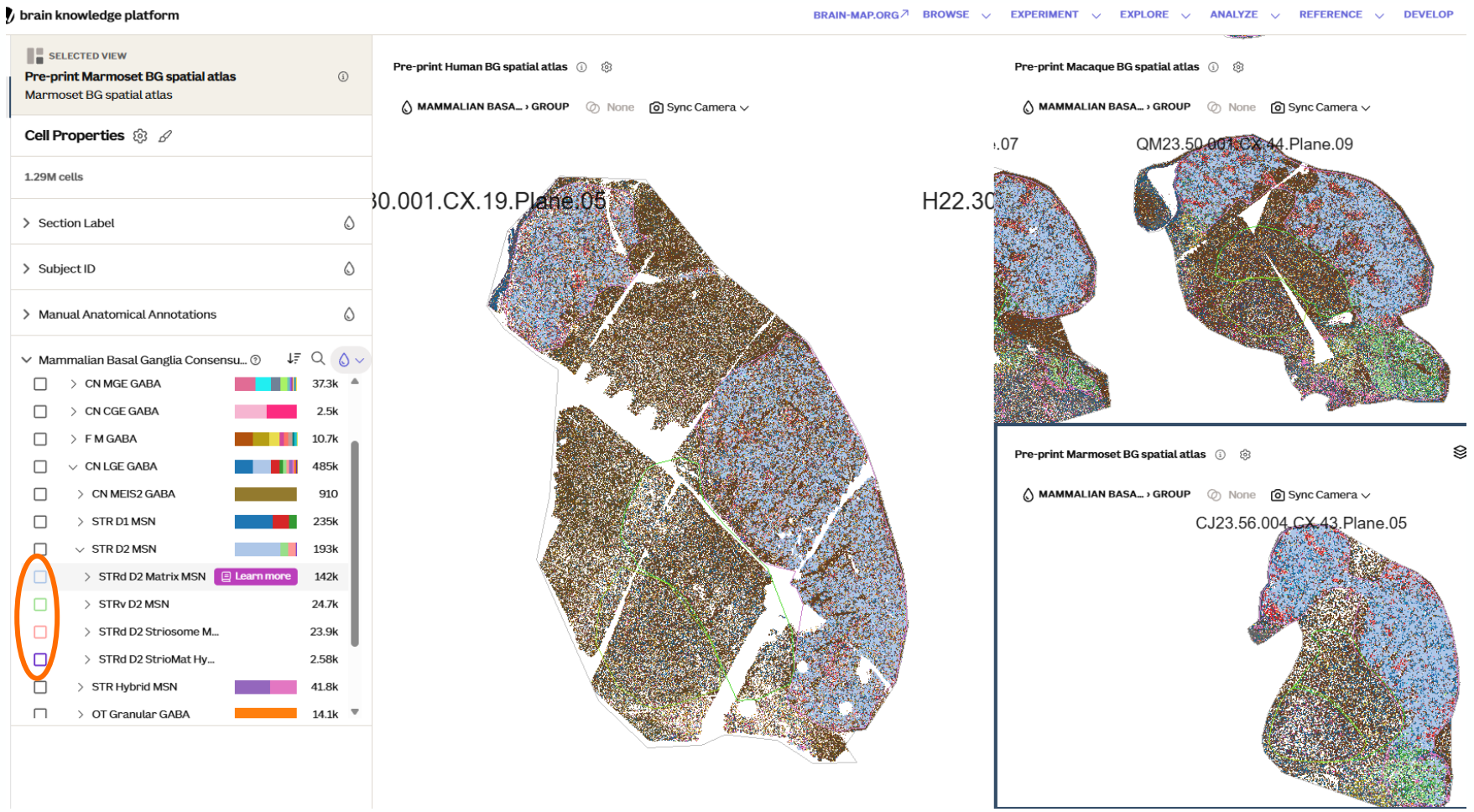
ABC Atlas: species-specific spatial tx data

A cross-species spatial transcriptomic atlas of the human and non-human primate basal ganglia Hewitt and Turner et al., 2025

YouTube tutorial at “Mapping the Cellular Architecture of the Mammalian Basal Ganglia”

ABC Atlas: species-specific spatial tx data

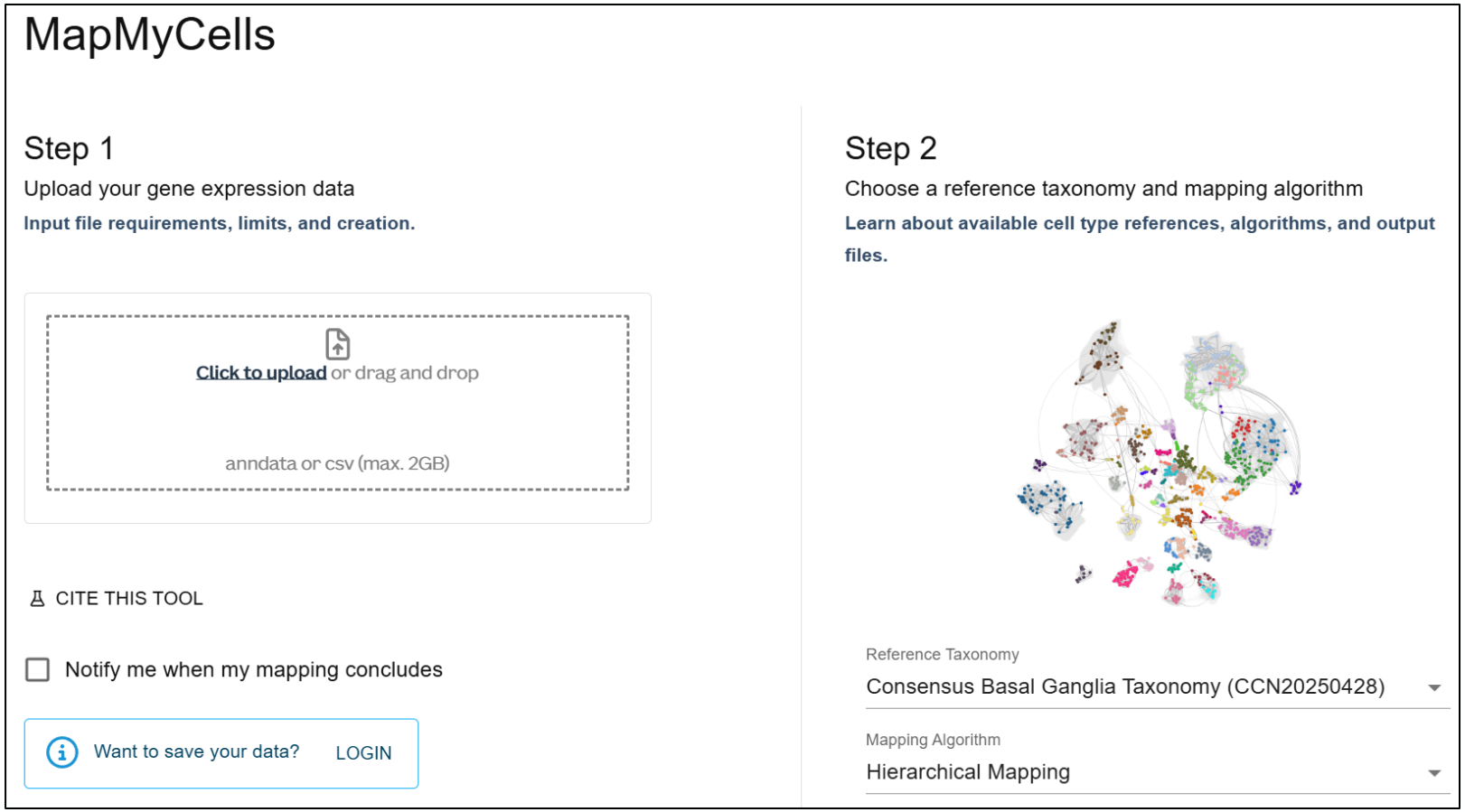


A cross-species spatial transcriptomic atlas of the human and non-human primate basal ganglia Hewitt and Turner et al., 2025

YouTube tutorial at “Mapping the Cellular Architecture of the Mammalian Basal Ganglia”

MapMyCells

- Mapping single cell data to the consensus basal ganglia taxonomy
- Accepts .h5ad or .csv files
- 2GB limit in GUI
- Maps at each level of taxonomy with confidence score



***MapMyCells:* High-performance mapping of unlabeled cell-by-gene data to reference brain taxonomies** Daniel et al., 2026

YouTube tutorial at “MapMyCells | Tutorial”

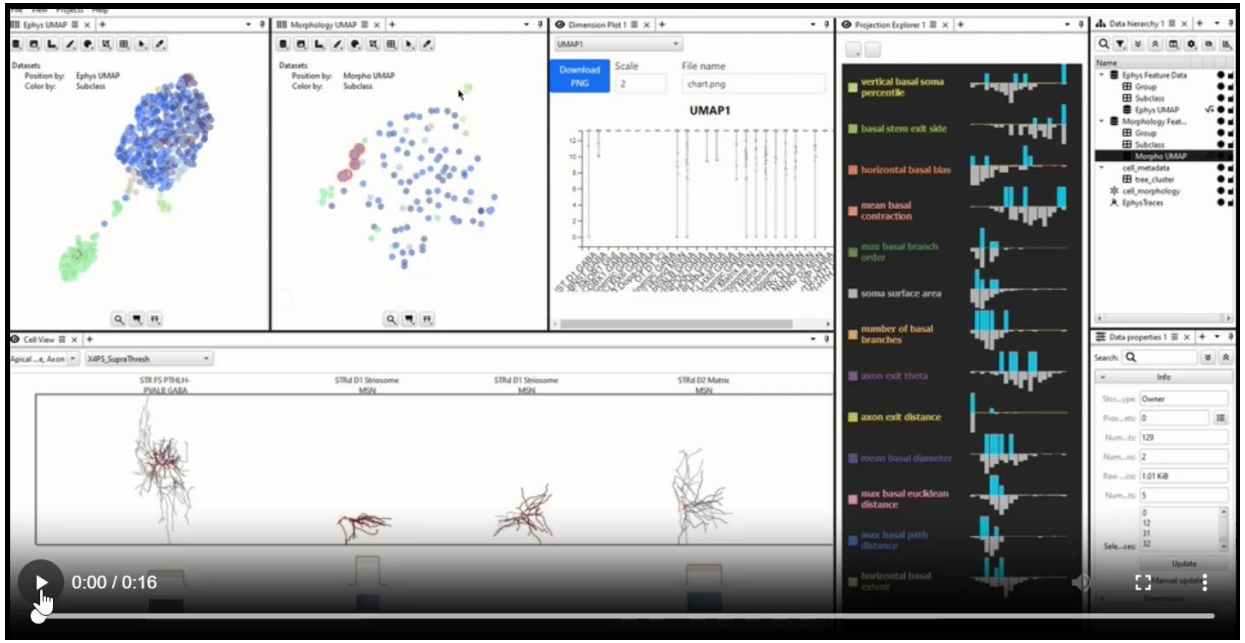
Cytosplore Viewer

Cytosplore Viewer is being developed in close collaboration between the **Allen Institute for Brain Science**, the **Leiden Computational Biology Center**, the **Division of Image Processing** at the **Leiden University Medical Center**, and the **Computer Graphics and Visualization Group** at the **TU Delft**. Data displayed in this viewer was downloaded and adapted from the **Allen Cell Types Database**.

HMBA data:

- Patch-seq human neocortex
- Patch-seq macaque basal ganglia (**pictured**)
- Cross-species snRNAseq taxonomy*
- Multi-species spatial transcriptomic*
- ATACseq + snRNAseq and spatial transcriptomic Marmoset data

*data also in ABC Atlas



viewer.cytosplore.org

Human Neocortical Glutamatergic Neurons Revealed Through Multimodal Profiling Dalley et al., 2026

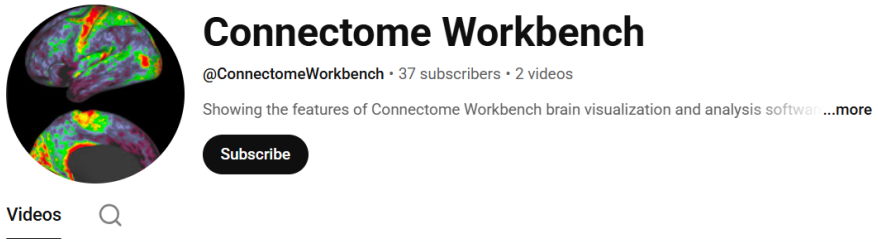
Morphoelectric Diversity and Specialization of Neuronal Cell Types in the Primate Striatum Liu et al., 2026

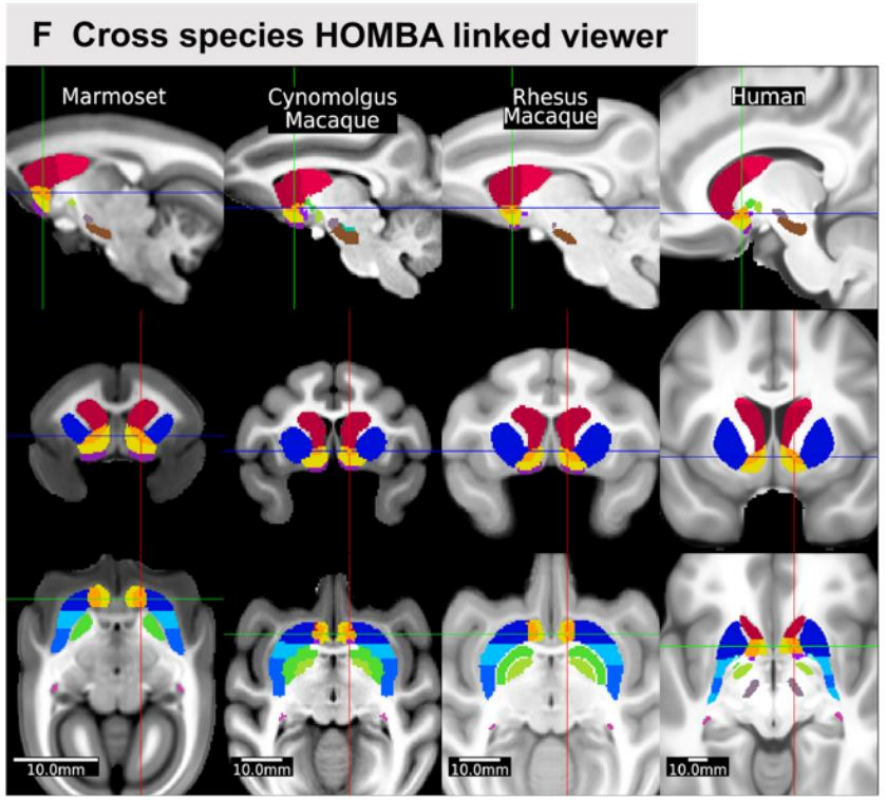
Spatial patterning of transcriptional and regulatory programs in the primate subcortex Dan, Turner, and DeBerardine et al., 2026

YouTube tutorial at “Mapping the Cellular Architecture of the Mammalian Basal Ganglia”

Connectome Workbench

- Visualization tool of MRI-registered data
- harmonized ontology of the mammalian brain anatomy (HOMBA) (**pictured**)
- Human basal ganglia spatial tx data
- Human connectivity data (human connectome project)





CONNECTOME COORDINATION FACILITY

Studies Software ConnectomeDB

CCF Staff

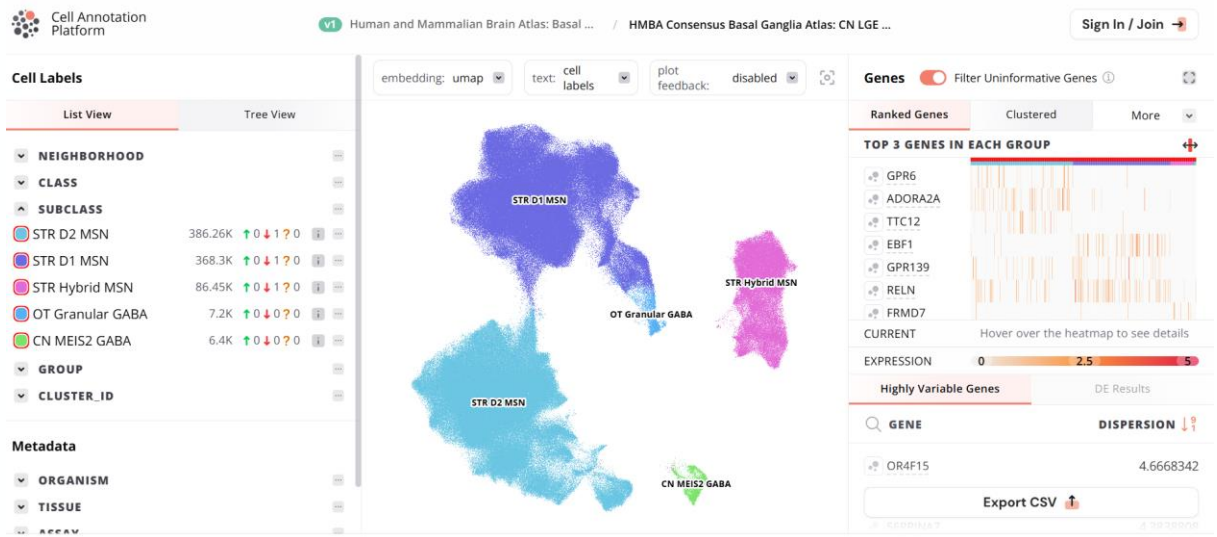
David Van Essen, Ph.D., Principal Investigator

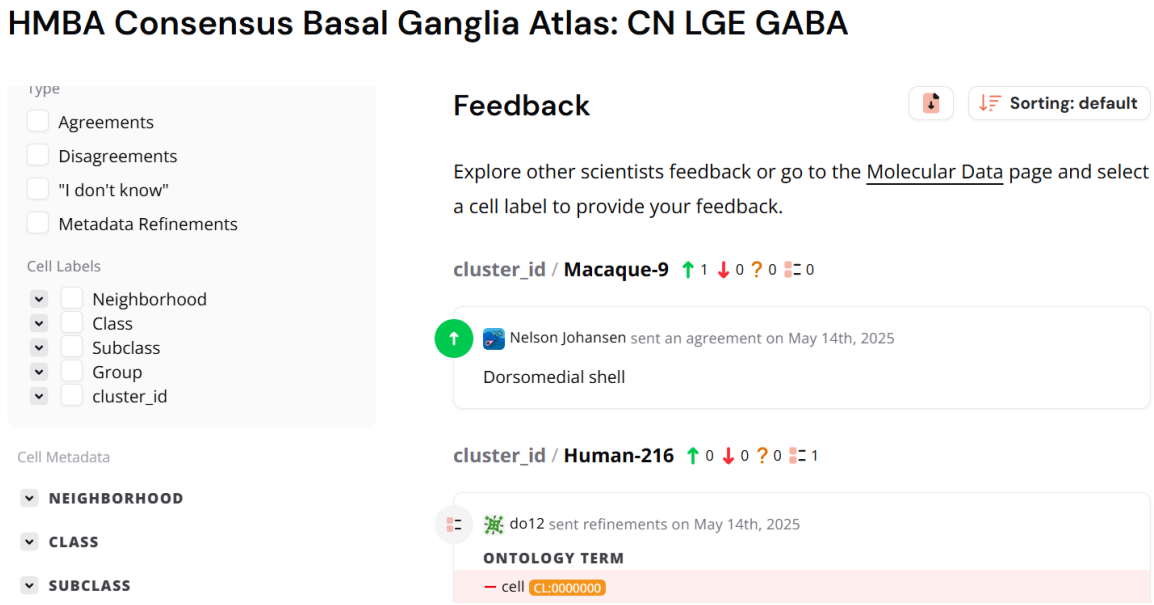
Daniel Marcus, Ph.D. – Principal Investigator

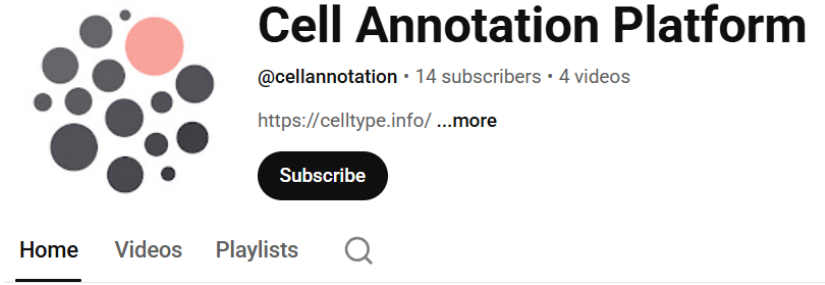
YouTube tutorial at “Mapping the Cellular Architecture of the Mammalian Basal Ganglia”

Cell Annotation Platform

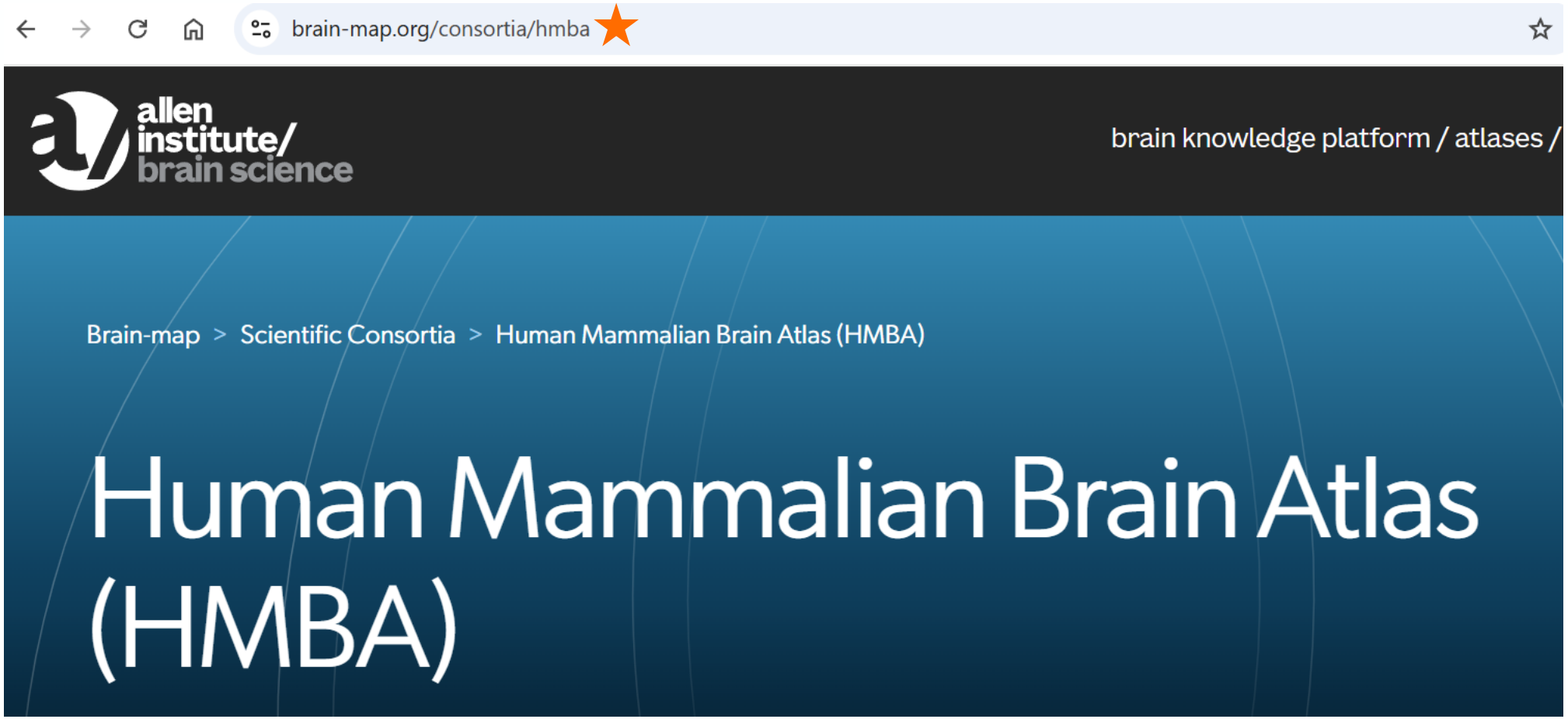
- Visualization tool of snRNAseq taxonomy data
- Provide comments and input on cell type annotations



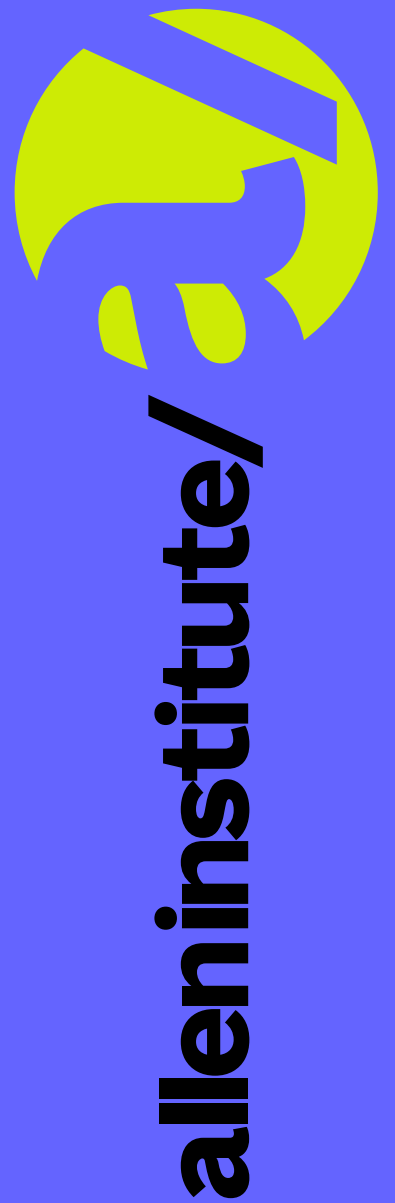




All tools accessible through brain-map

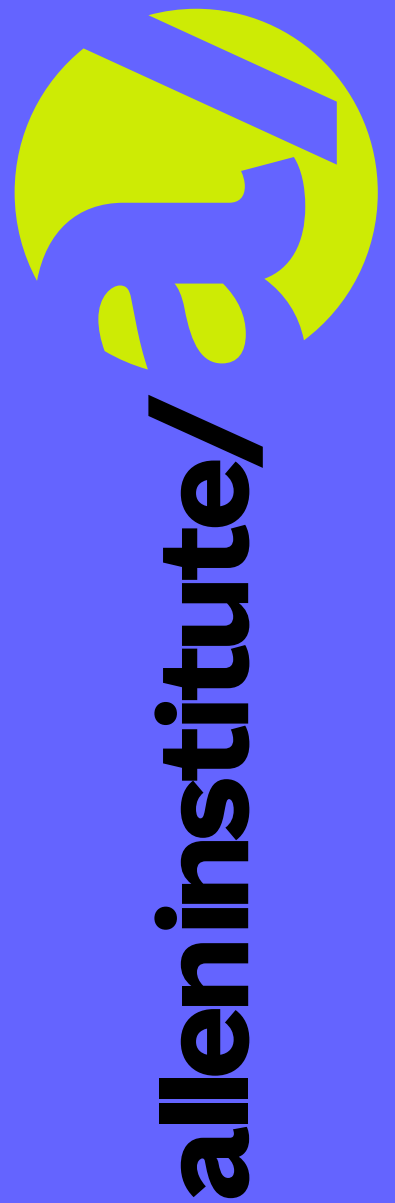


Questions?

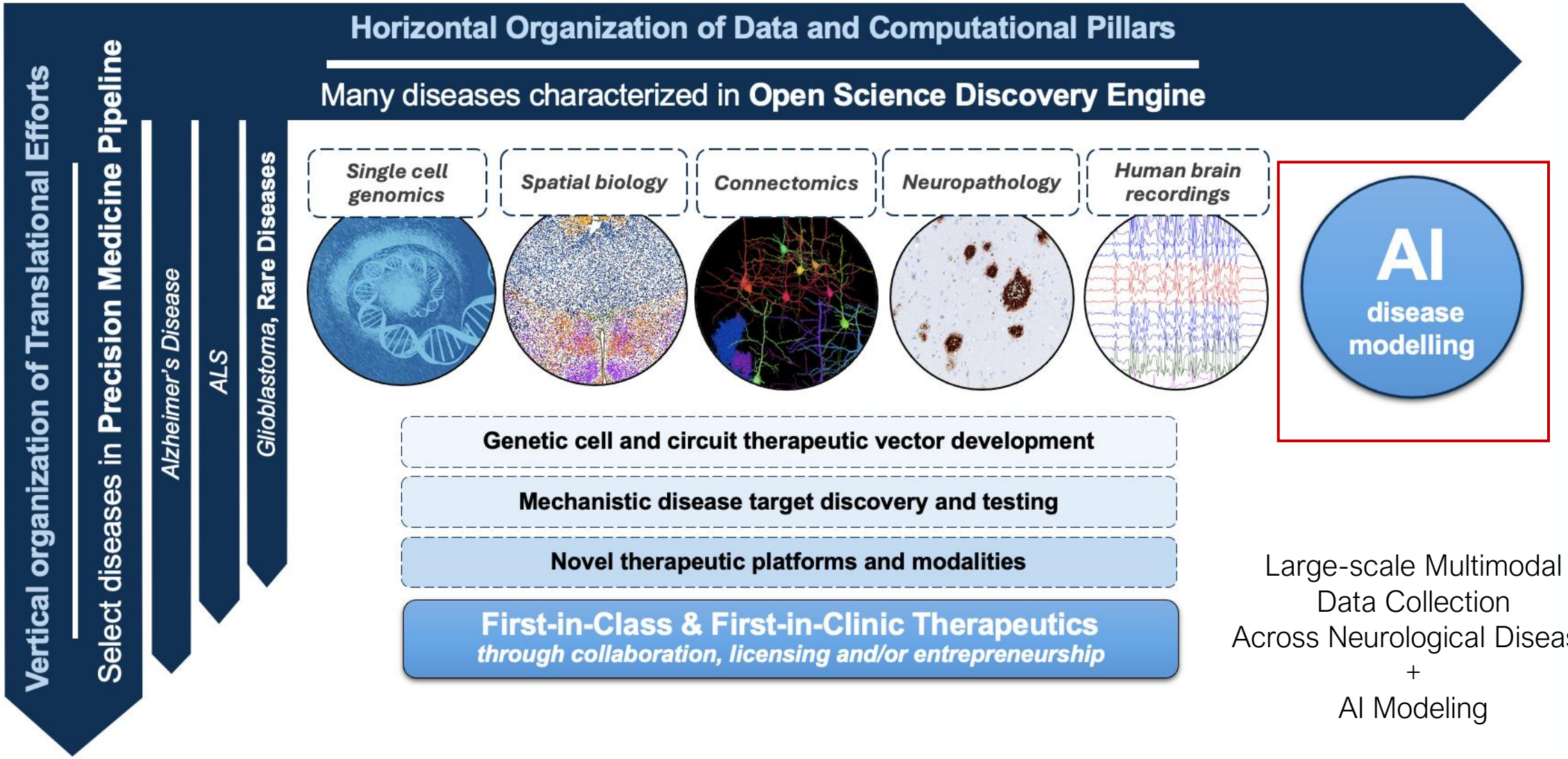


alleninstitute/BrainHealth

Mariano Gabitto



Brain Health Accelerator/



Large-scale Multimodal
Data Collection
Across Neurological Disease
+
AI Modeling

Brain Health Accelerator/AI

BHA Data / Community Data

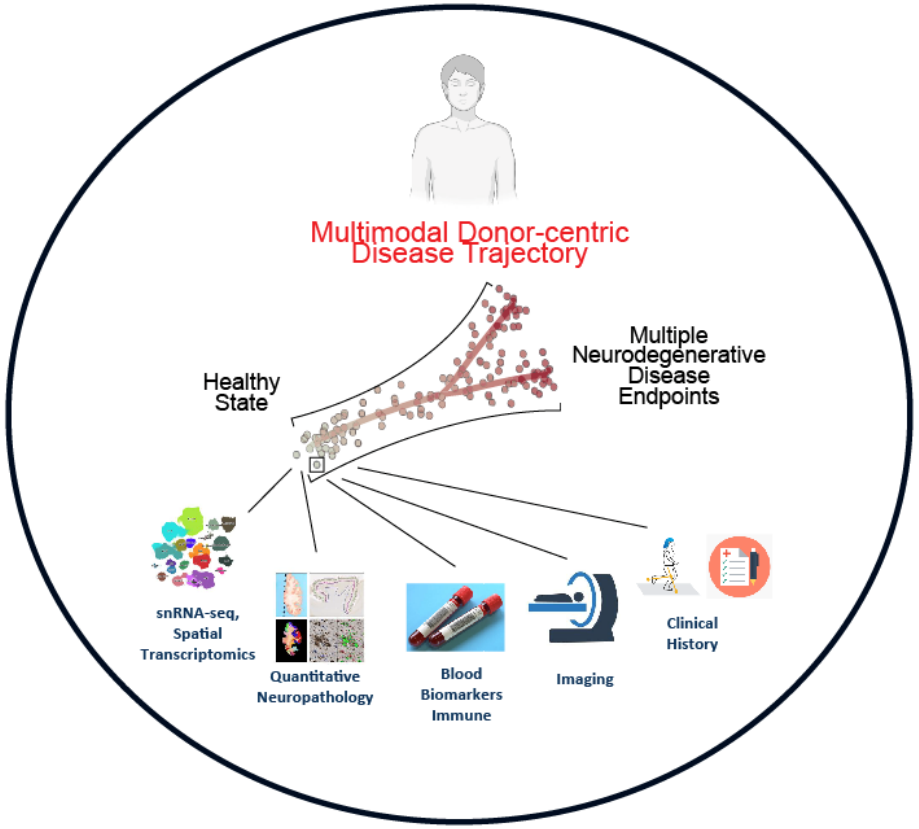
AD

HD

PD

ALS

...



Knowledge

Reference Taxonomy

Integration Across Diseases

Cell Type Mapping

Disease Trajectories

In vivo/ex vivo Predictions

Dissemination Tools

Goal: To create multimodal disease trajectories, integrating omics, the genome, pathological and clinical information.

Brain Health Accelerator/AI

BHA Data / Community Data

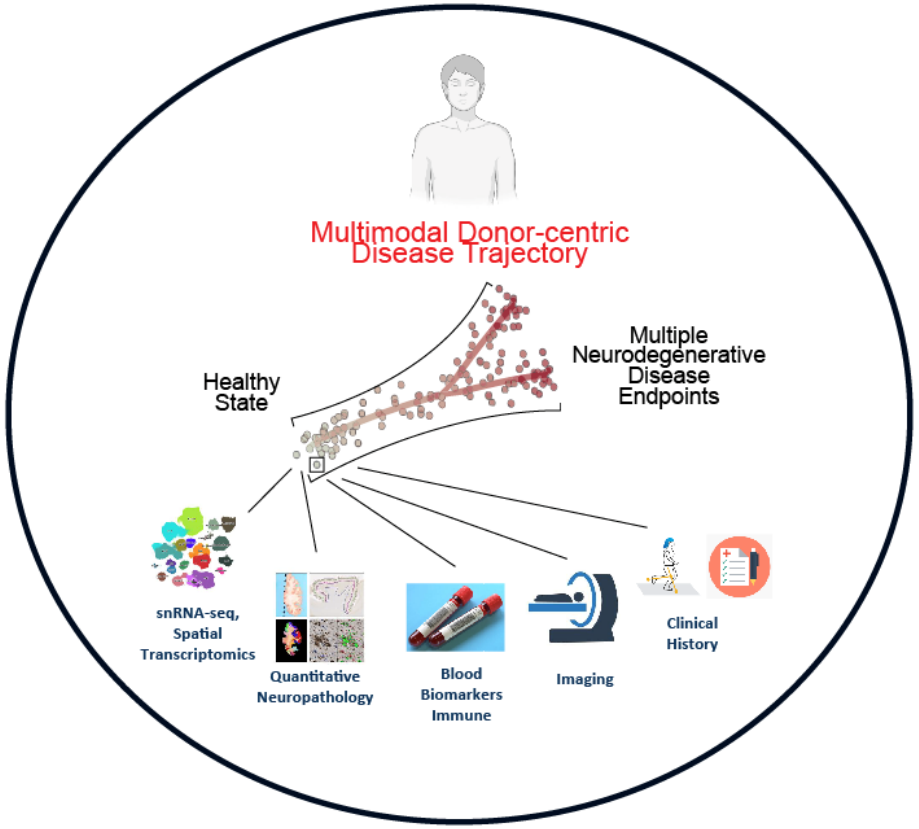
AD

HD

PD

ALS

...



Knowledge

Reference Taxonomy

Integration Across Diseases

Cell Type Mapping

Disease Trajectories

In vivo/ex vivo Predictions

Dissemination Tools

To bridge *post-mortem* ground truth to *in vivo* disease biomarkers and identify disease phenotypes and their underlying molecular and cellular consequences.

Understanding the brain and the consequences of neurodegenerative diseases present enormous complexity.

Diverse cellular system

100 billion
Neurons

>5,000
Cell Types, based on genetics

Highly connected circuitry

100 trillion
Synapses between cells

10x-100x
More complex than other organs

Single-cell technologies have been instrumental in mapping the cells in the human brain both in health and disease

Photo credit: Ivana Kapustová and Sten Linnarsson

ALLEN INSTITUTE

ALLEN INSTITUTE for BRAIN SCIENCE

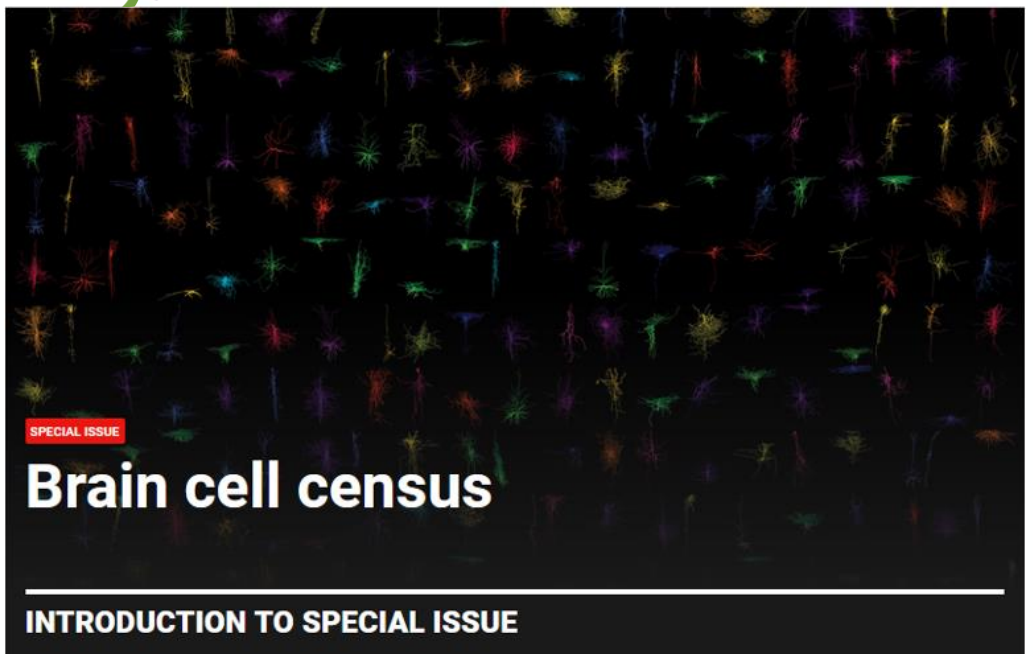
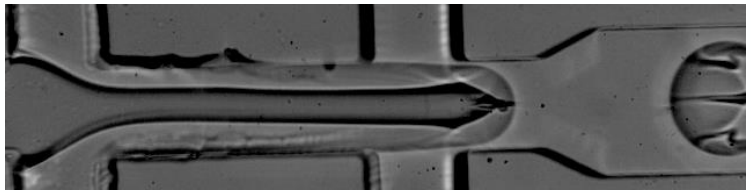
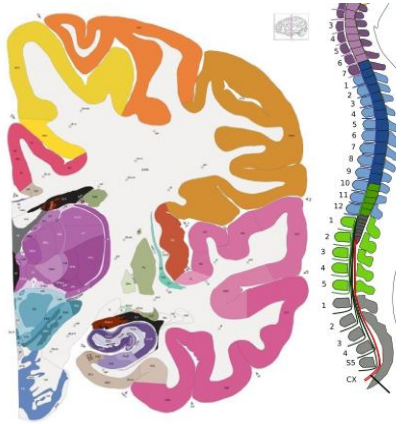


Photo Credit: Rachel Dalley, Sarah Walling-Bell and Brian Lee



MacCarrol Lab

Human and cross-species Brain Cell Atlases as Foundational References: Building on investment from the NIH Brain Initiative



>3000 cell types: Survey of cell type diversity in ~100 regions of the human brain and spinal cord in 3 individuals

Siletti, Hodge…Lein, Linnarsson (Science, 2023)

HMBA

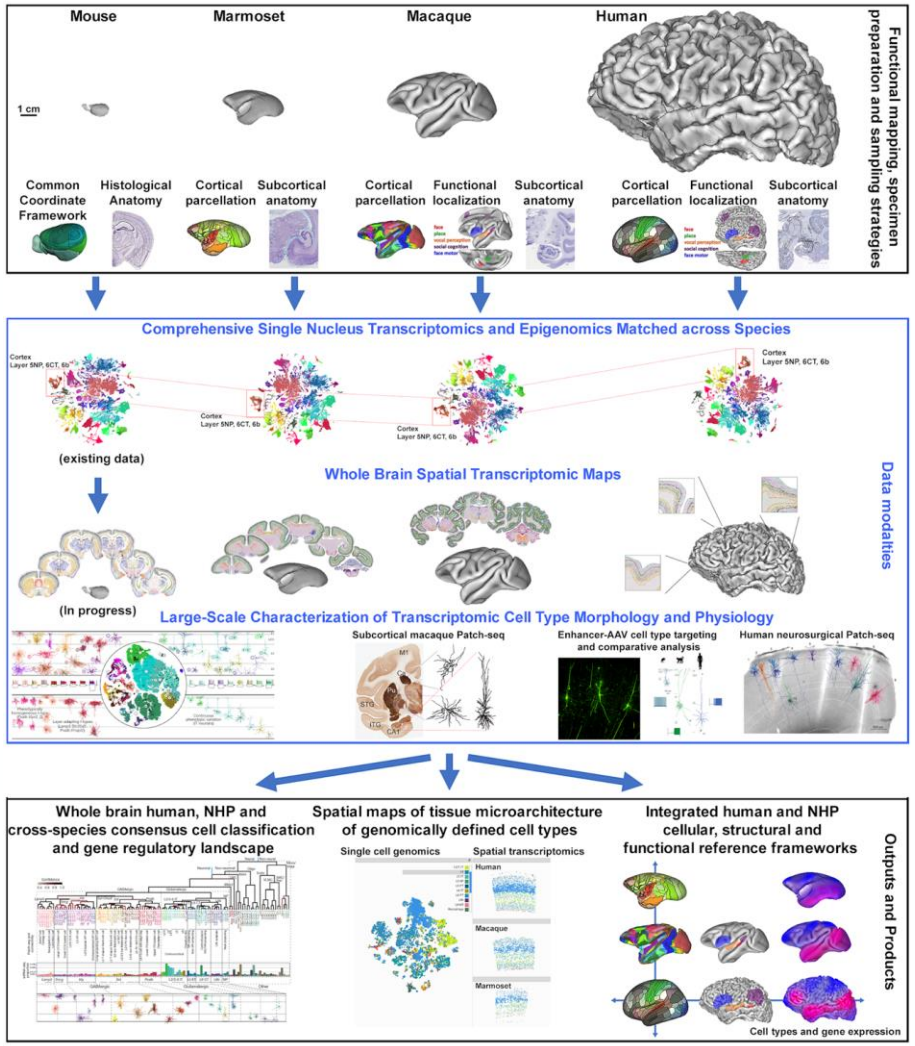
fMRI

Single cell genomics

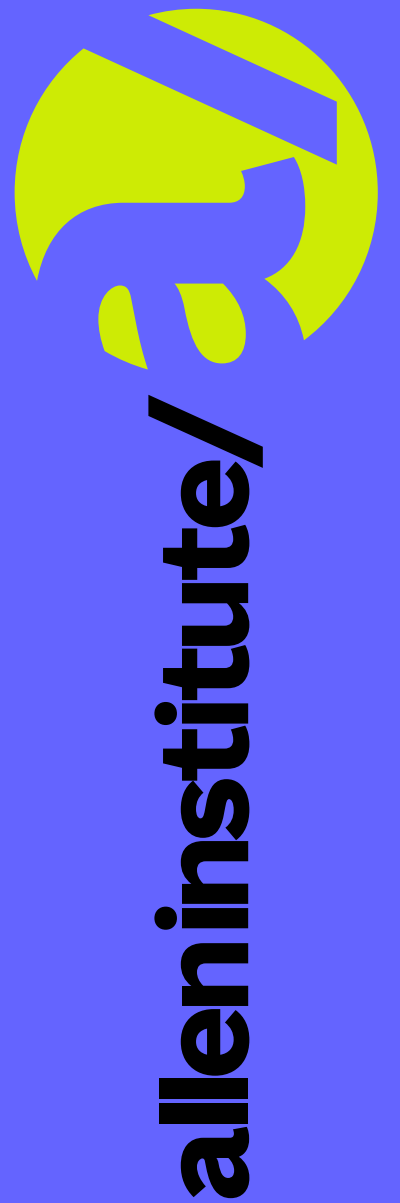
Spatial transcriptomics

Patch-seq

Products (Tools/Data for scientist)

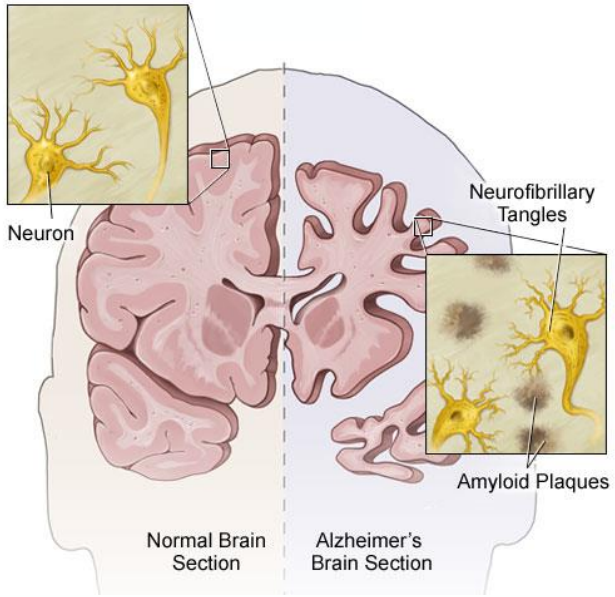


Seattle Alzheimer's Disease Brain Cell Atlas/



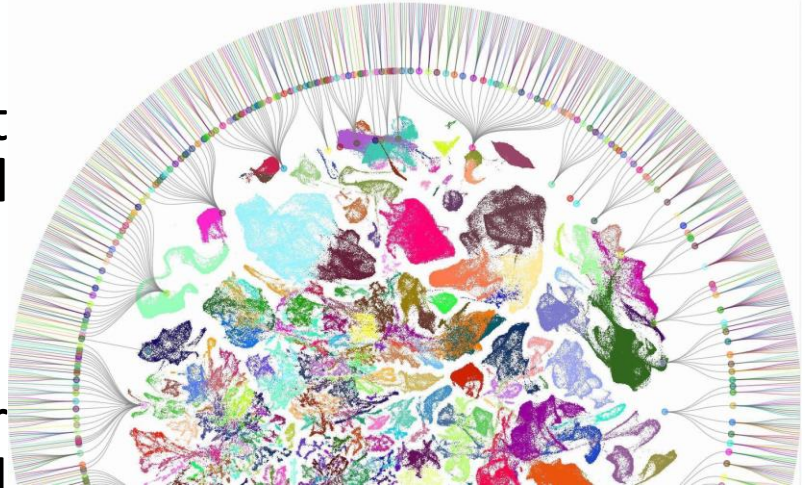
what is SEA-AD?

(Seattle Alzheimer's Disease Brain Cell Atlas)



- A multisite consortium project employing cutting- edge single cell and spatial genomics coupled with quantitative neuropathology to create a **high-resolution brain cell atlas** of Alzheimer's disease.

- Overarching goal: map the cellular and molecular events that occur over the course of disease to **identify vulnerable cell types** across the brain.



- Share this data and knowledge by creating a **data resource** for the scientific community to accelerate future diagnostic and therapeutic strategies.

SEA-AD consortium = team science

The Seattle Alzheimer's Disease Brain Cell Atlas (SEA-AD) consortium is supported by the National Institutes on Aging (NIA) grant 2U19AG060909.

NIH National Institute on Aging

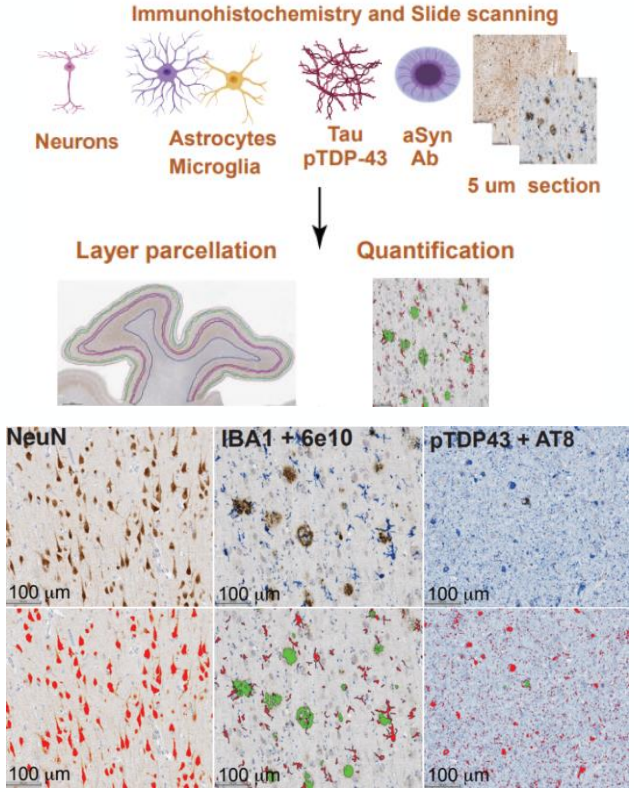
Shanahan Family Foundation

data generation, analysis, and the search for vulnerable cell types in AD

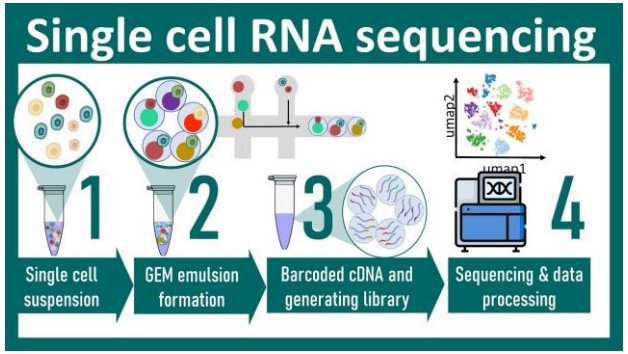
Donor Metadata/ Clinical History

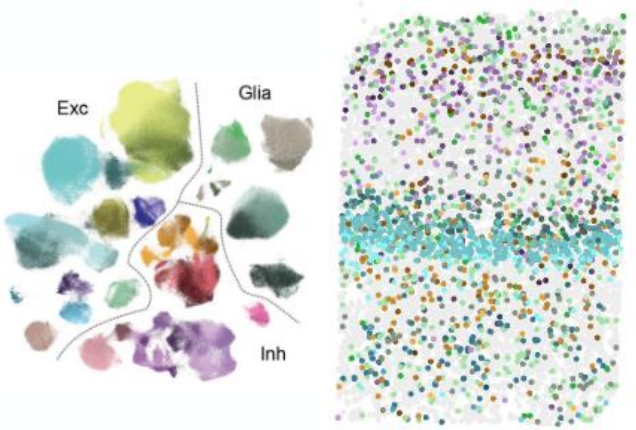
Donor ID	Age at d...	Sex	APOE ge...	Cognitive status	Years o...	Continuous ...	ADNC	Braak stage	Thal phase
H20.33.040	98	Male	3/3	Dementia	13	0.439	Intermediate	Braak IV	Thal 4
H20.33.036	100	Female	2/3	No dementia	15	0.844	High	Braak V	Thal 5
H21.33.019	75	Male	2/3	No dementia	15	0.517	Low	Braak 0	Thal 1
H21.33.040	83	Male	3/4	No dementia	17	0.745	High	Braak V	Thal 4
H20.33.029	91	Female	3/3	Dementia	13	0.613	High	Braak V	Thal 4
H20.33.015	88	Male	3/3	Dementia	18	0.879	Intermediate	Braak V	Thal 3
H21.33.012	93	Female	2/4	Dementia	21	0.486	Intermediate	Braak IV	Thal 3
H21.33.003	78	Male	3/3	No dementia	16	0.238	Not AD	Braak 0	Thal 0
H20.33.002	97	Female	2/3	No dementia	12	0.207	Not AD	Braak IV	Thal 0
H21.33.030	89	Male	3/4	No dementia	17	0.252	Intermediate	Braak III	Thal 3
H20.33.046	94	Male	3/3	Dementia	21	0.868	High	Braak VI	Thal 5
H20.33.031	87	Female	3/3	Dementia	12	0.778	High	Braak VI	Thal 4
H20.33.032	98	Male	3/3	No dementia	17	0.818	High	Braak V	Thal 5
H21.33.027	92	Male	3/4	Dementia	18	0.687	High	Braak V	Thal 5
H19.33.004	80	Female	3/3	No dementia	17	0.302	Not AD	Braak IV	Thal 0
H20.33.028	94	Female	3/3	Dementia	12	0.815	High	Braak V	Thal 4
H20.33.008	92	Female	3/4	No dementia	18	0.754	High	Braak V	Thal 4

Quantitative Neuropathology

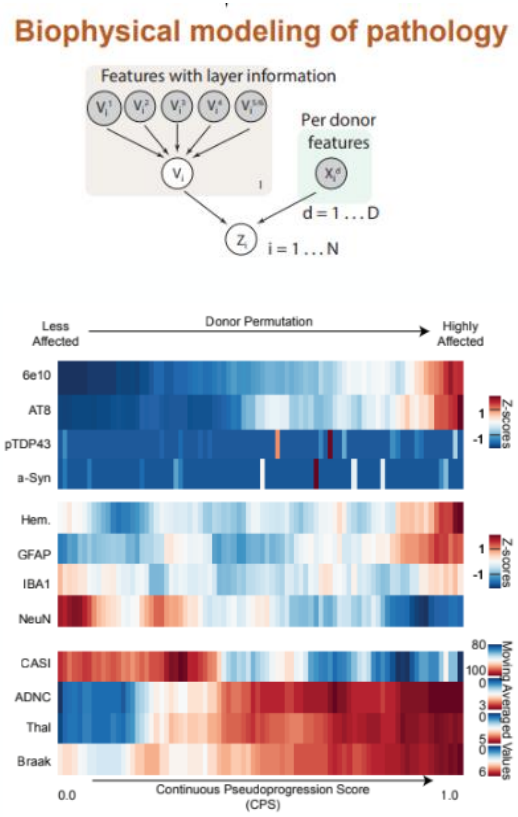


Single Cell and Spatial Genomics



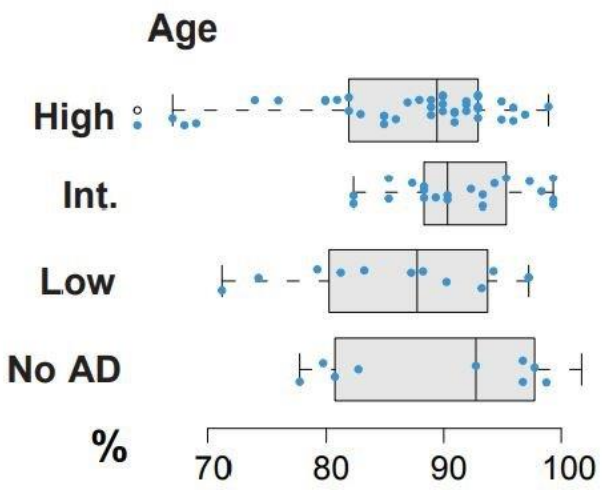


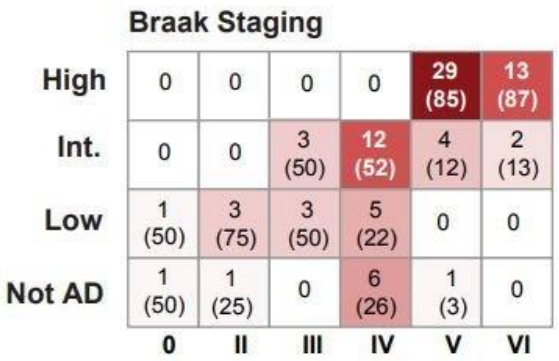
Modeling

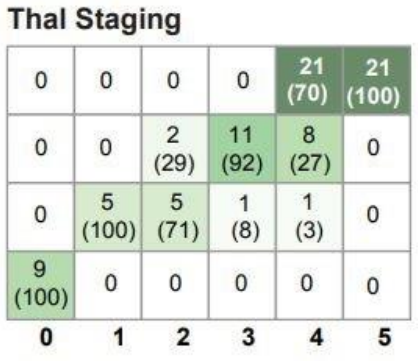


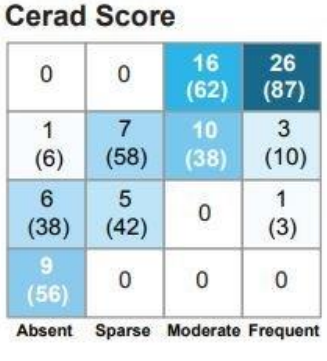
Gabitto et. al., Nature Neuroscience 2024

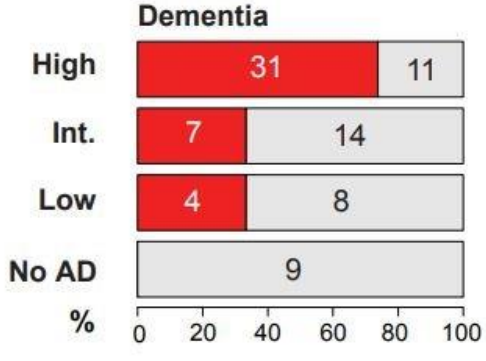
SEA-AD cohort: moving beyond case-control to capture spectrum of AD pathology

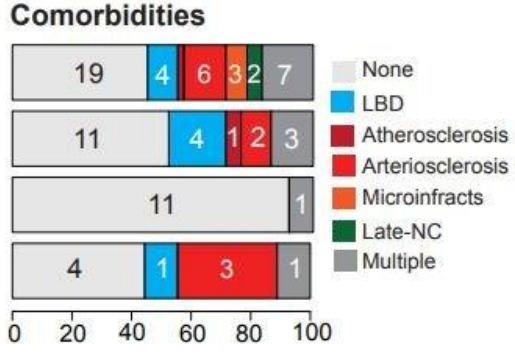












HMBA gave us an initial cellular taxonomy to understand single-cell information

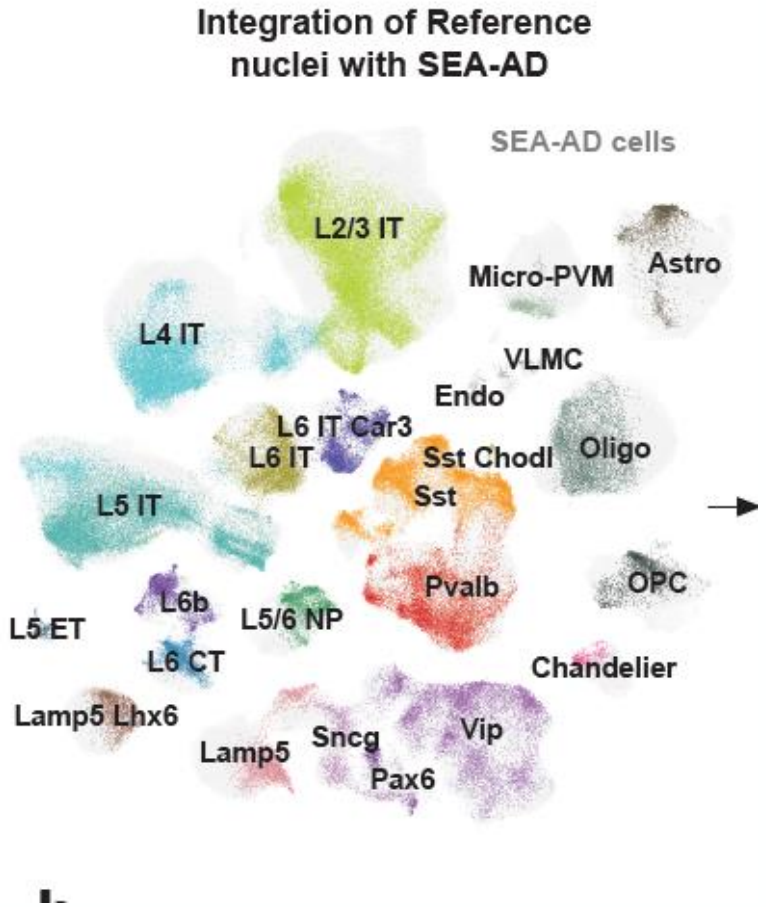
We profiled

Single-nucleus

RNA & ATAC –seq

in each of the SEA-AD

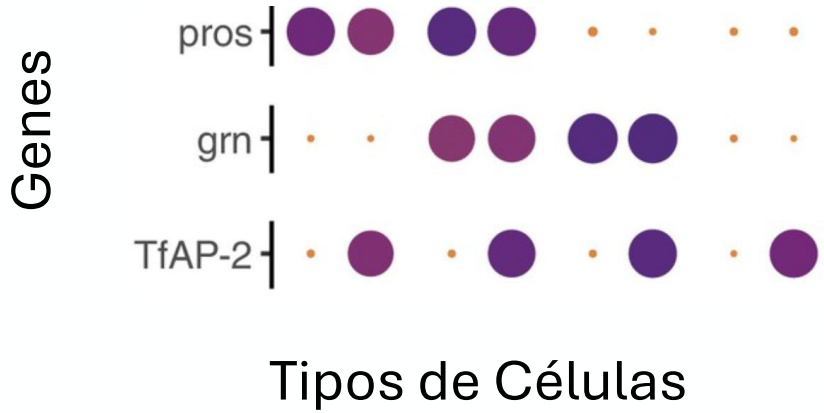
donors



Gabitto et. al., Nature Neuroscience 2024

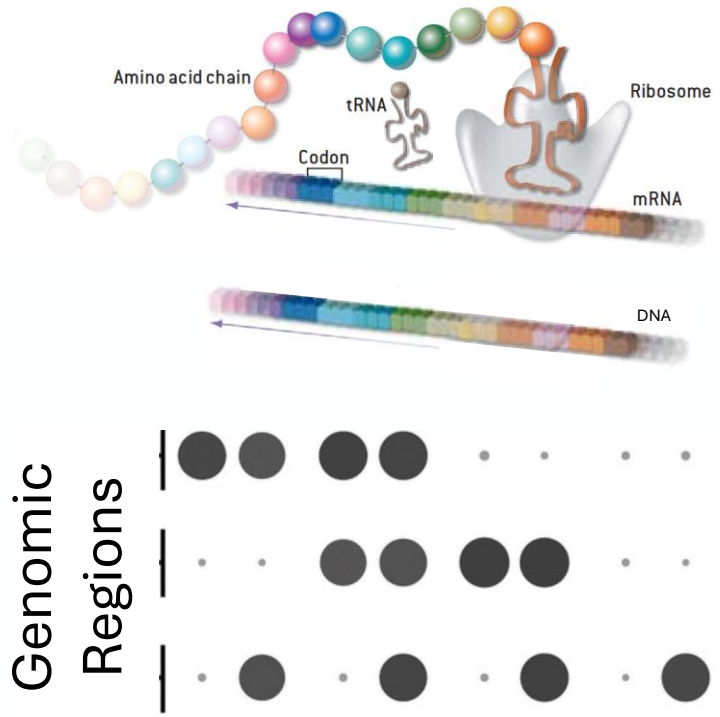
Multiple single-cell Technologies profile different aspect of the cell

Transcriptional Landscape



Chromatin Landscape

Shapiro & Benenson 2006



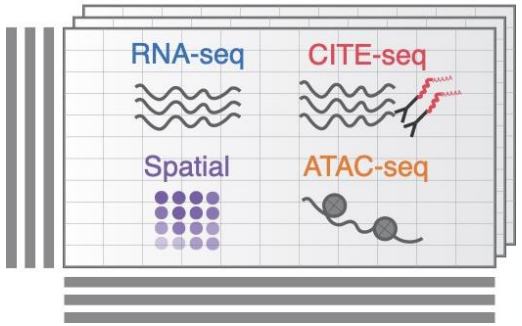
Enabling Multi- and Single-omic Studies

Condense into a single low dimensional representation

Solving the challenges when analyzing single-nucleus genomics with Probabilistic Modeling

Observations

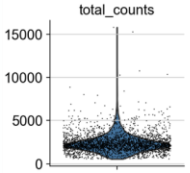
Data (X)



Covariates (S)

Genotype

Batch

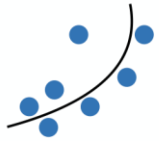


Total counts

Cell type

Not observed

Latent (Z)

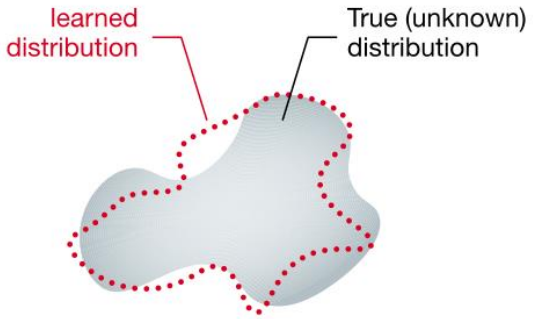


Cell state

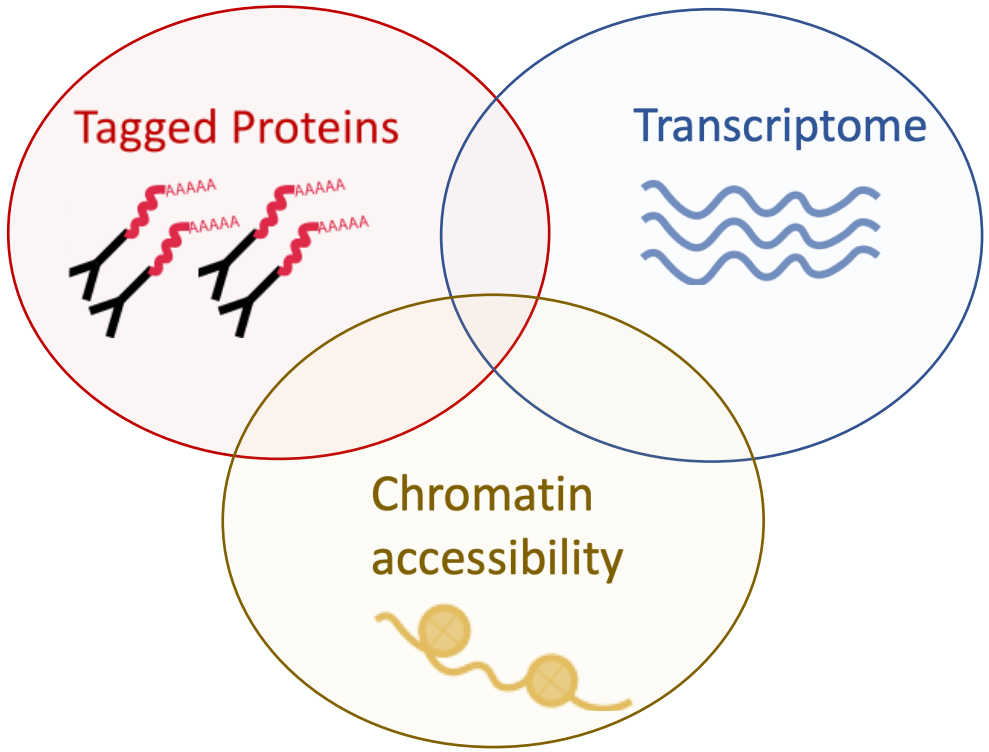
Technical factors

Cell type

Model: P(X, Z | S)



Enabling Multi- and Single-omic Studies



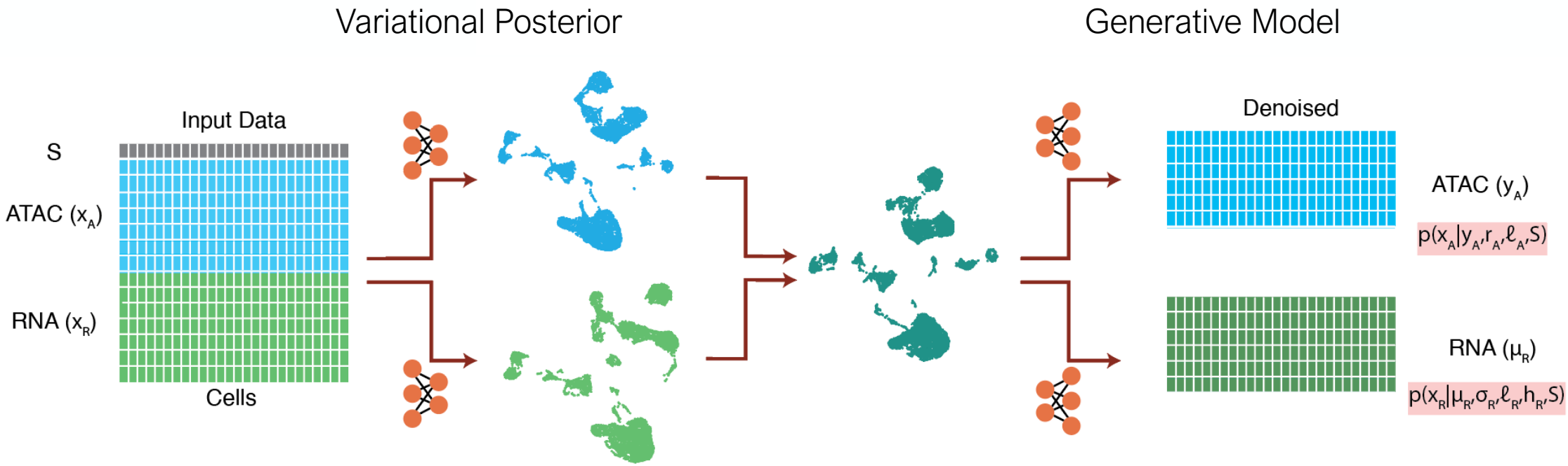
MultiVI

(Ashuach*, Gabitto* et al., Nat. Methods 2023)

- Enhanced notion of cellular identity
- Hypotheses testing for all modalities
- Enhancement of single- modality data

scVI-tools: Python-based scalable Tools for single-cell analysis.

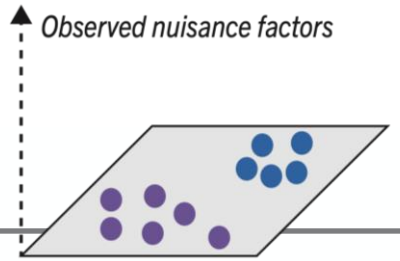
MultiVI for the Joint Analysis of RNA-seq and ATAC-seq



Low dimensional representation:

- Stratification into subpopulations, gradients
- Accelerated annotation
- Visualization

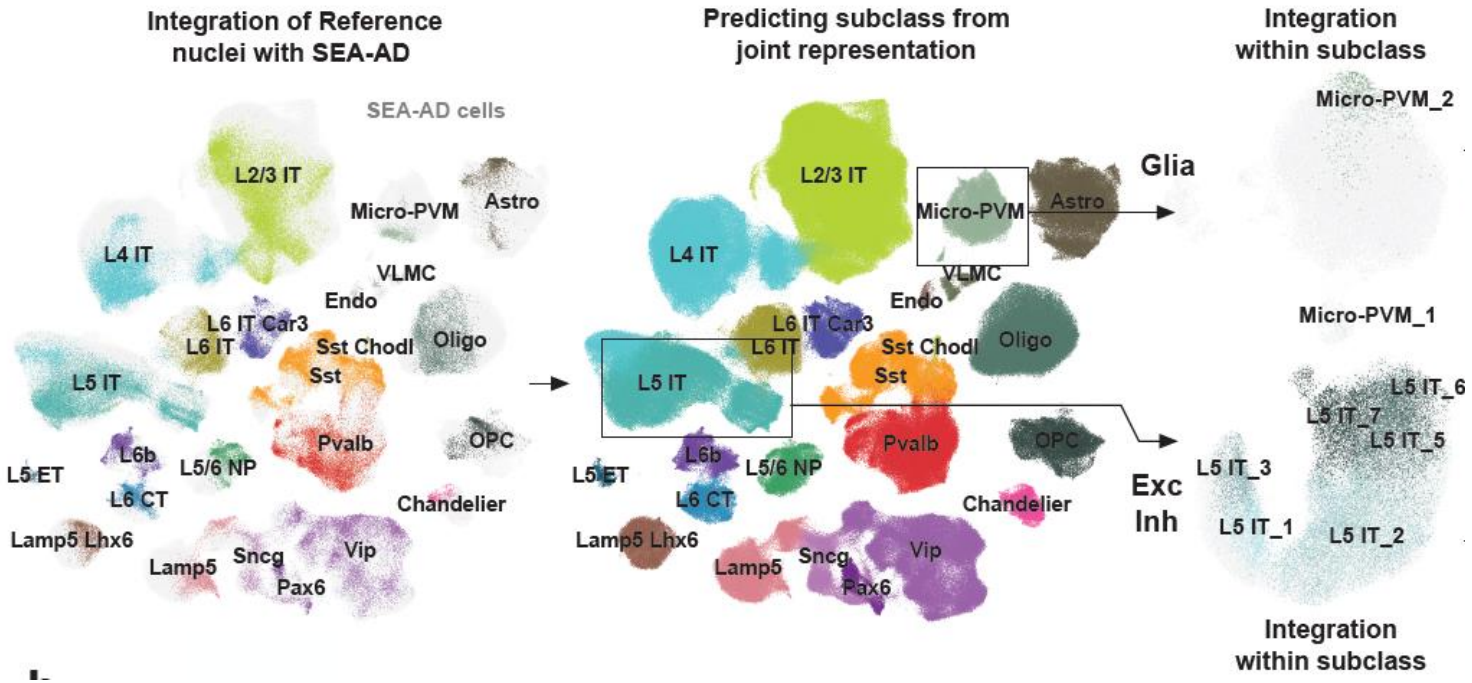
Removal of unwanted variation
(Dataset integration)

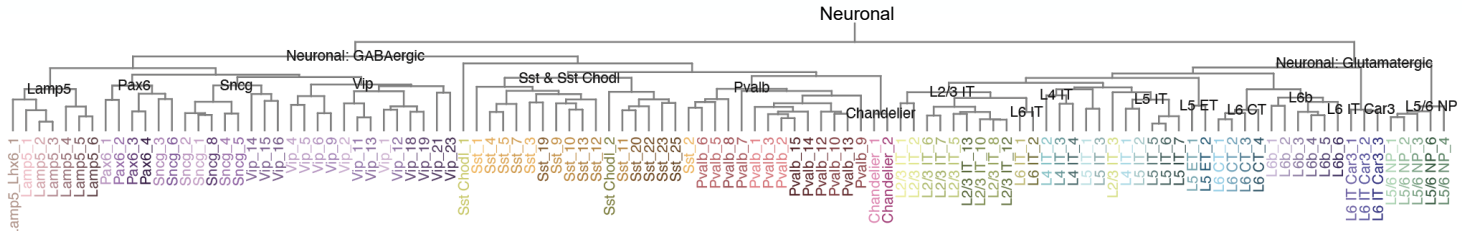


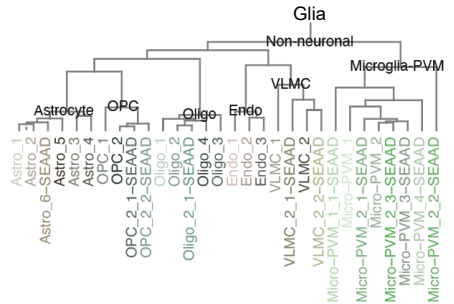
High dimensional representation:

- Differential analysis
- Accelerated annotation
- Dependencies between molecules

initial cell type taxonomy extends healthy adult MTG cell types aged donors







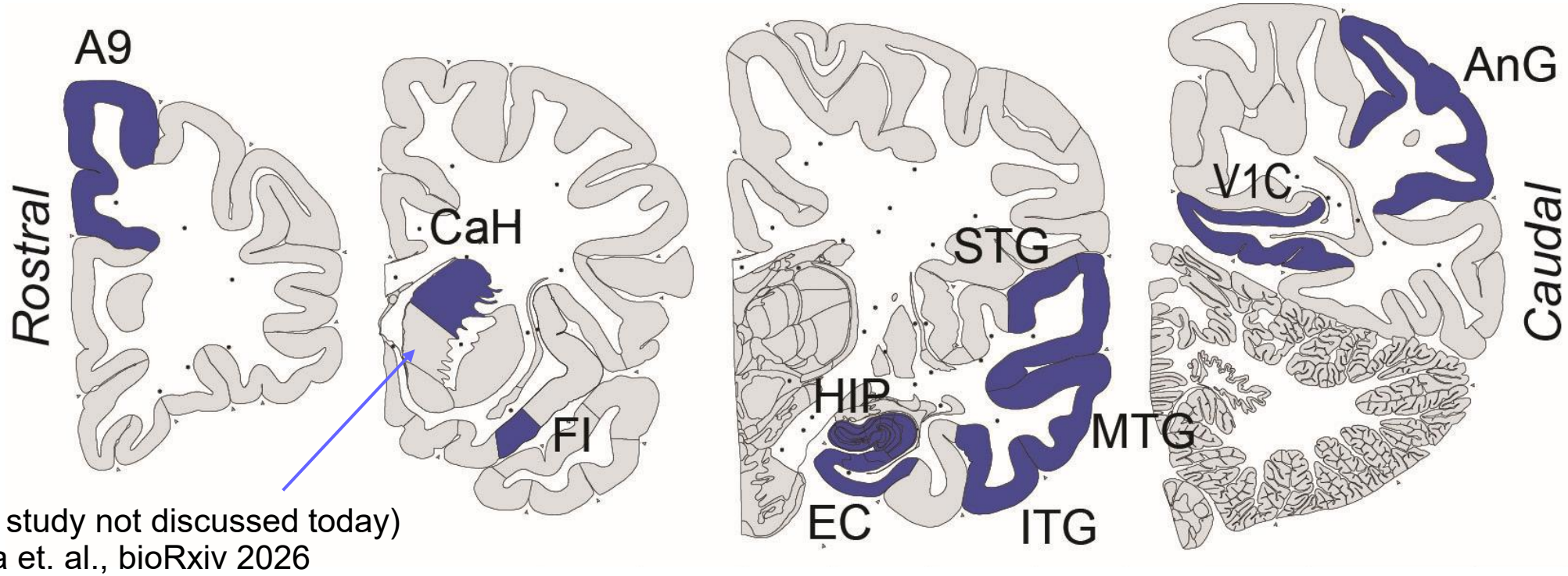
Class (3)

Subclass (24)

Supertype (139)

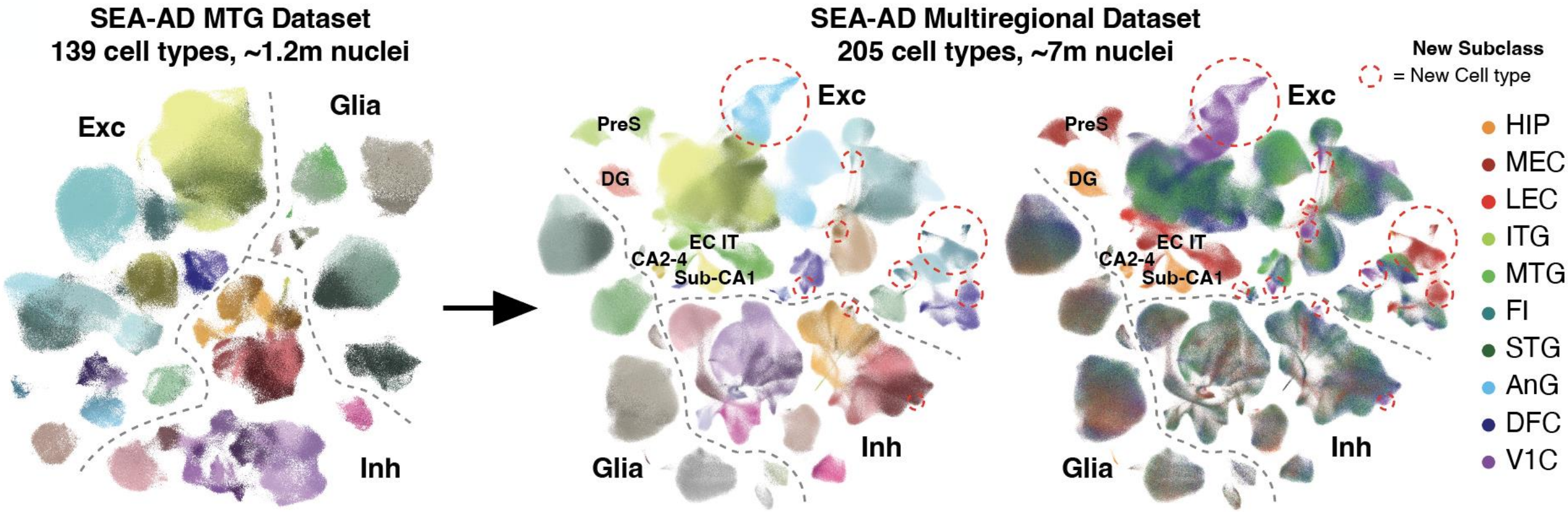
Gabitto et. al., Nature Neuroscience 2024

current data sets span 11 brain structures in the aged brain, across modalities



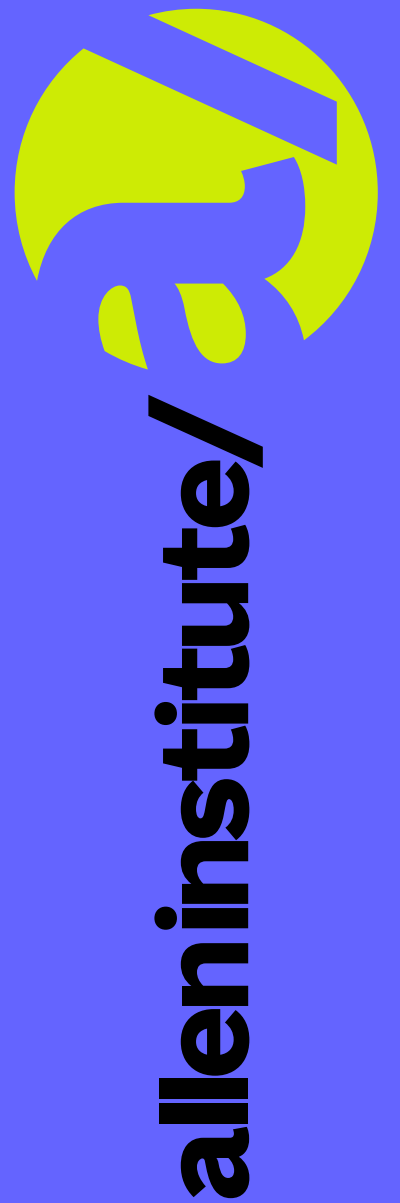
(Separate study not discussed today)
Kana et. al., bioRxiv 2026

all new data expanded MTG taxonomy further to preserve cellular types, but account for additional regional cell types



Travaglini et. al., bioRxiv 2026

cell type and molecular changes in disease



An integrated multimodal cell type Atlas of Alzheimer's Disease

Brief Intro to AD

Analysis & Algorithms

Neuropathology

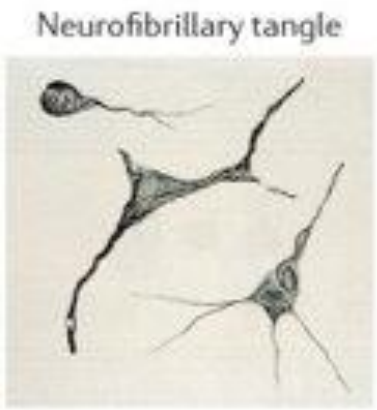
Analysis & Algorithms

Cellular Vulnerabilities

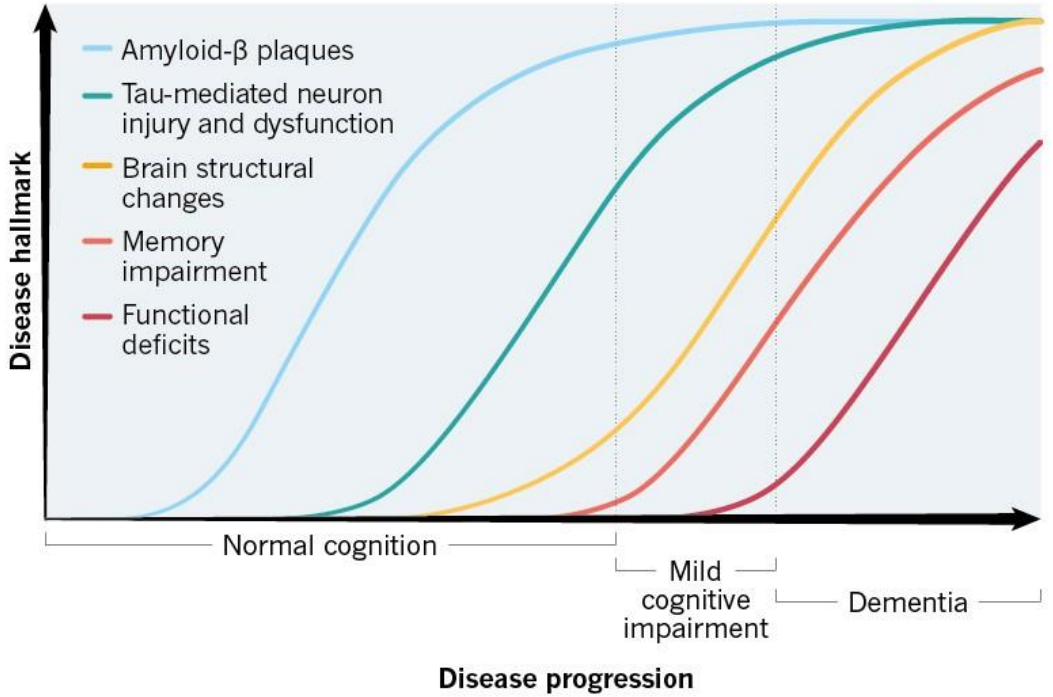
1 brain region

Following the Trajectory of AD

AD is characterized by the progressive accumulation of proteinopathies

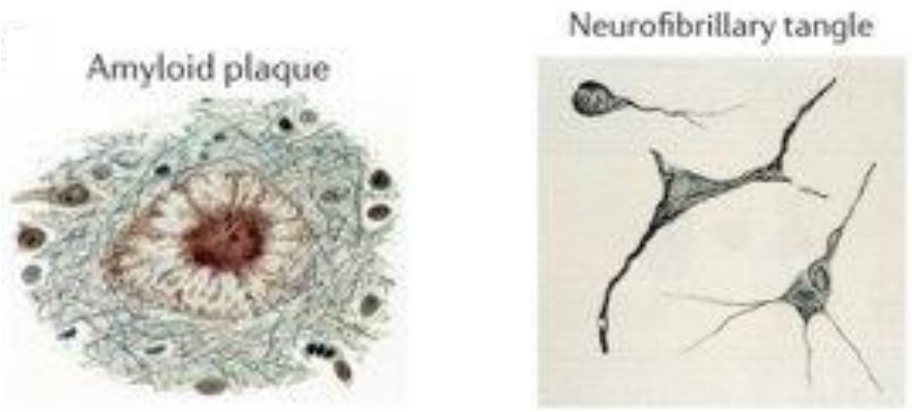


Accumulation of neuropathological peptides
neurofibrillary tangles of tau and amyloid-beta plaques

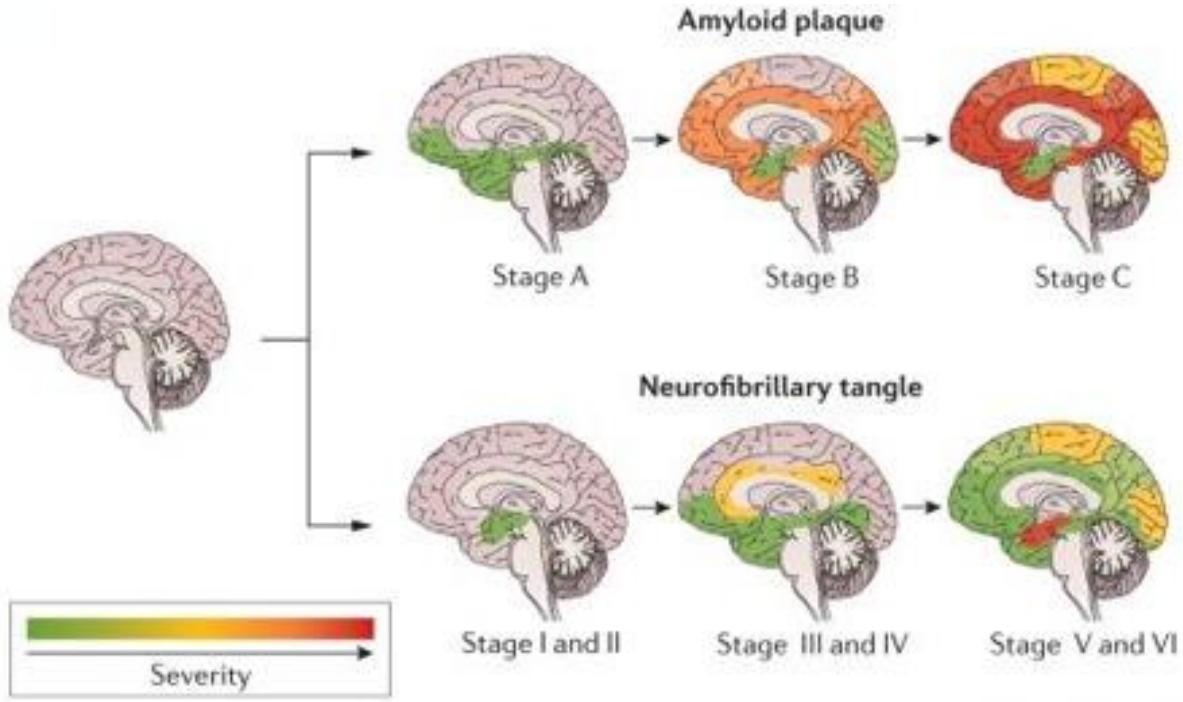


Cognitive decline is observed late in the disease process, proteinopathies accumulate for decades

AD is characterized by the progressive accumulation of proteinopathies



Accumulation of neuropathological peptides
neurofibrillary tangles of tau and amyloid-beta plaques



Masters. *et al Nat. Rev. Dis. Primers,* 1. (2015)

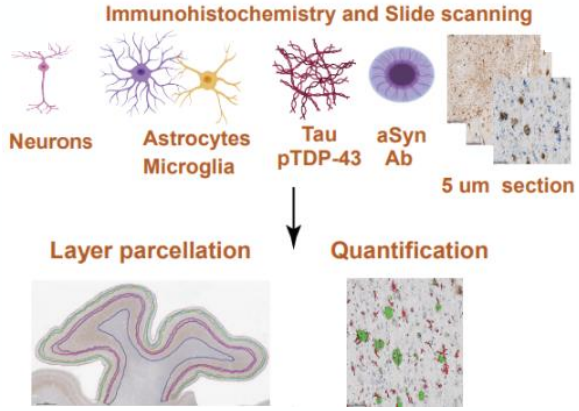
Disease progression follows a **stereotype spatiotemporal** trajectory

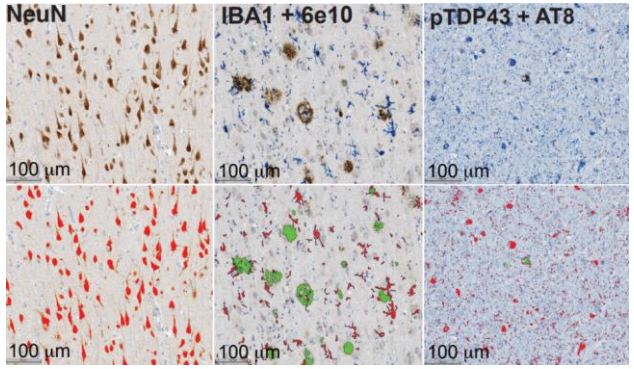
How did we harness neuropathology to study AD progression/

Donor Metadata/ Clinical History

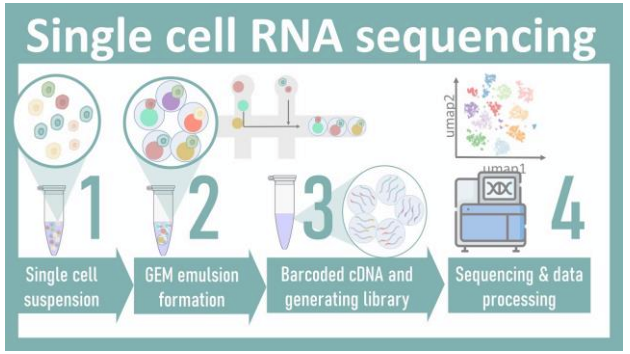
Donor ID	Age at d...	Sex	APOE ge...	Cognitive status	Years o...	Continuous ...	ADNC	Braak stage	Thal phase
H20.33.040	98	Male	3/3	Dementia	13	0.439	Intermediate	Braak IV	Thal 4
H20.33.036	100	Female	2/3	No dementia	15	0.844	High	Braak V	Thal 5
H21.33.019	75	Male	2/3	No dementia	15	0.517	Low	Braak 0	Thal 1
H21.33.040	83	Male	3/4	No dementia	17	0.745	High	Braak V	Thal 4
H20.33.029	91	Female	3/3	Dementia	13	0.613	High	Braak V	Thal 4
H20.33.016	88	Male	3/3	Dementia	18	0.879	Intermediate	Braak V	Thal 3
H21.33.012	93	Female	2/4	Dementia	21	0.486	Intermediate	Braak IV	Thal 3
H21.33.003	78	Male	3/3	No dementia	16	0.236	Not AD	Braak 0	Thal 0
H20.33.002	97	Female	2/3	No dementia	12	0.207	Not AD	Braak IV	Thal 0
H21.33.030	89	Male	3/4	No dementia	17	0.252	Intermediate	Braak III	Thal 3
H20.33.046	94	Male	3/3	Dementia	21	0.868	High	Braak VI	Thal 5
H20.33.031	87	Female	3/3	Dementia	12	0.778	High	Braak VI	Thal 4
H20.33.032	98	Male	3/3	No dementia	17	0.816	High	Braak V	Thal 5
H21.33.027	92	Male	3/4	Dementia	18	0.687	High	Braak V	Thal 5
H19.33.004	80	Female	3/3	No dementia	17	0.302	Not AD	Braak IV	Thal 0
H20.33.028	94	Female	3/3	Dementia	12	0.815	High	Braak V	Thal 4
H20.33.008	92	Female	3/4	No dementia	18	0.754	High	Braak V	Thal 4

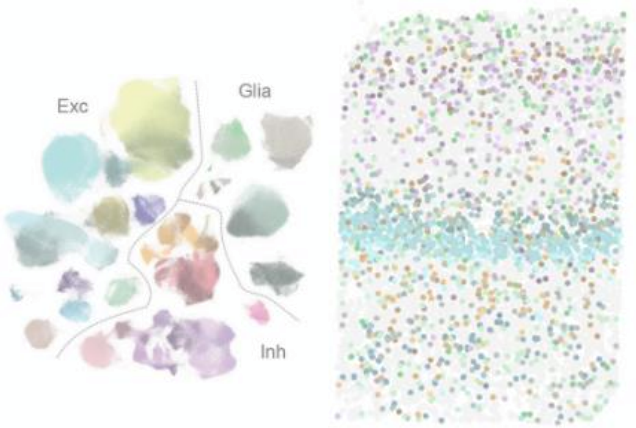
Quantitative Neuropathology



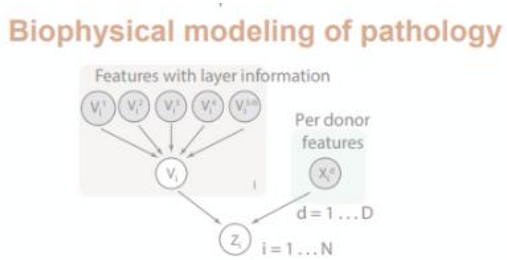


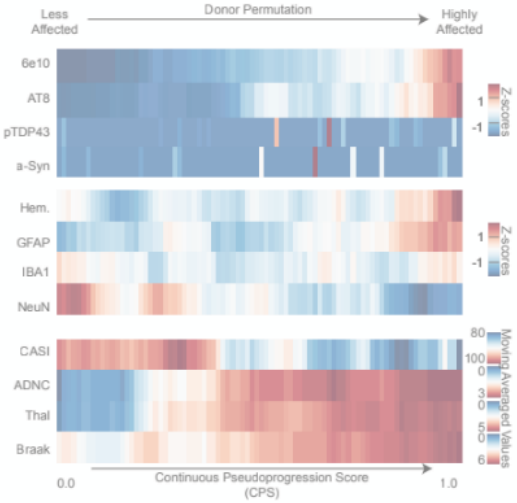
Single Cell and Spatial Genomics





Modeling





Gabitto et. al., Nature Neuroscience 2024

Quantitative neuropathology (QNP) describes detailed pathological burden

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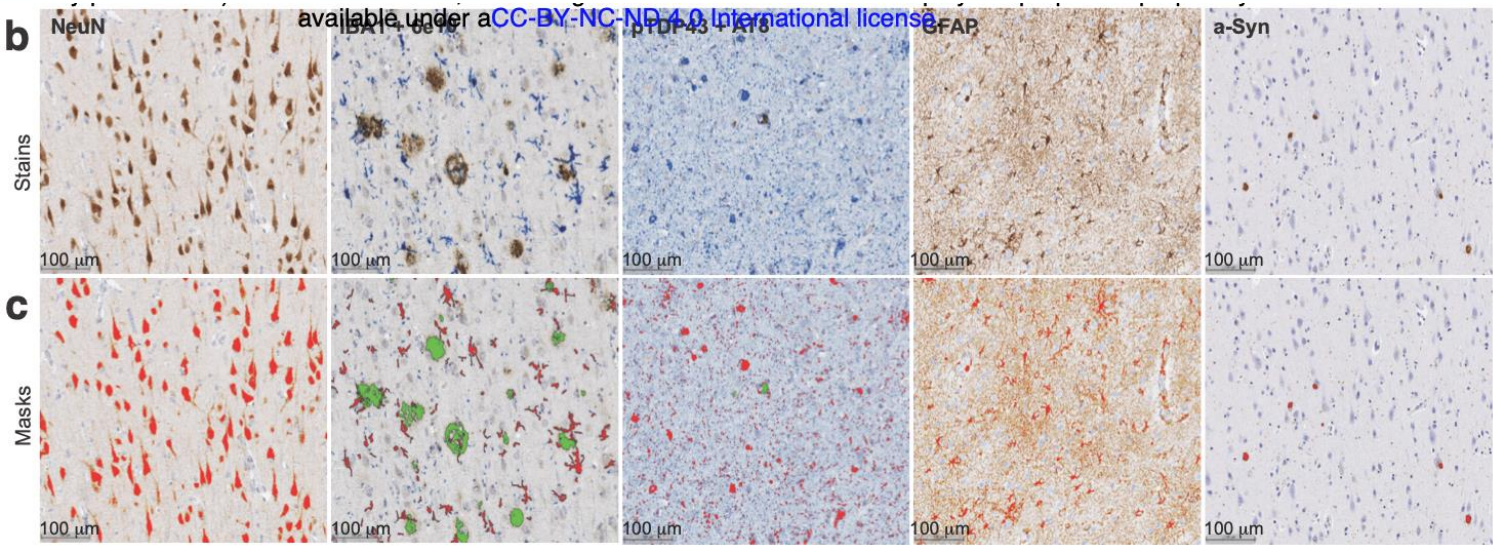
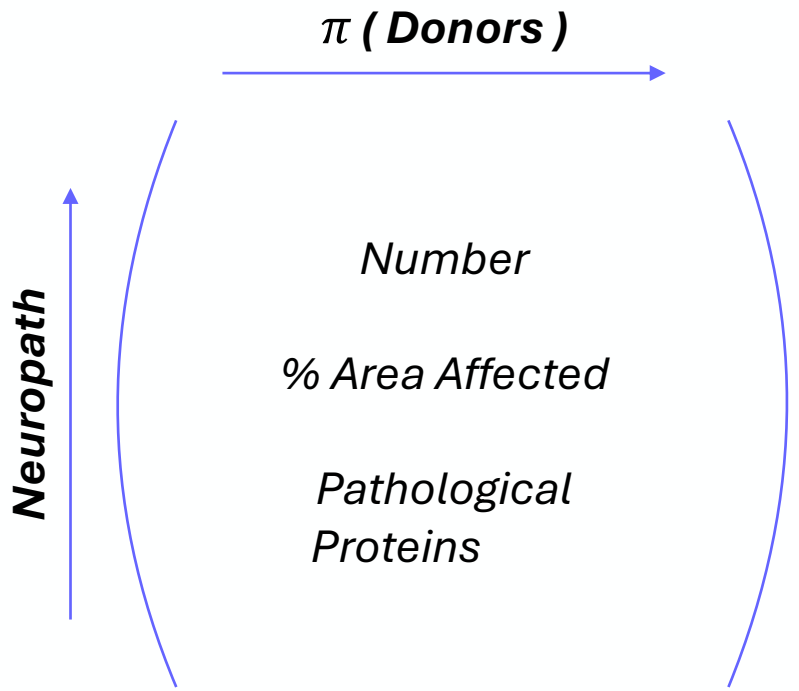


Quantitative description of brain pathology

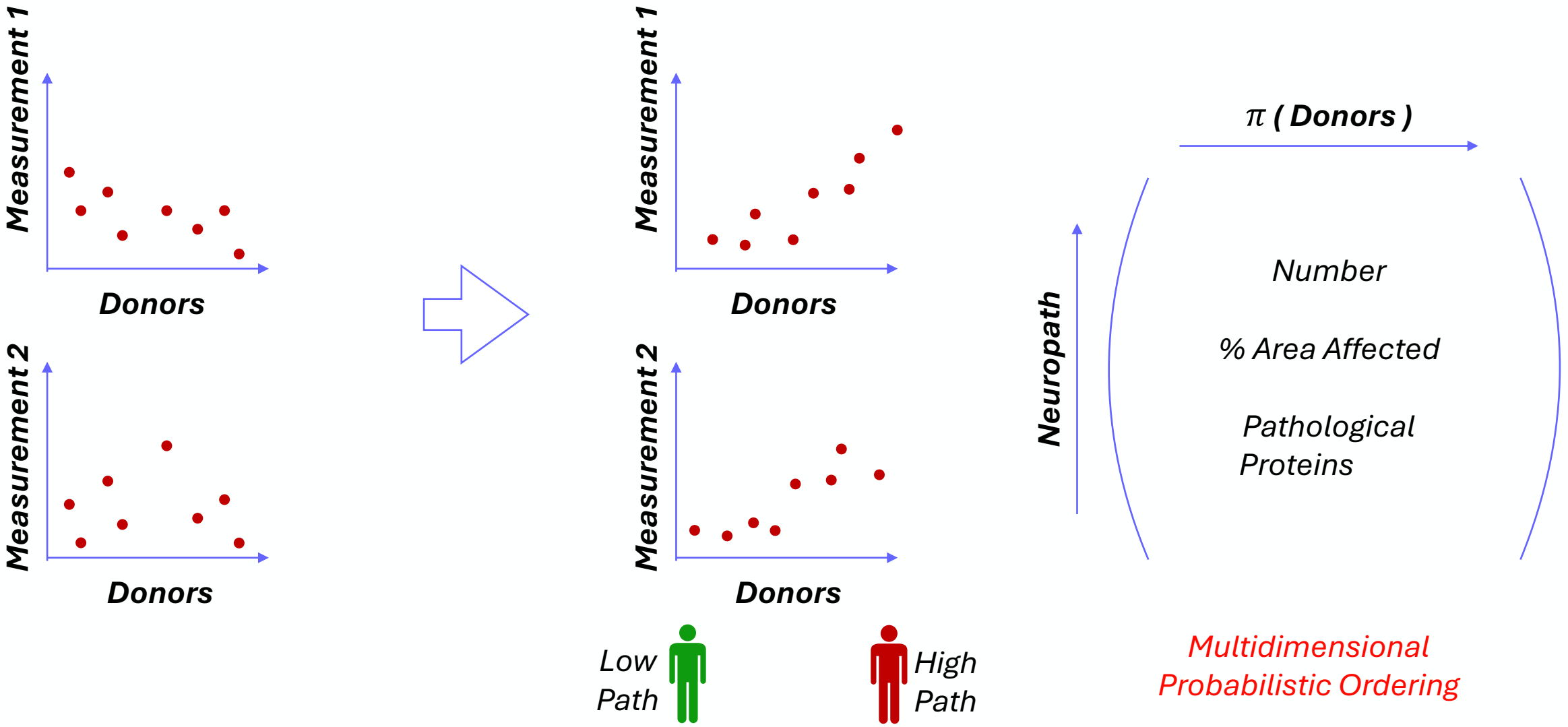


Protein Pathologies

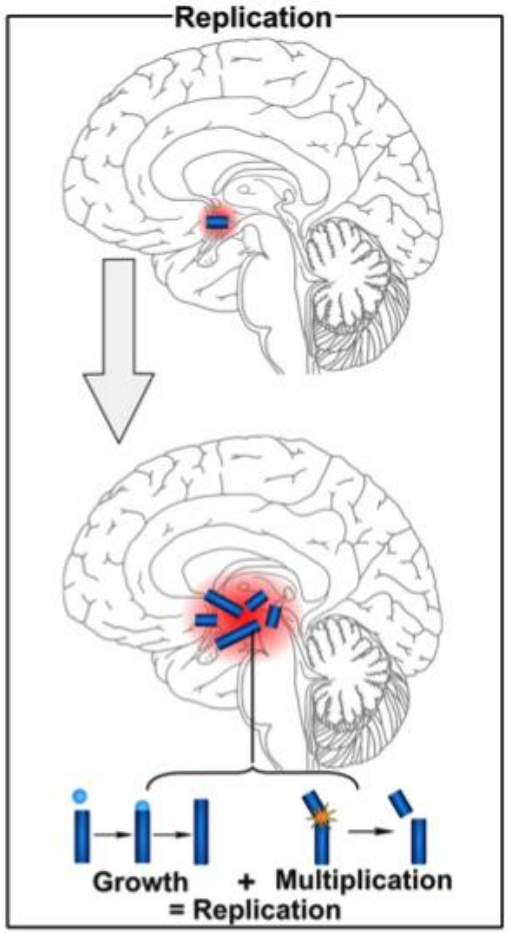
Cellular Populations

Can we harness detailed neuropathology to map pathology progression ?

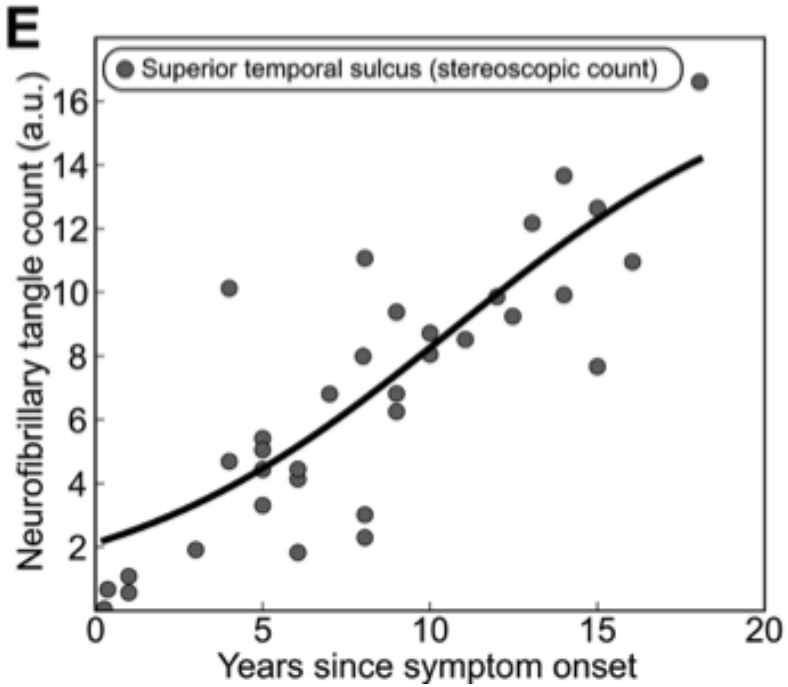
Harness QNP to order pathological progression



Exponential Accumulation of Pathological Proteins Locally



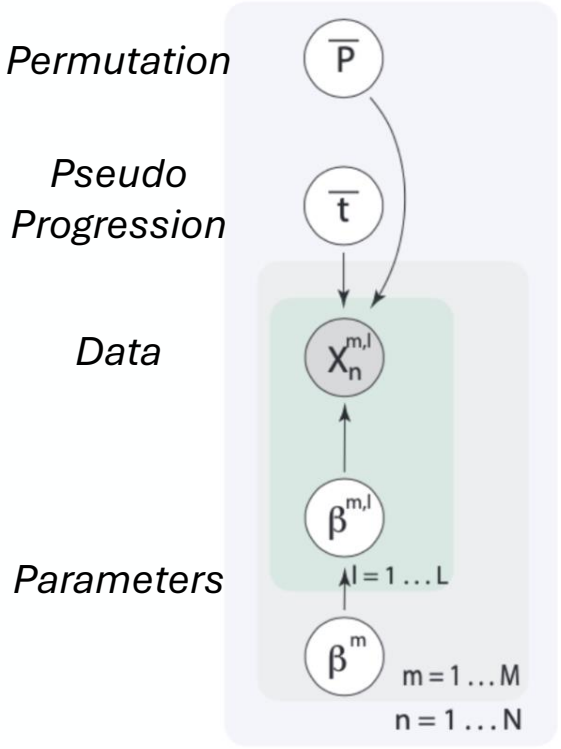
Tau aggregates can self-replicate



The accumulation of Tau follows an exponential process

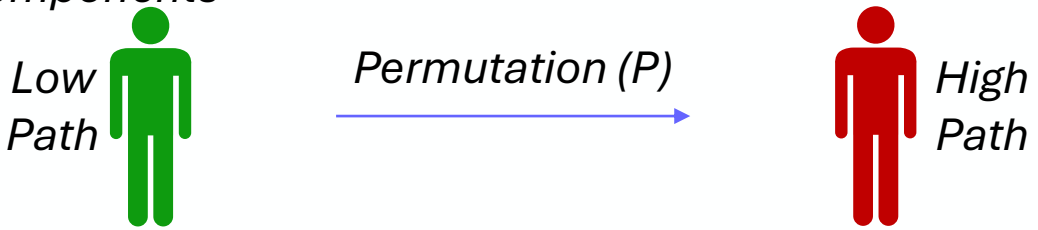
Meisl et al. Science Advances 7(44) 2021.

Biophysically inspired statistical models to understand disease progression



Arrange the donors according to disease severity using (3) components

$$\pi \sim U[Permutations]$$



$$t_d \sim Beta(2,2)$$

PseudoProgression (t)

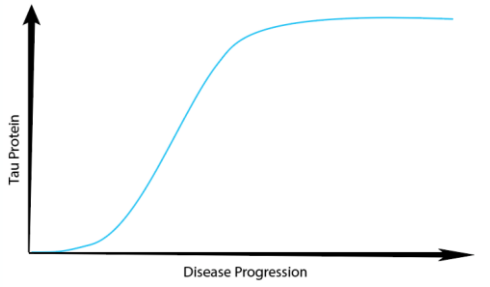
0 → *1*

$$X_{m,l}^d | t, \beta \sim Poisson(\exp^{\beta_1 t + \beta_0})$$

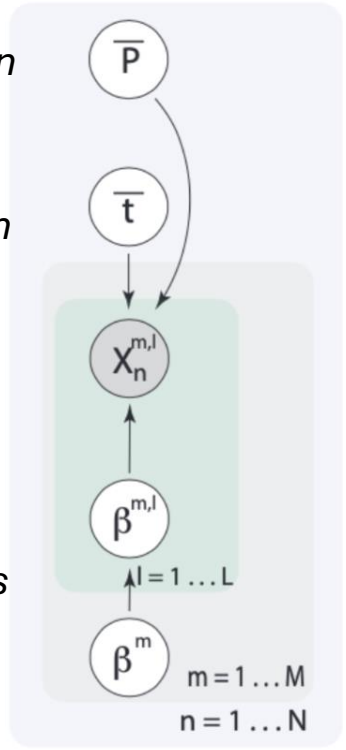
Monotonic Sigmoidal / Exponential Process

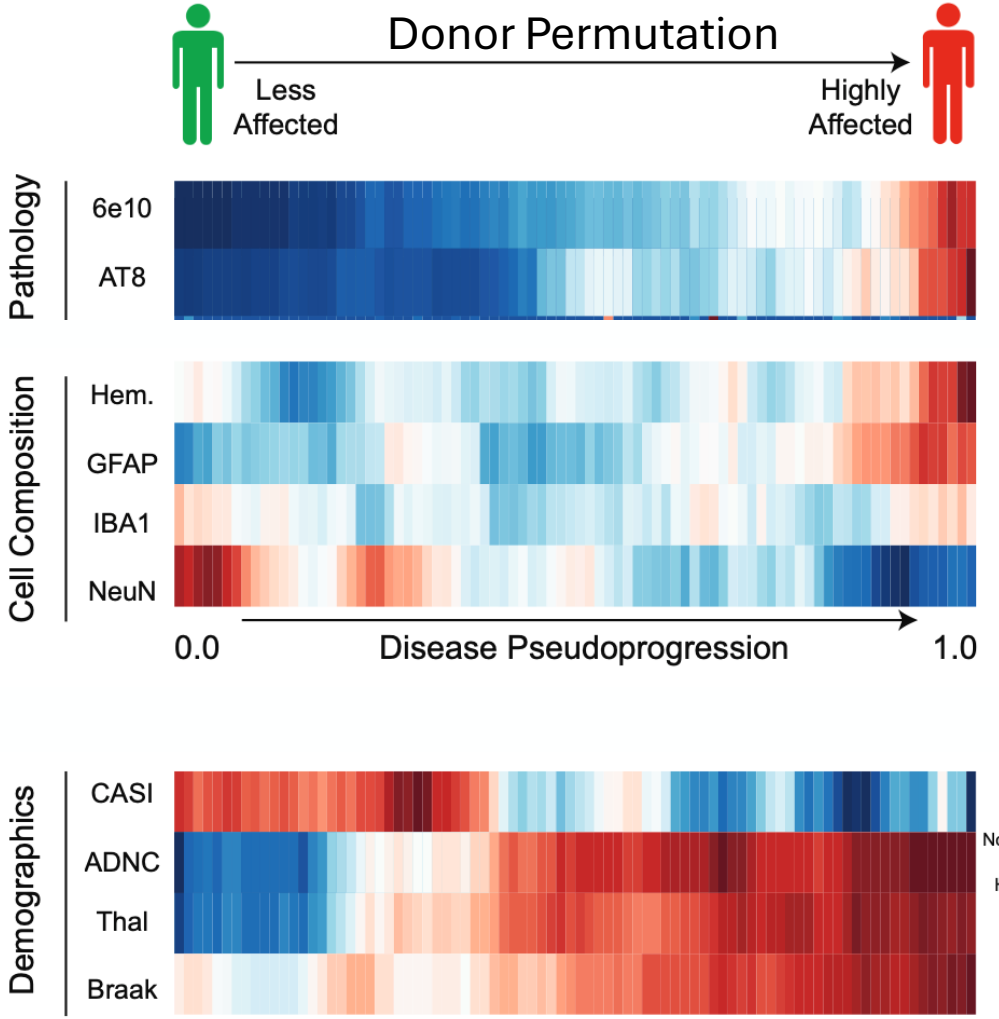
$$\beta^{m,l} | \beta^m \sim TNormal(\beta^m, 1)$$

$$\beta^m \sim TNormal(0,1)$$



Statistical models to understand disease progression





Orthogonal Data

Local Pathology Recapitulates Brain-Wide

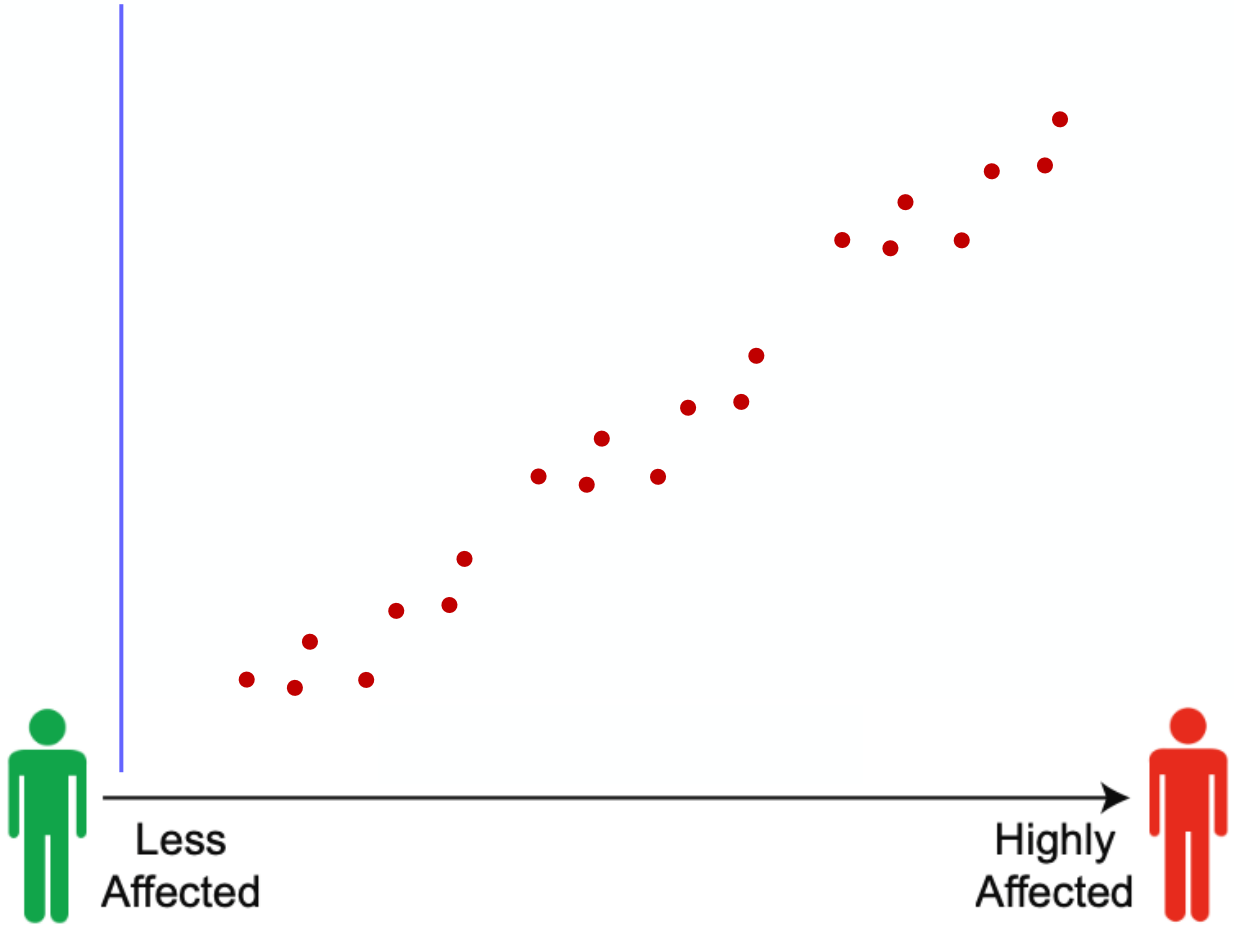
using neuropathology as a ground truth for comparisons

Cell type abundances

Gene expression changes

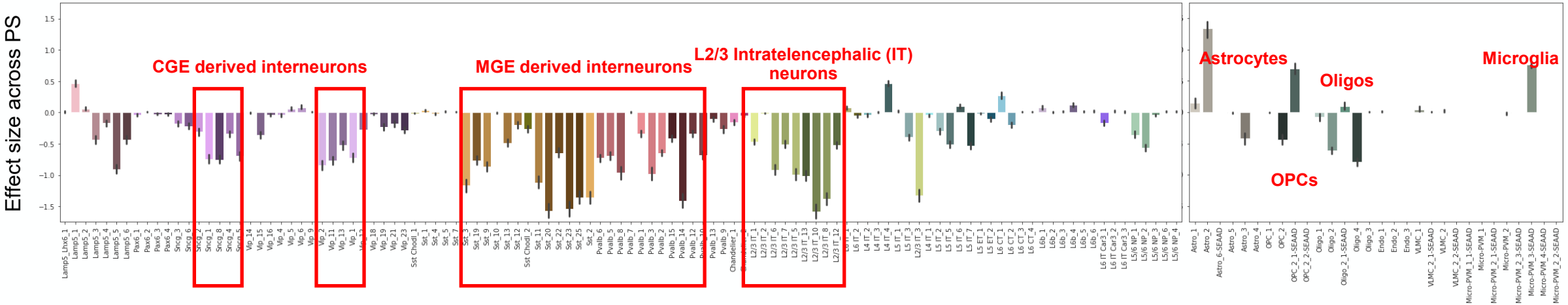
Spatial localization

...

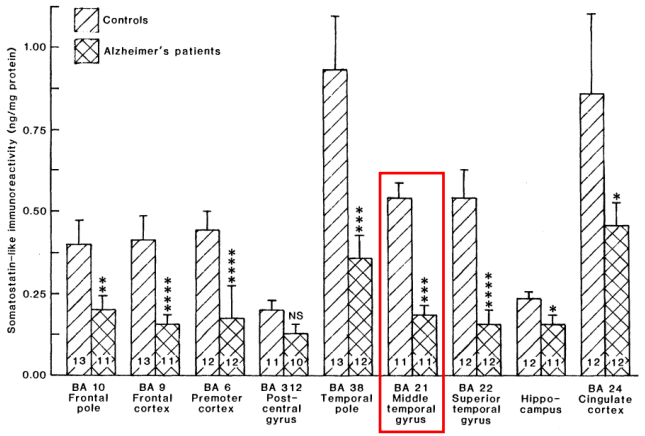


pseudoprogression

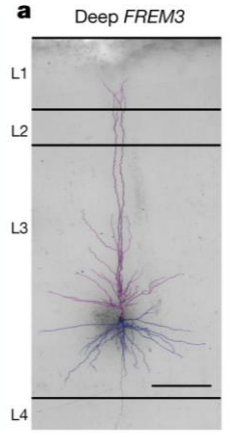
MTG affected populations span excitatory, inhibitory neurons and nonneuronal populations



Donor Proportions ~ CPS + Age at death + Sex + Race + 10x Chemistry

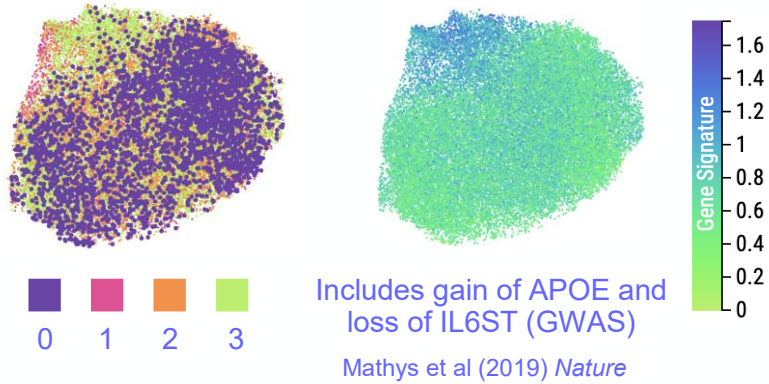


Beal et al (1985) *Science*
Roberts et al (1985) *Nature*



Berg et al (2021) *Nature*
Hoff et al (1990) *JCN*

Microglia-PVM_1 AD Pathology

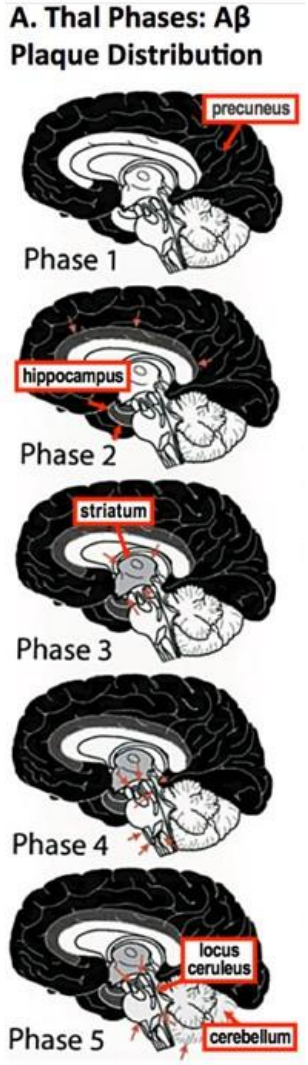


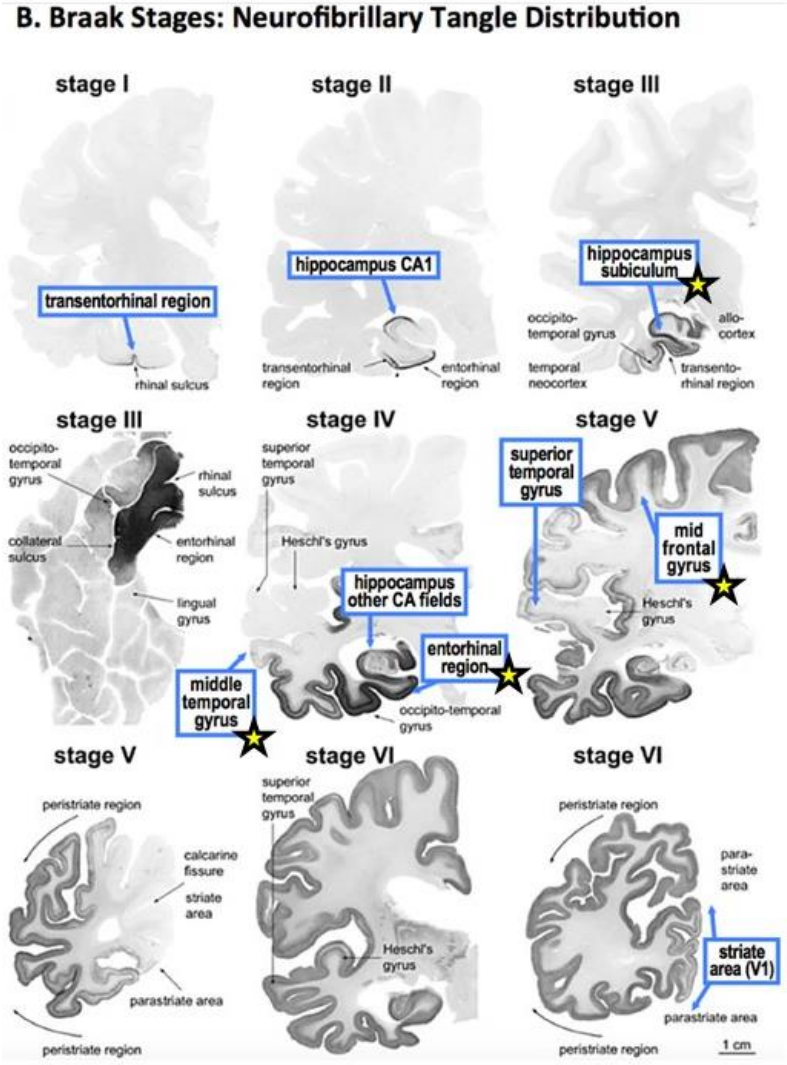
Includes gain of APOE and loss of IL6ST (GWAS)

Mathys et al (2019) *Nature*

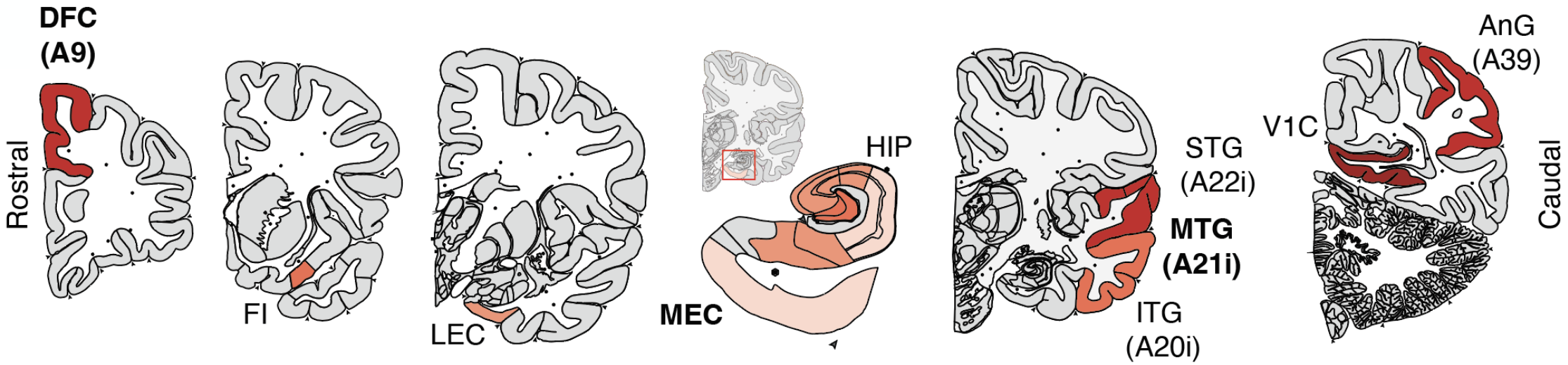
Alzheimer's disease affects multiple brain regions with different pathologies in a stereotyped manner

★ = phase 2 region





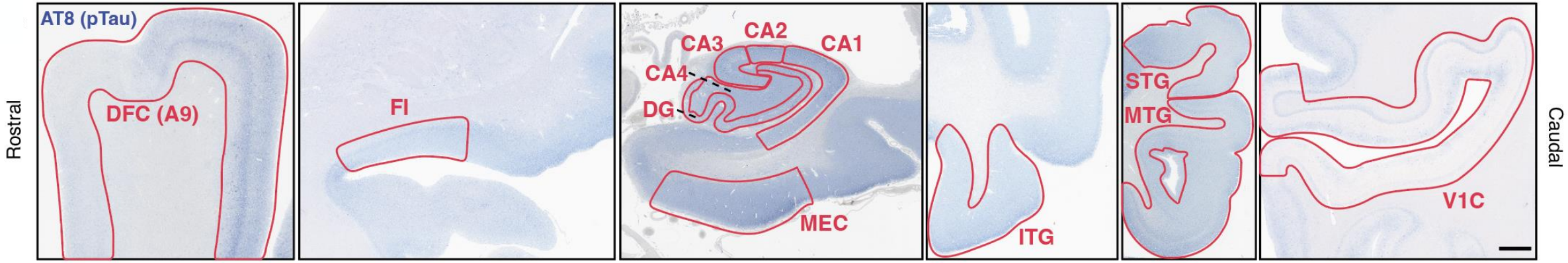
SEA-AD has now profiled brain regions affected at different Braak (pTau) and Thal (Amyloid Beta) stages

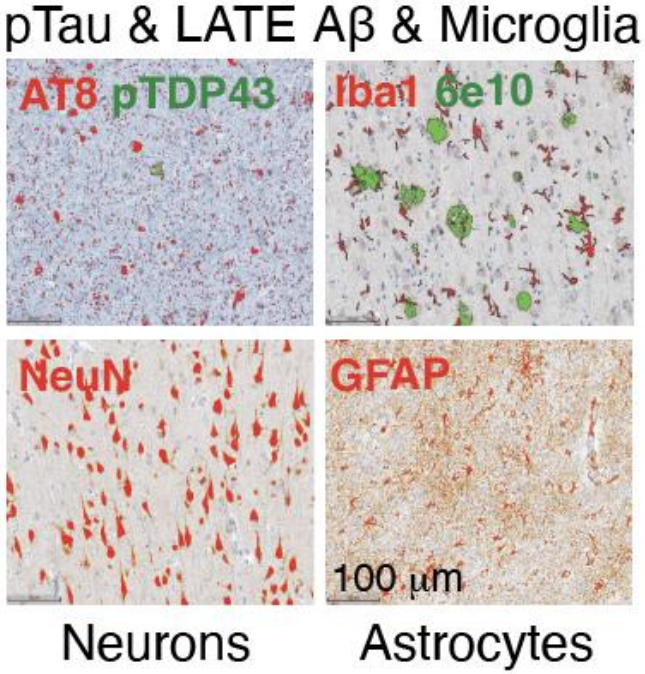


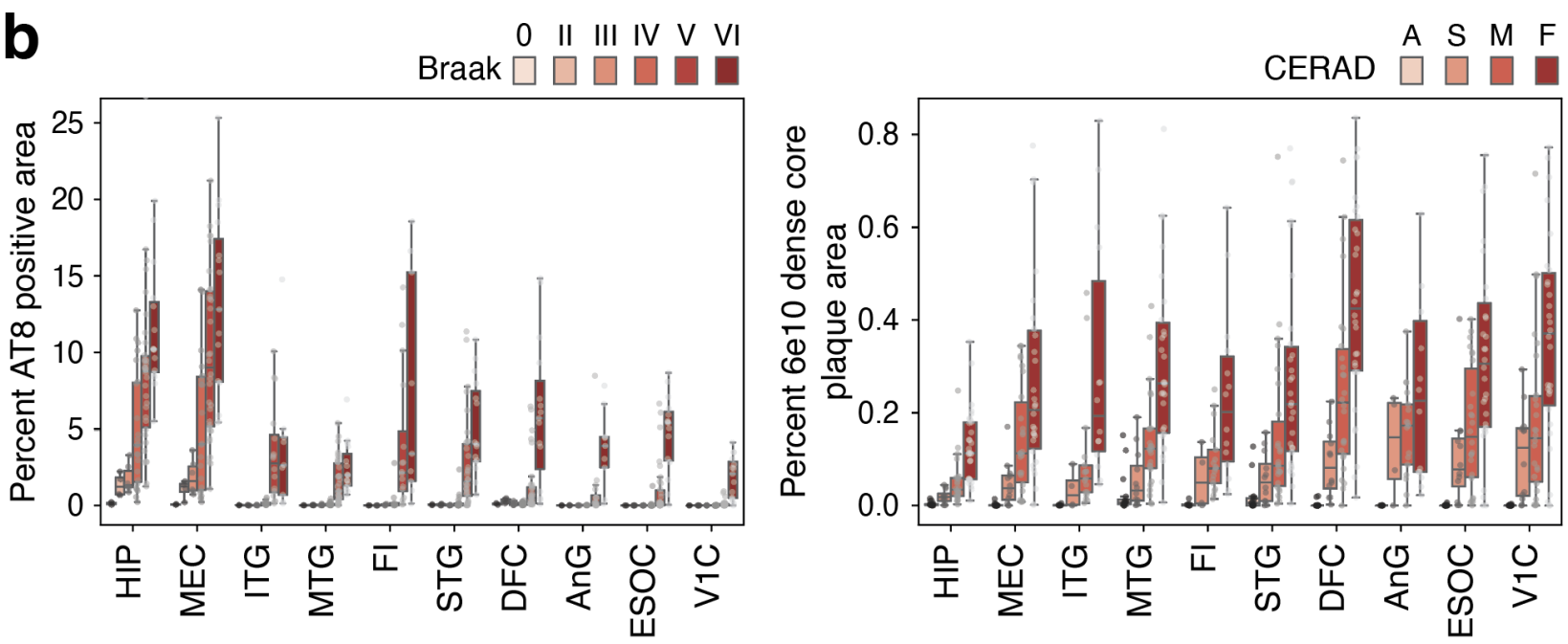
Braak I/II: MEC, CA1 (HIP) Braak III: Sub (HIP), CA2-3 (HIP), LEC Braak IV: CA4 (HIP), ITG, MTG, FI

Braak V: STG, DFC, AnG Braak VI: V1C AD spectrum only (n=43) / **AD spectrum and co-morbidities (n=84)**

Quantitative neuropathology across brain regions

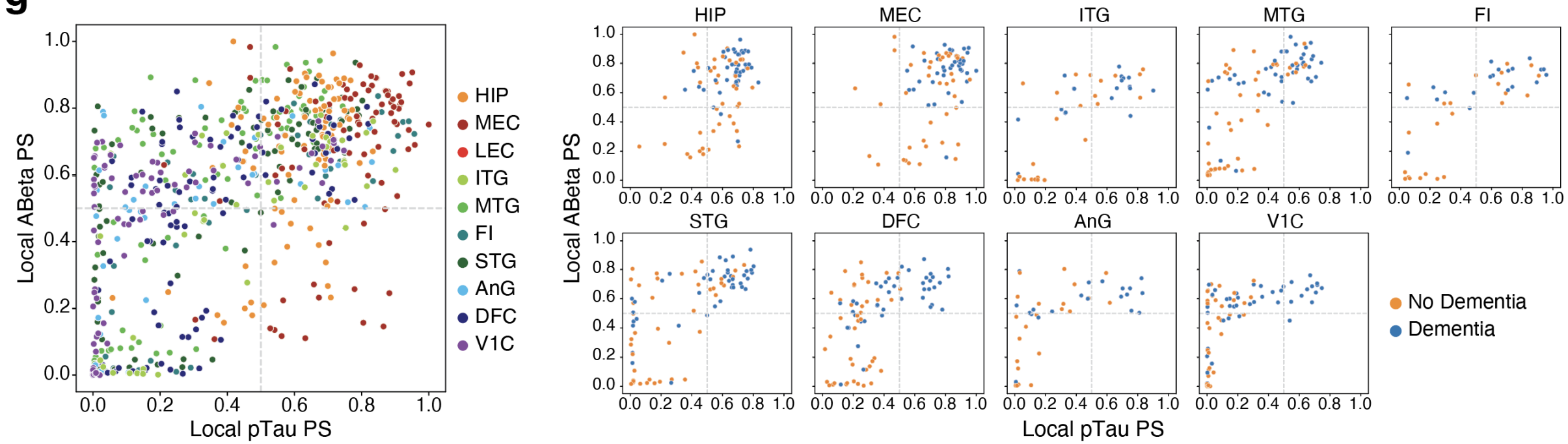




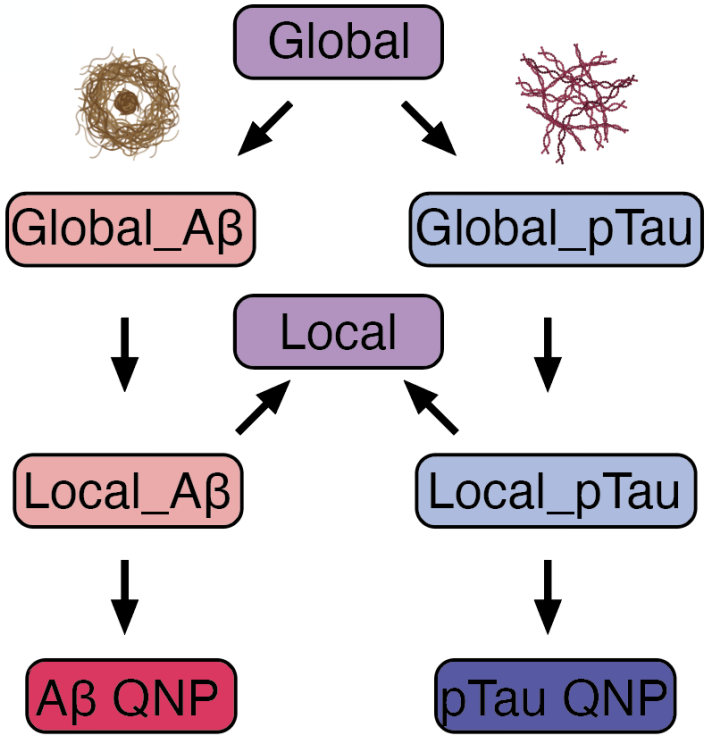


Distinct accumulation patterns of amyloid and tau in the neo- versus allo-cortex

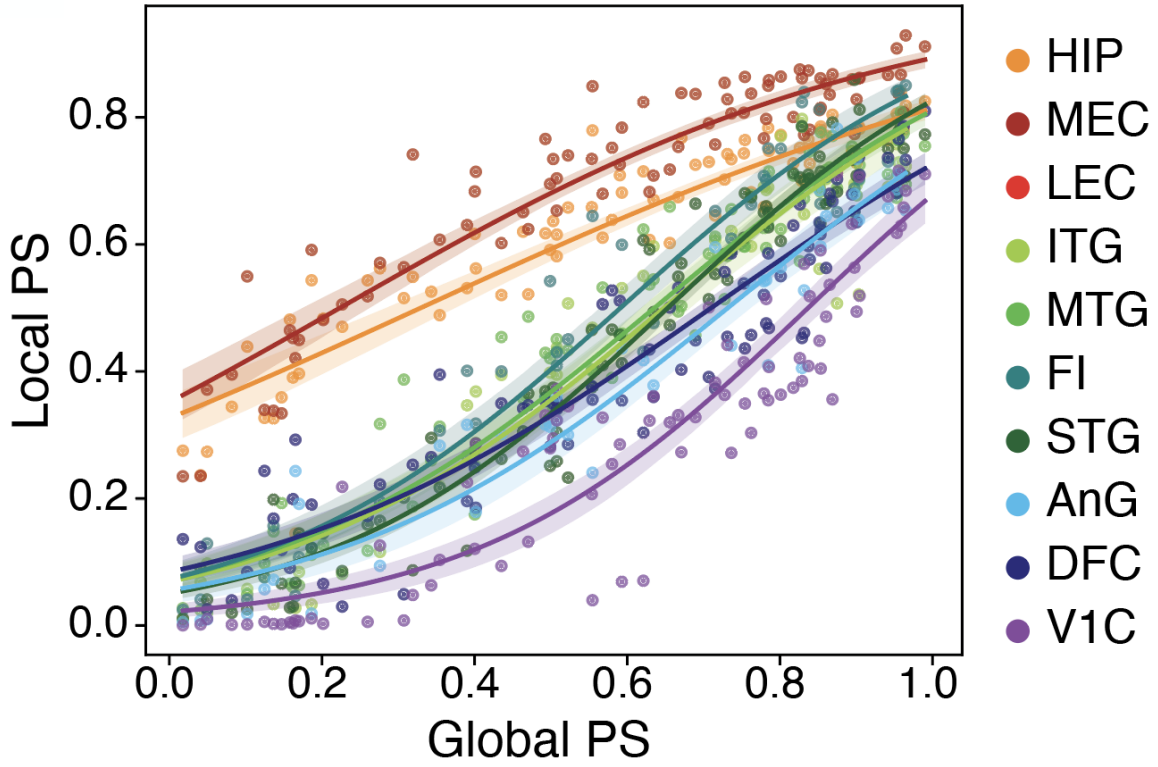
g



Biophysically modeling local pathological accumulation of amyloid plaques and pTau tangles as a latent global process



QNP ~ sigmoid(Local_Path)



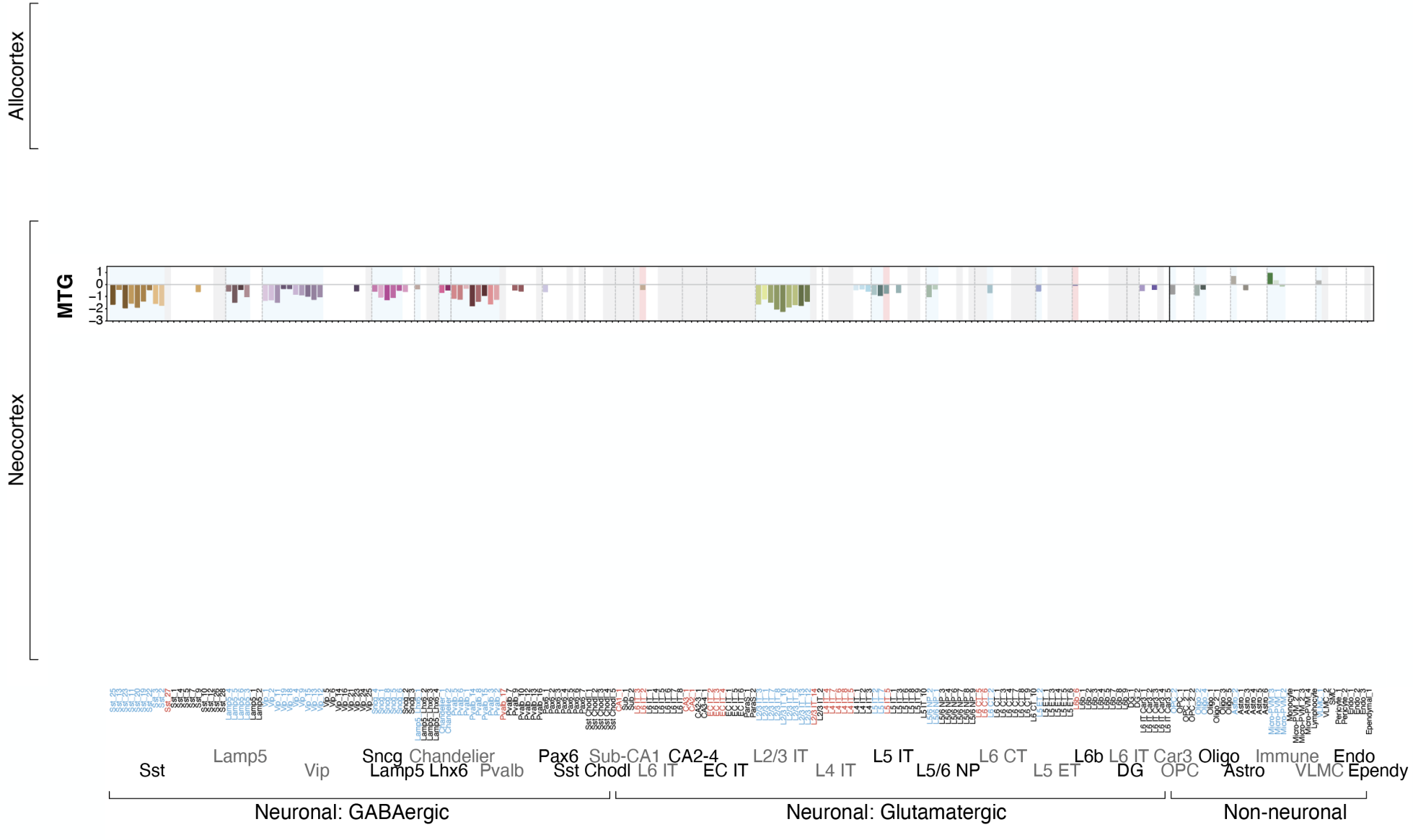
Common changes to cell types shared across brain regions

Cellular Abundances ~ **CPS_Local** + Sex + Age at Death + APOE4 Status + PMI

AD Spectrum+
AD Spectrum only

type not in region
regional change
shared change

Effect size Local CPS of each cell type across brain regions

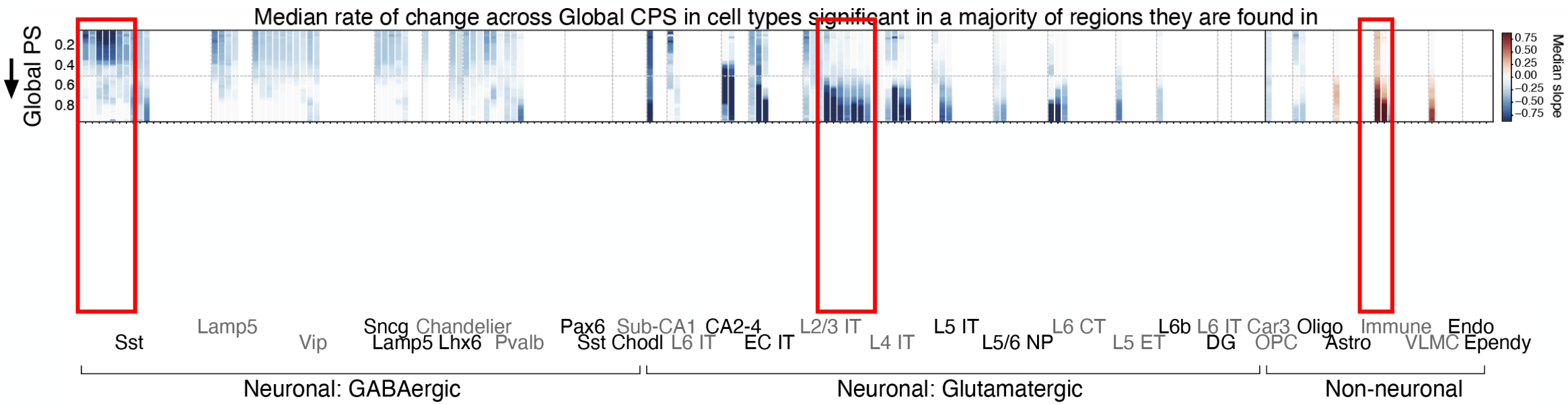


Common changes to cell types shared across brain regions

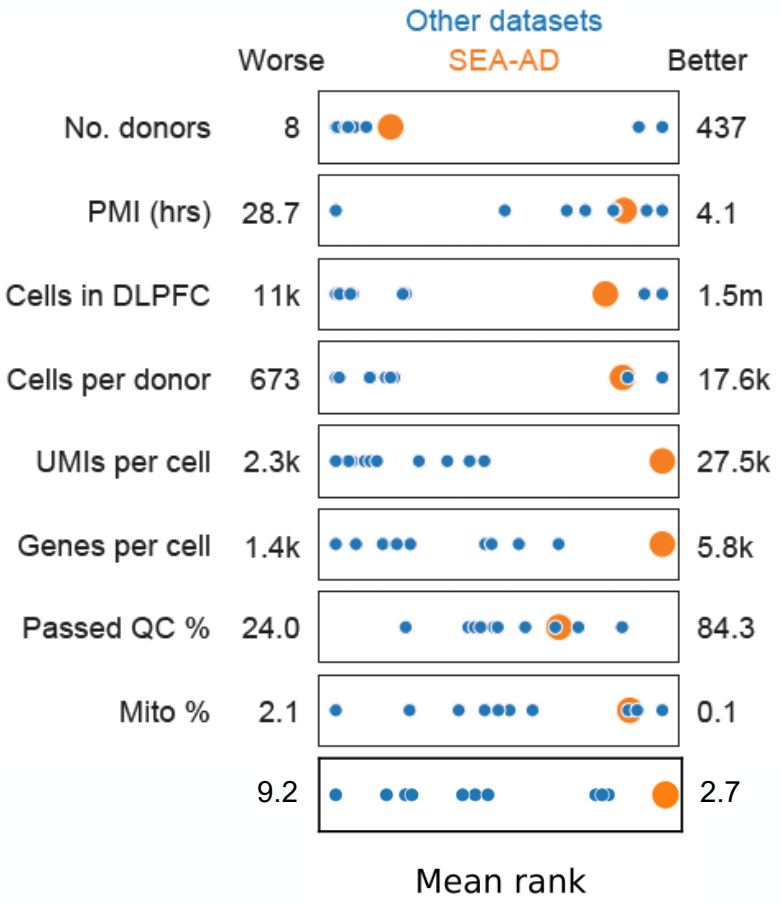
Cellular Abundances ~ **CPS_Local** + Sex + Age at Death + APOE4 Status + PMI

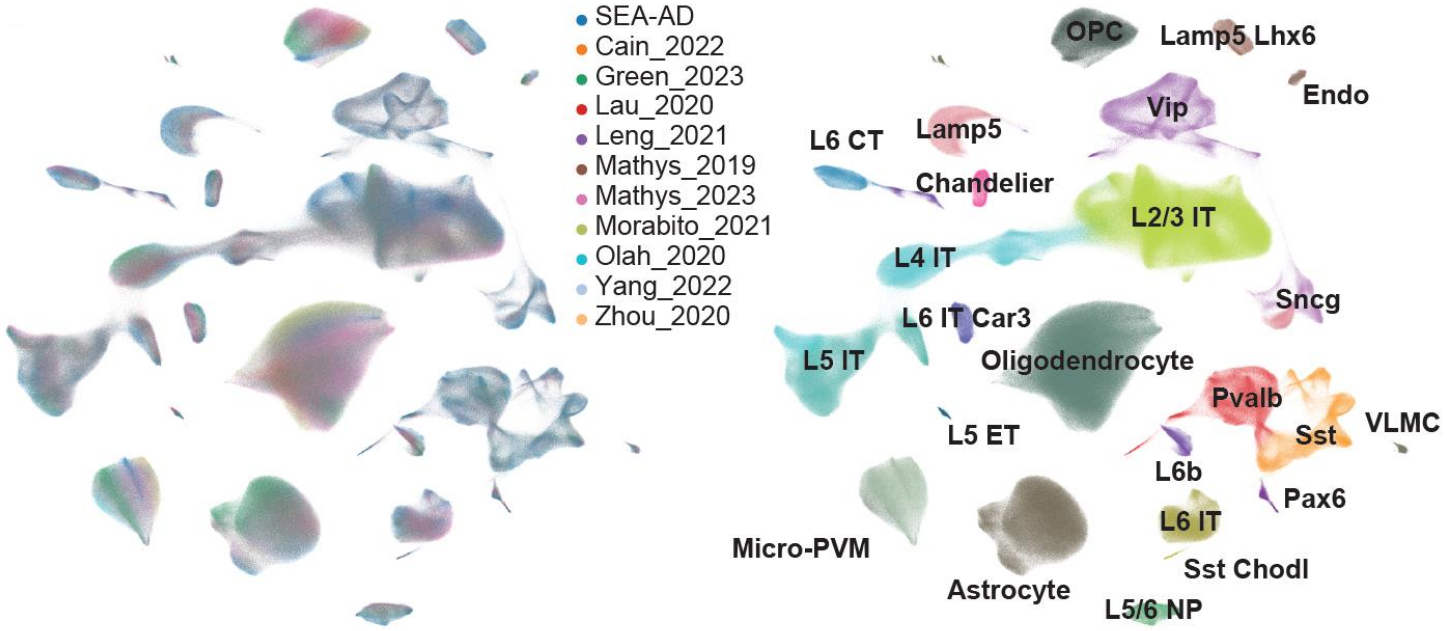
Certain regionally specialized neurons are also vulnerable

Median timing of changes along Global PS and association with plaques and tangles

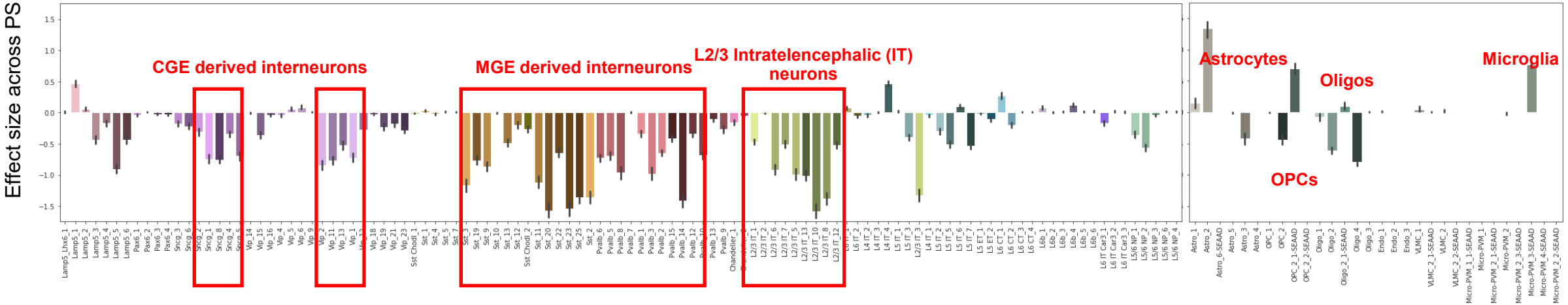


Integration of 10 publicly available datasets spanning 707 donors and 4.5 million cells from the DLPFC





MTG affected populations span excitatory, inhibitory neurons and nonneuronal populations



Donor Proportions ~ CPS + Age at death + Sex + Race + 10x Chemistry

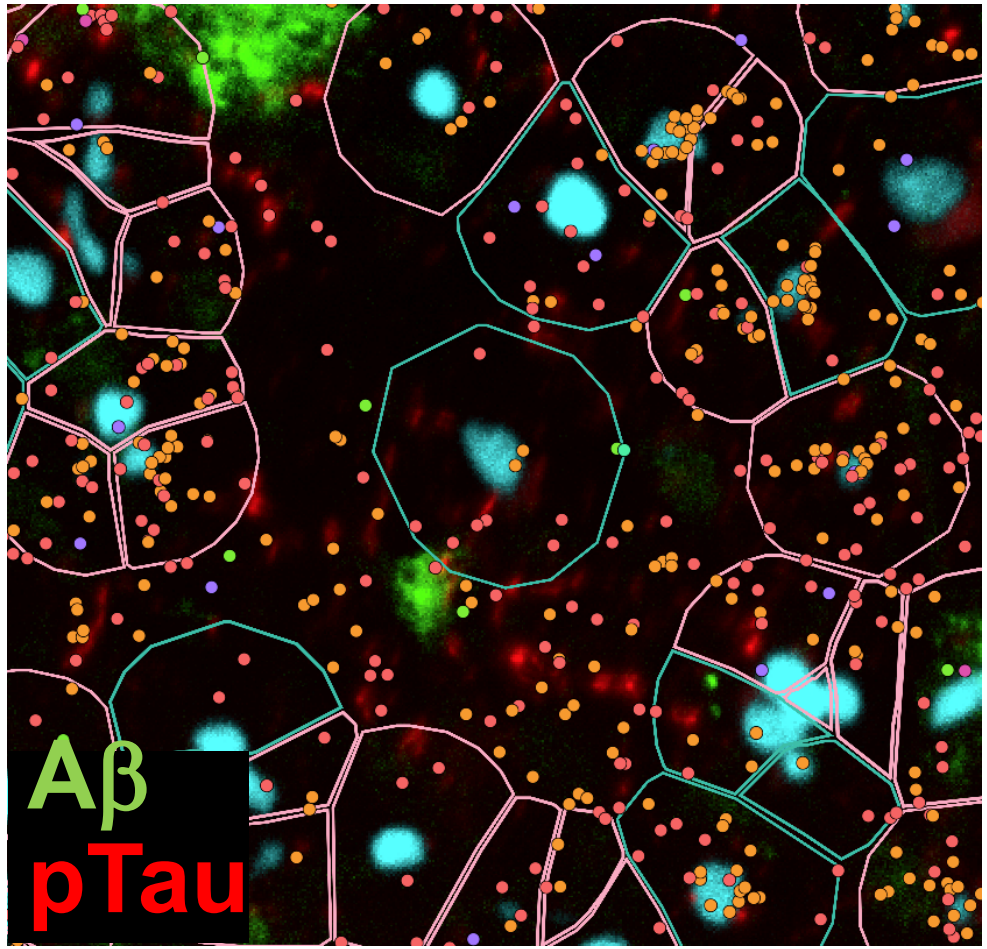
This allowed us to identify the types of neuron vulnerable to AD,

How susceptible they are to AD pathology, and

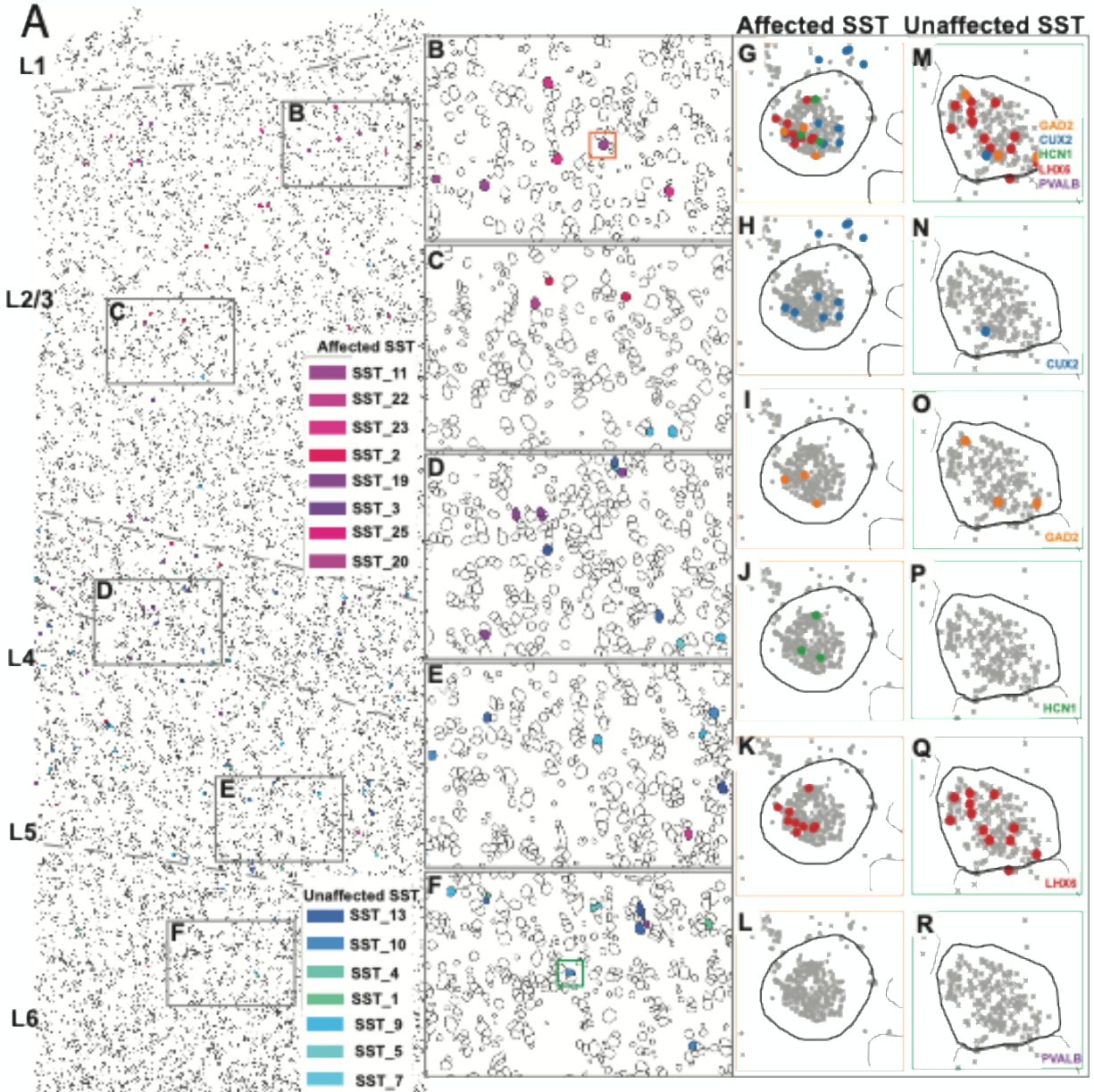
changed the paradigm from neuropath to neurons.

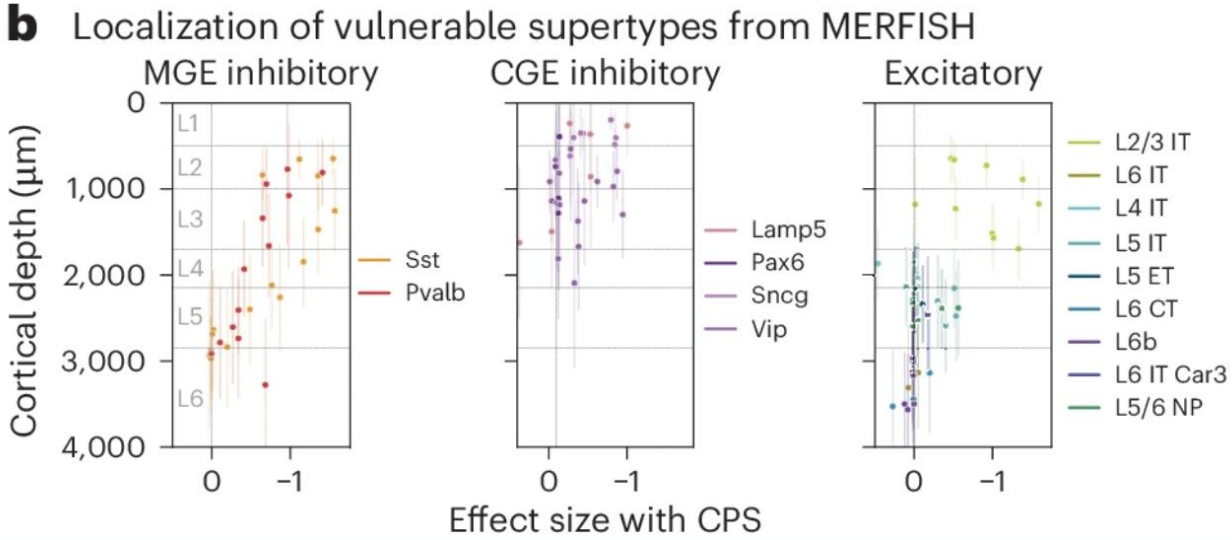
Spatial data allows us to determine where and how cells are responding to pathology

- What is the spatial relationship of affected cell types to neuropathology?
- What are the spatial relationships between pathological protein aggregates and expression of genes that are differentially expressed in AD progression?
- How are tissue structure and functional architecture altered with disease progression?



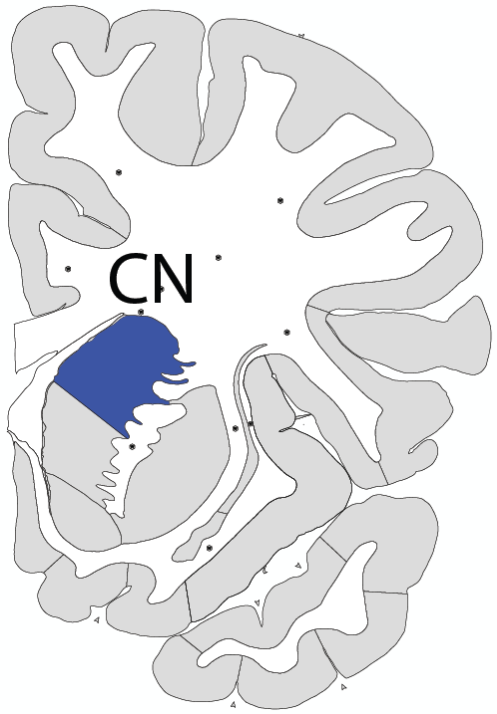
MTG dataset provides supertype and vulnerable cell type locations and proportions

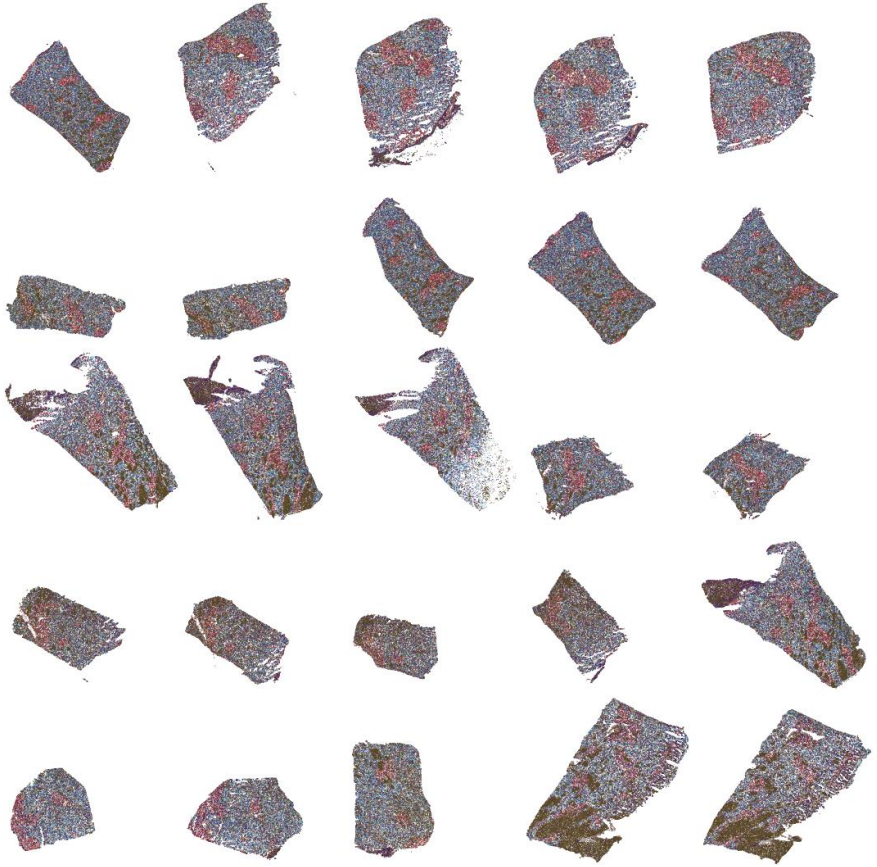




- Confirmed that vulnerable neurons localize to upper layers
- Determined that proportions of these types were significantly reduced with disease progression

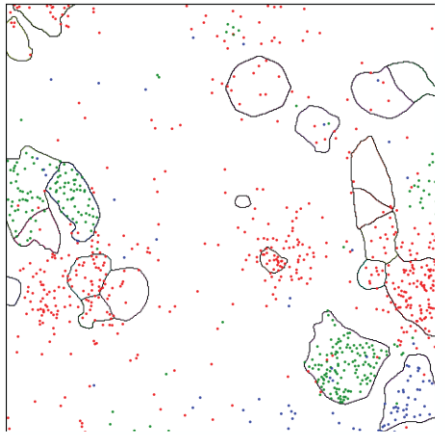
we also studied how caudate changes in AD; read our recent preprint to learn more/



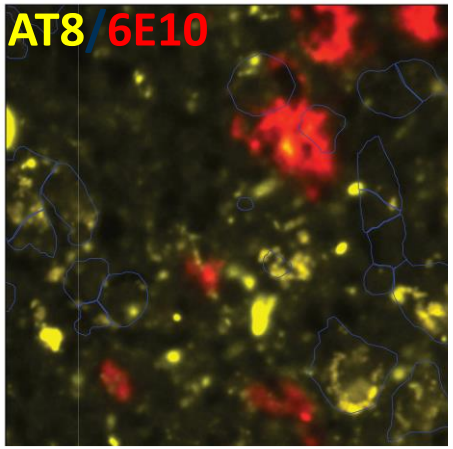


Xenium 454 gene panel + AT8 and 6E10 staining for AD pathology allows us to determine which cell types in a region are changing in response to AD pathology.

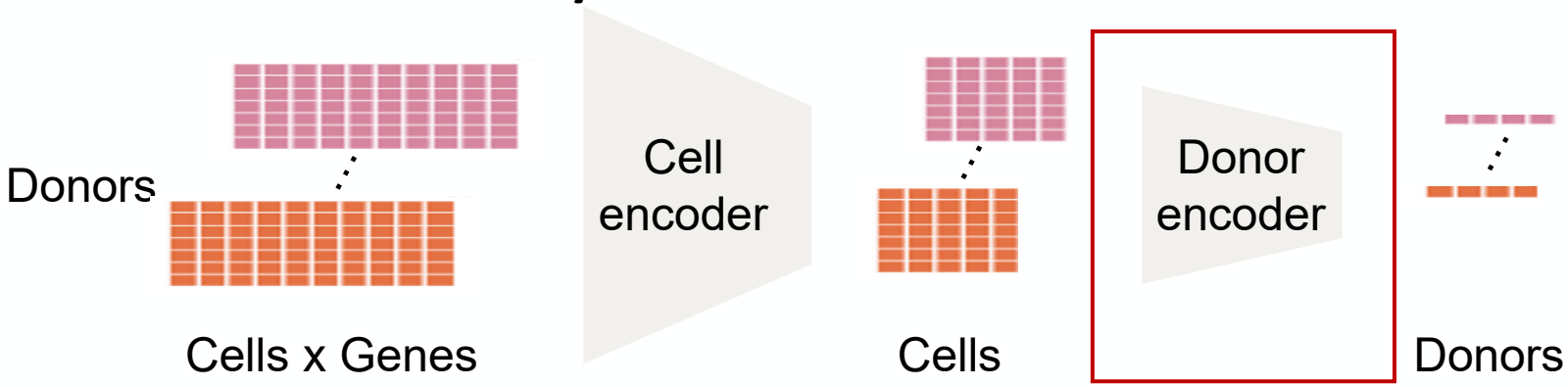
Transcript / Segments

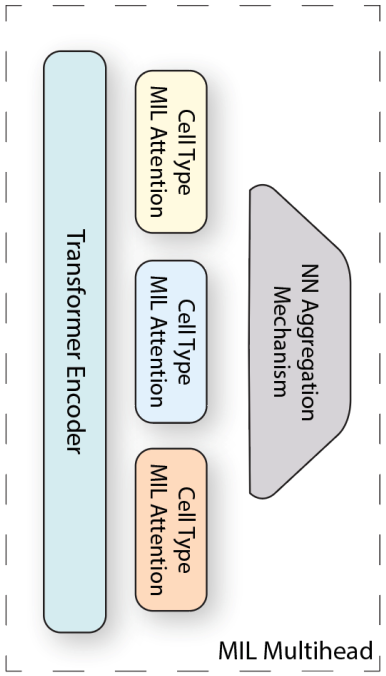


Post hoc IF



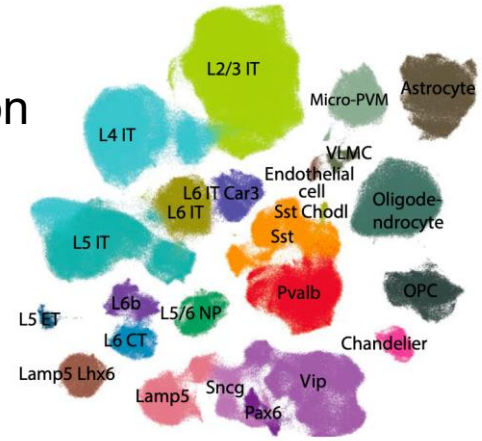
omics data can extend donor representations beyond AD neuropathology to other diseases and cohorts/



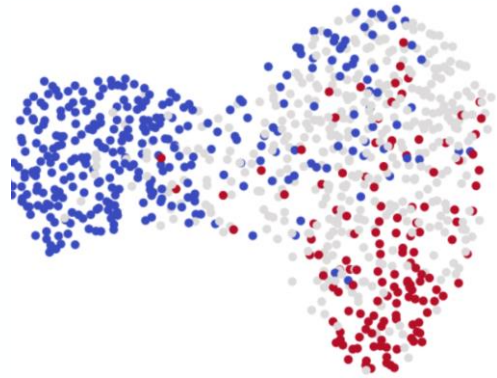


Novel multihead self- attention

The encoder harnesses disease-associated changes in each cell type to learn a donor's latent representation

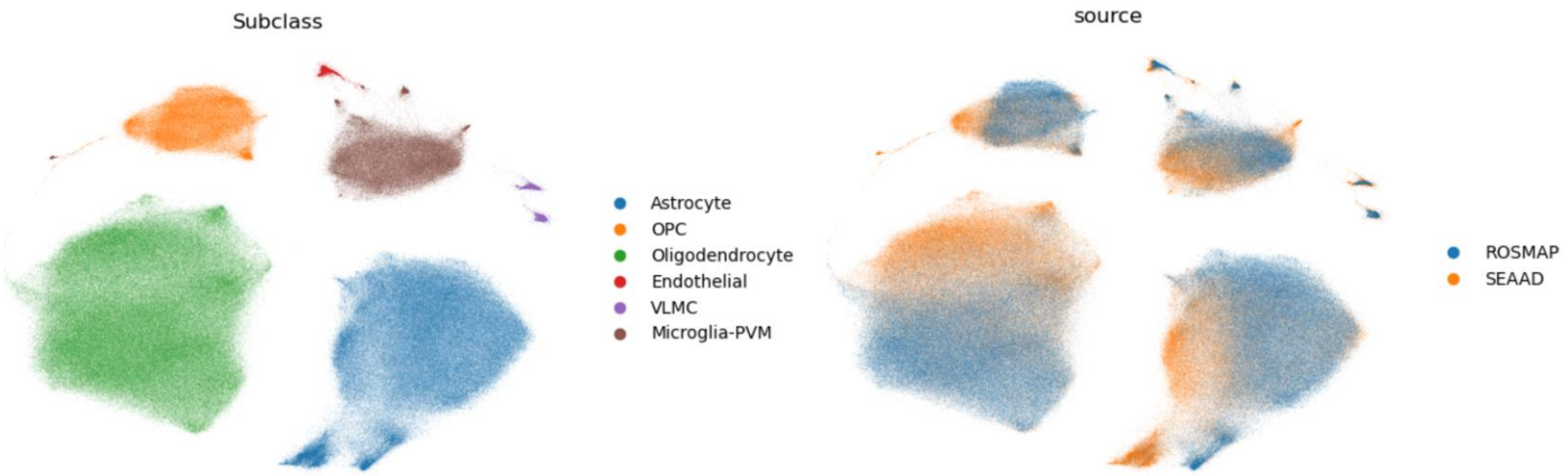


Cell representations
(Each dot is one cell. The colors are the different cell types.)



Donor representations
(Each dot is one donor. The colors are the ADNC scores.)

scVIP defines matched latent representations on SEA-AD and ROSMAP data



Datasets:
ROSMAP: 0.8M glia cells, 5k genes, 424 donors, DLPFC
SEA-AD: 0.2M glia cells, 5k genes, 84 donors, MTG + A9

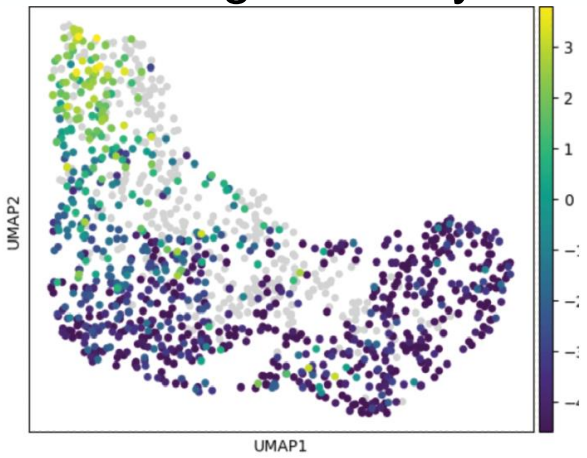
Phenotypes:
- Shared: Braak, Thal, ADNC, CERAD
- ROSMAP: neuritic plaque, diffuse plaque, tau density
- SEA-AD: percent 6e10/ AT8

Goals:
- Understand the model's ability to learn phenotype, cell-type, genotype info
- Show harmonization efforts

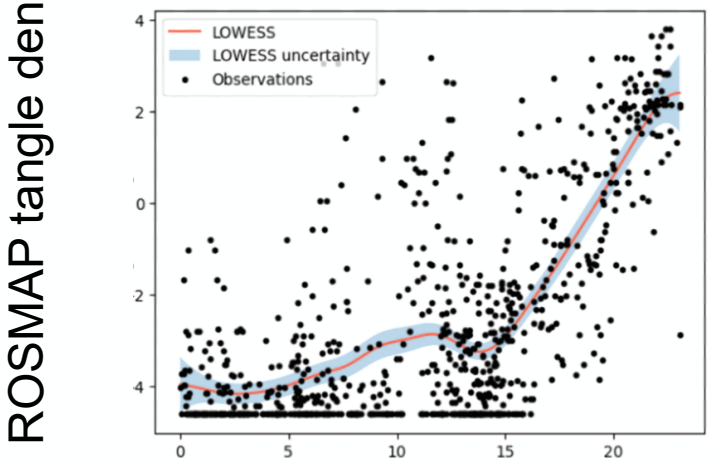
scVIP defines matched latent representations on SEA-AD and ROSMAP data, consistent with pseudoprogression

ROSMAP donors and measurements

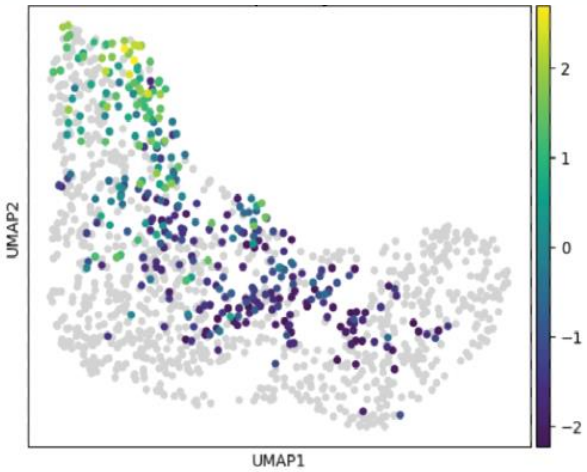
Donor UMAP colored with tangle density

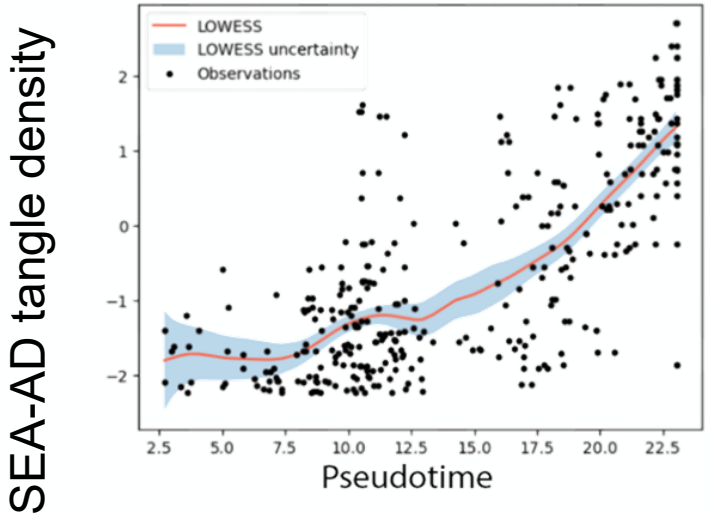


Tangle density over pseudotime learned from donor UMAP



SEA-AD donors and measurements





access these papers and more at sea-ad.org/our-science

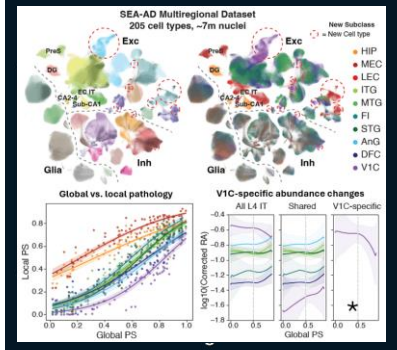
Our Science (SEA-AD)

The Seattle Alzheimer's Disease Brain Cell Atlas (SEA-AD) consortium aims to provide scientific insight on the pathogenesis of Alzheimer's disease using data generated by SEA-AD and others. Access our latest published work describing cellular and molecular changes in AD progression.

Note: Code for reproducing analyses from some of these papers is now available via the "Our Code" landing page.

Platform papers

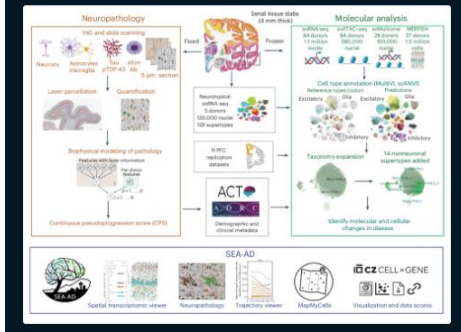
Large studies from the whole SEA-AD consortium describing major data generation efforts and associated knowledge from these data sets.



Multiregional single-cell profiling reveals shared andspecialized cellular vulnerability in Alzheimer's disease (Travaglini,Gabitto, Ding et al., bioRxiv, 2026)

This paper extends the SEA-AD atlas across ten neo- and allocortical brain regions to determine whether the cellular changes associated with Alzheimer's disease are shared across the brain. By analyzing the molecular profiles of roughly seven million cells and relating them to regional pathology, the researchers found that vulnerability is remarkably consistent across regions, with the same subsets of inhibitory neurons, oligodendrocytes, and glia changing in a stereotyped sequence that begins before symptoms emerge. The study also uncovered unexpected vulnerability in specialized cell populations, including primary visual cortex layer 4 excitatory neurons previously thought to be resilient, and identified neuronal hyperexcitability as a potential common mechanism linking both early- and late-vulnerable cell types.

READ THE PAPER | Download the data



Integrated multimodal cell atlas of Alzheimer's disease (Gabitto, Travaglini, et al, Nature Neuroscience, 2024)

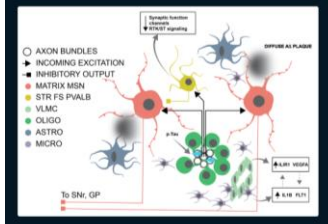
This paper describes the creation of a massive molecular and cellular atlas of middle temporal gyrus to better frame and understand how Alzheimer's disease progresses. By analyzing the genetic activity of millions of individual cells, the researchers discovered that the disease actually unfolds in two distinct phases: an early, slow-moving stage where specific Sst+ interneurons are lost and inflammation begins, followed by a later stage of accelerated neuropathology and loss of excitatory neurons and Pvalb+ and Vip+ inhibitory neuron subtypes. These findings were replicated in prefrontal cortex in SEA-AD and other major AD studies.

READ THE PAPER | Download the data

Explore interactive resources

Focused studies

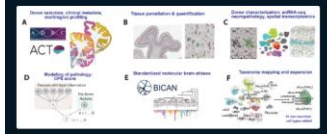
Other papers with SEA-AD researchers as primary authors focused on SEA-AD data sets or novel computational algorithms.



The Caudate Nucleus Exhibits Distinct Pathology and Cell Type-Specific Responses Across Alzheimer's Disease (Kana et al, bioRxiv, 2026)

This paper represents the characterization of the Alzheimer's pathological spectrum within the caudate nucleus. Utilizing digital pathology, researchers were able to confirm a less pronounced phosphorylated tau and amyloid-β burden when compared to other cortical areas. Phosphorylated tau and amyloid-β were also shown to localize separately into white and gray matter, respectively. Both, in part, explain the lack of neuronal cell death observed in the caudate even in high burden parts of the spectrum. However, researchers were able to identify glimmers of a global phosphorylated tau response in microglia.

READ THE PAPER | Download the data

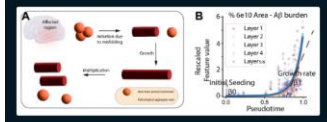


SEA-AD is a multimodal cellular atlas and resource for Alzheimer's disease (Hawrylycz et al, Nature Aging, 2024)

This commentary on Gabitto, Travaglini, et al 2024 presents SEA-AD as a multifaceted open community resource designed to identify cellular and molecular pathologies that underlie Alzheimer's disease. Integrating neuropathology, single cell and spatial genomics, and longitudinal clinical metadata, SEA-AD is a unique resource for studying the pathogenesis of Alzheimer's and related dementias.

READ THE PAPER | Download the data

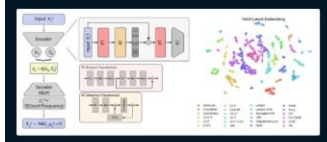
Explore interactive resources



B-BIND: Biophysical Bayesian Inference For Neurodegenerative Dynamics (Agrawal et al, bioRxiv, 2024, accepted at the Annals of Applied Statistics)

This paper introduces B-BIND, a new mathematical framework designed to track how Alzheimer's disease progresses by analyzing the buildup of various pathological proteins. Because researchers often only have "snapshots" of the brain from different individuals, this model uses a "pseudotime" scale to rank patients along a continuous timeline of disease severity. Ultimately, this framework lays the groundwork for identifying the hidden biological stages of neurodegeneration and the specific cellular changes that drive the disease over time.

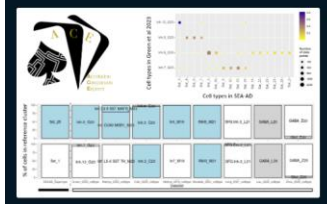
READ THE PAPER



TabVI: Leveraging Lightweight Transformer Architectures to Learn Biologically Meaningful Cellular Representations (Chandrashekar et al, bioRxiv, 2026)

This paper introduces TabVI, a lightweight AI model designed to analyze single-cell genomic data more effectively than standard language-based AI. By adapting "transformer" technology—the same tech behind ChatGPT—to better fit the non-sequential, hierarchical nature of genes, TabVI can more accurately identify cell types and biological patterns. This tool provides a more efficient and interpretable way for researchers to study how diseases like Alzheimer's affect individual cells across large datasets.

READ THE PAPER



Annotation Comparison Explorer (ACE): connecting brain cell types across studies of health and Alzheimer's Disease (Miller et al, bioRxiv, 2025)

This paper introduces Annotation Comparison Explorer (ACE), a web-based tool developed by the Allen Institute to address the challenge of comparing cell-type classifications across different brain studies. ACE allows researchers to map their own single-cell data to established taxonomies, such as the Seattle Alzheimer's Disease Brain Cell Atlas (SEA-AD), and compare annotations like donor demographics and disease-related changes. By applying ACE to multiple published Alzheimer's datasets, the authors identified consistent signatures of the disease, such as a decrease in specific somatostatin interneurons, demonstrating the tool's utility in unifying diverse datasets to advance the understanding of brain health and neurodegeneration.

READ THE PAPER | Explore ACE

Collaboratory studies

Projects led by other folks in the AD community, but with significant SEA-AD input or using SEA-AD data resulting in co-authorship by SEA-AD researchers.

Somatic cancer driver mutations are enriched and associated with inflammatory states in Alzheimer's disease microglia (Huang et al, bioRxiv, 2024, accepted at Cell)

Using transcriptomic datasets (including SEA-AD), this study reveals that somatic mutations—genetic changes occurring after birth—are twice as common in Alzheimer's brains than in healthy, age-matched controls. These mutations are concentrated in the brain's immune cells (microglia) and often affect "cancer driver" genes, causing the cells to multiply and trigger inflammation. This suggests that Alzheimer's may be driven by a tumor-like process where mutated immune cells expand throughout the brain and accelerate neurodegeneration.

Read the paper | Use RNA-MosiacHunter

Single Cell Landscape of Sex-specific Drivers of Alzheimer's Disease (Wu et al, Alzheimer's & Dementia, 2025)

This study analyzed single nucleus transcriptomics data (including from SEA-AD) to understand how Alzheimer's disease affects men and women differently at a genetic level. Researchers found female-specific gene associations providing "protective" effects in neurons of dorsolateral prefrontal cortex, while specific inflammatory signals in microglia may increase risk for protein buildup in women. These findings highlight that the molecular path to cognitive decline is sex-specific, pointing toward new, targeted genes for future treatment research.

Read the paper | Access the SEA-AD data | Access the ROS/Map data

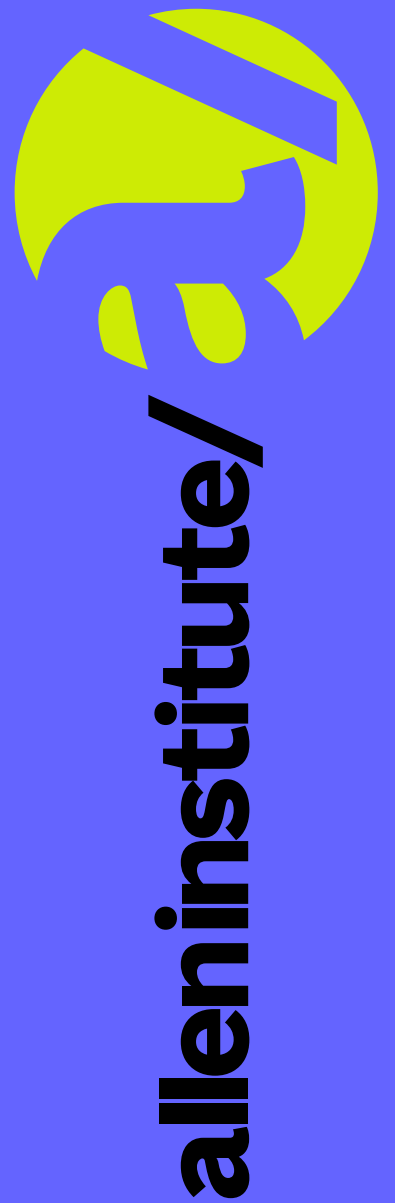
CRISPR screens in iPSC-derived neurons reveal principles of tau proteostasis (Samelson et al, bioRxiv, 2025)

This study utilizes genome-wide CRISPRi screens in human iPSC-derived neurons to identify cellular factors regulating the accumulation of tau aggregates, a primary driver of Alzheimer's disease. The researchers identified the ubiquitin ligase CUL5-SOCS4, alongside pathways such as UFMylation and mitochondrial function, as critical modifiers of tau stability and potential therapeutic targets. Furthermore, the team leveraged SEA-AD data to validate these findings, demonstrating that these tau-regulating genes exhibit altered expression within the specific cell types most vulnerable to Alzheimer's in human donors.

Read the paper | Explore CRISPR resources

Conclusions and resources/

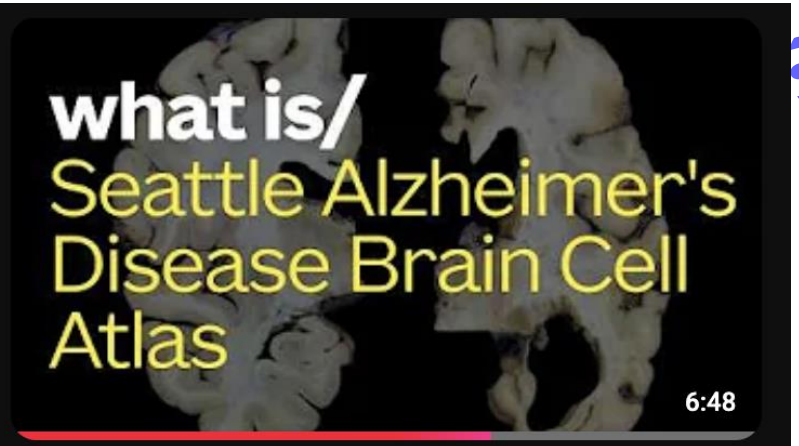
Lauren Alfiler



YouTube Channel

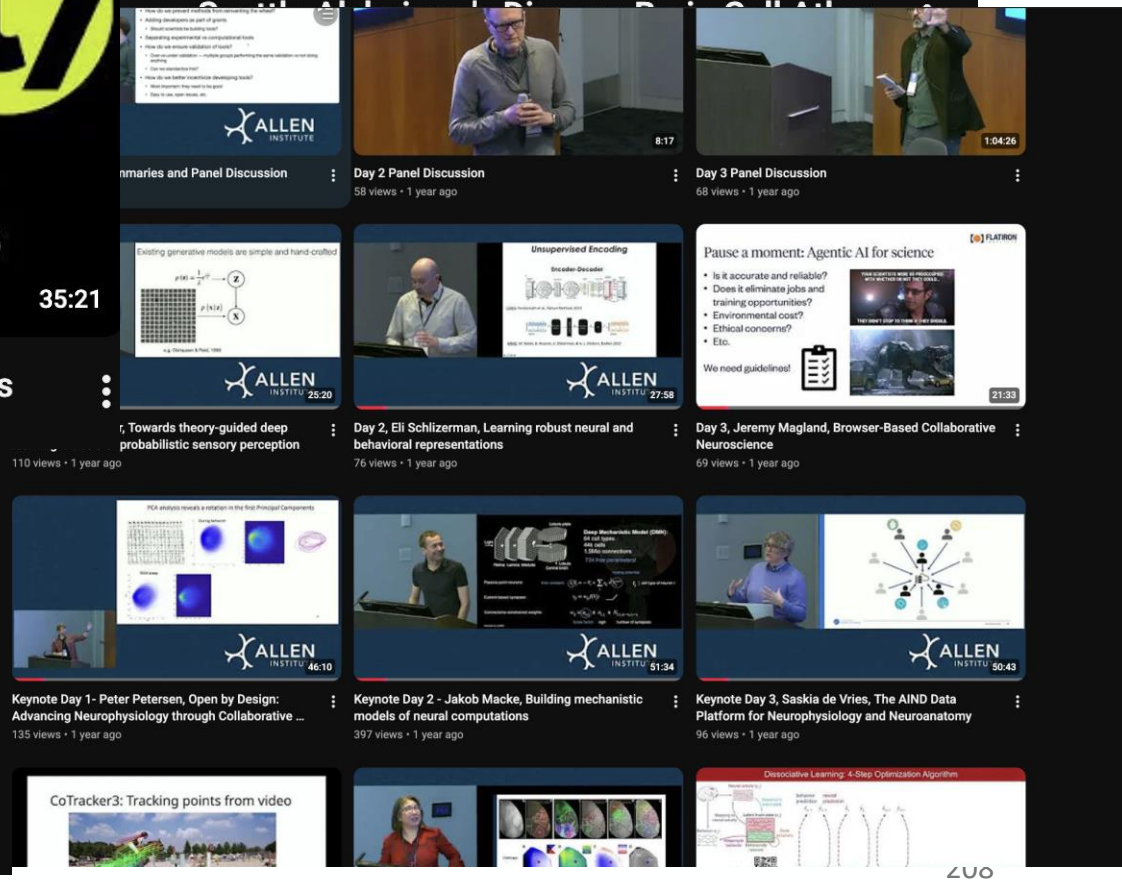
@AllenInstitute

2026 cell types workshop/ Genetic Tools



2026 cell types workshop/ cross-species analysis

2026 cell types workshop/ ABC Atlas Access
54 views • 2 months ago



Our YouTube channel has plenty of seminars, webinars, office hours, recorded videos.

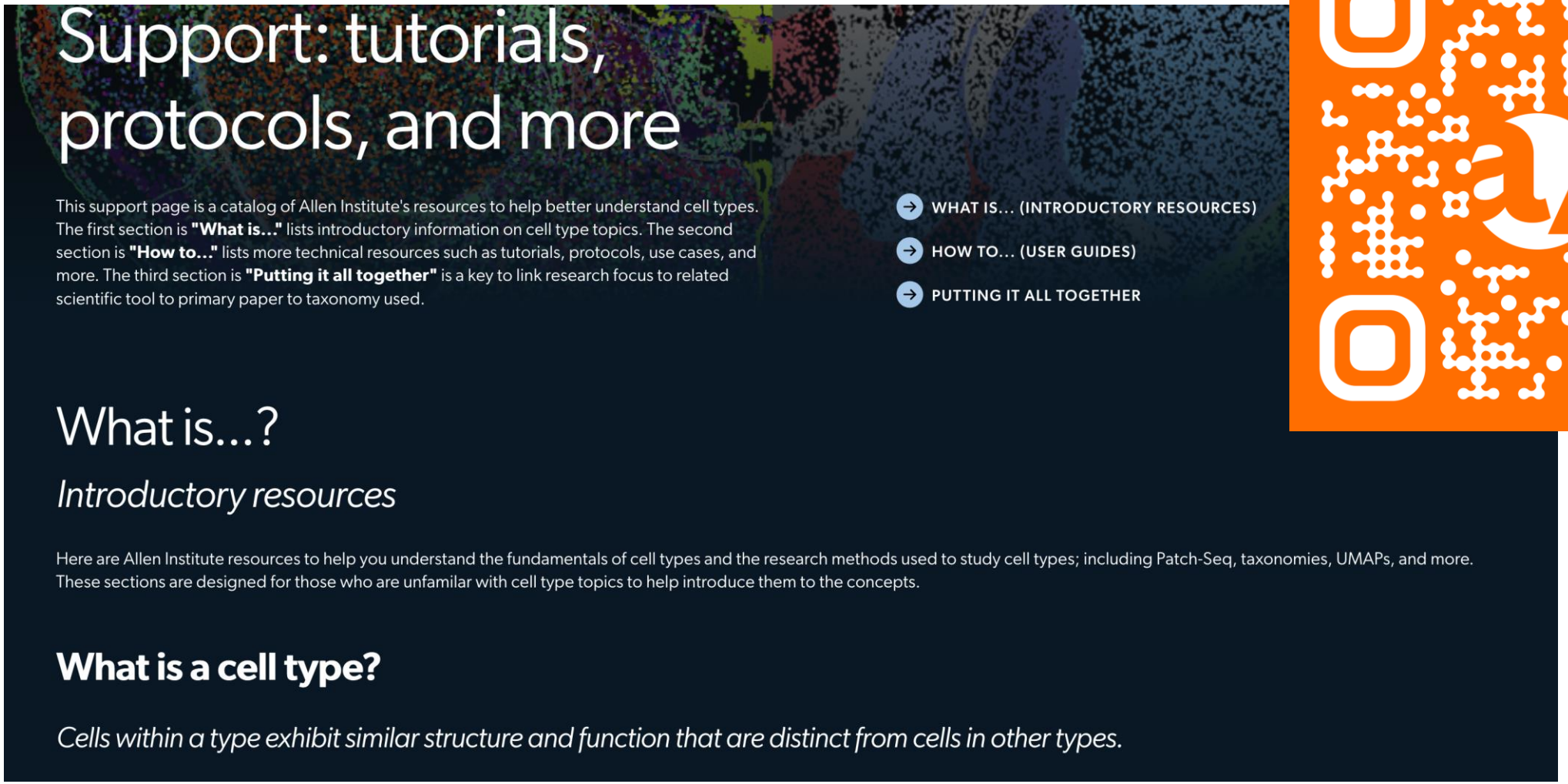
Protocols & GitHub

Allen Institute shares a lot of our protocols on **protocols.io** and a lot code on our **GitHub** repos. Check them out!

protocols.io/workspaces/allen-institute-for-brain-science/publications

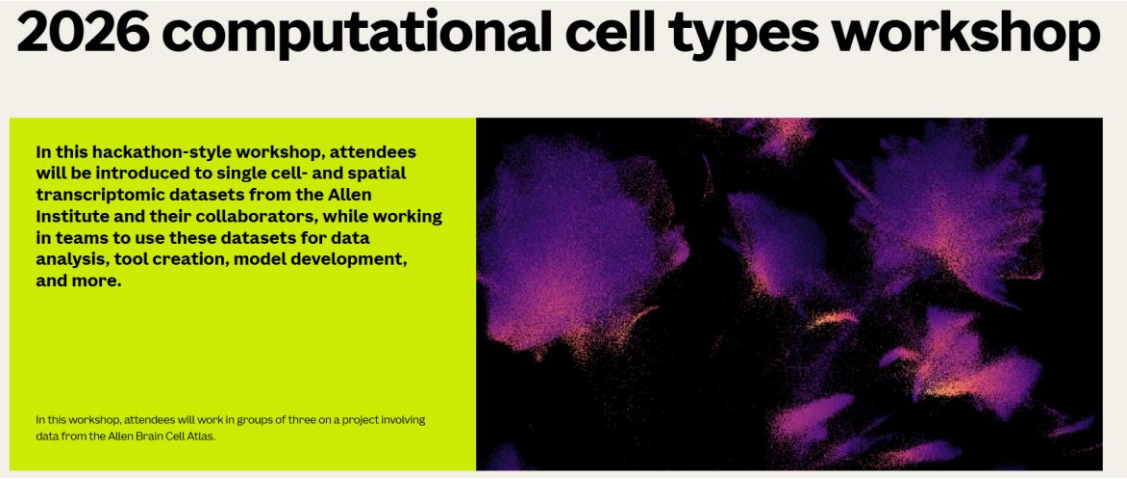
github.com/alleninstitute

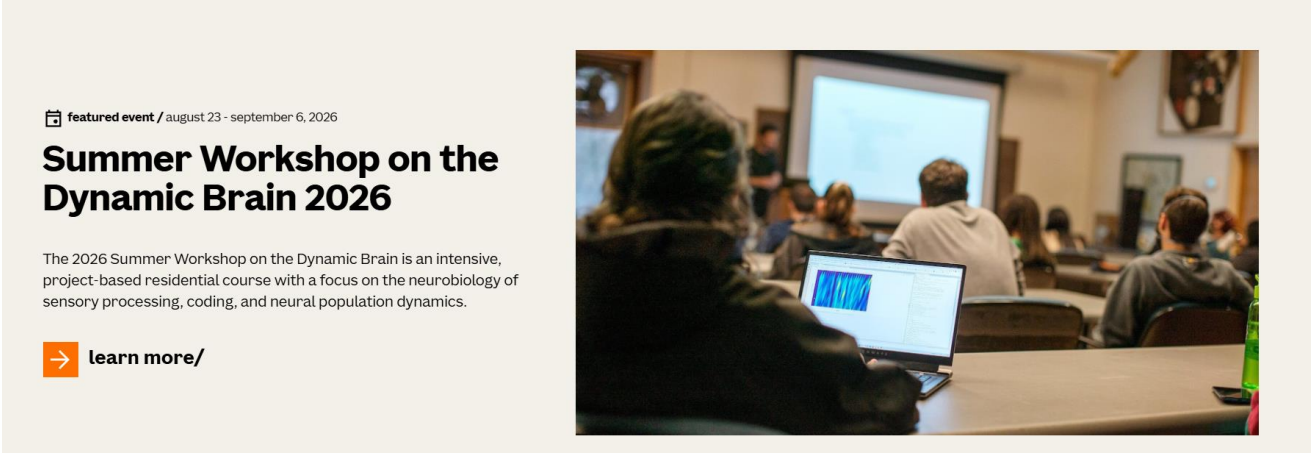
Cell Types 101 Support



Training for Early Career Scientists

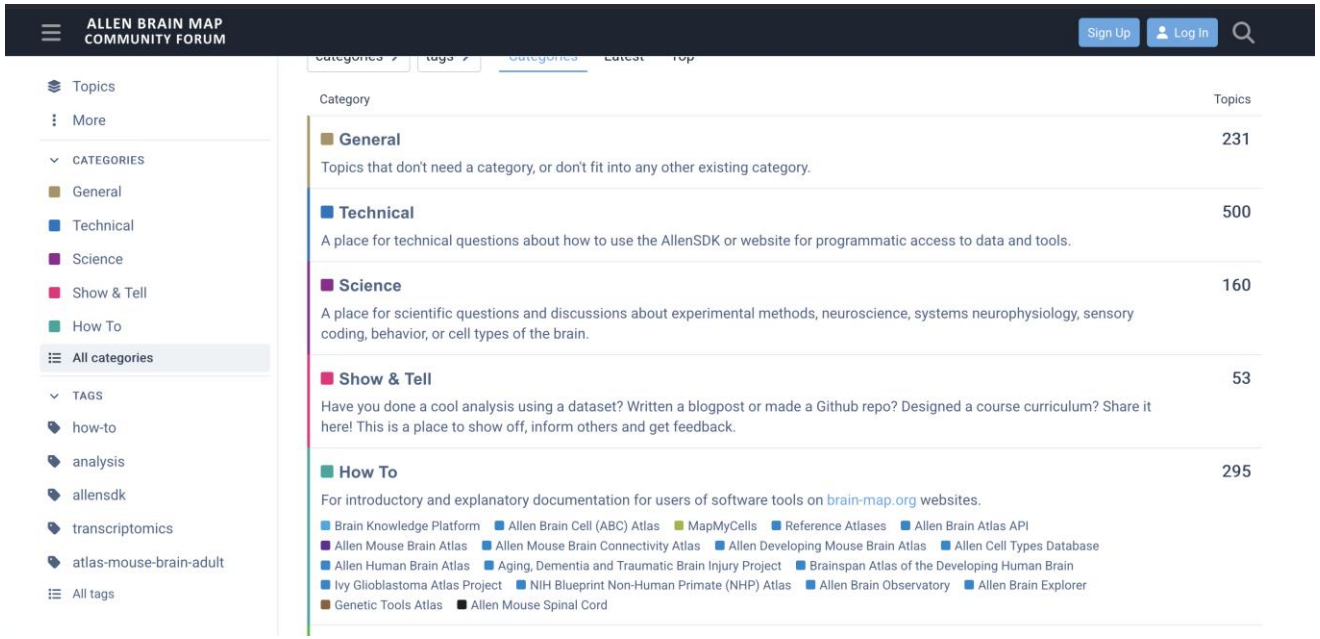
- We host multiple workshops focused on select datasets & tools to help early career-scientists learn how to use the data in their work – see the Allen Institute in person & meet the teams behind the work
- Workshops are by application to make sure it is good fit
- Workshops & applications never cost – some have travel support funding
- Applications tend to open near or after SfN!





Community Forum

- Have technical question about any of the brain science data? Post it here and one of our staff will try answer it! (Tons of previous posts as well to check)



LinkedIn & Newsletter

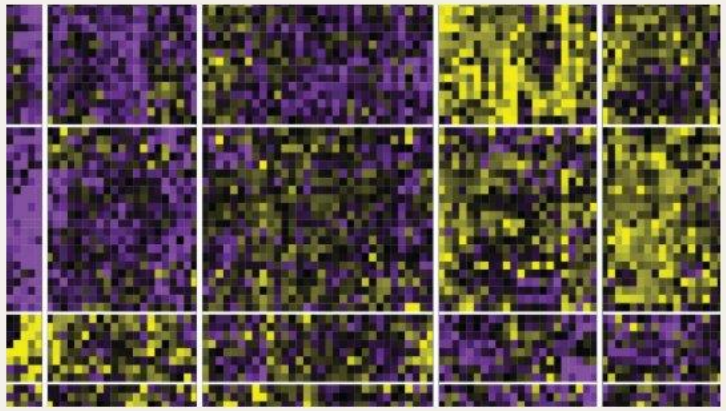
Linkedin: @AllenInstitute

Sign up for our Newsletter

Educator Development

- We host workshops for teaching professors to form a community and develop curriculum around various datasets & open science principles.
- Key takeaways for participants:
  - integrate open science practices and values into courses
  - produce resources that extend that integration to peers
  - develop the capacity to design new resources and workshops independently

Educator Resources



explainer

What is omics?

Explore the branch of research known as “omics“– including genomics, transcriptomics, and proteomics

database guide

Database Guide | Allen Brain Cell (ABC) Atlas

An in-depth guide on how to use the ABC Atlas to ask and evaluate research questions

interactive activity

Teaching and learning with python

Explore resources to introduce undergraduate students in the life sciences to Python programming

All freely available on:

alleninstitute.org/education/resources

Careers

- We hire straight out of graduate school – no post-doc required!
  - This is typically a Scientist I
  - The average “Scientist I” salary is about $80k-$100k USD
  - Each position will be clear if it supports Visa or not - it is position dependent
- Hiring is year-round, although sometimes more open spots during the start of the new year
- We often look for specific skill sets to fill open positions; the skill set depends on the position.

- *NOTE: We don’t have any undergraduate internships available for international students, and as of now, we do not have any PhD-level internships*

All careers will be posted on:
alleninstitute.org/careers

Shanahan Fellowship

- **3-year program for recent Ph.D.s** to explore complex data in neuroscience at the Allen Institute and University of Washington
- **Neuroscience experience not needed**—we're seeking applicants with expertise in computer science, data science, engineering, mathematics, physics, and allied fields
- **Designed to provide freedom** to explore novel computational research areas and datasets of interest, through the Allen Institute's vast **data banks**. Includes a discretionary fund for each fellow.
- **Participation in the 2-week Summer Workshop on the Dynamic Brain** course in the San Juan Islands
- **Mentorship from leading Allen Institute and UW investigators.** Fellows are hired as an Allen Institute employee with a starting salary of $106,780 annually, immigration support and benefits package

- Applications will open in mid-September

Any questions about anything you heard in the past 2 days?

Feedback Survey

This workshop is supported by NIH grant.

This is a new style of workshop to us where we come to you.

thankyou

allen institute/

August 14th afternoon agenda/

- **2:00-3:00 |** Guided exploration of data and tools – Rachel
- **3:00-3:15 |** Coffee break
- **3:15-4:00 |** Open exploration time – hands-on help from us
- **4:00-4:30 |** Group examples of data/tool use
- **4:30-5:00 |** Group discussion and Q&A